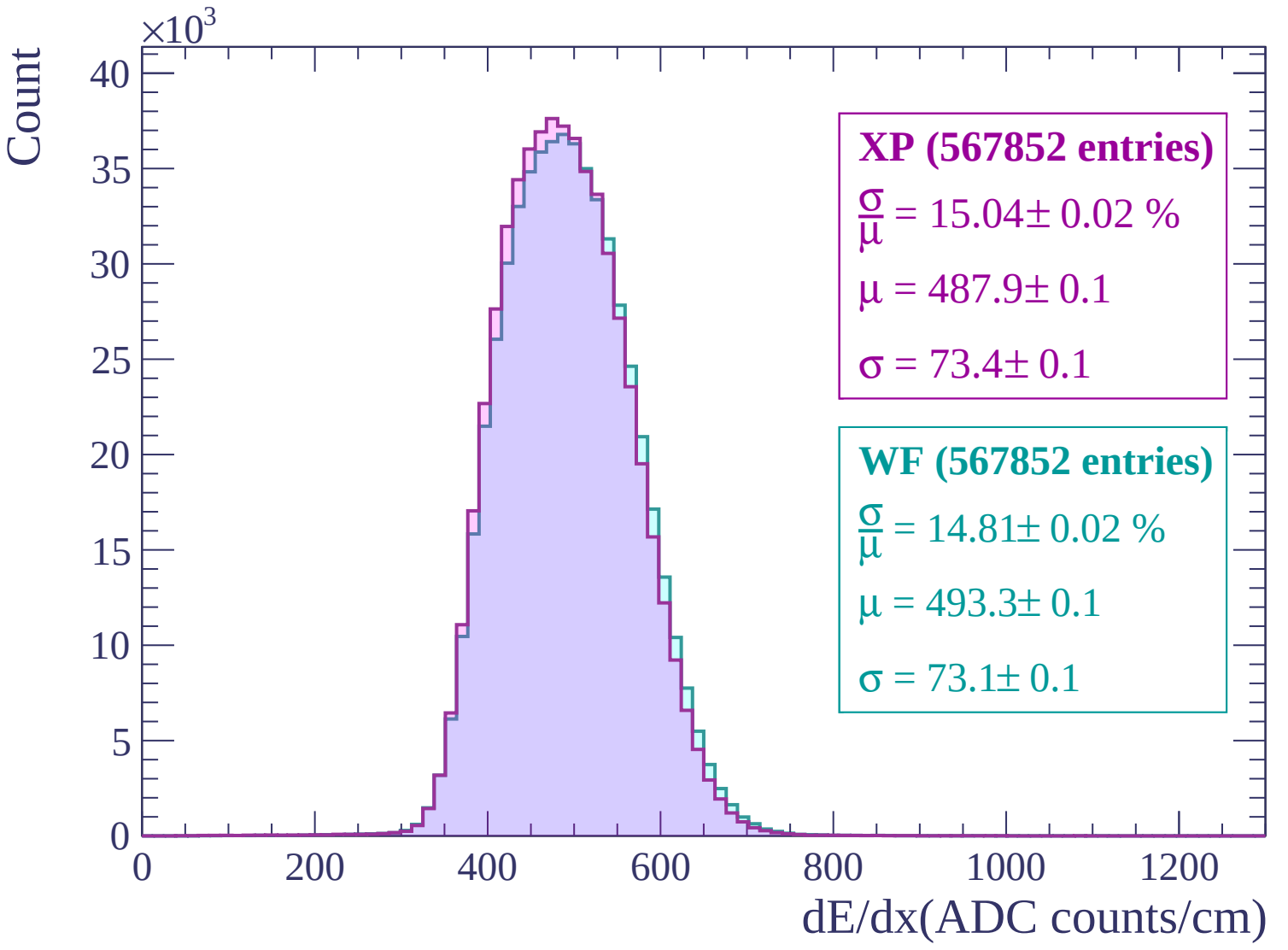
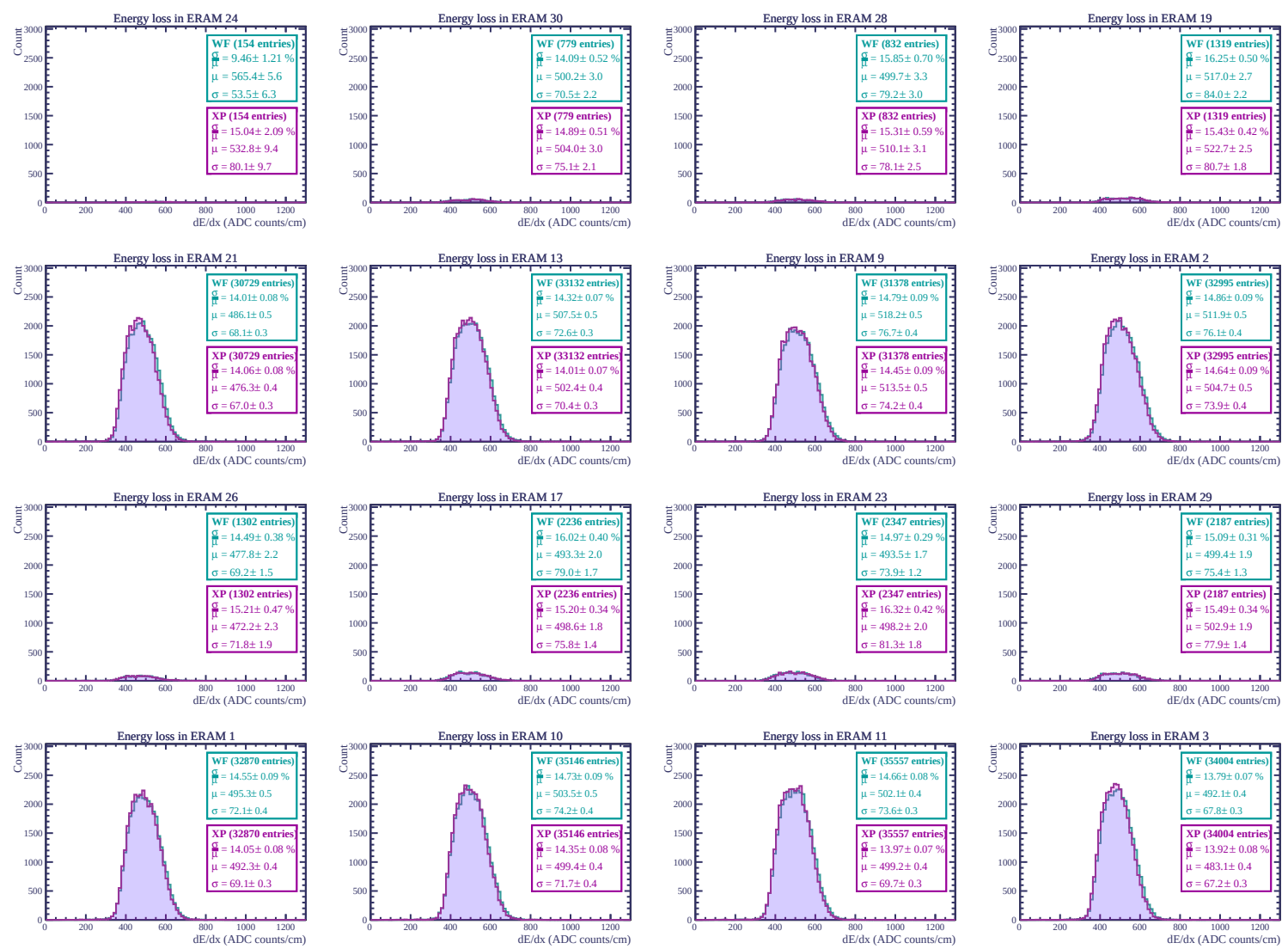
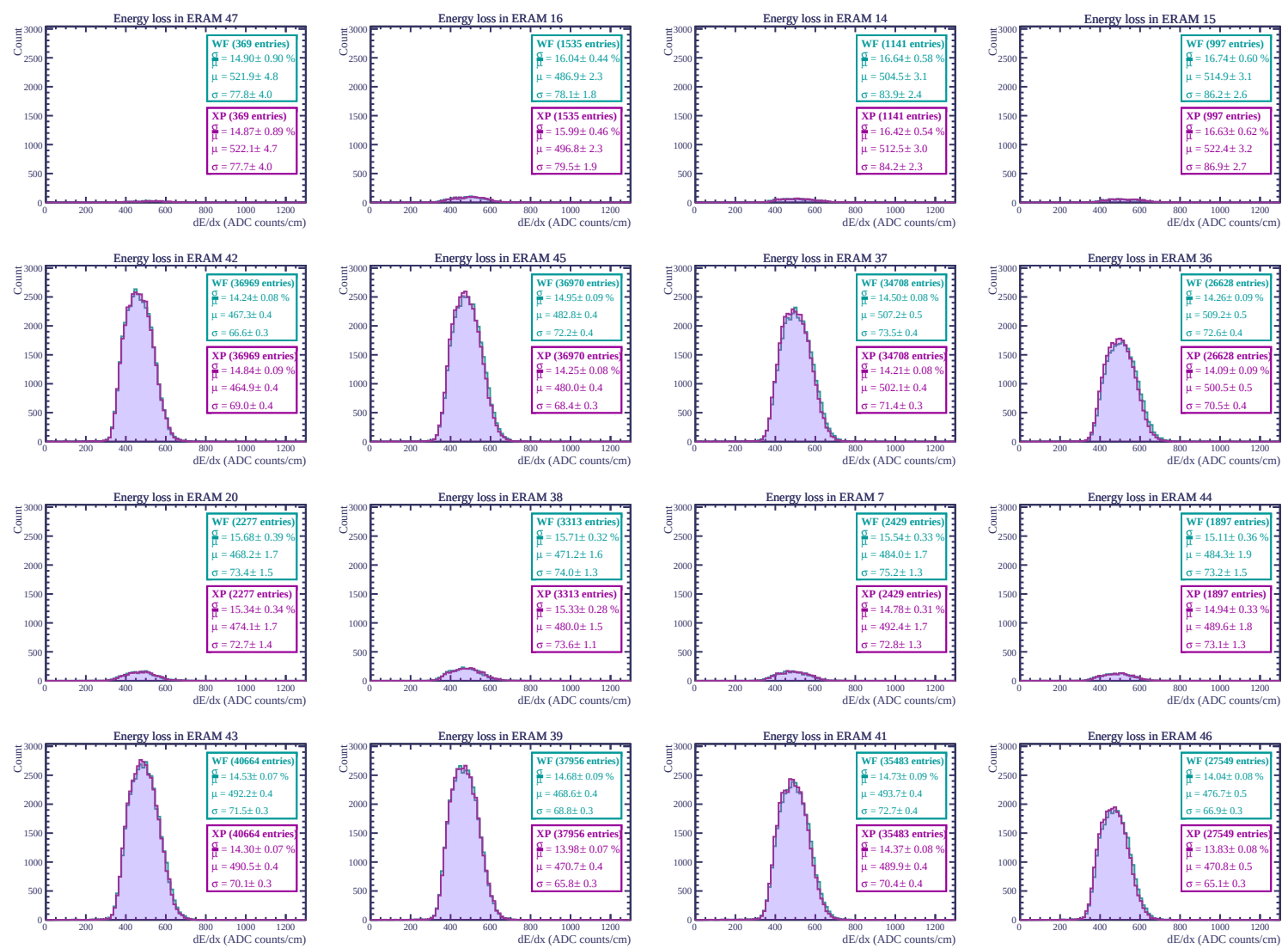
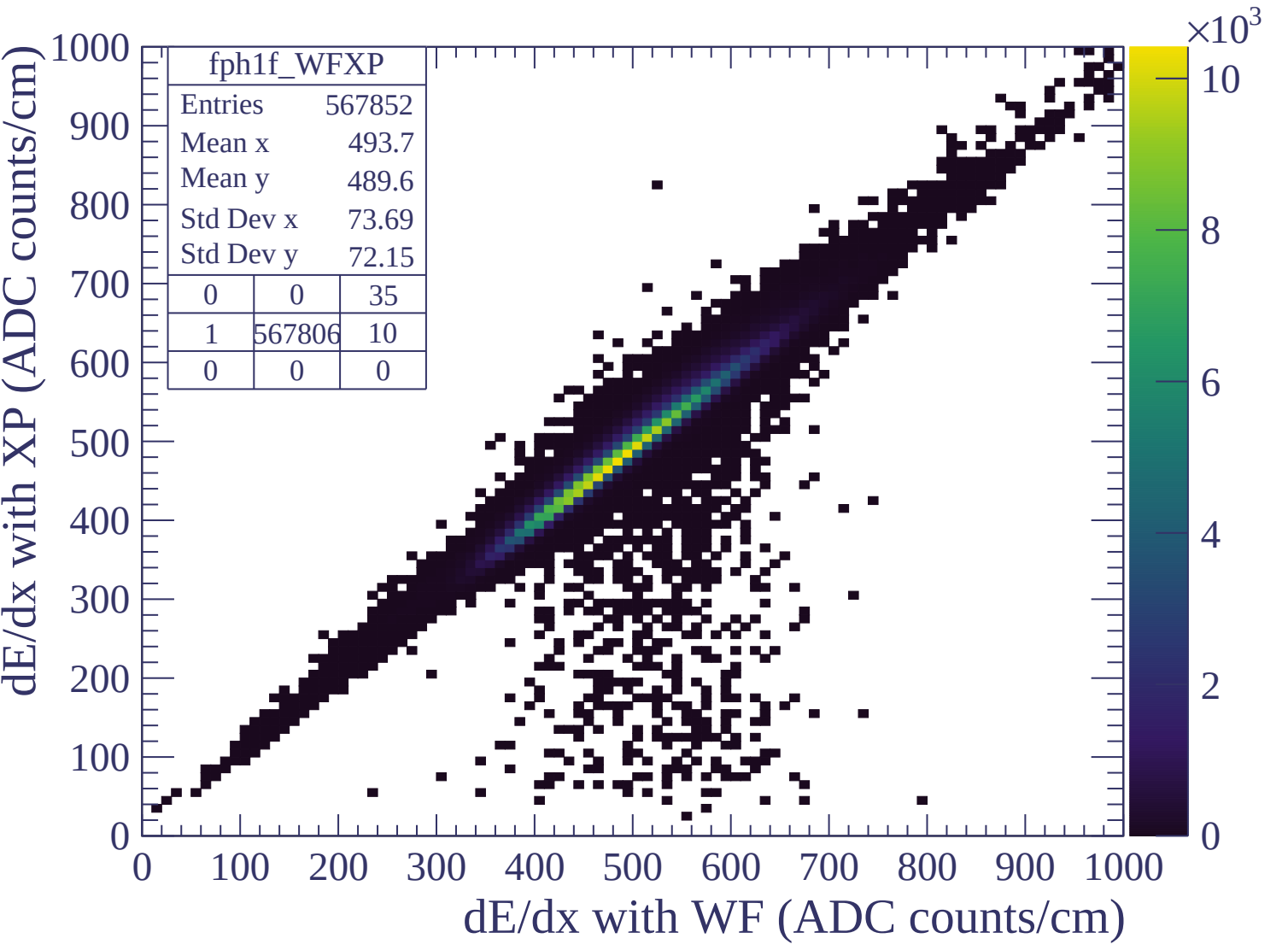
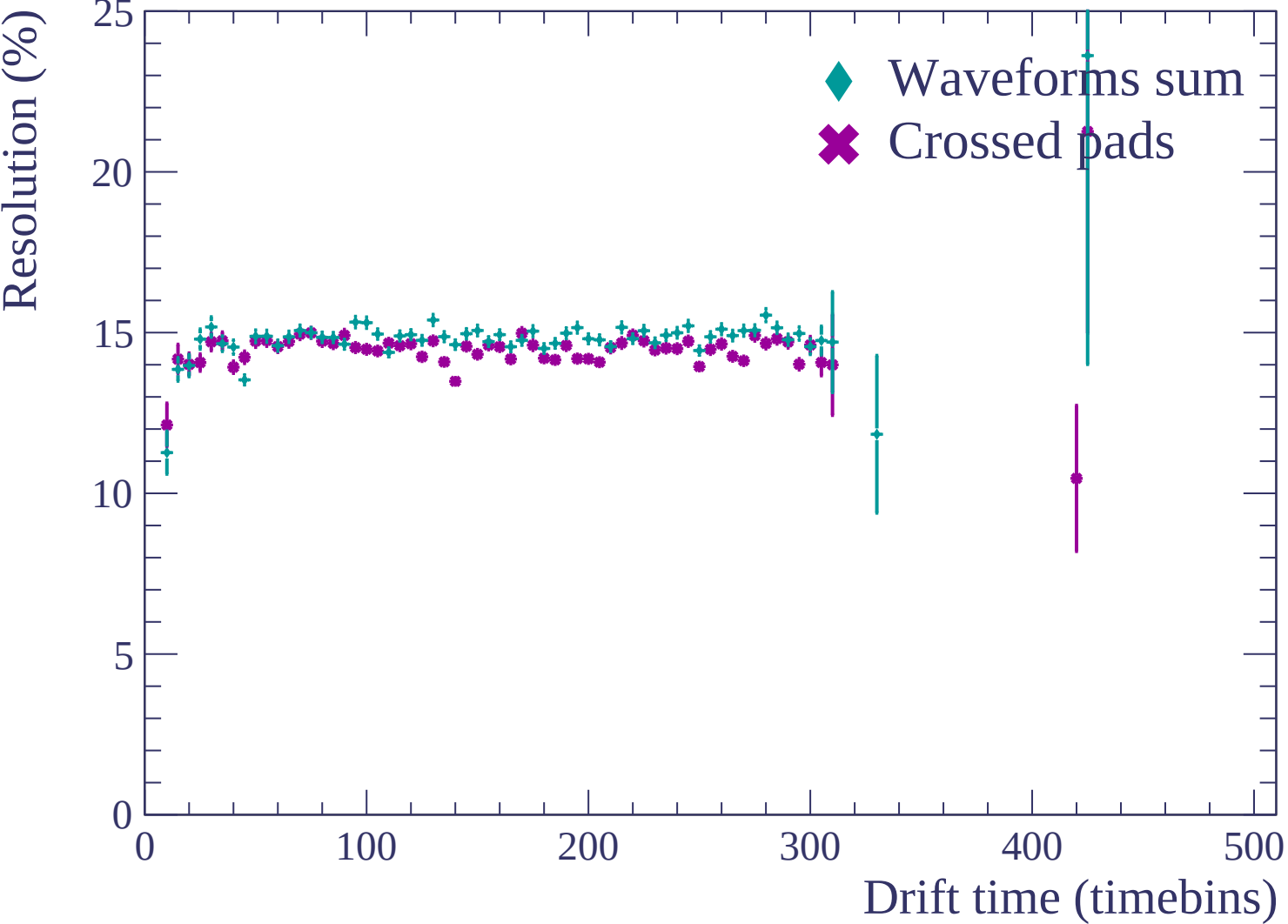


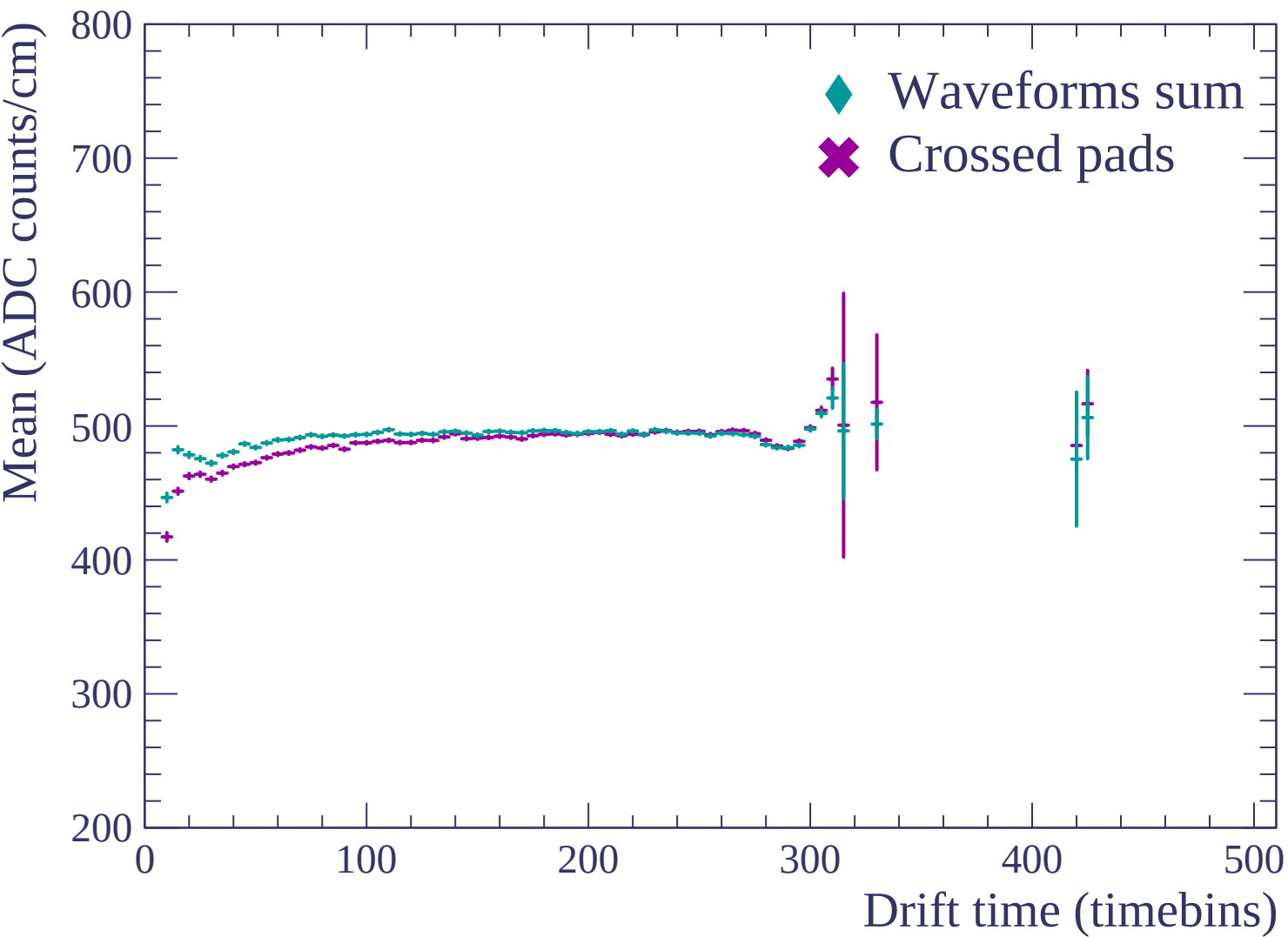


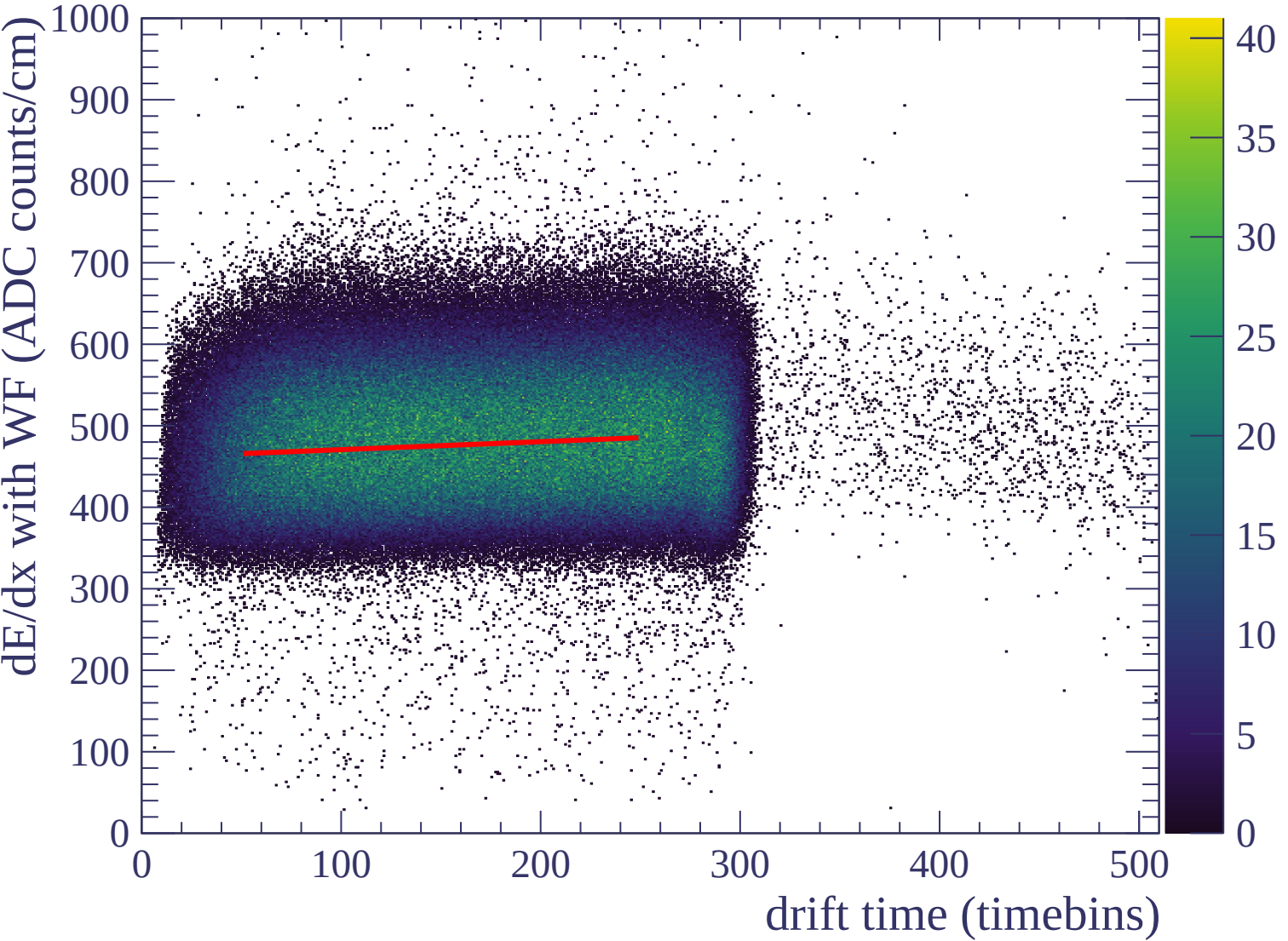


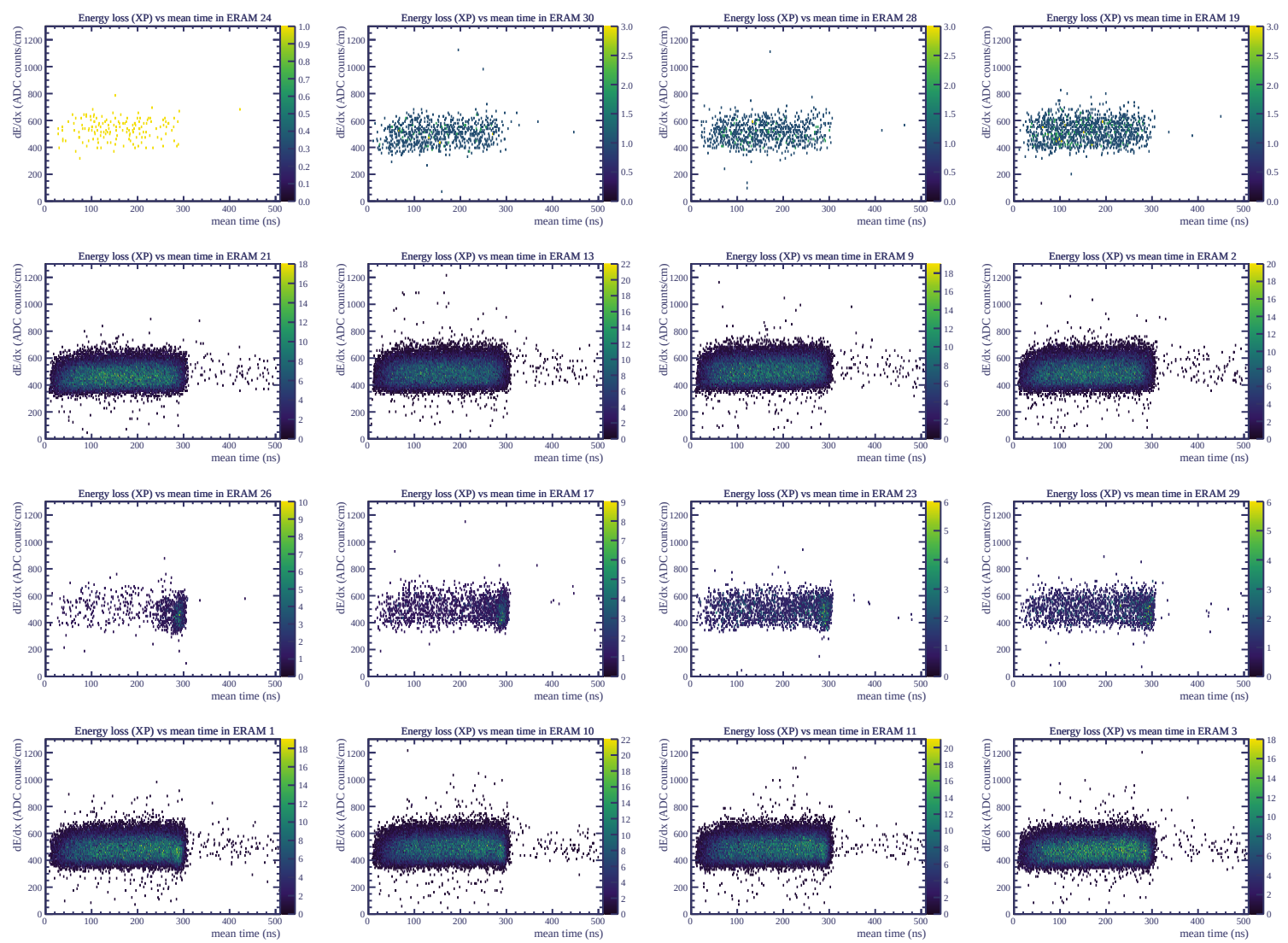


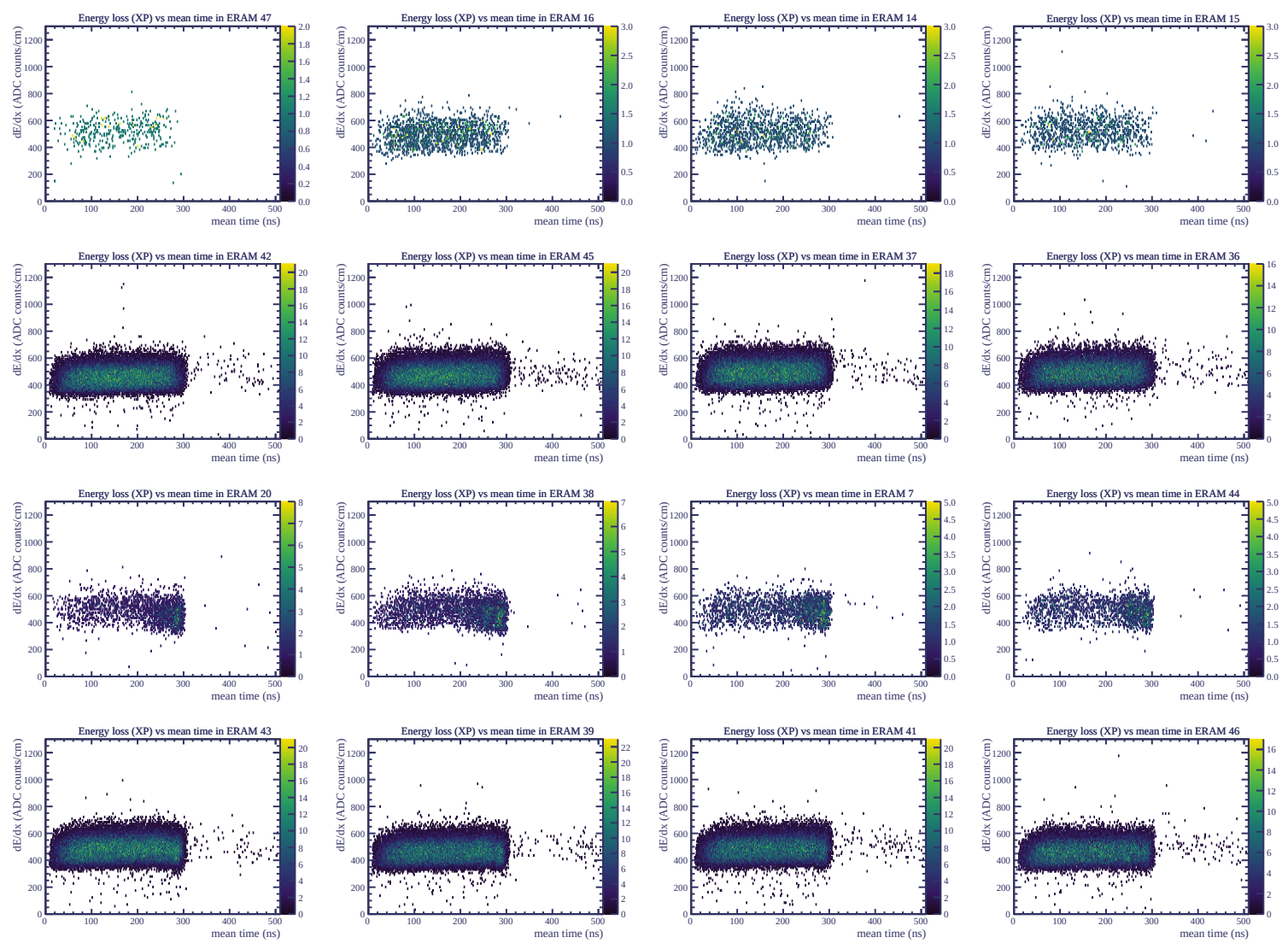


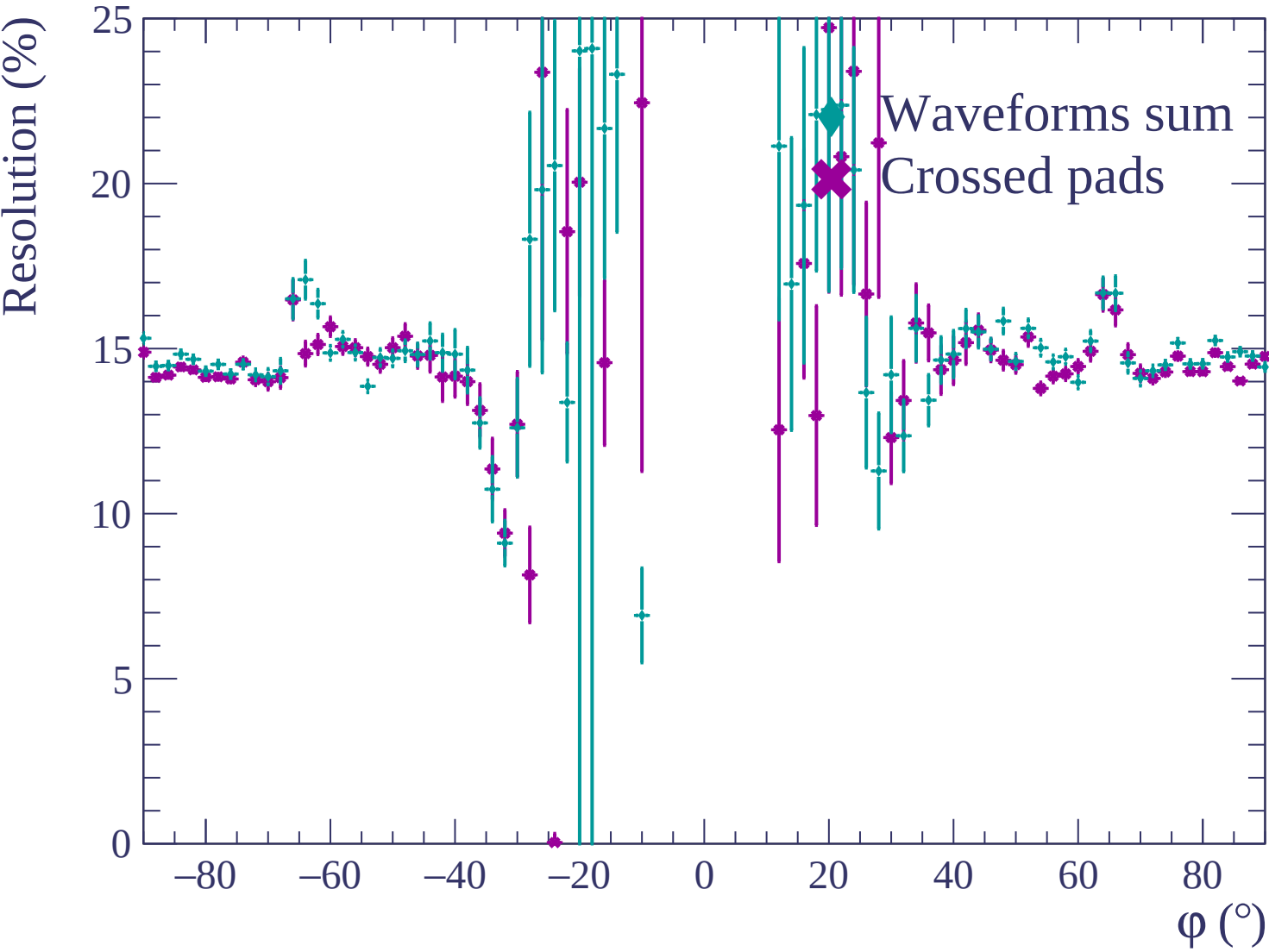


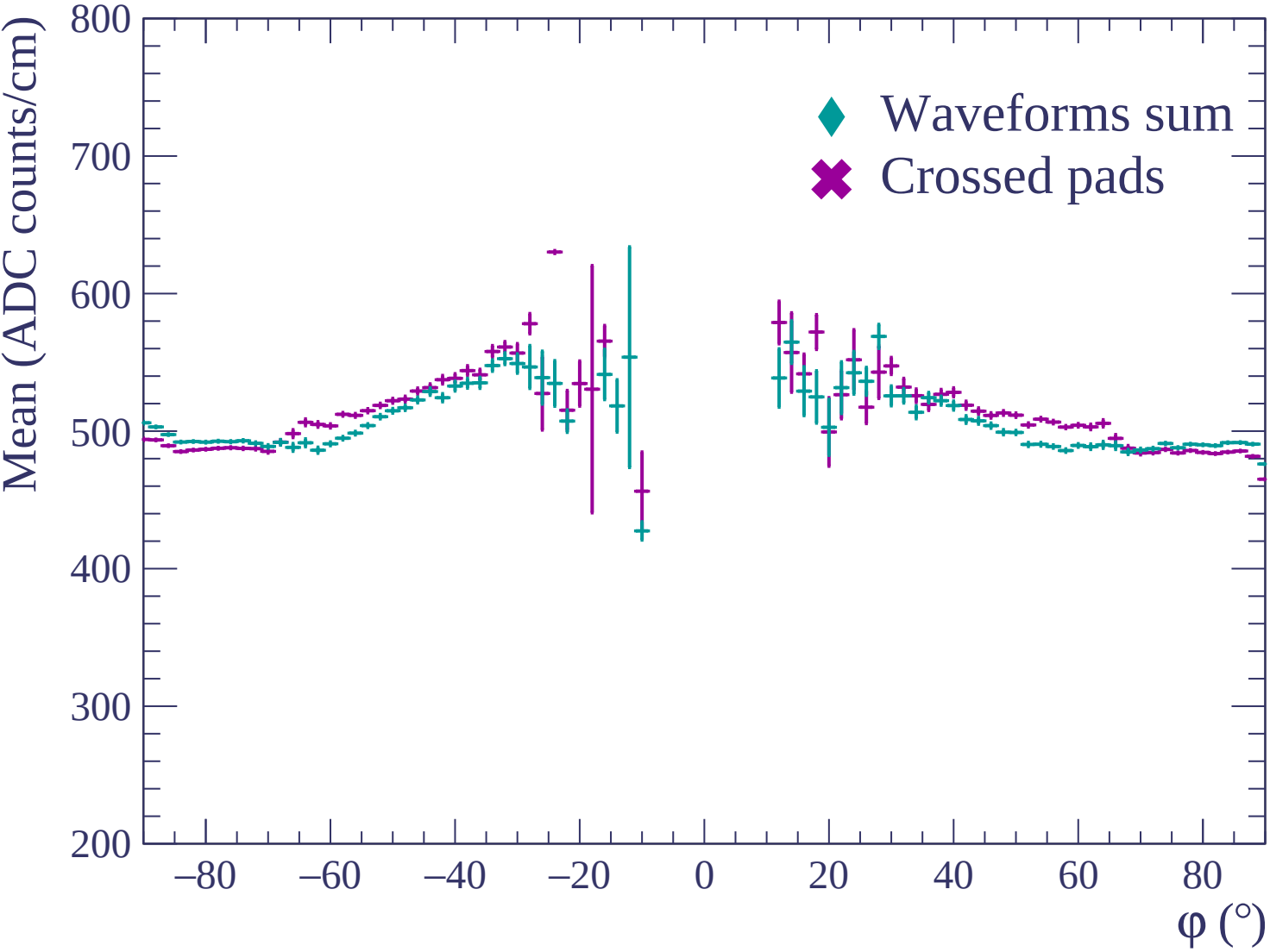


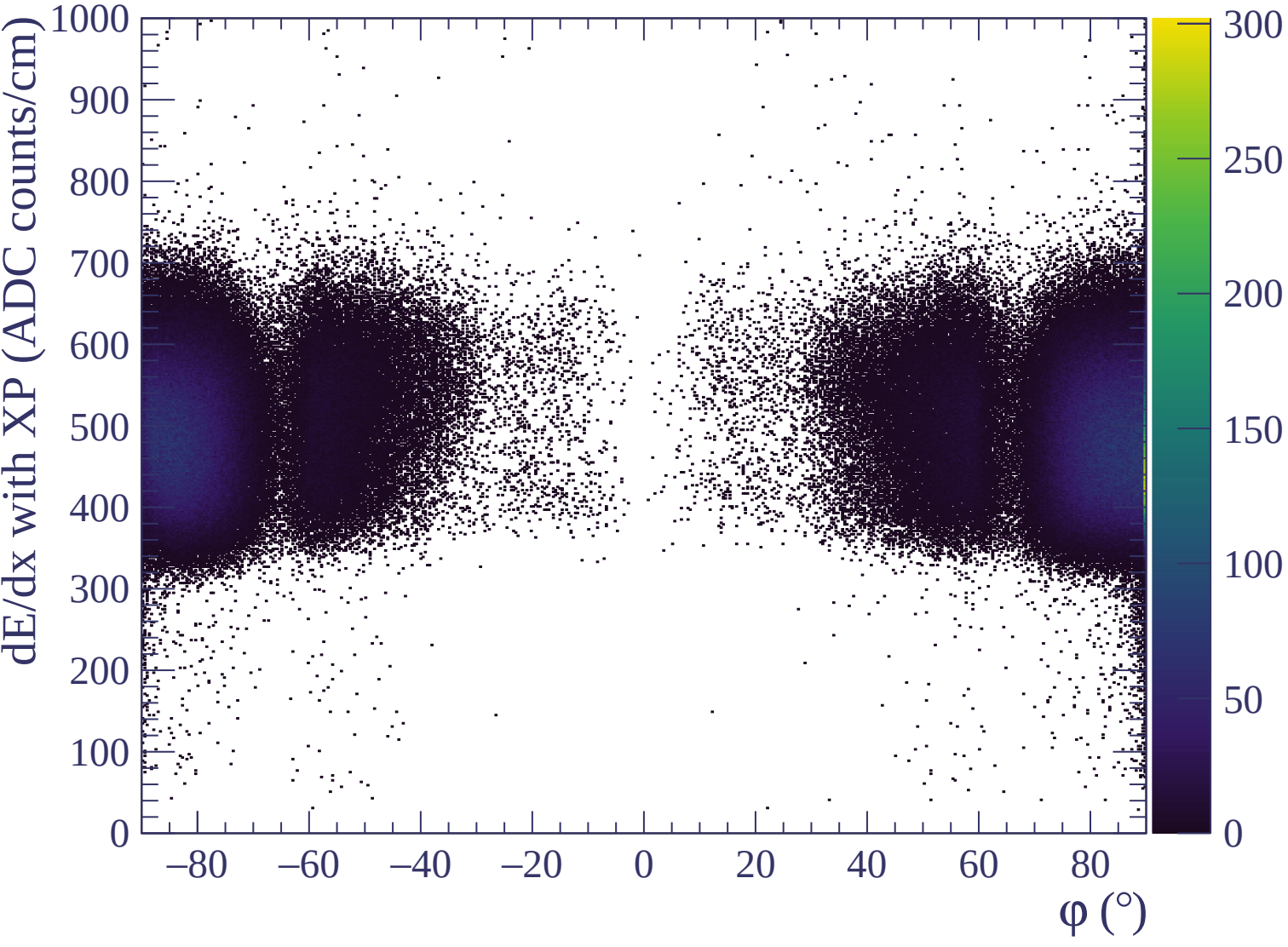


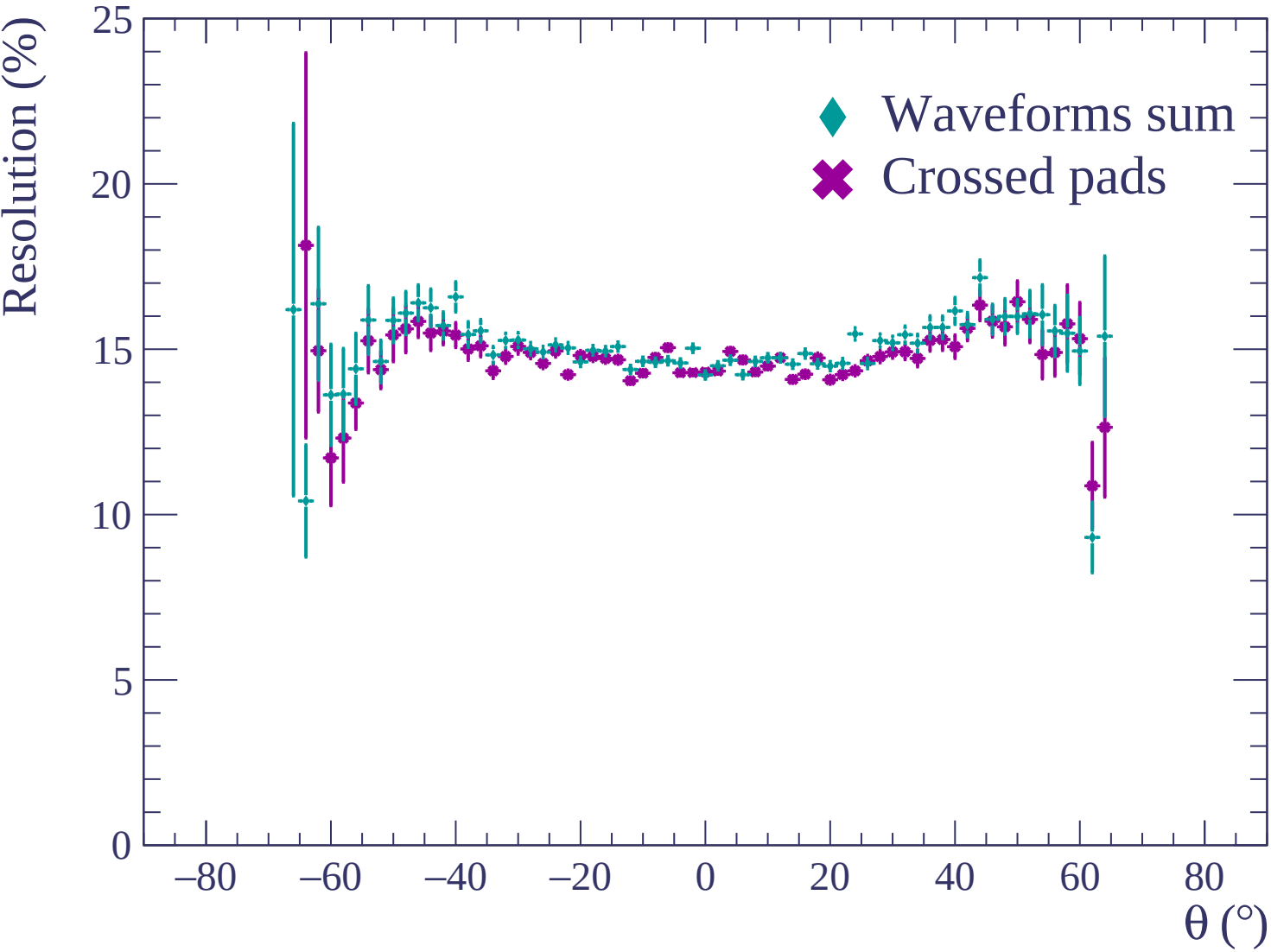


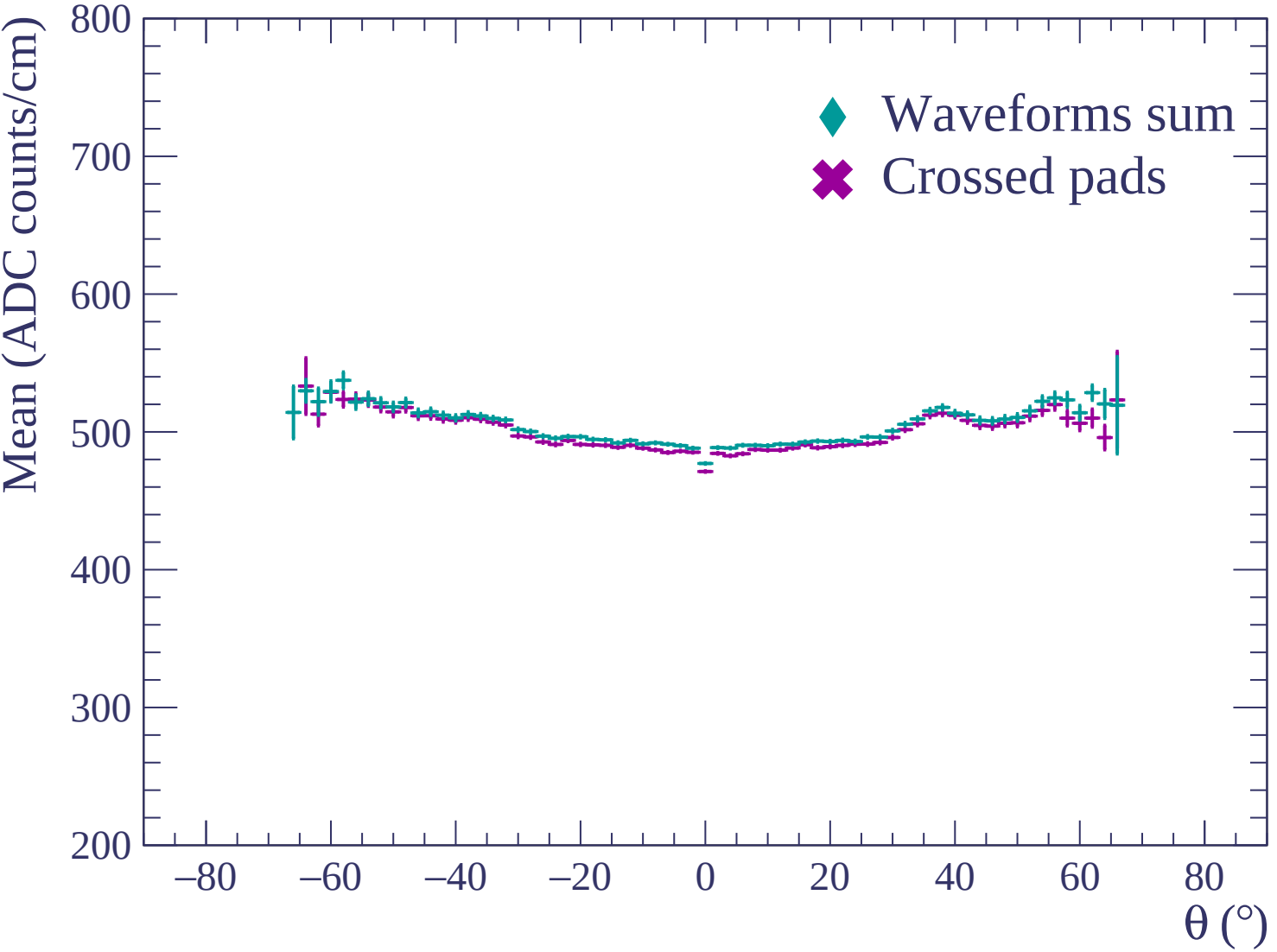


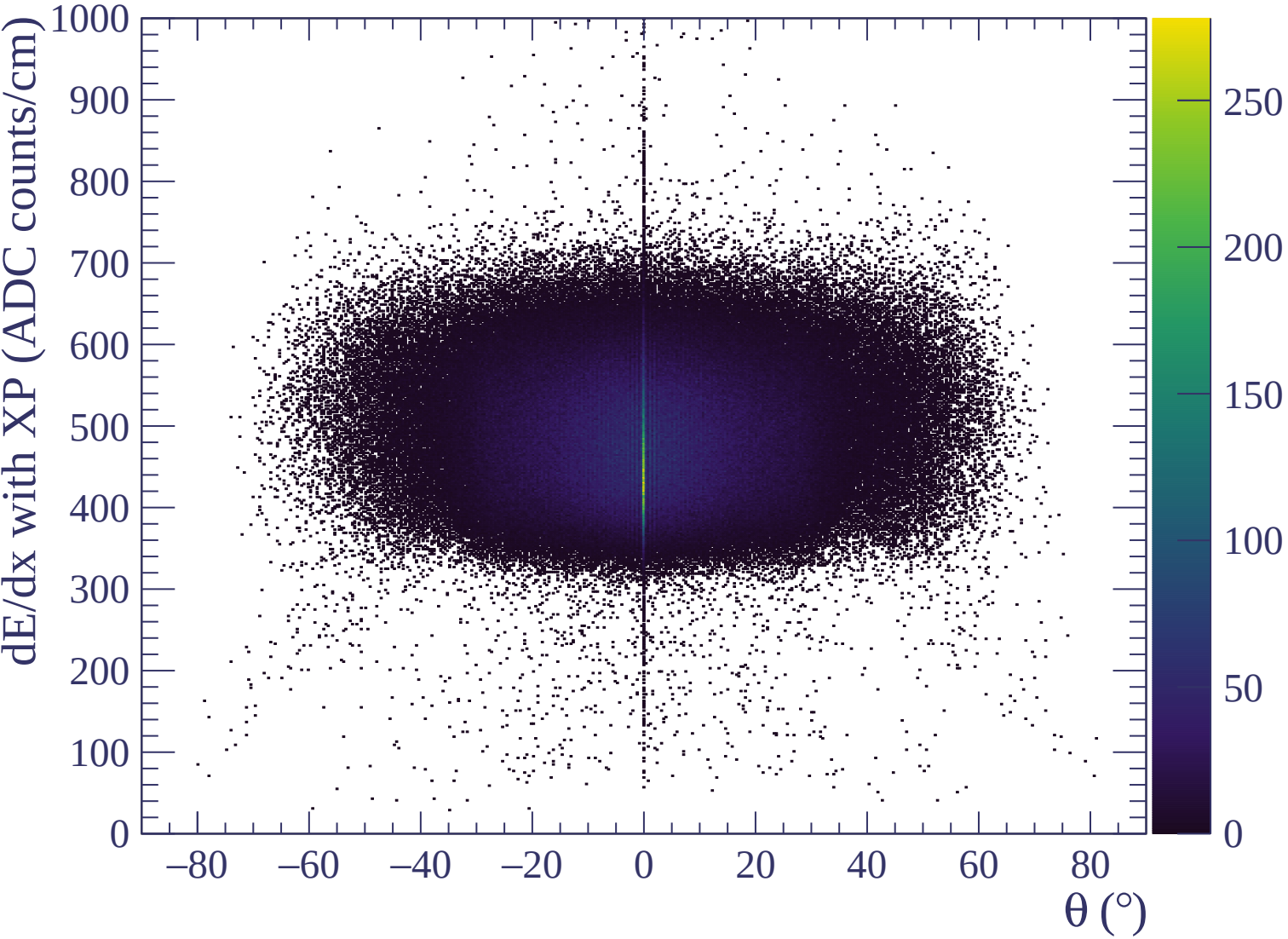


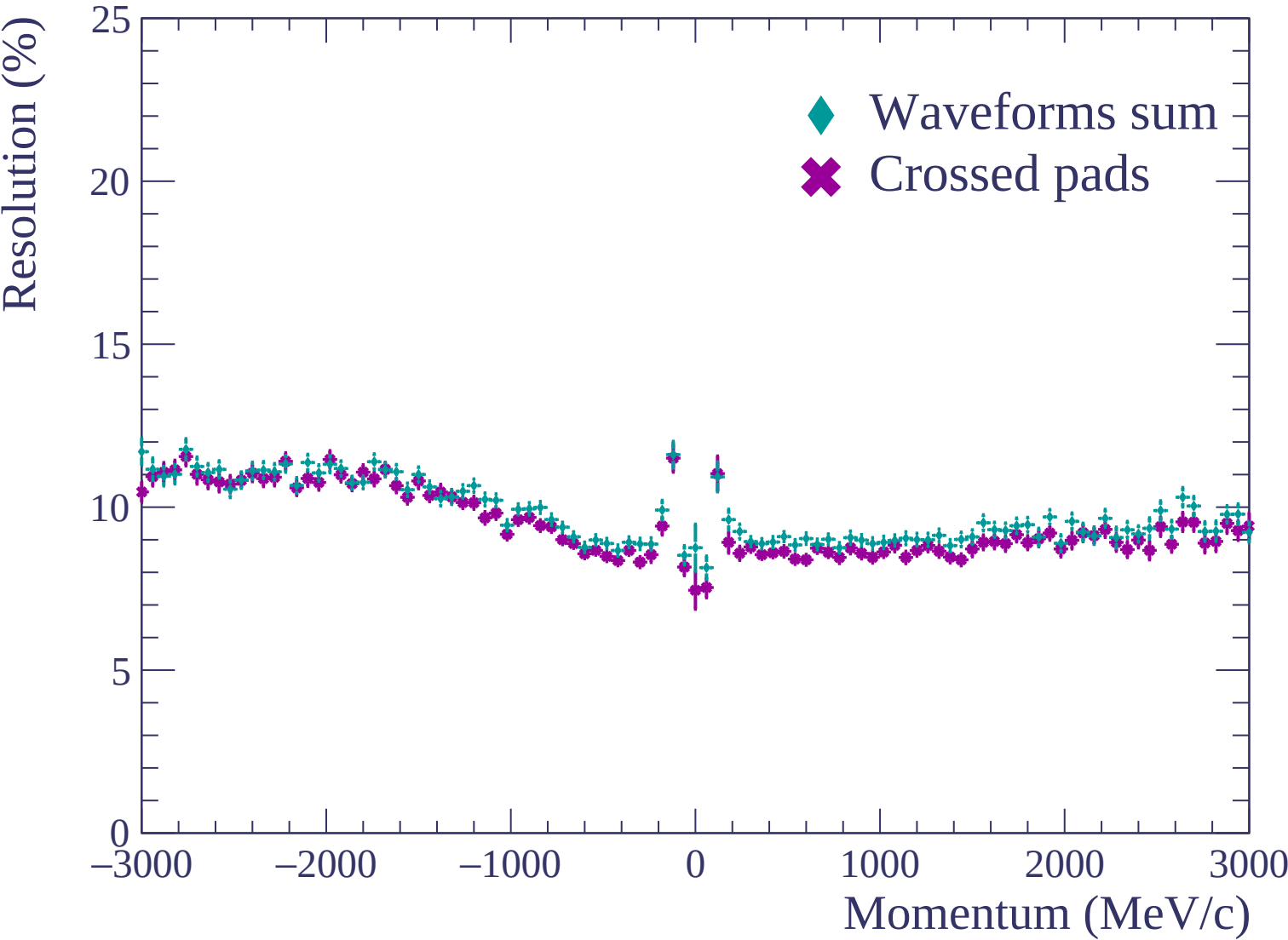


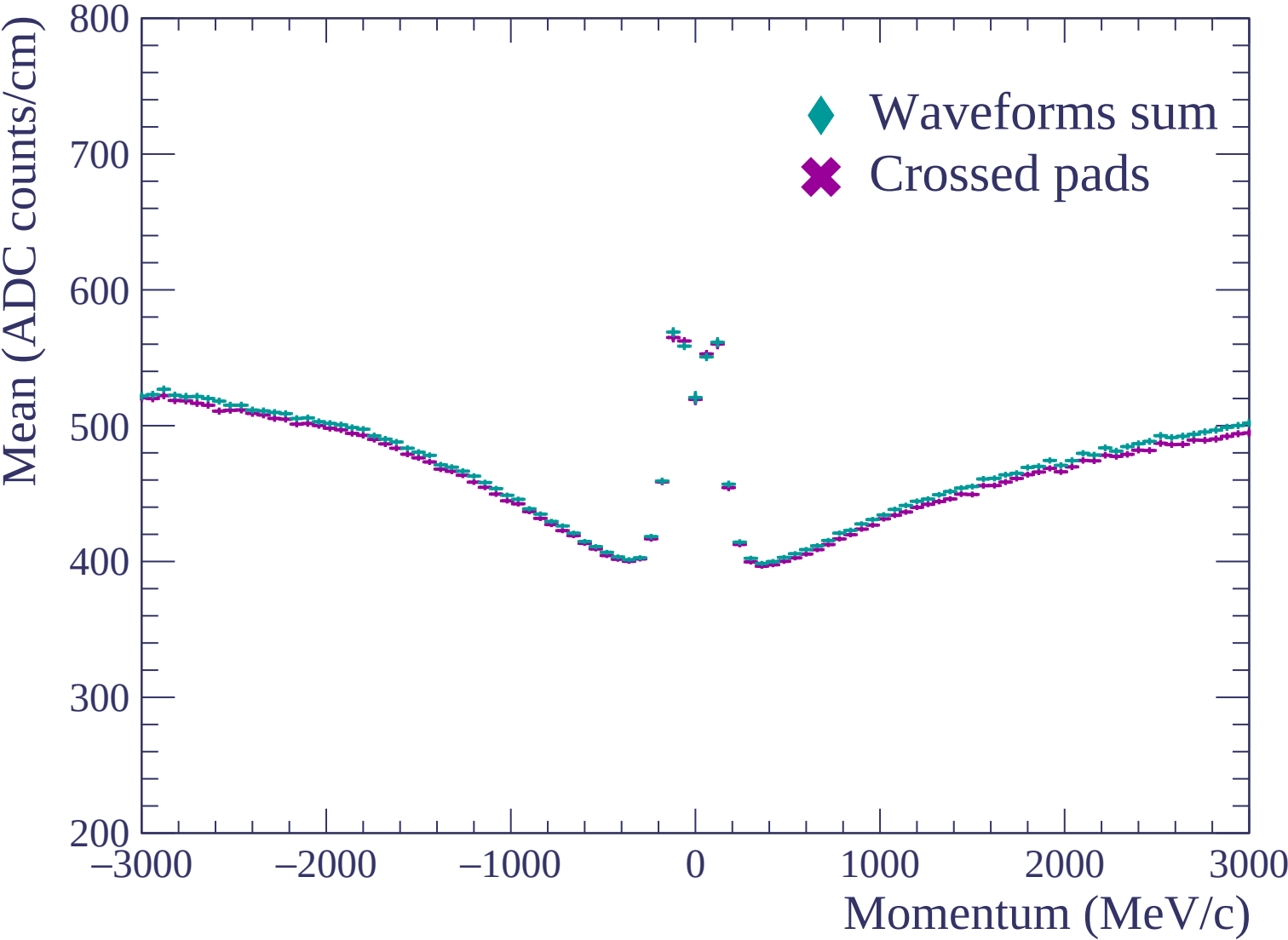


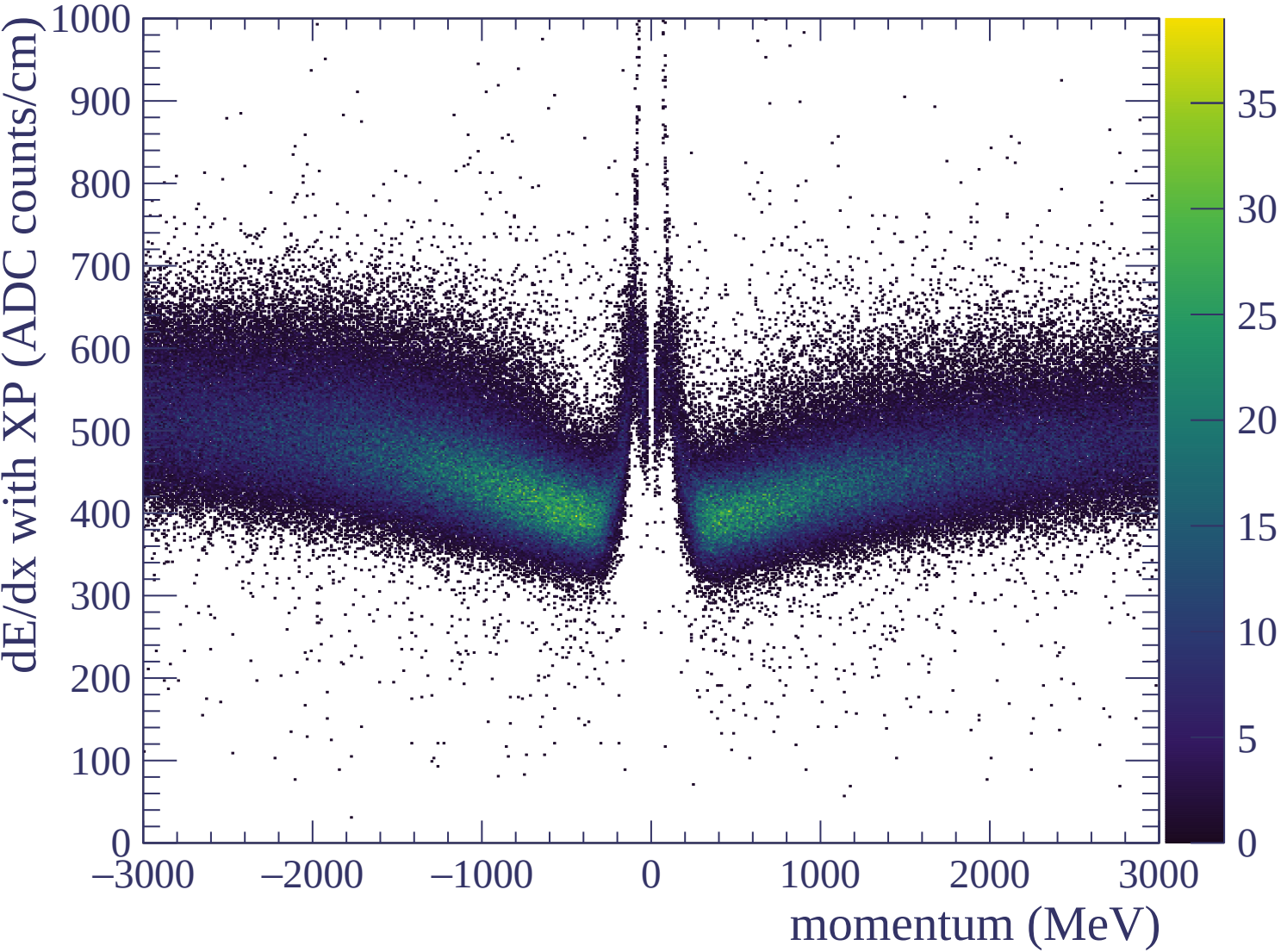


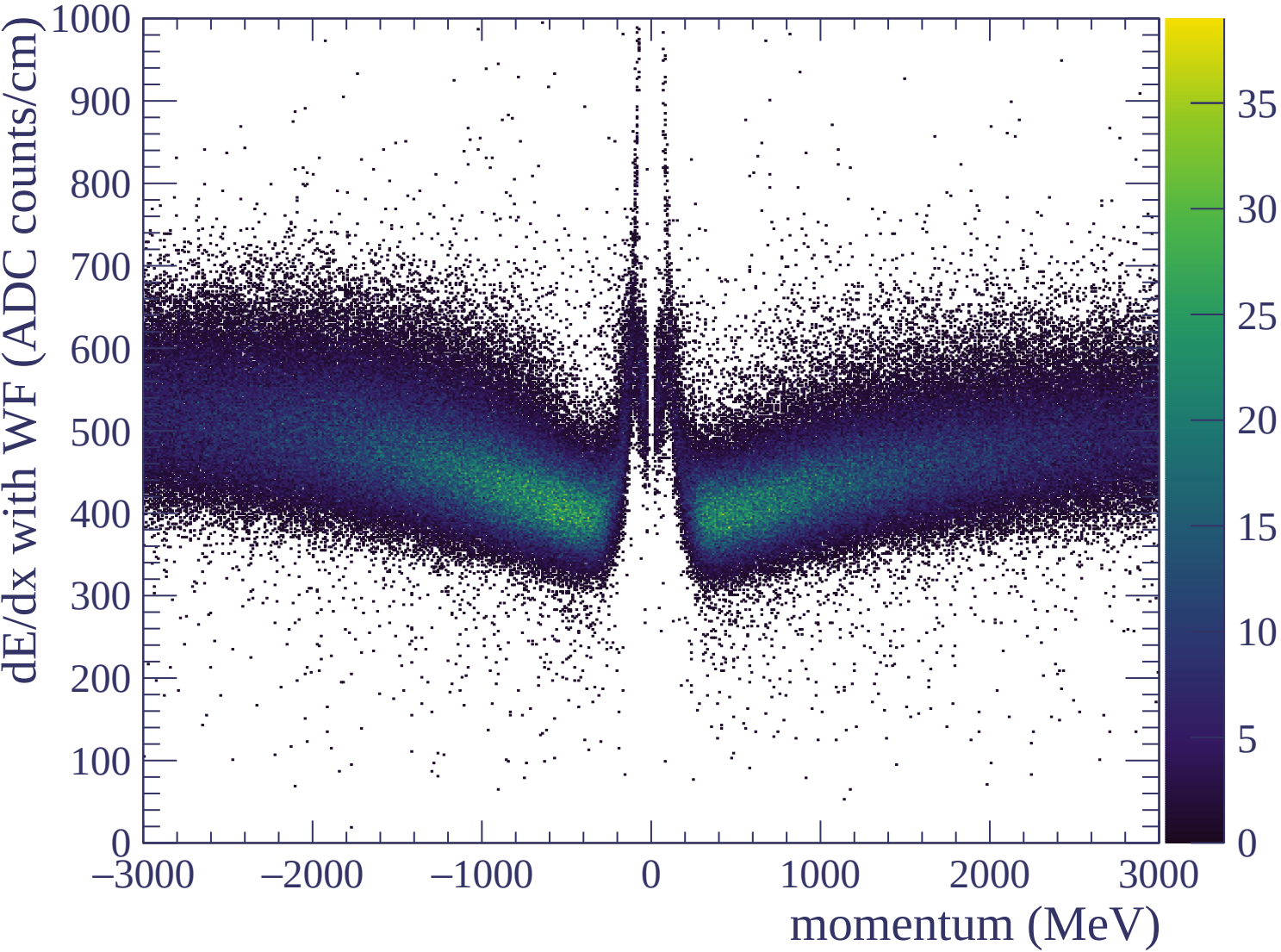


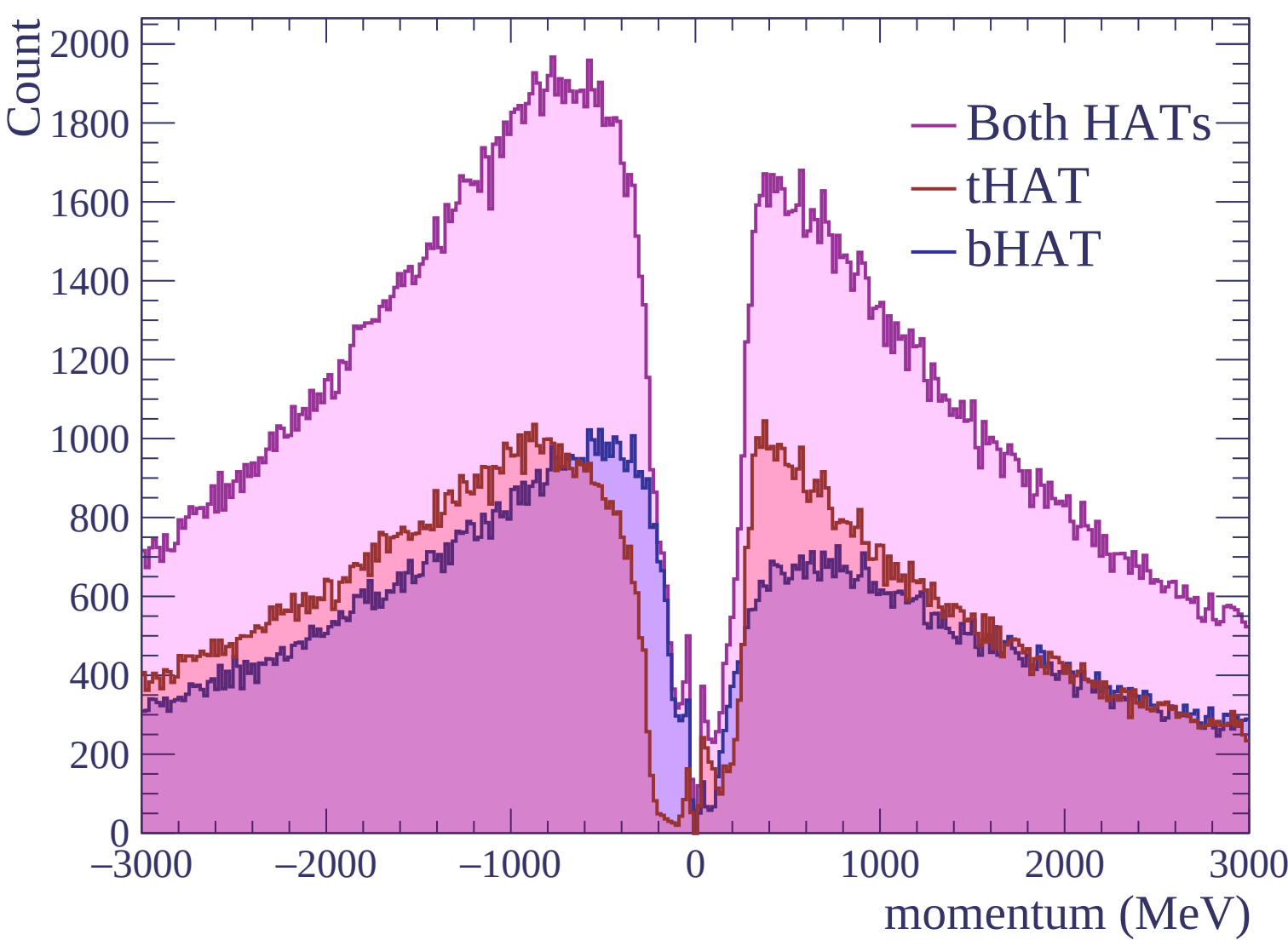


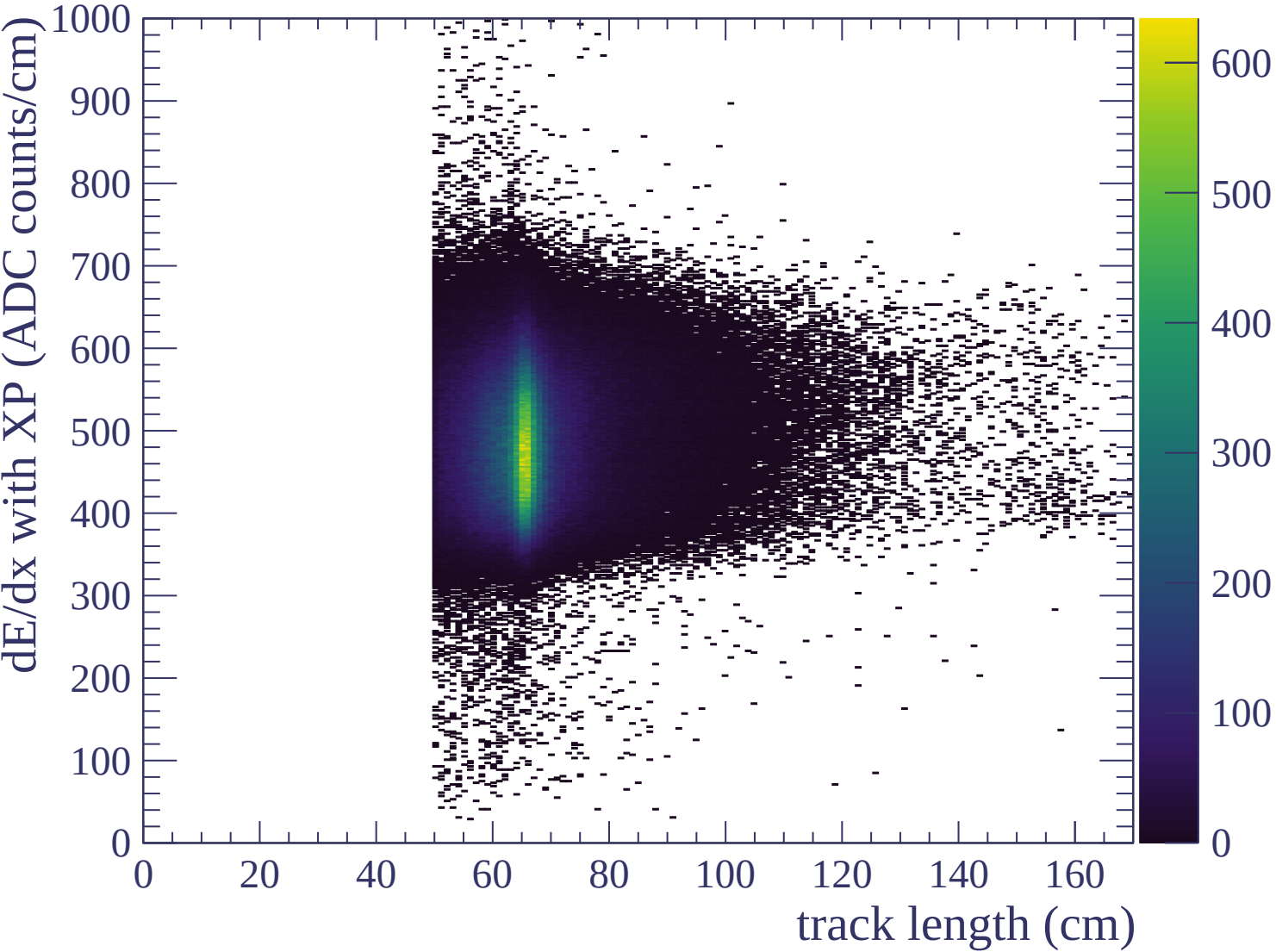


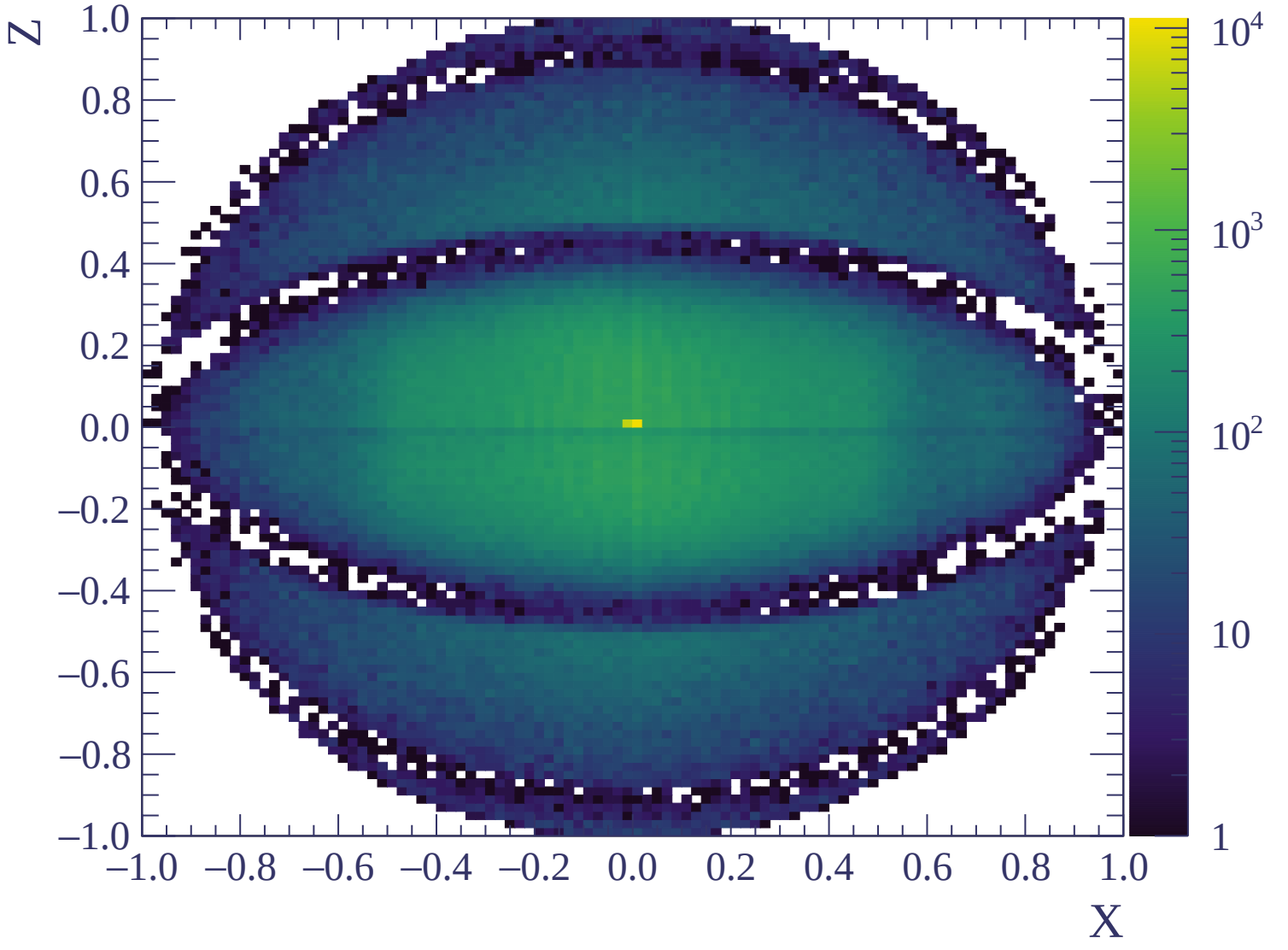


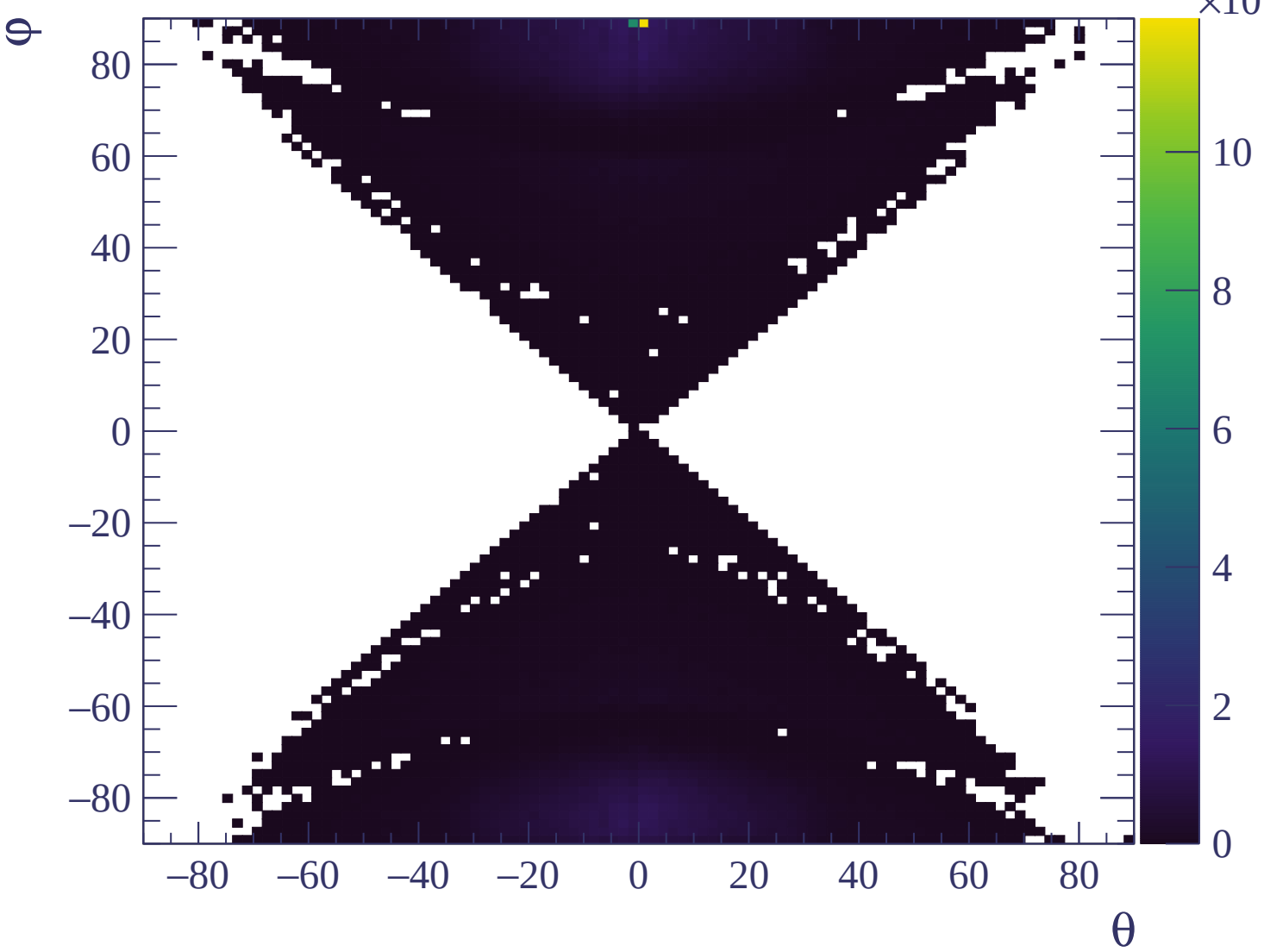


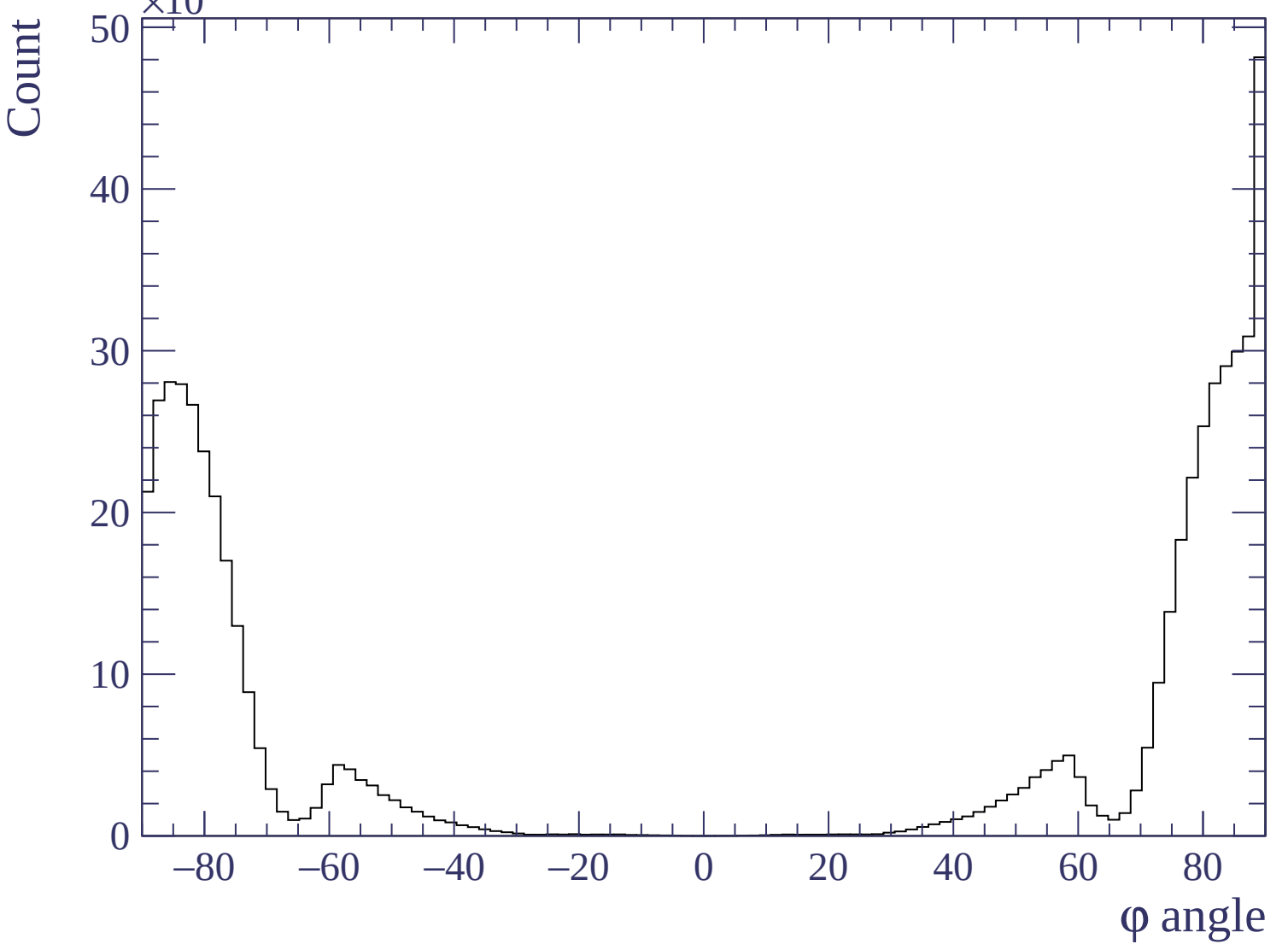


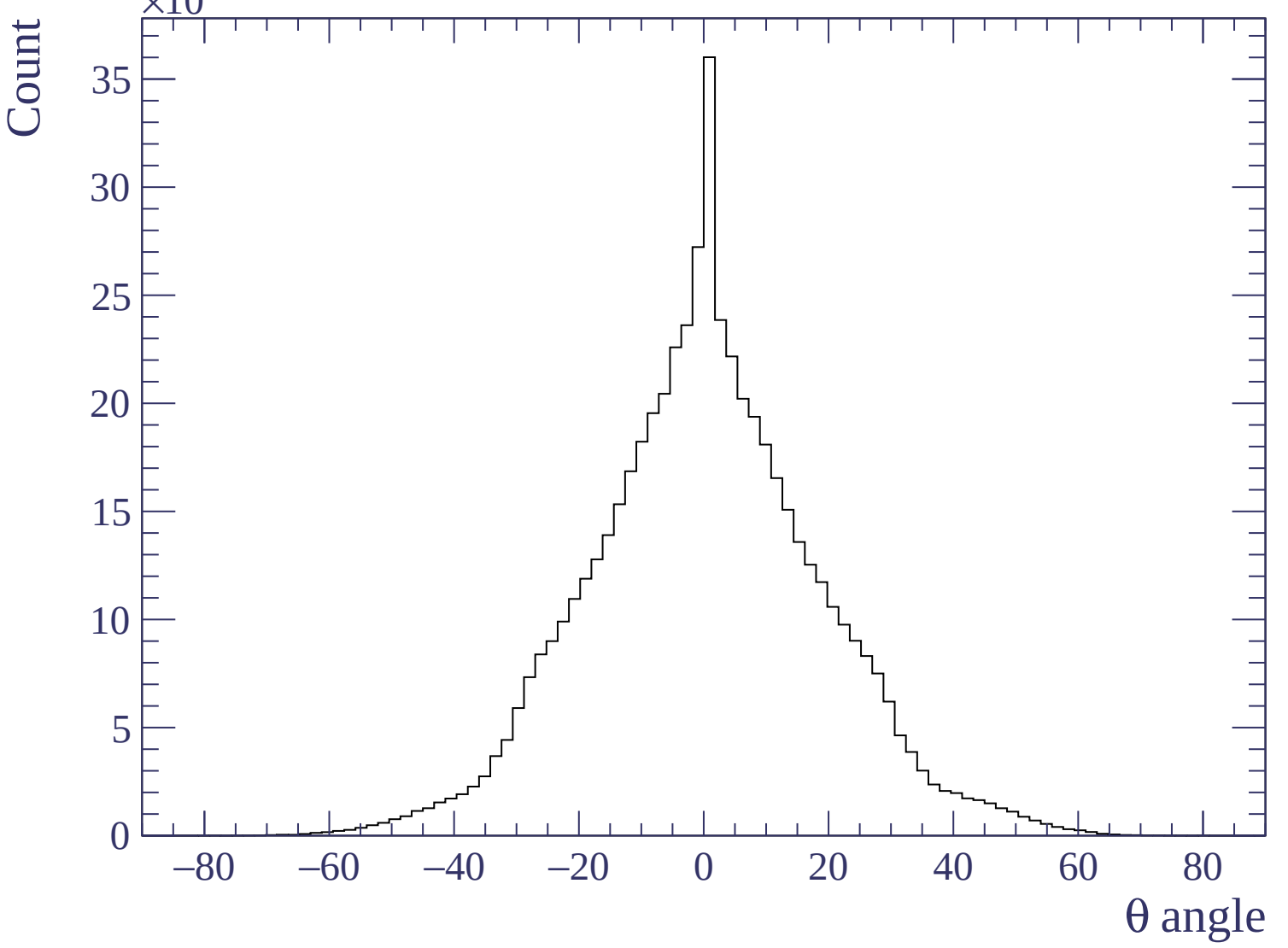


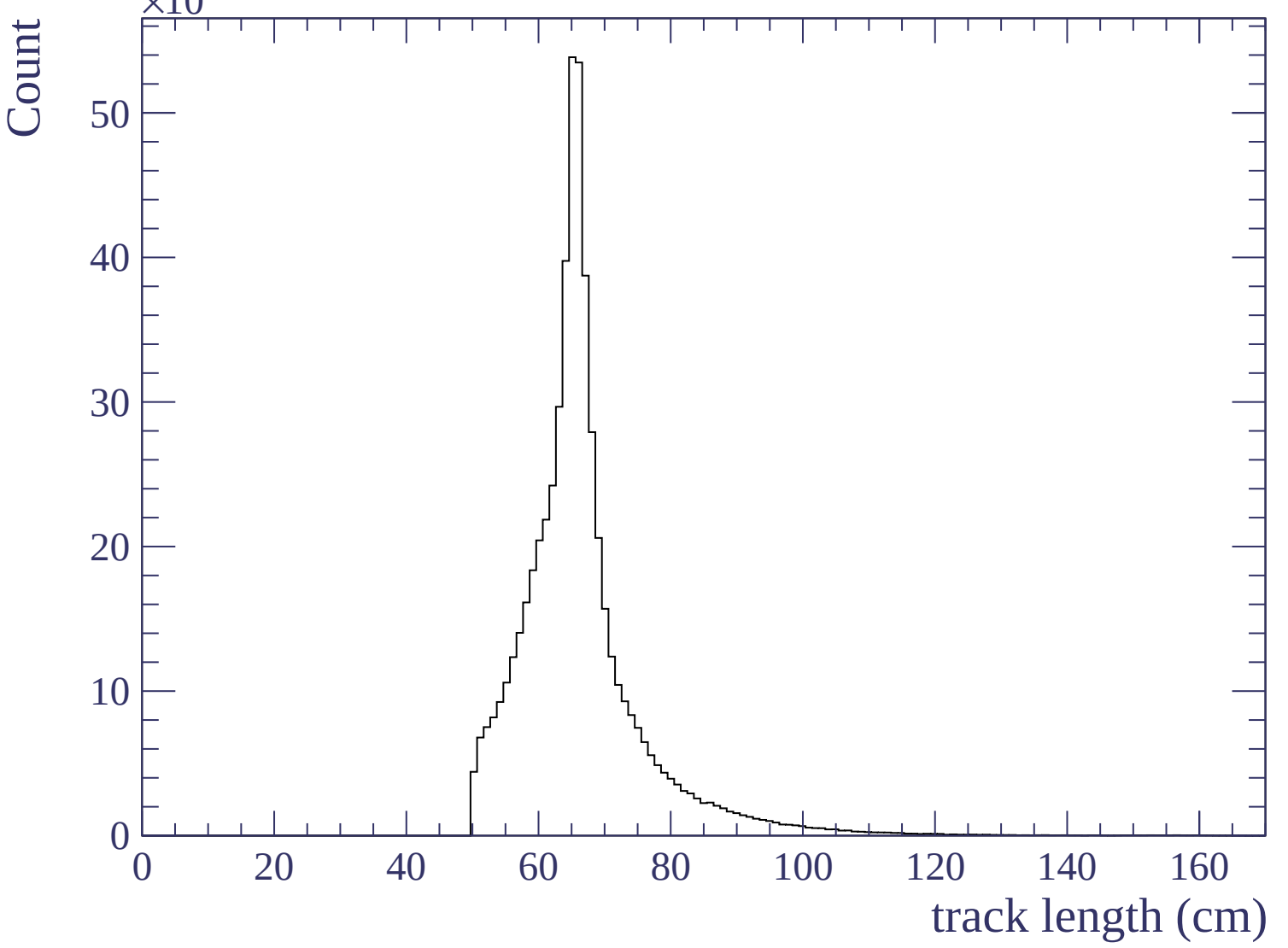


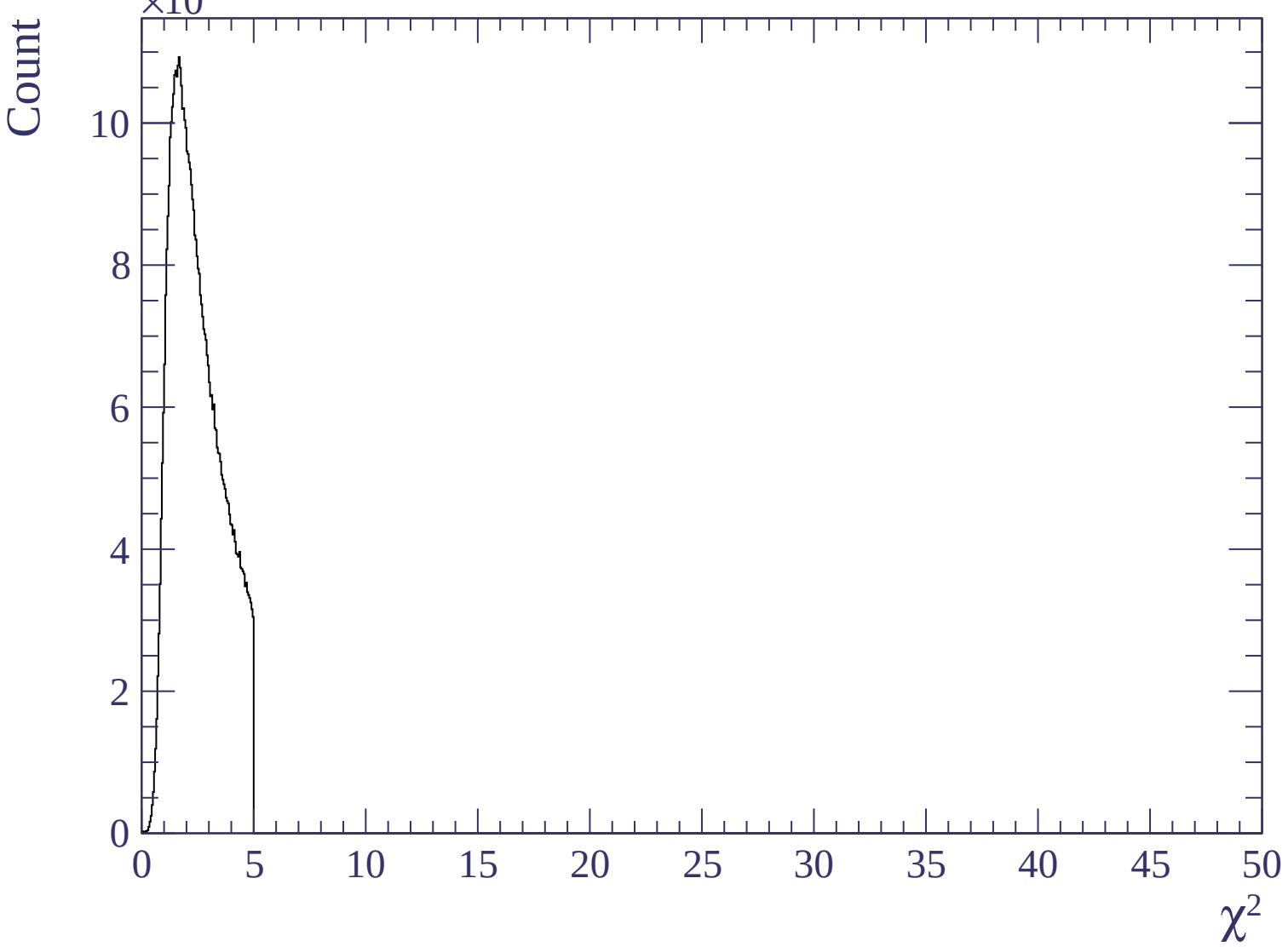


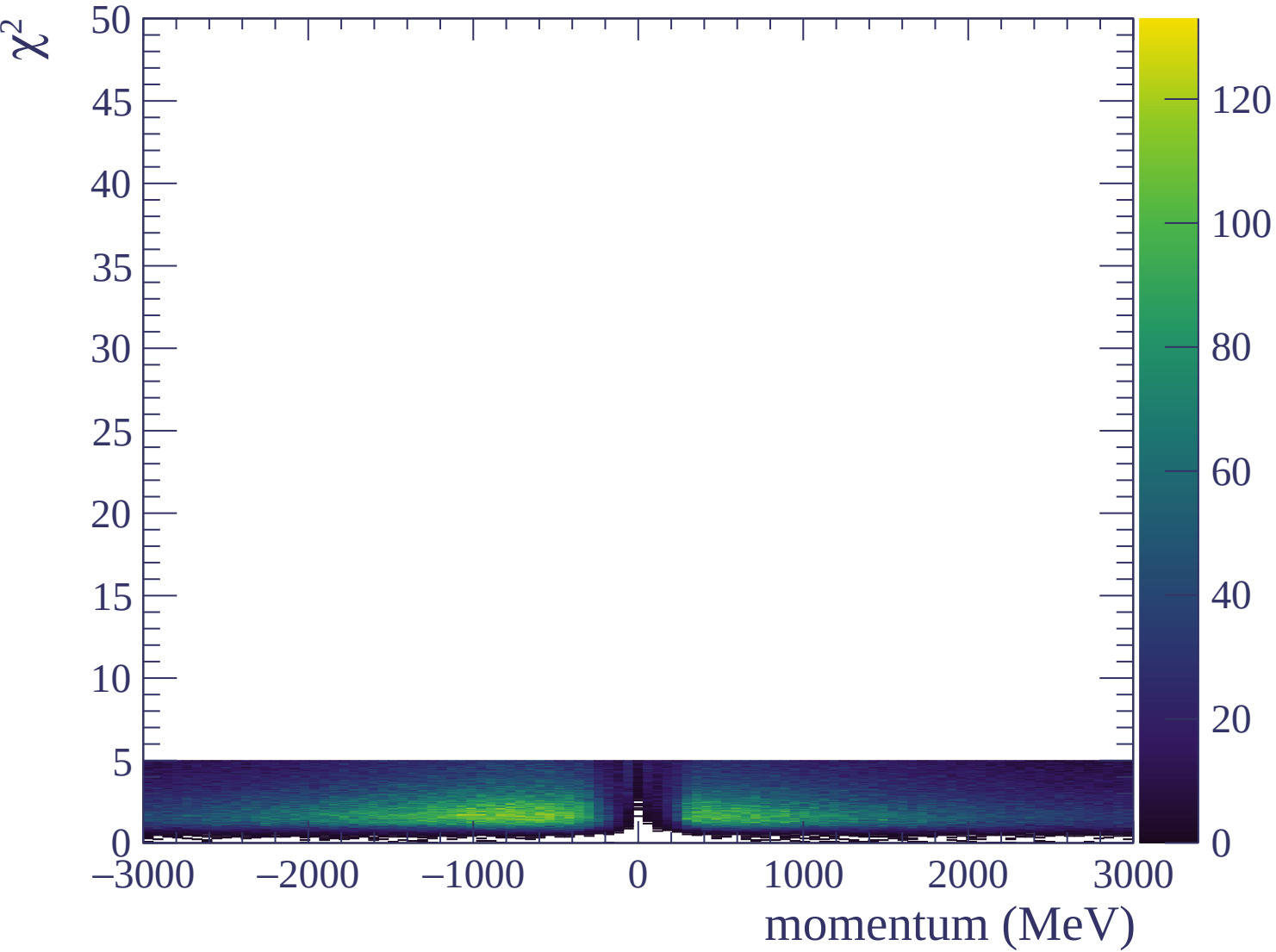


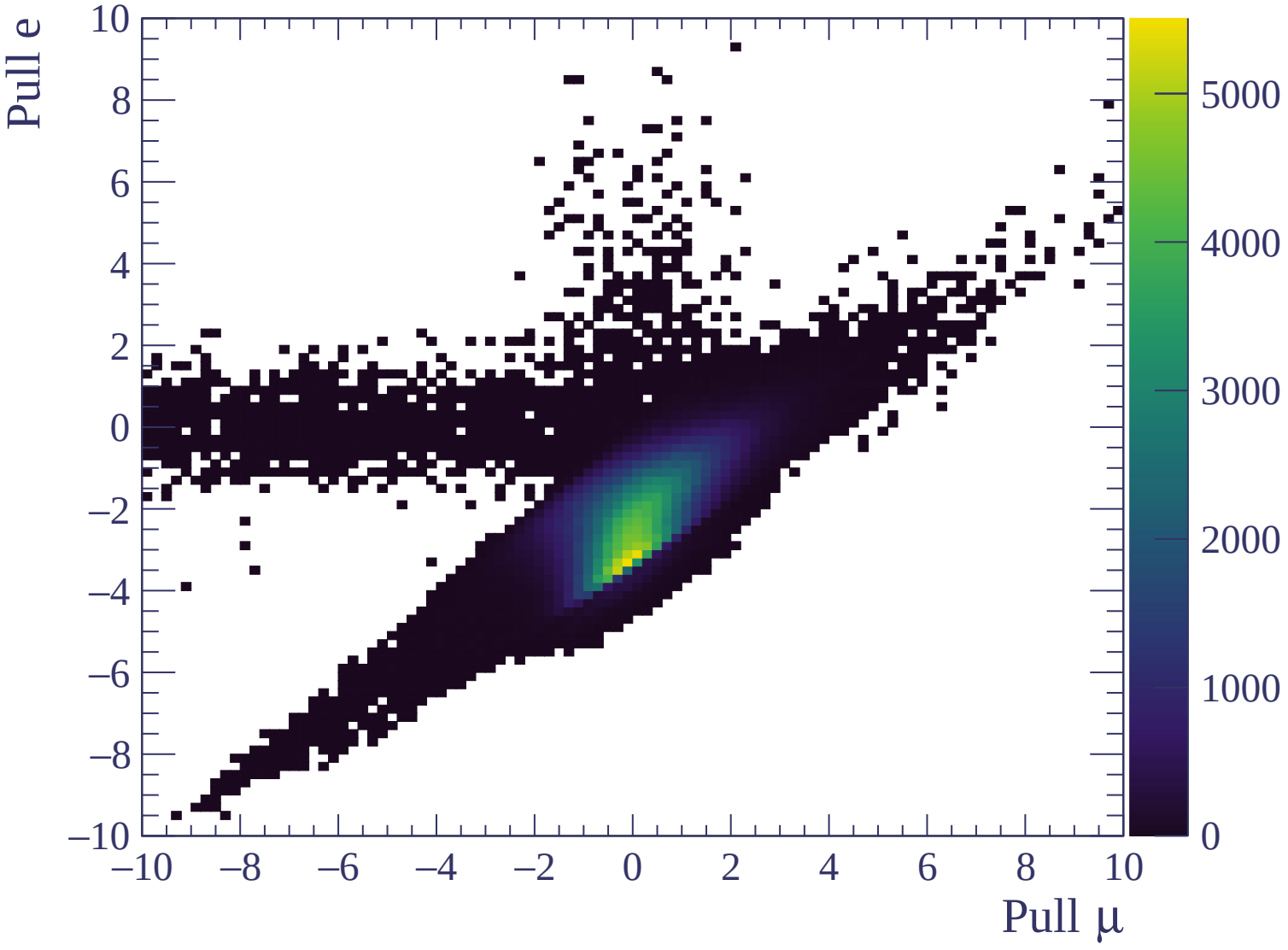


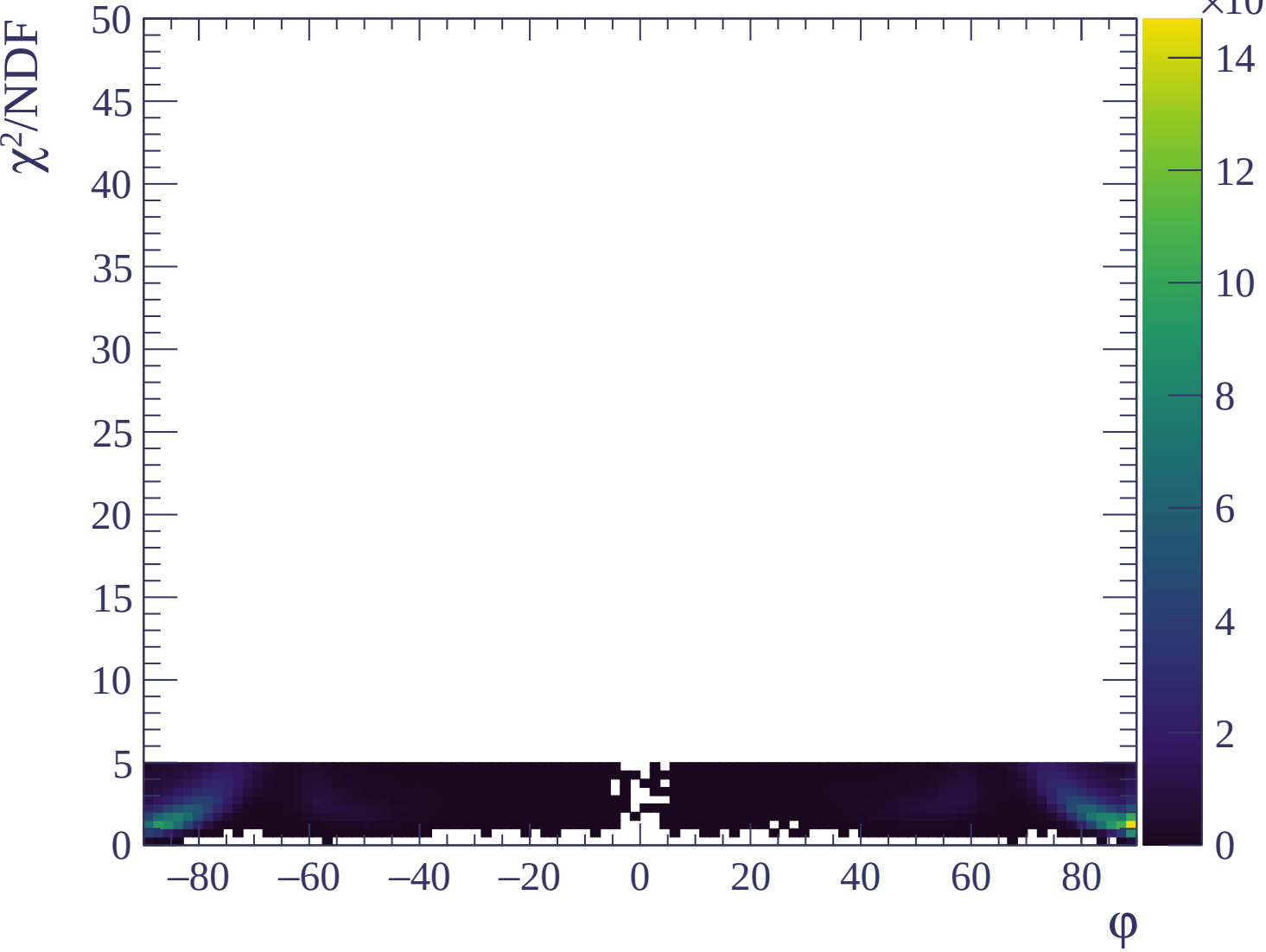


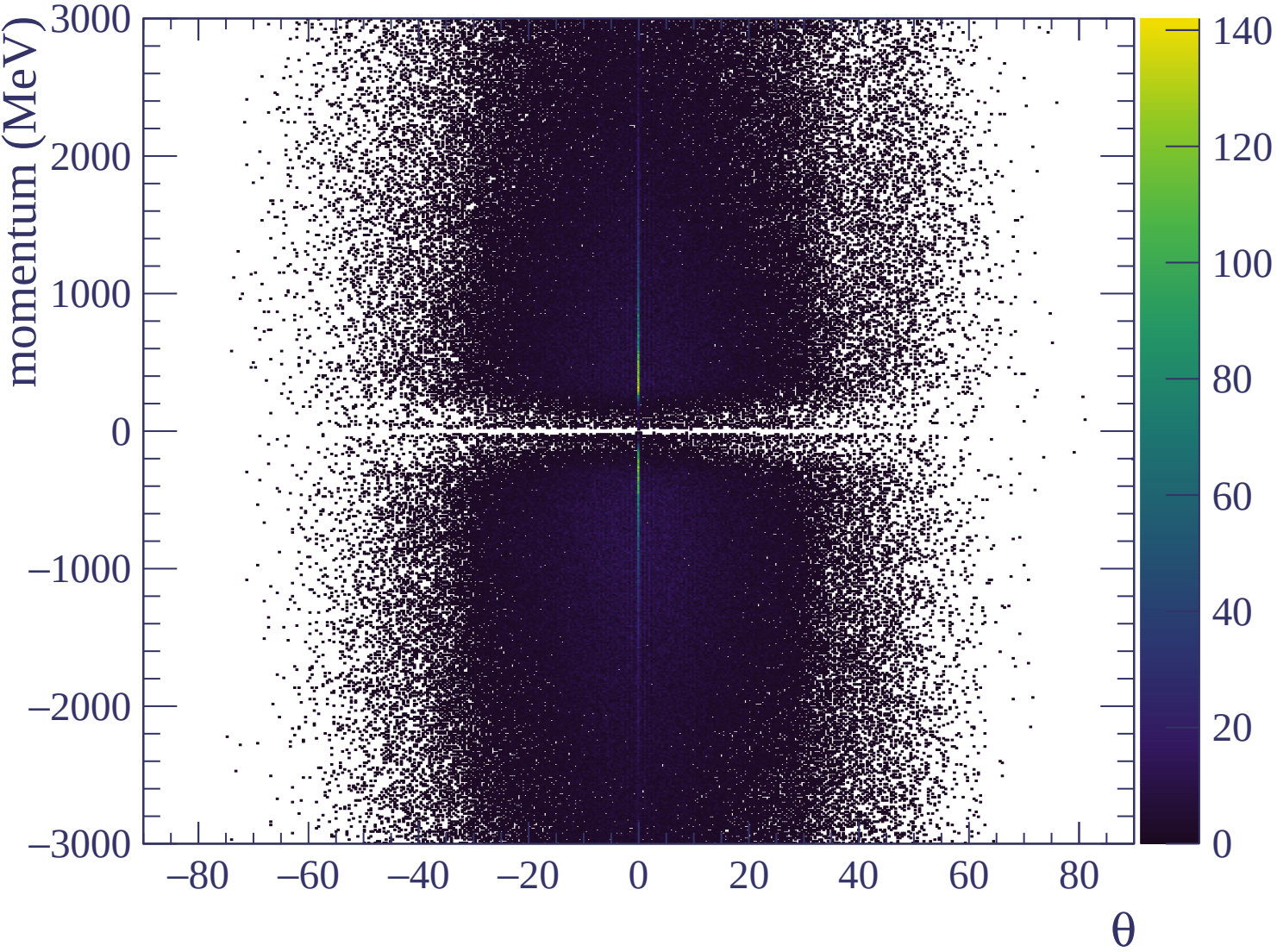


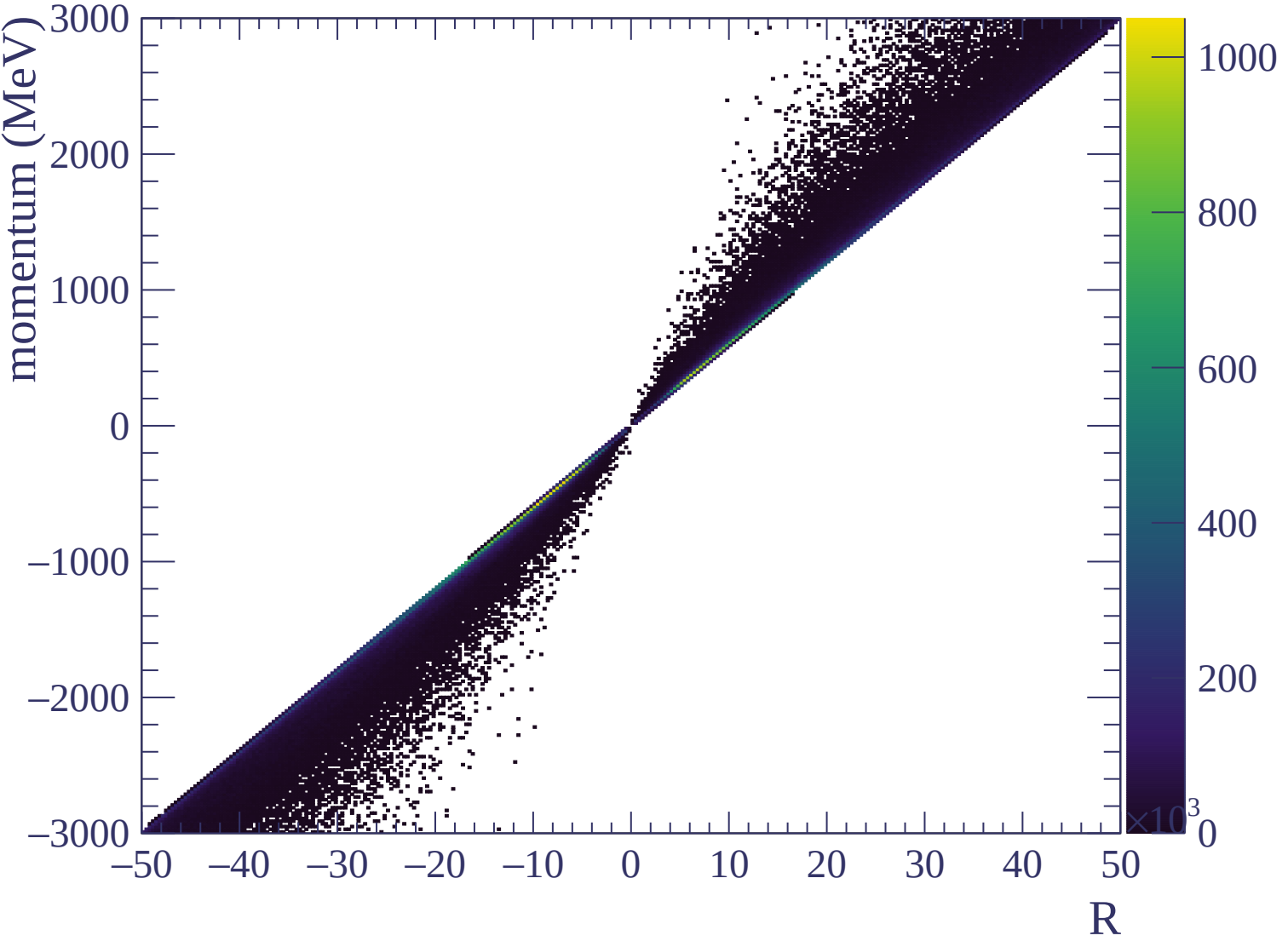


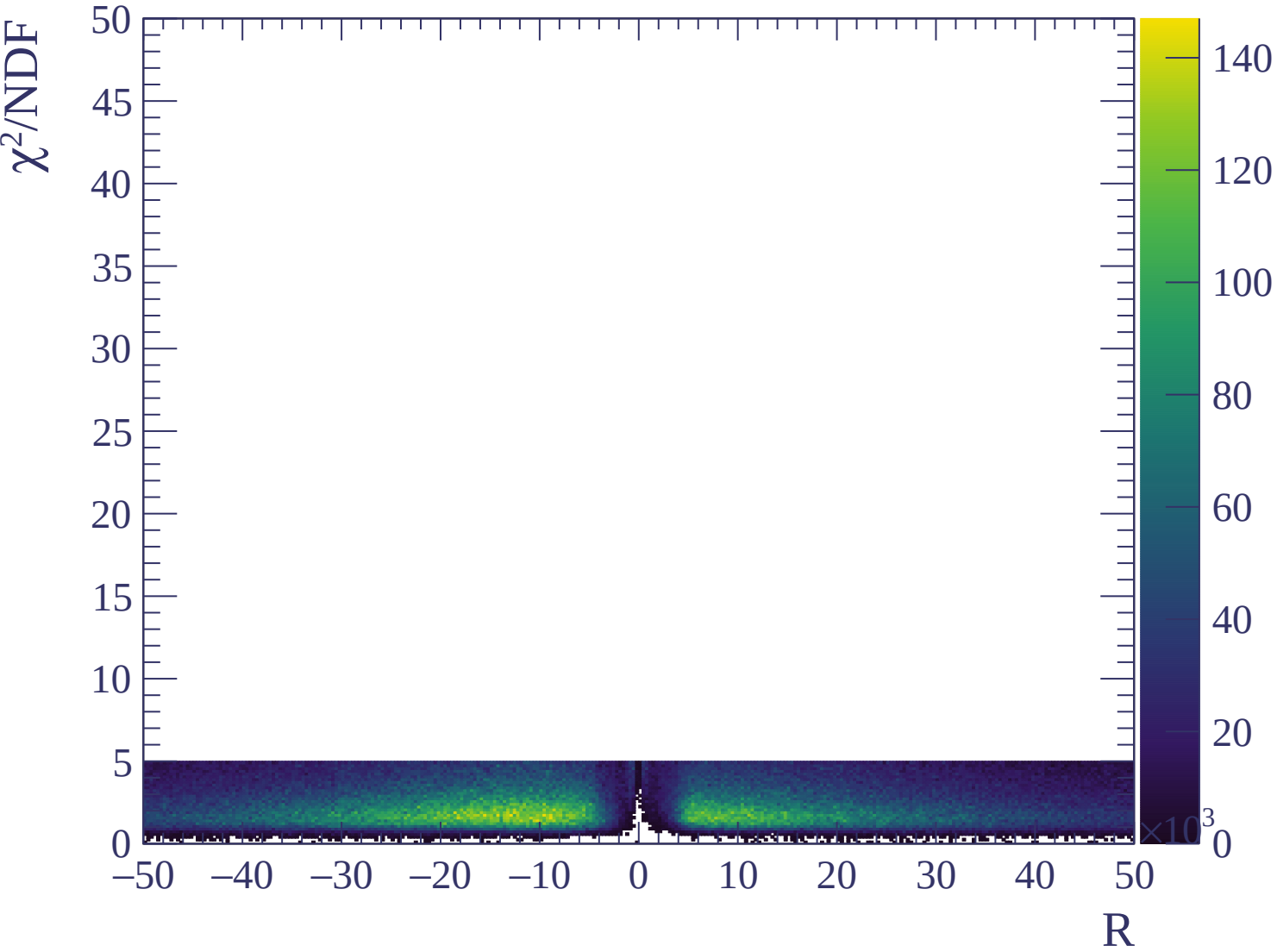


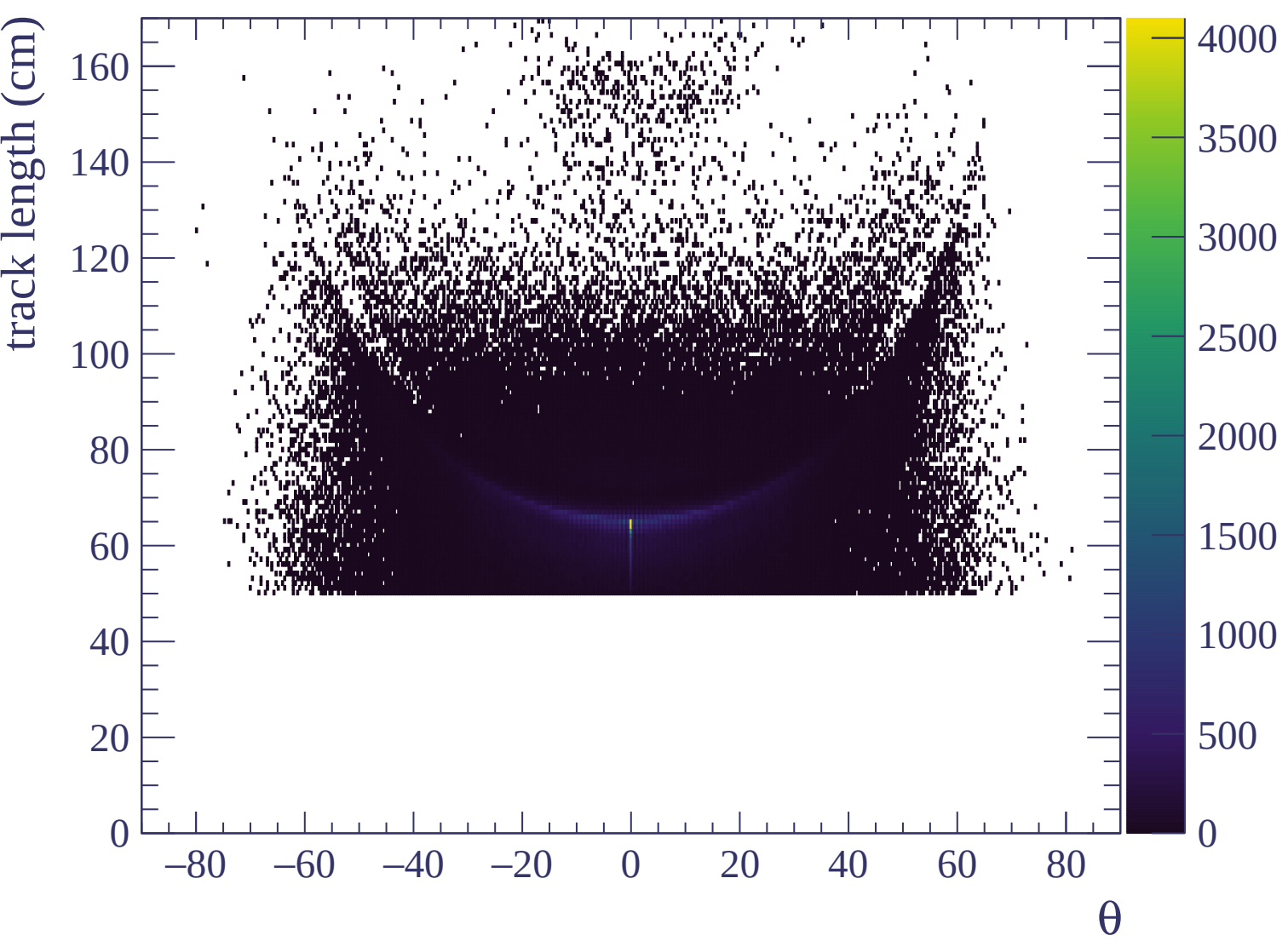


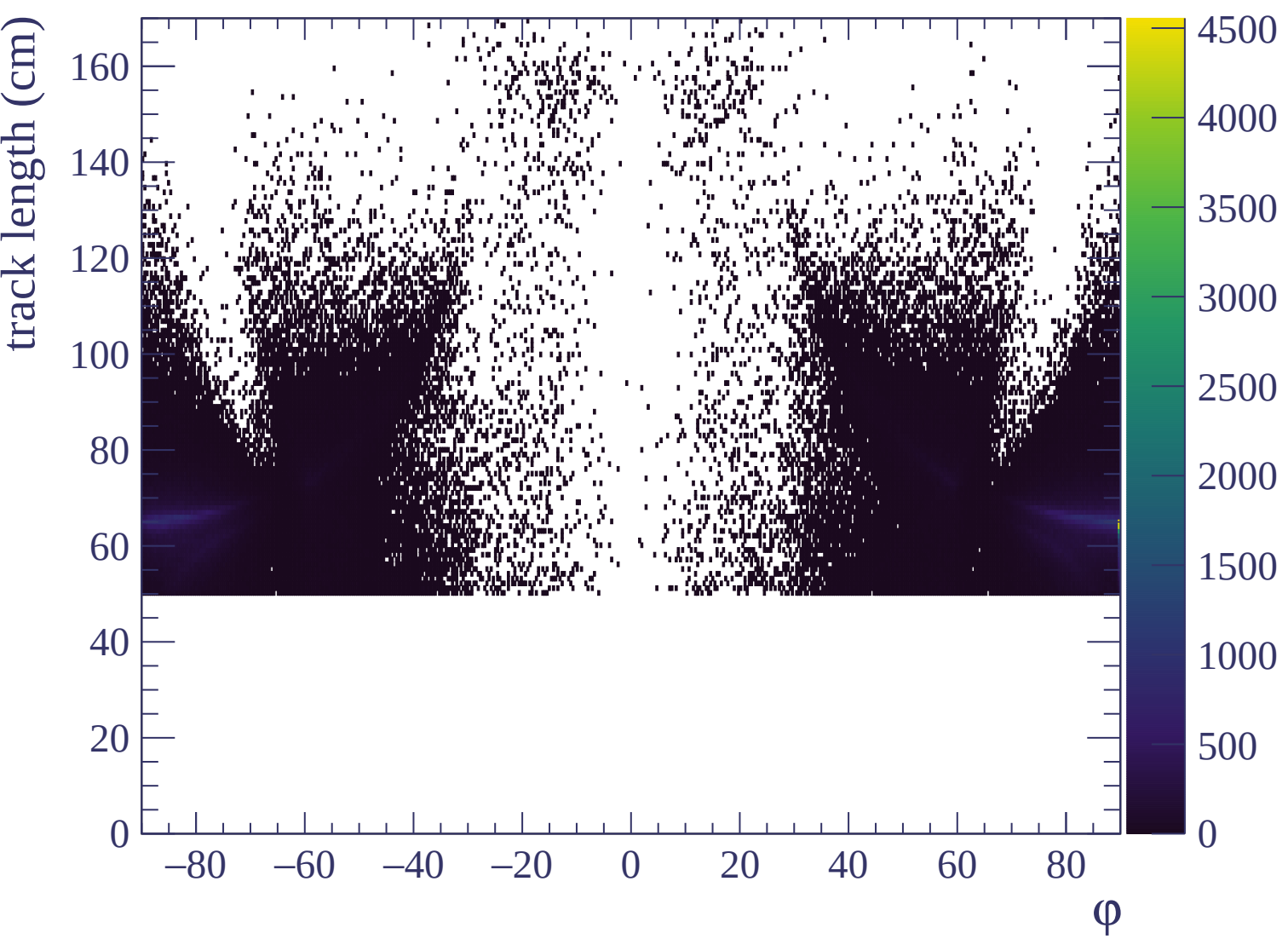






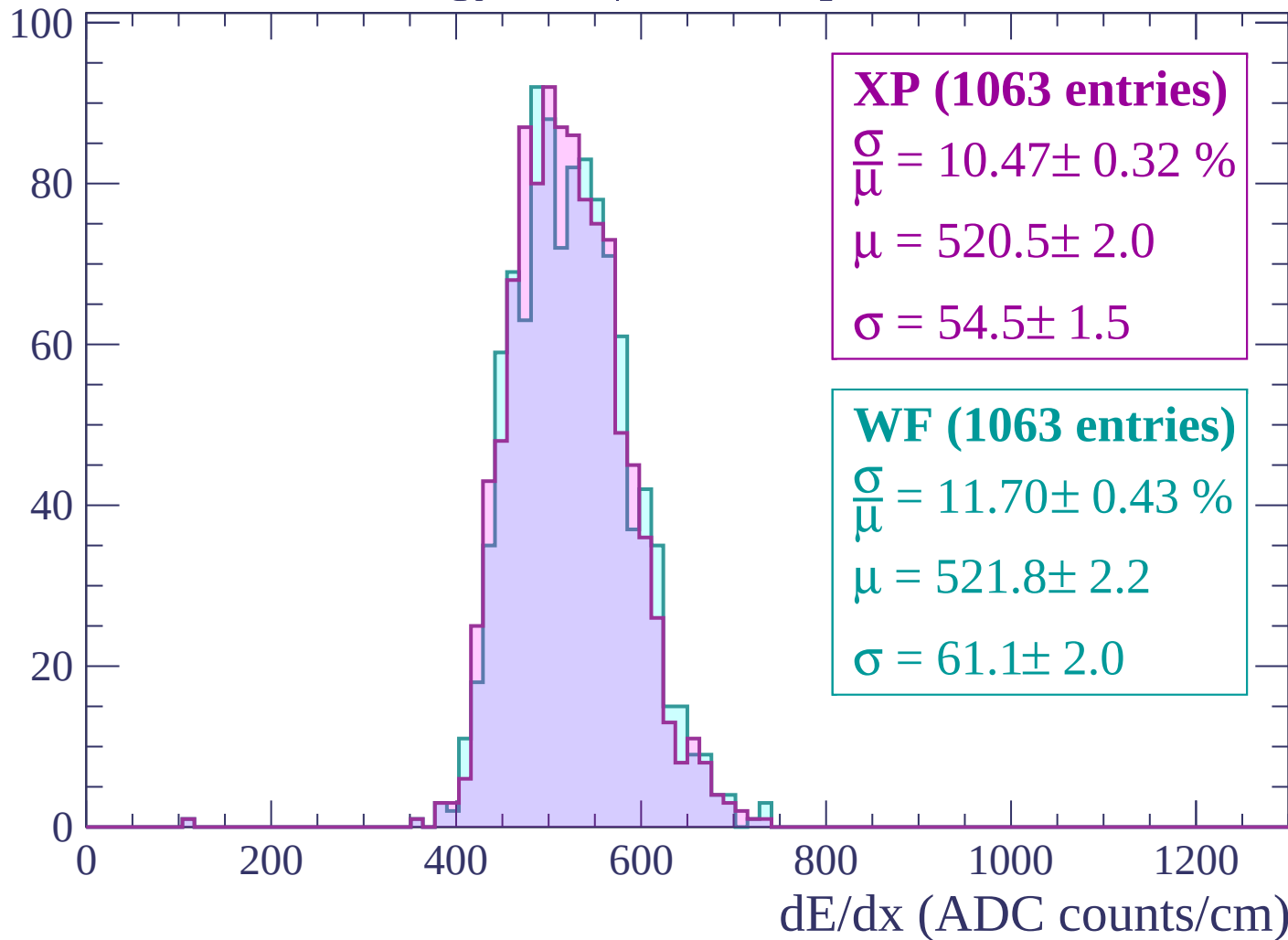






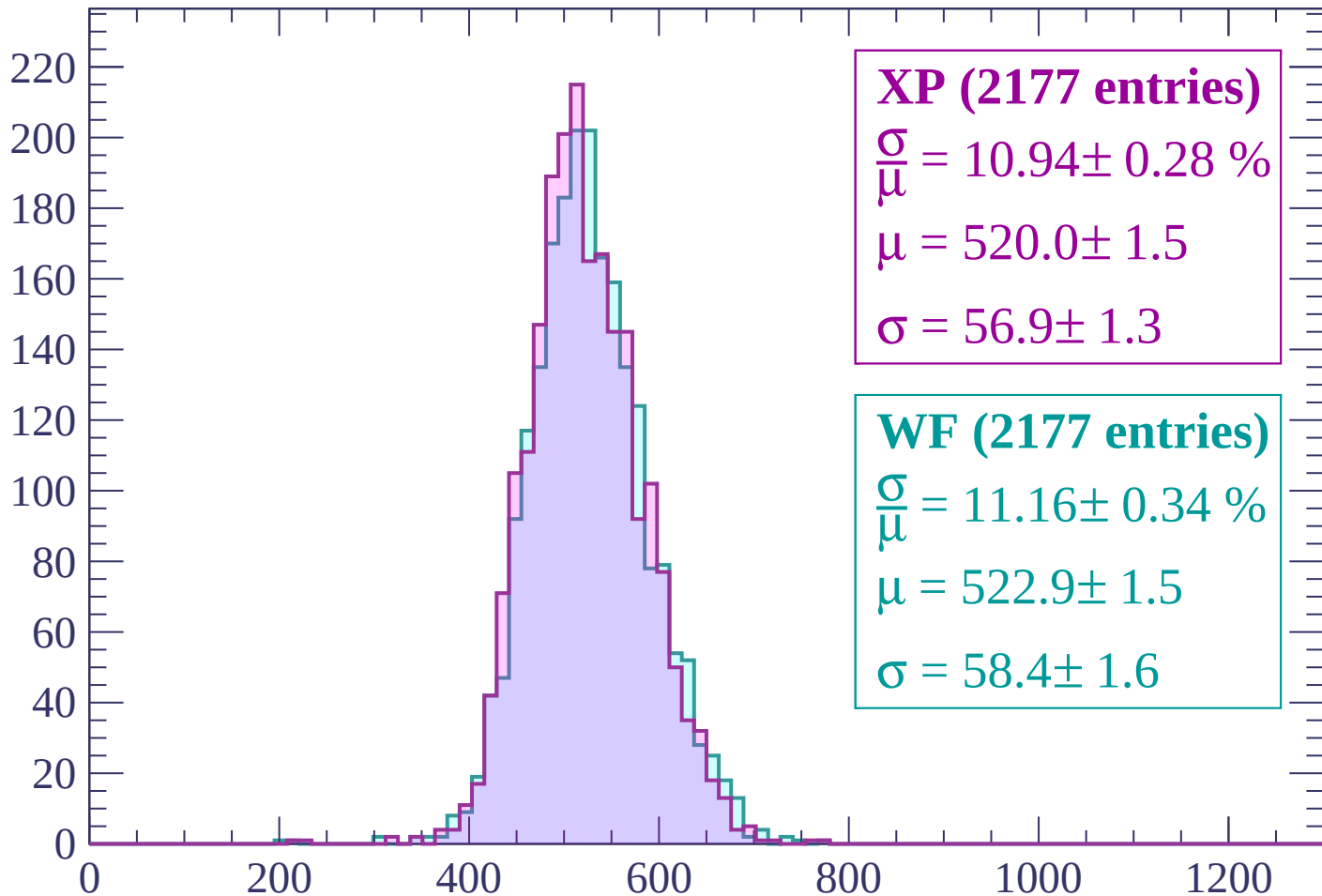
Energy loss | -3000 < p < -2940

Count



Energy loss | $-2940 < p < -2880$

Count



XP (2177 entries)

$$\frac{\sigma}{\mu} = 10.94 \pm 0.28 \%$$

$$\mu = 520.0 \pm 1.5$$

$$\sigma = 56.9 \pm 1.3$$

WF (2177 entries)

$$\frac{\sigma}{\mu} = 11.16 \pm 0.34 \%$$

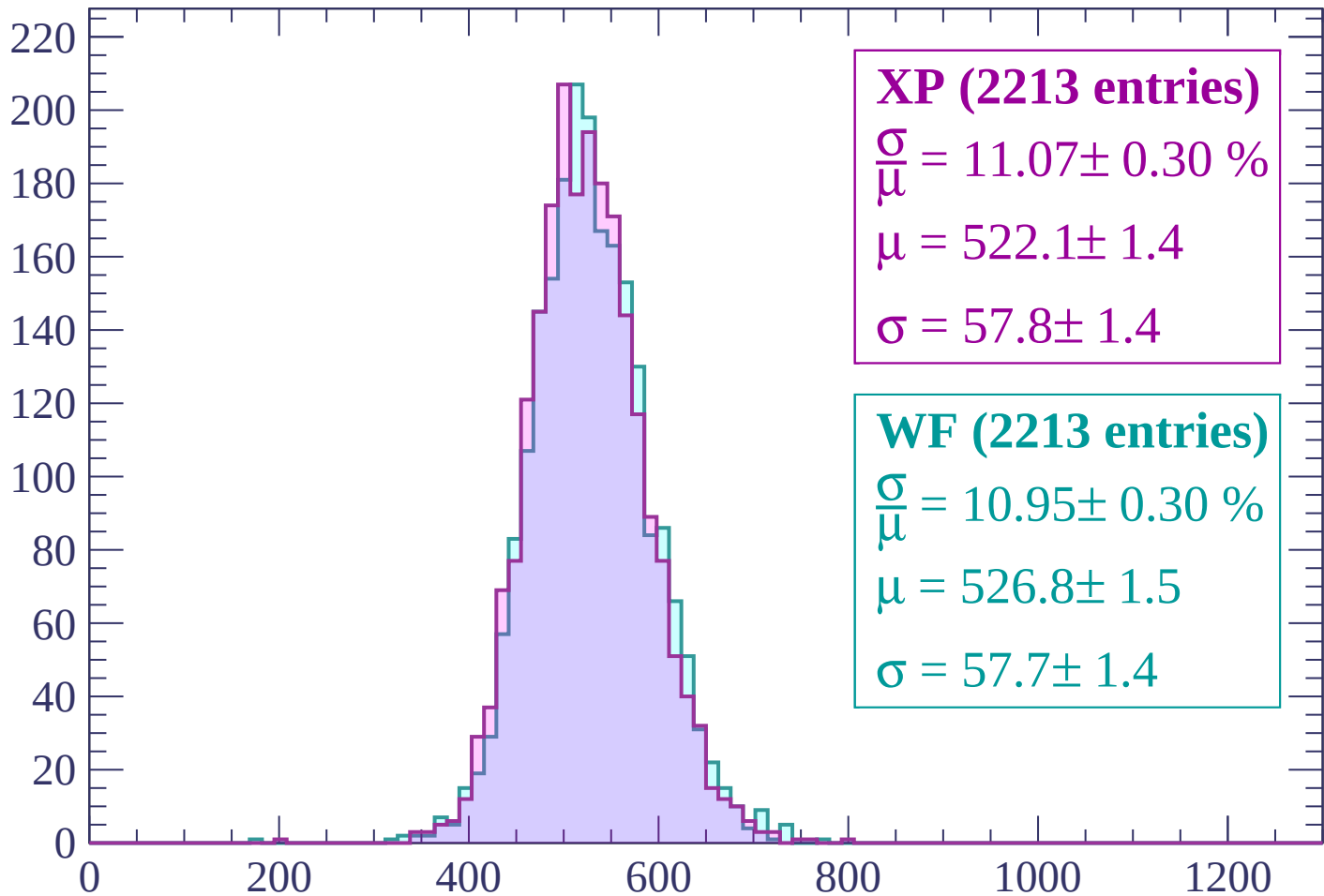
$$\mu = 522.9 \pm 1.5$$

$$\sigma = 58.4 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-2880 < p < -2820$

Count



XP (2213 entries)

$$\frac{\sigma}{\mu} = 11.07 \pm 0.30 \%$$

$$\mu = 522.1 \pm 1.4$$

$$\sigma = 57.8 \pm 1.4$$

WF (2213 entries)

$$\frac{\sigma}{\mu} = 10.95 \pm 0.30 \%$$

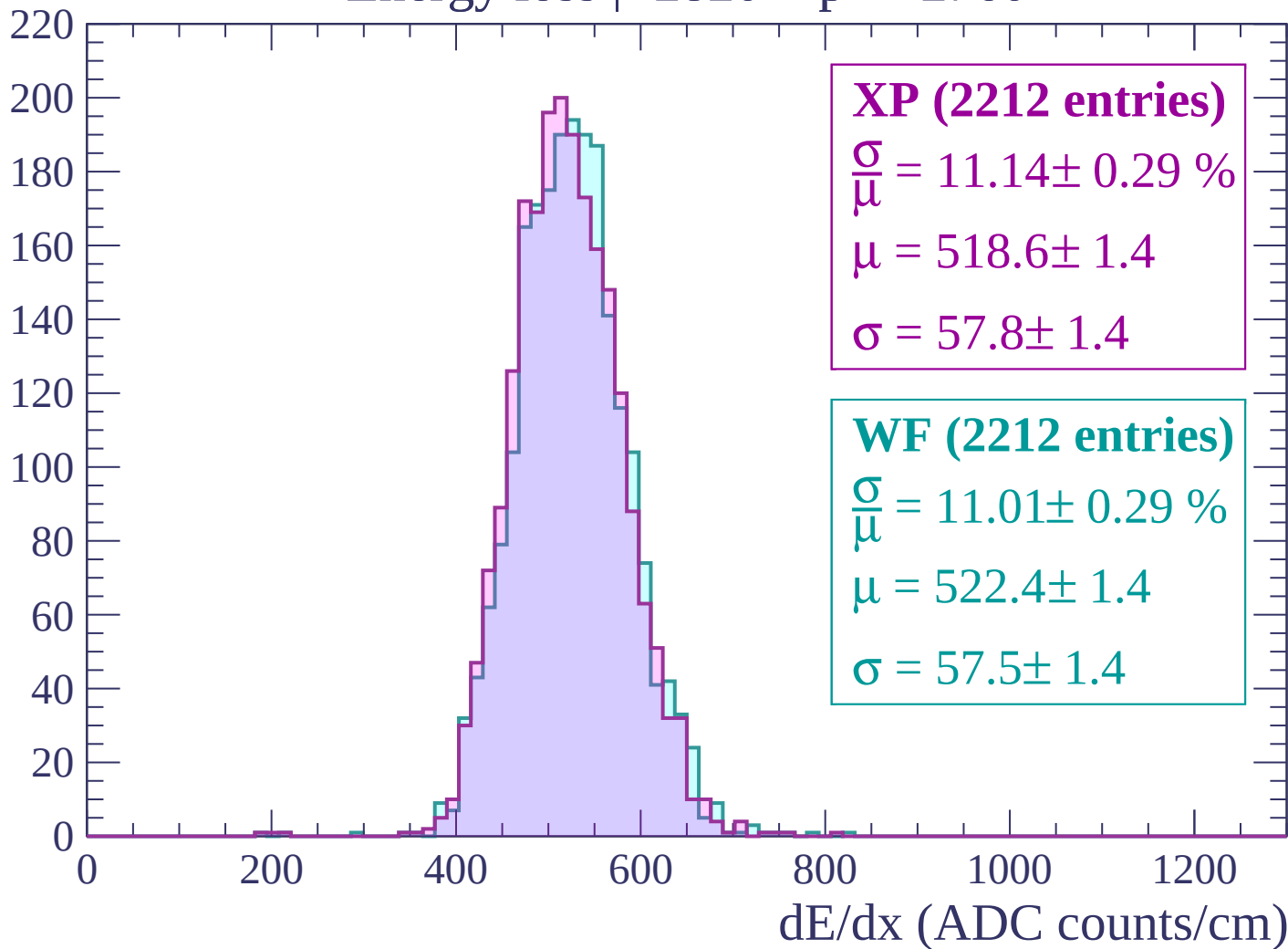
$$\mu = 526.8 \pm 1.5$$

$$\sigma = 57.7 \pm 1.4$$

dE/dx (ADC counts/cm)

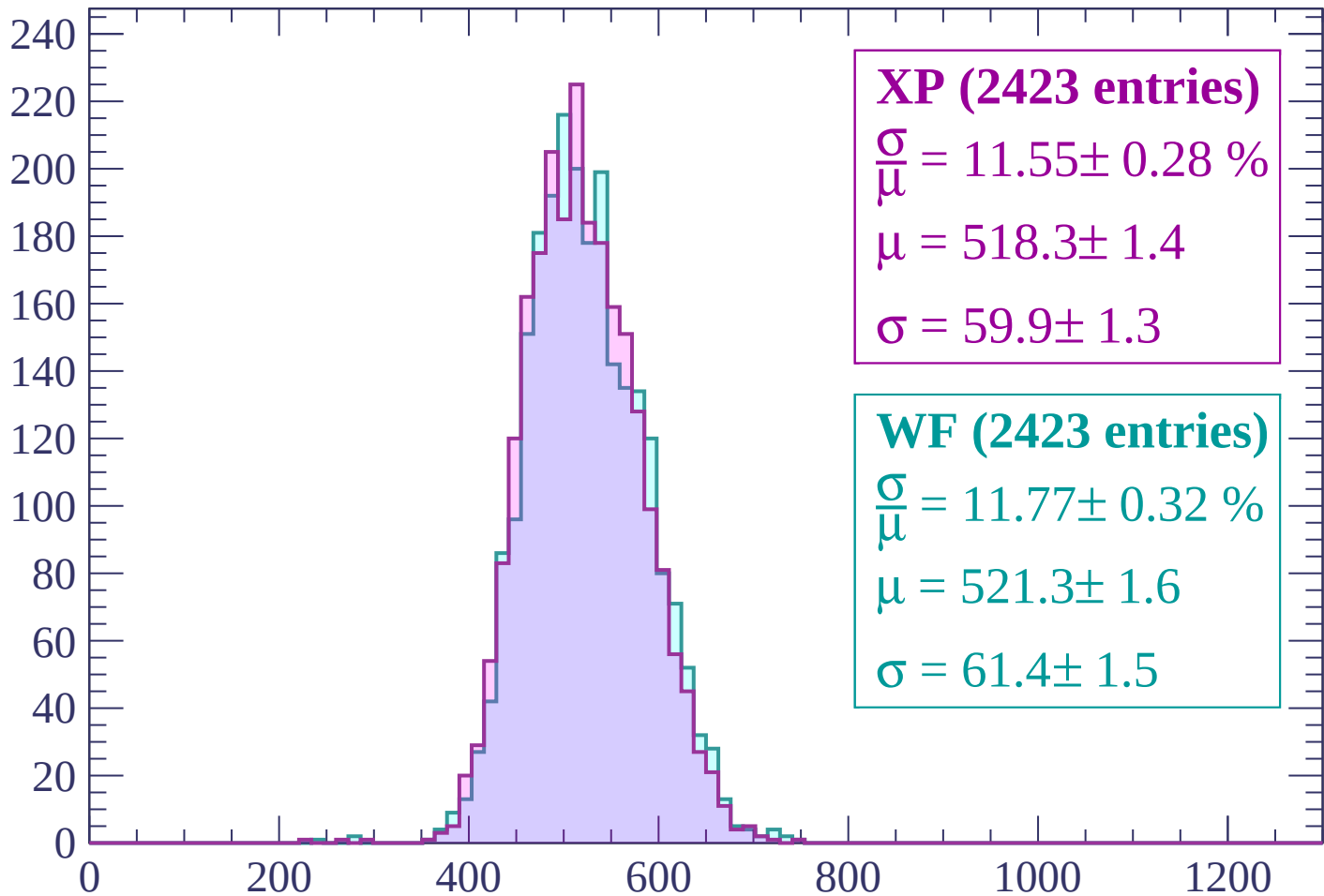
Energy loss | $-2820 < p < -2760$

Count



Energy loss | $-2760 < p < -2700$

Count



XP (2423 entries)

$$\frac{\sigma}{\mu} = 11.55 \pm 0.28 \%$$

$$\mu = 518.3 \pm 1.4$$

$$\sigma = 59.9 \pm 1.3$$

WF (2423 entries)

$$\frac{\sigma}{\mu} = 11.77 \pm 0.32 \%$$

$$\mu = 521.3 \pm 1.6$$

$$\sigma = 61.4 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | -2700 < p < -2640

Count

250

200

150

100

50

0

0

200

400

600

800

1000

1200

XP (2477 entries)

$\frac{\sigma}{\mu} = 11.01 \pm 0.30 \%$

$\mu = 516.5 \pm 1.4$

$\sigma = 56.9 \pm 1.4$

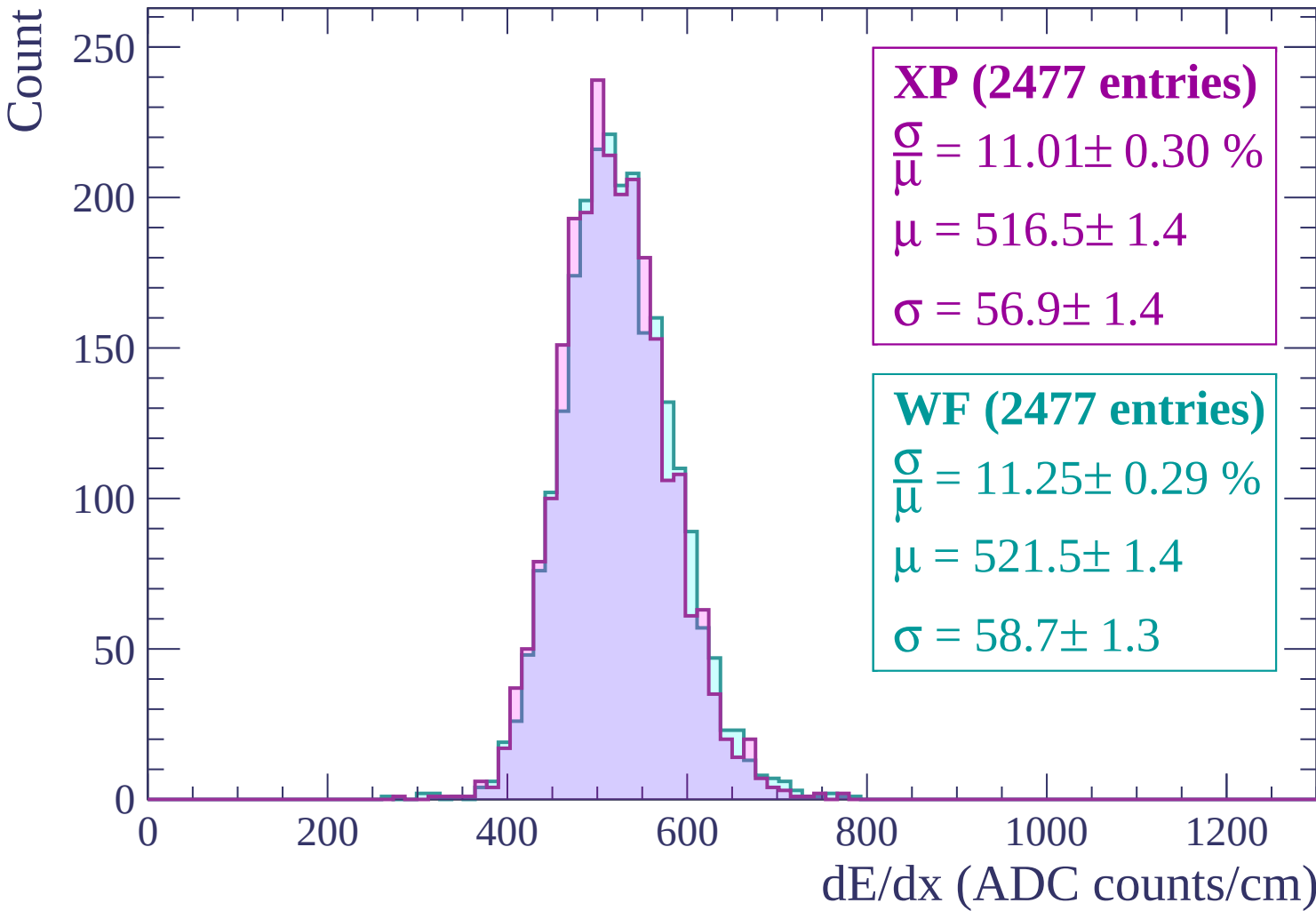
WF (2477 entries)

$\frac{\sigma}{\mu} = 11.25 \pm 0.29 \%$

$\mu = 521.5 \pm 1.4$

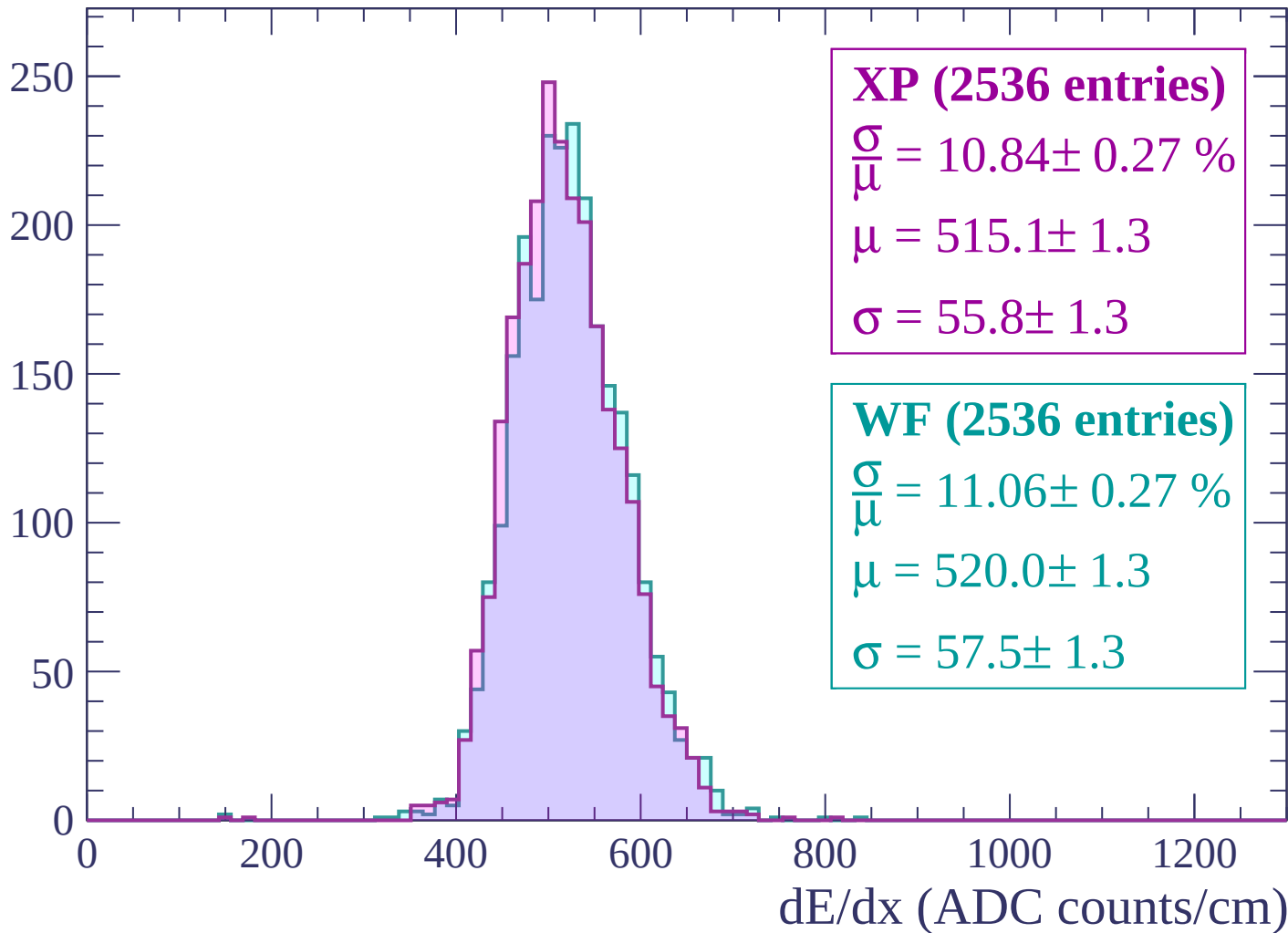
$\sigma = 58.7 \pm 1.3$

dE/dx (ADC counts/cm)



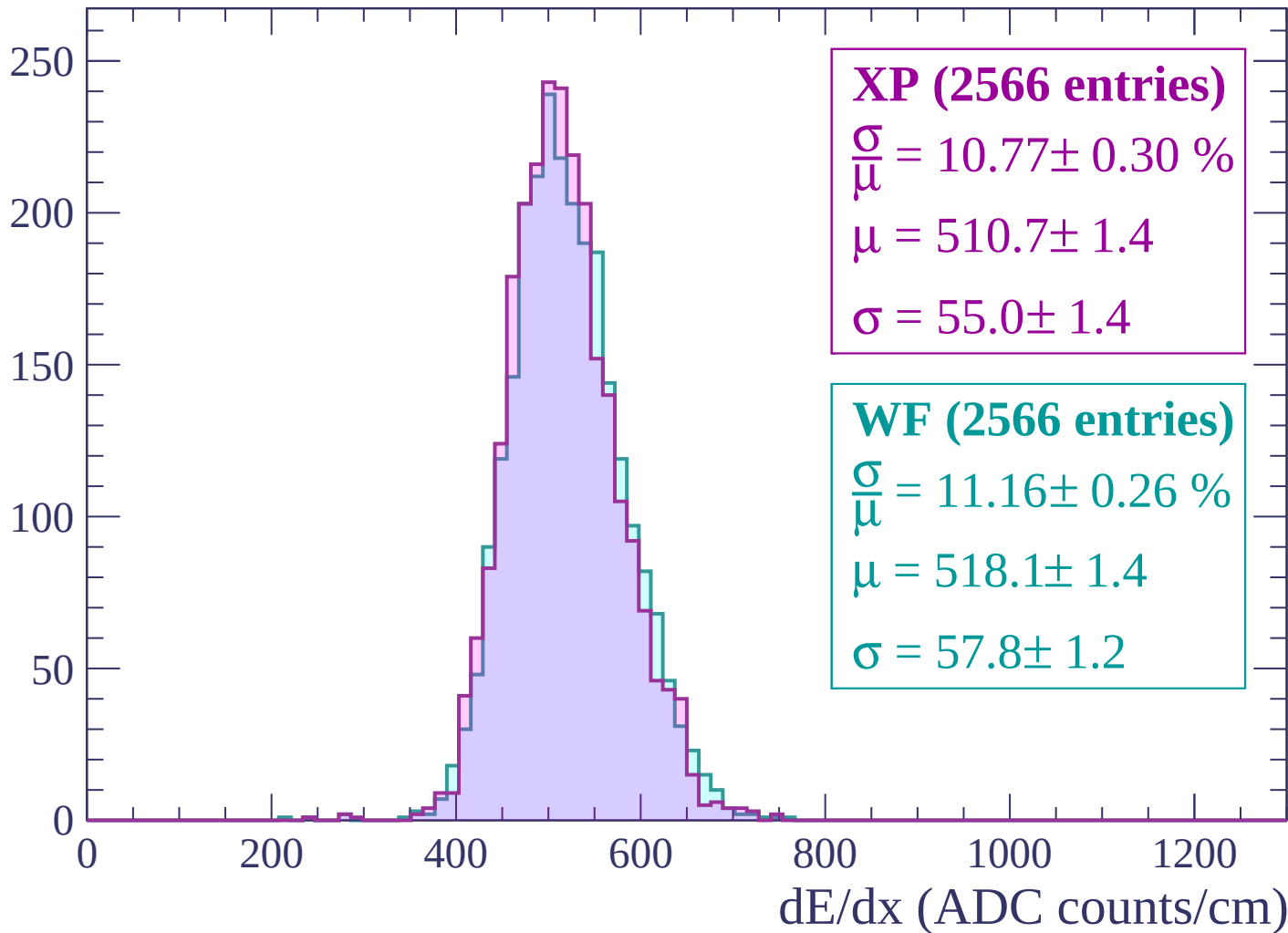
Energy loss | $-2640 < p < -2580$

Count



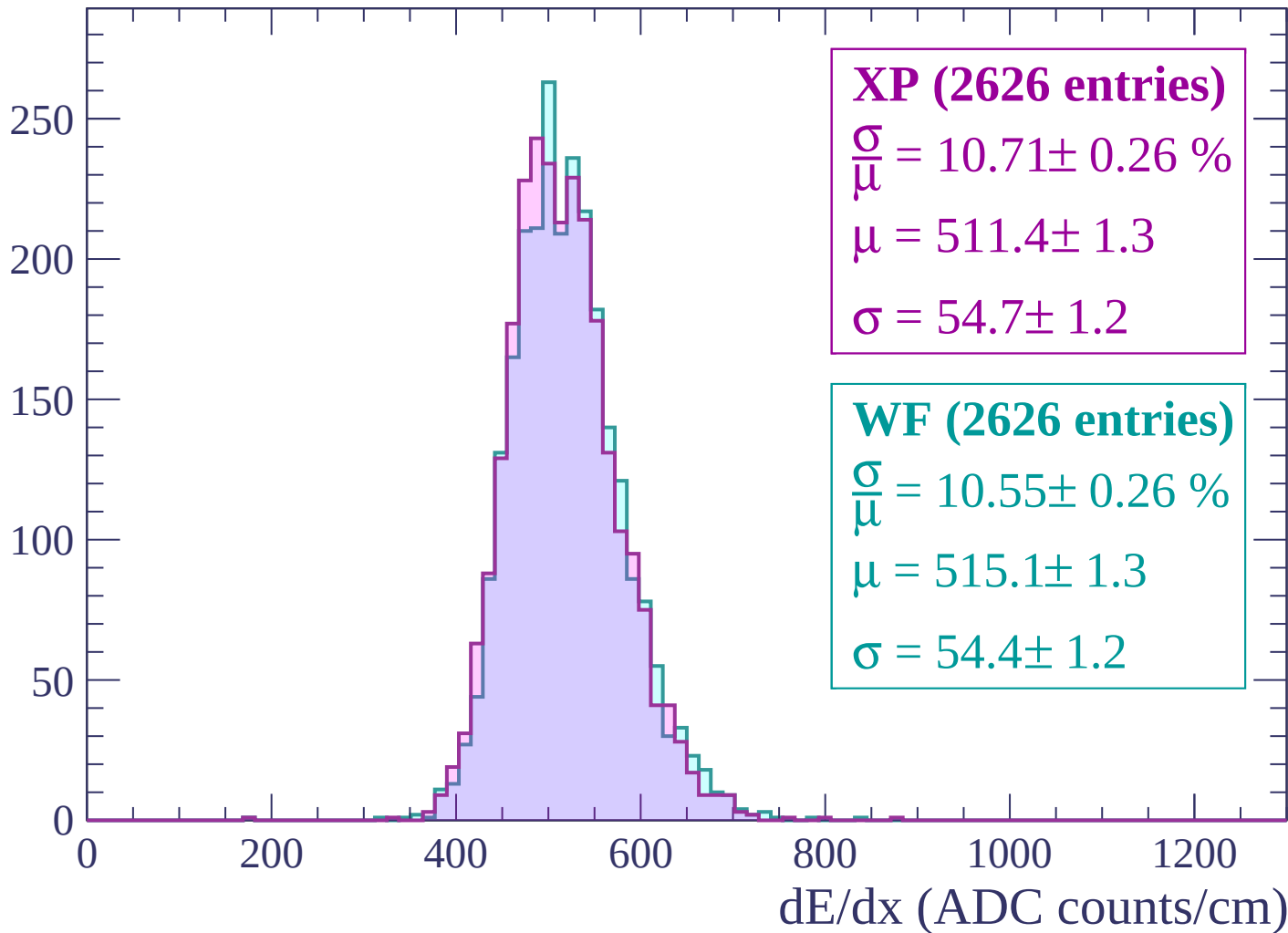
Energy loss | -2580 < p < -2520

Count



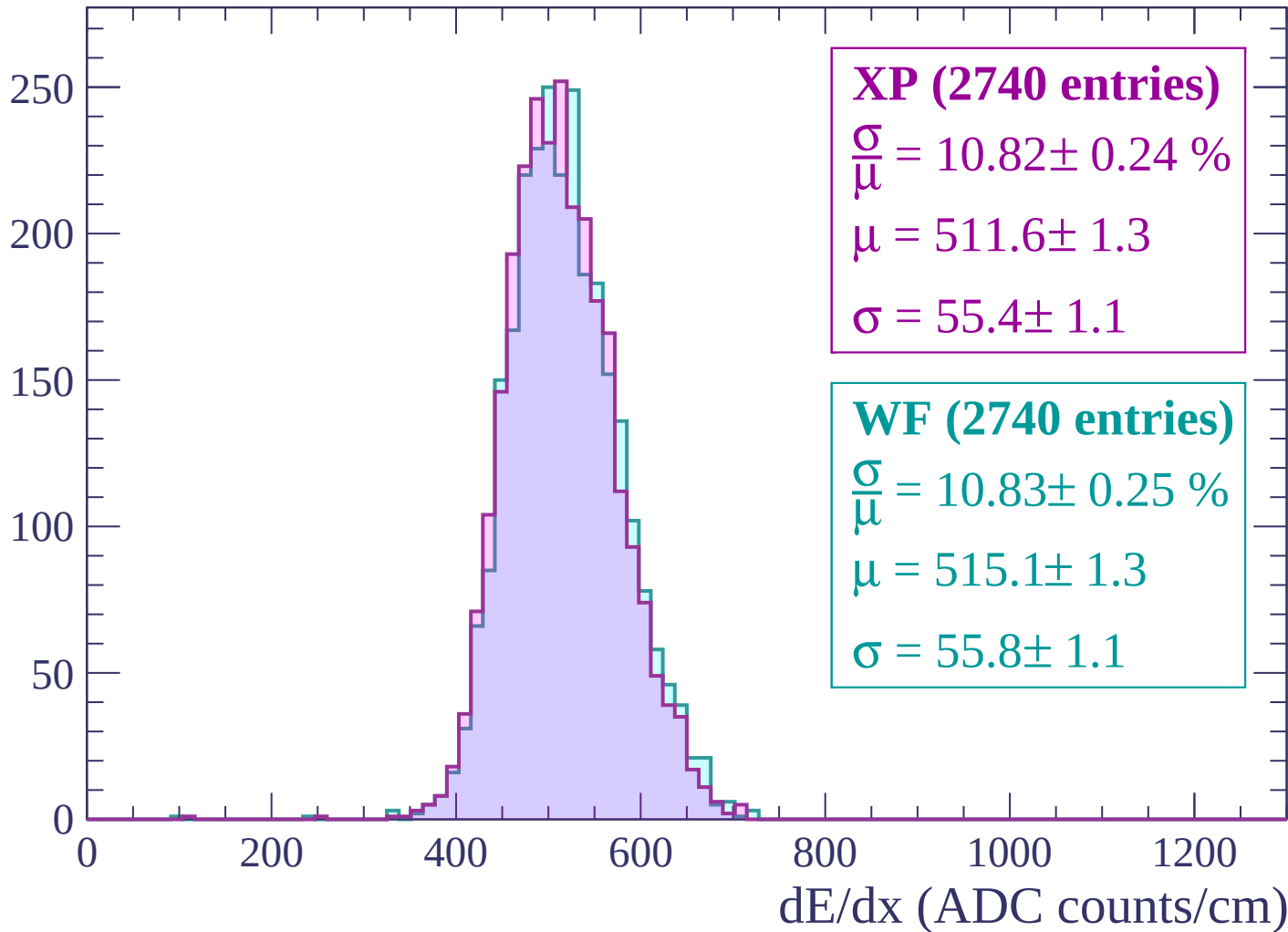
Energy loss | -2520 < p < -2460

Count



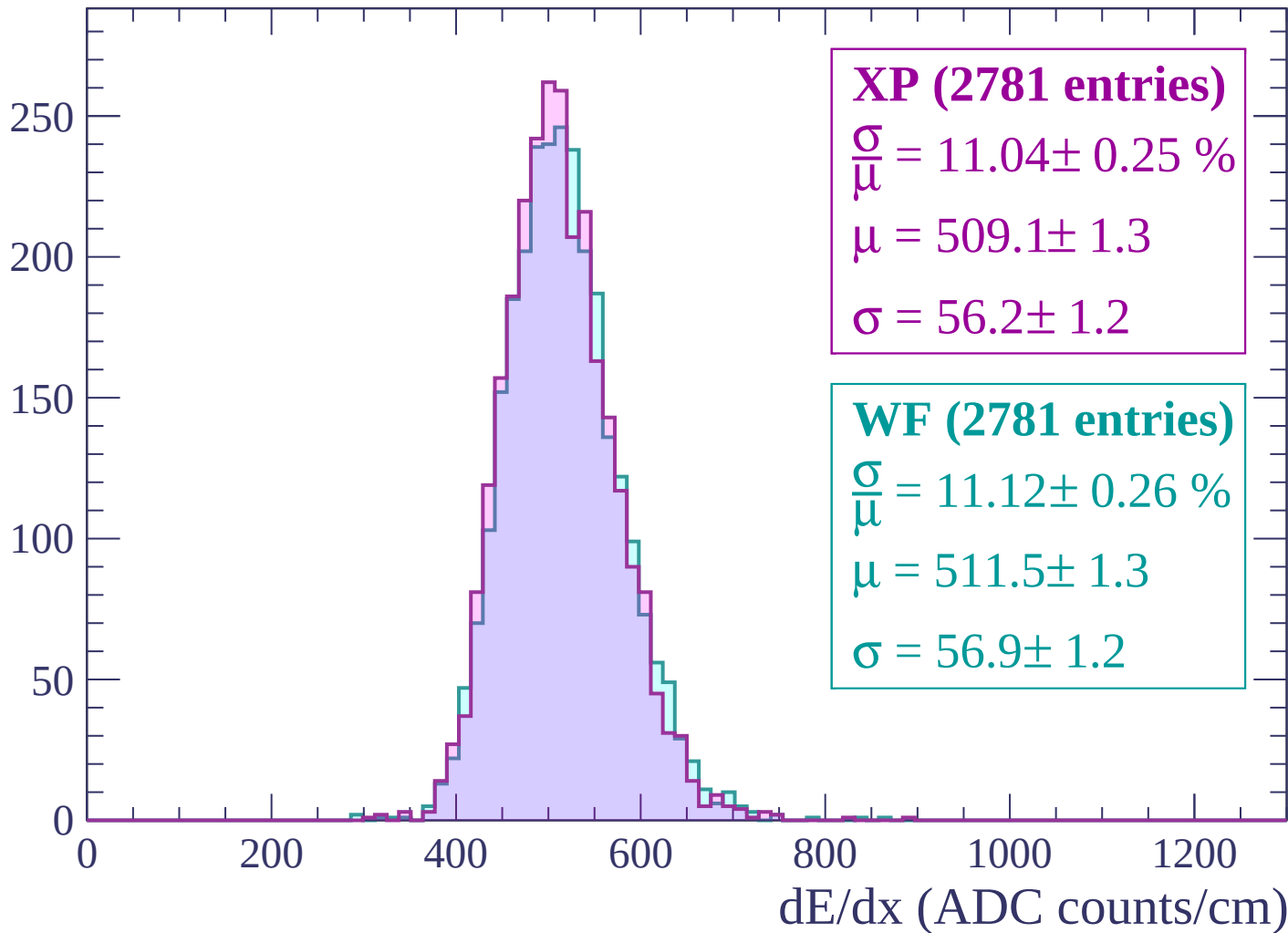
Energy loss | $-2460 < p < -2400$

Count



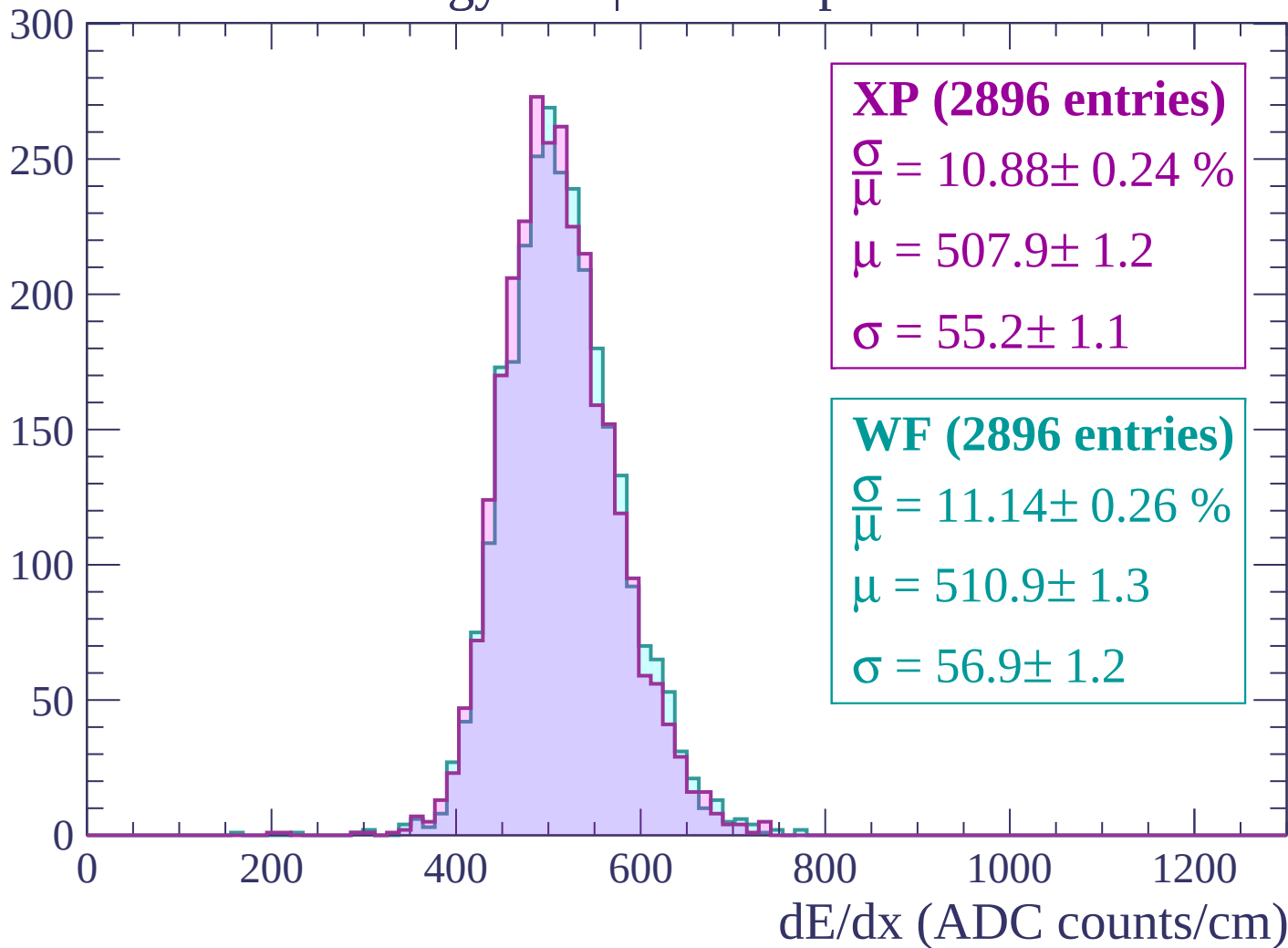
Energy loss | -2400 < p < -2340

Count



Energy loss | $-2340 < p < -2280$

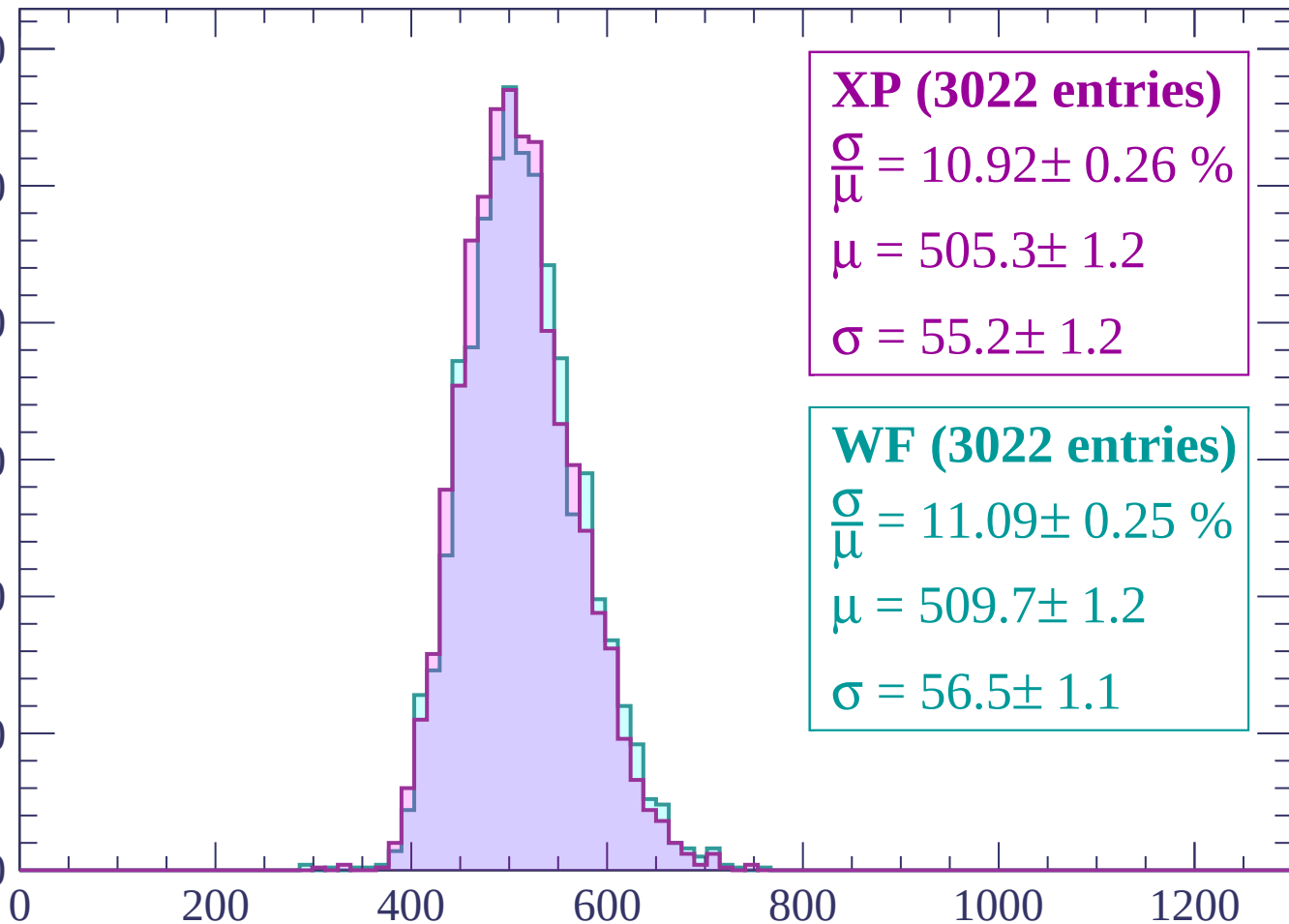
Count



Energy loss | $-2280 < p < -2220$

Count

300
250
200
150
100
50
0



XP (3022 entries)

$\frac{\sigma}{\mu} = 10.92 \pm 0.26 \%$

$\mu = 505.3 \pm 1.2$

$\sigma = 55.2 \pm 1.2$

WF (3022 entries)

$\frac{\sigma}{\mu} = 11.09 \pm 0.25 \%$

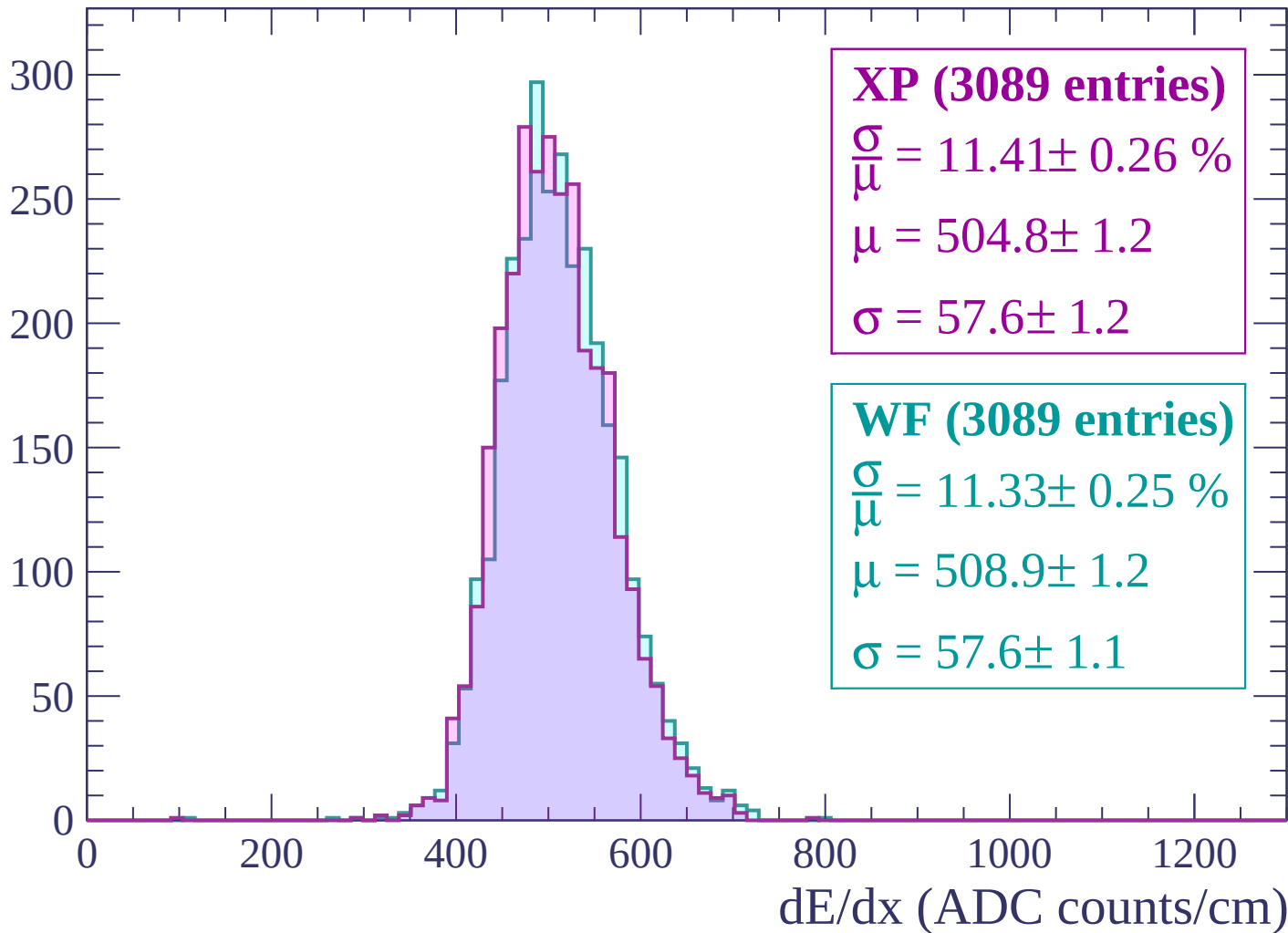
$\mu = 509.7 \pm 1.2$

$\sigma = 56.5 \pm 1.1$

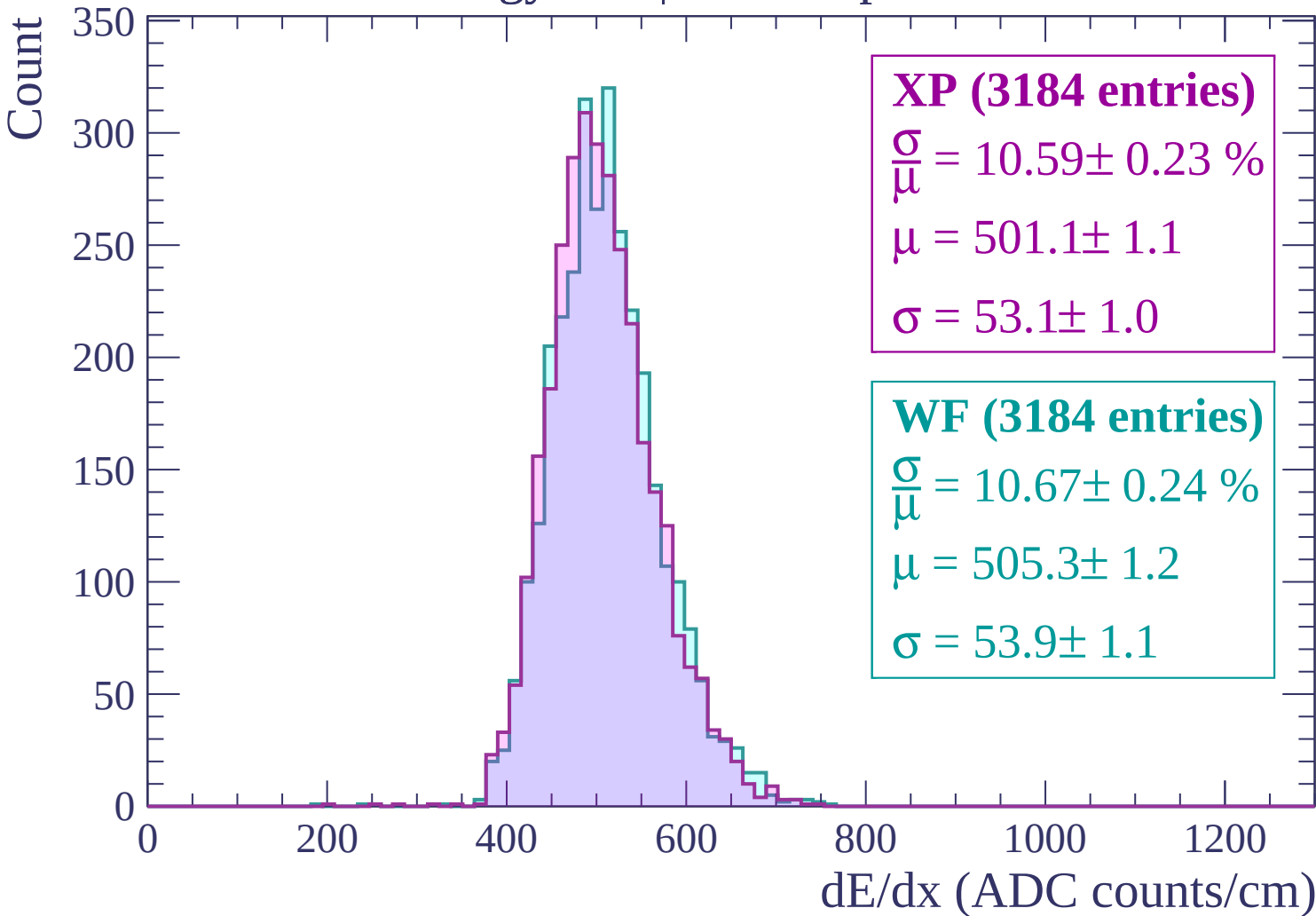
dE/dx (ADC counts/cm)

Energy loss | -2220 < p < -2160

Count

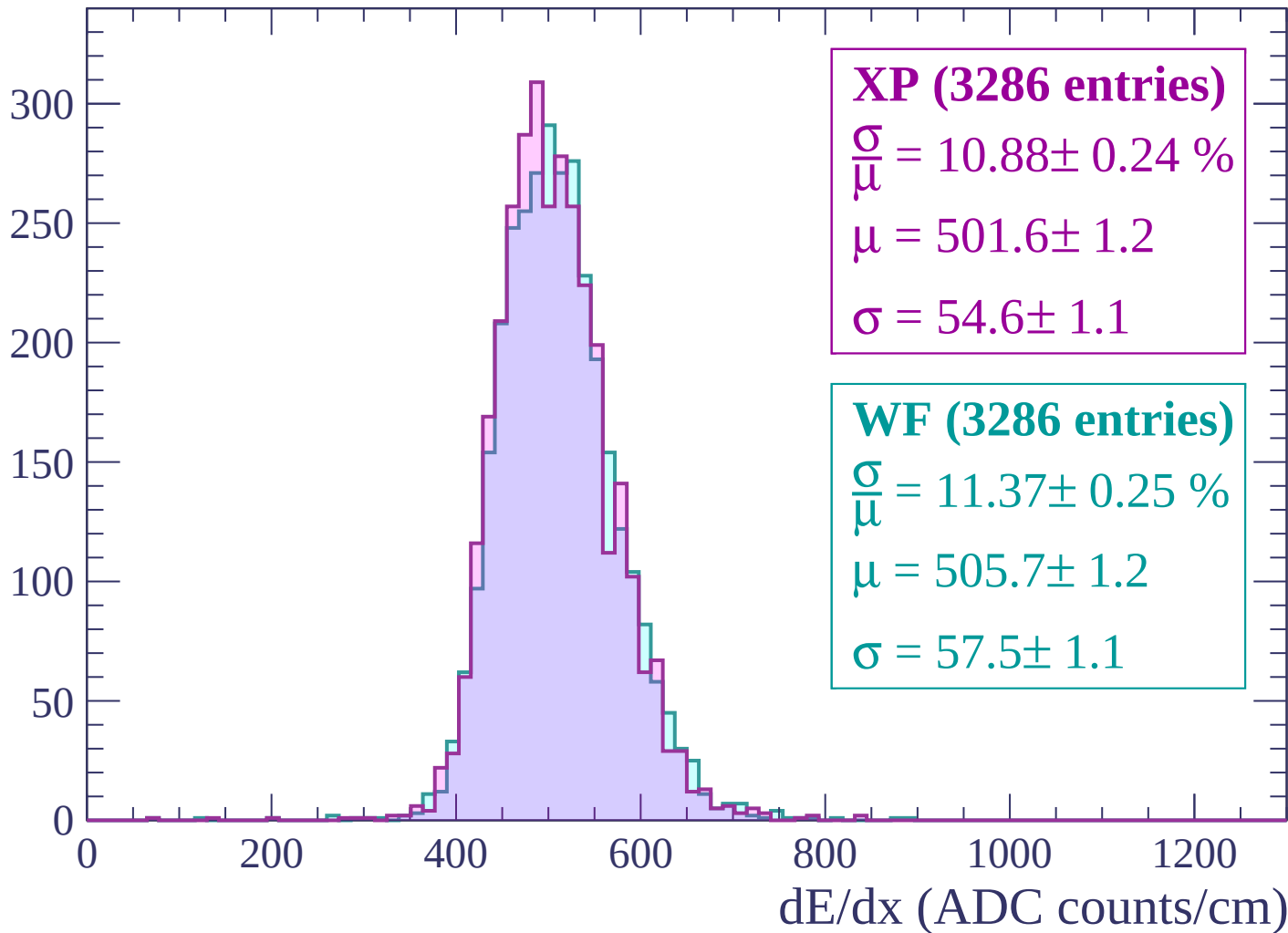


Energy loss | -2160 < p < -2100



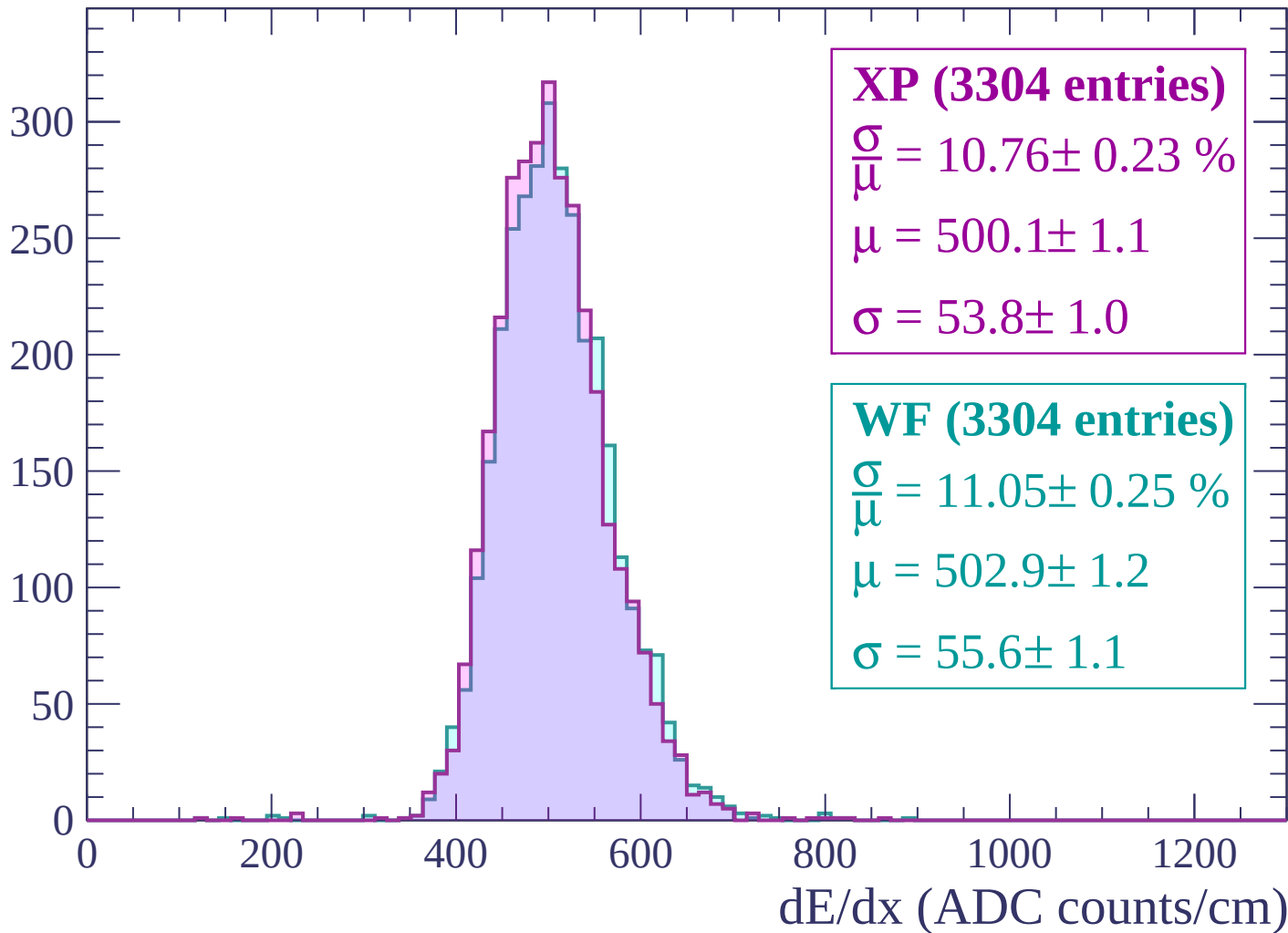
Energy loss | -2100 < p < -2040

Count

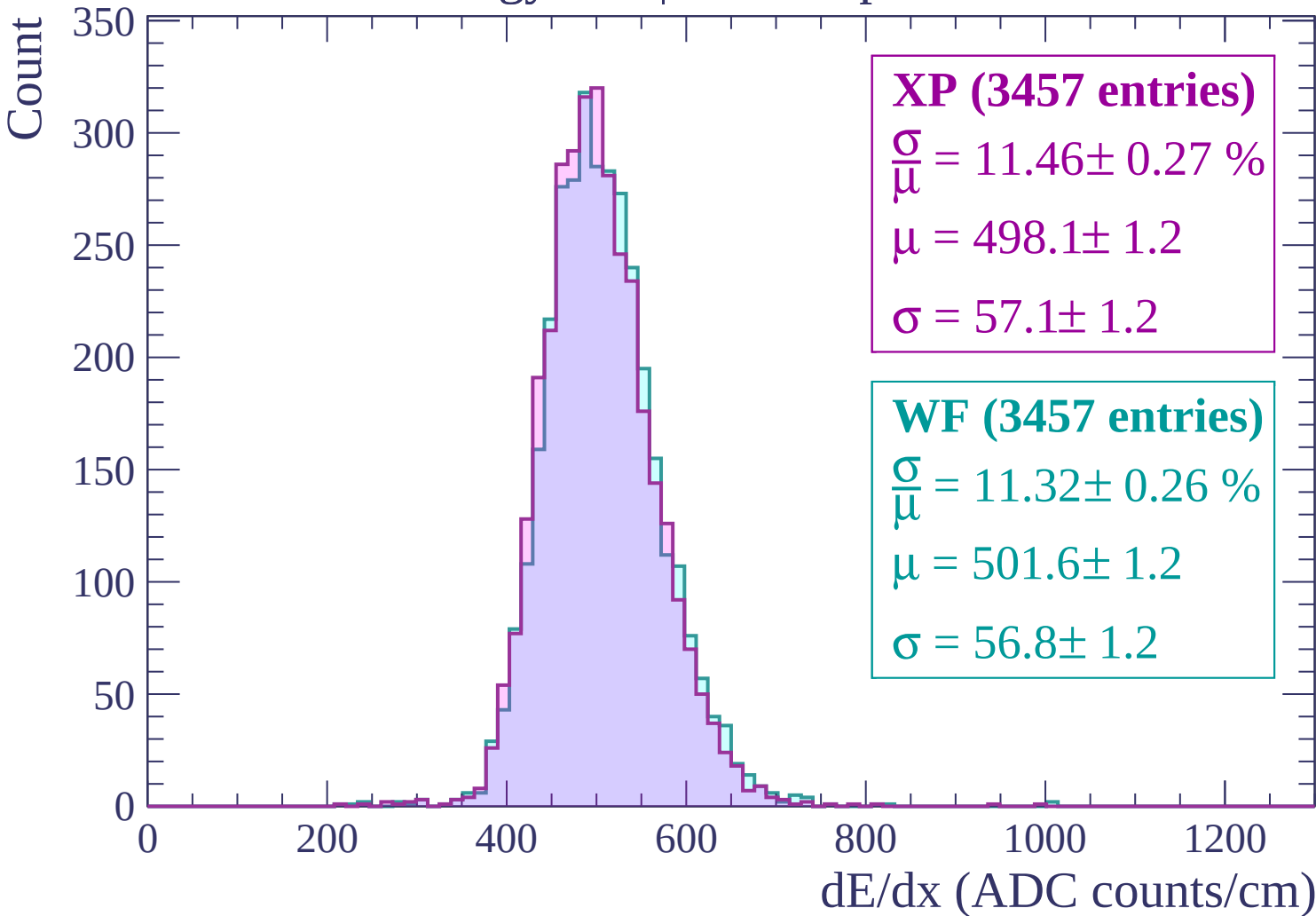


Energy loss | -2040 < p < -1980

Count

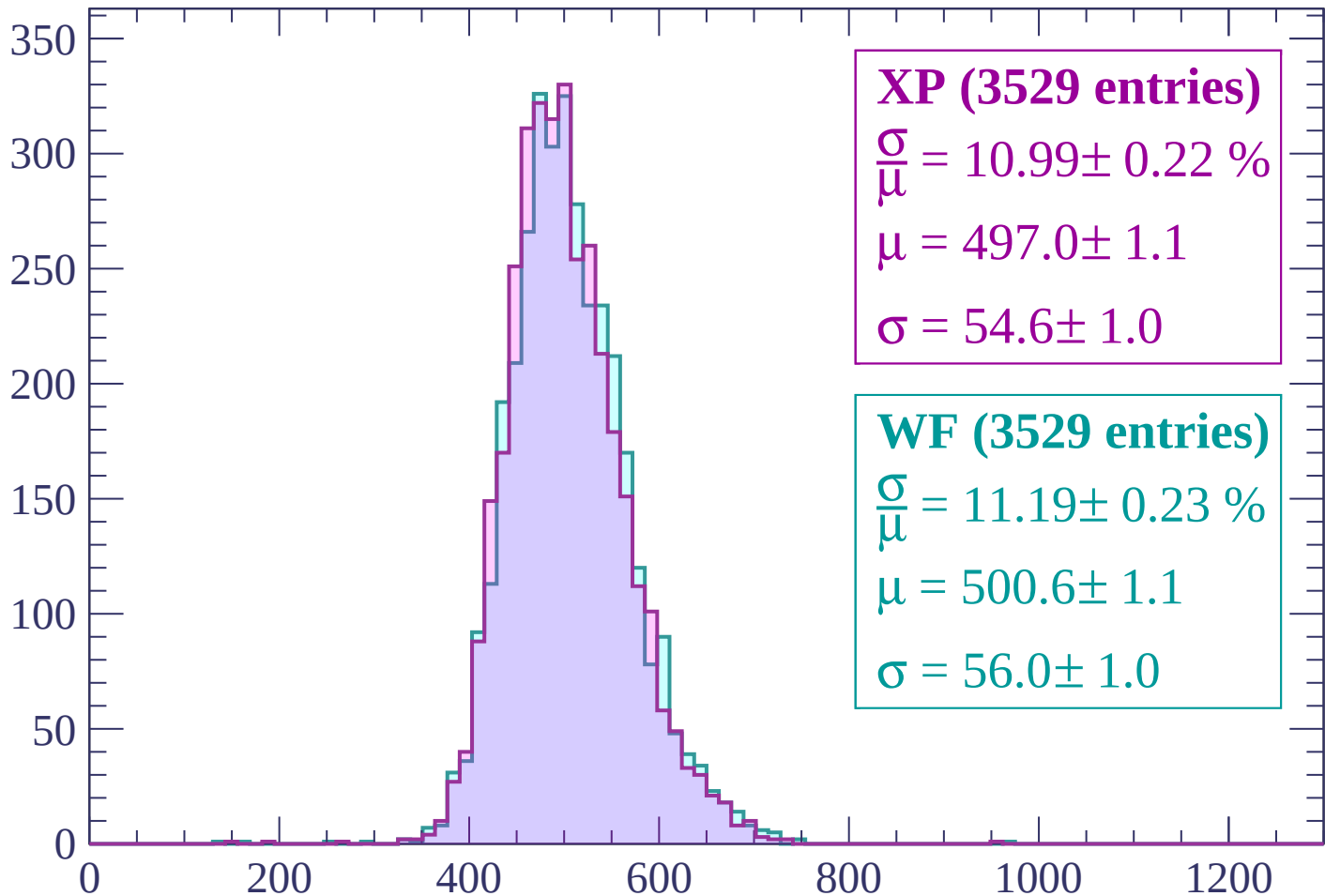


Energy loss | -1980 < p < -1920



Energy loss | -1920 < p < -1860

Count



XP (3529 entries)

$$\frac{\sigma}{\mu} = 10.99 \pm 0.22 \%$$

$$\mu = 497.0 \pm 1.1$$

$$\sigma = 54.6 \pm 1.0$$

WF (3529 entries)

$$\frac{\sigma}{\mu} = 11.19 \pm 0.23 \%$$

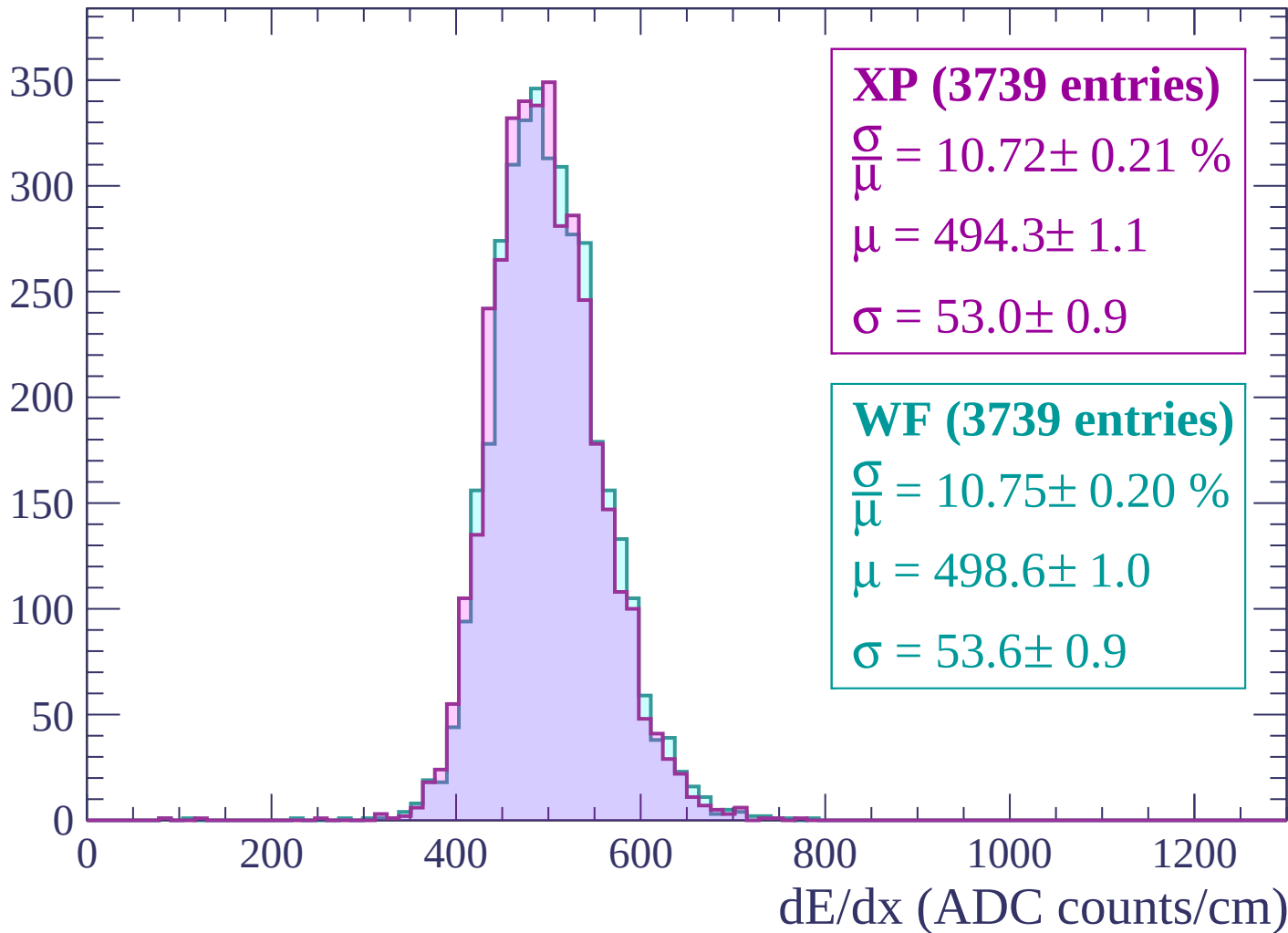
$$\mu = 500.6 \pm 1.1$$

$$\sigma = 56.0 \pm 1.0$$

dE/dx (ADC counts/cm)

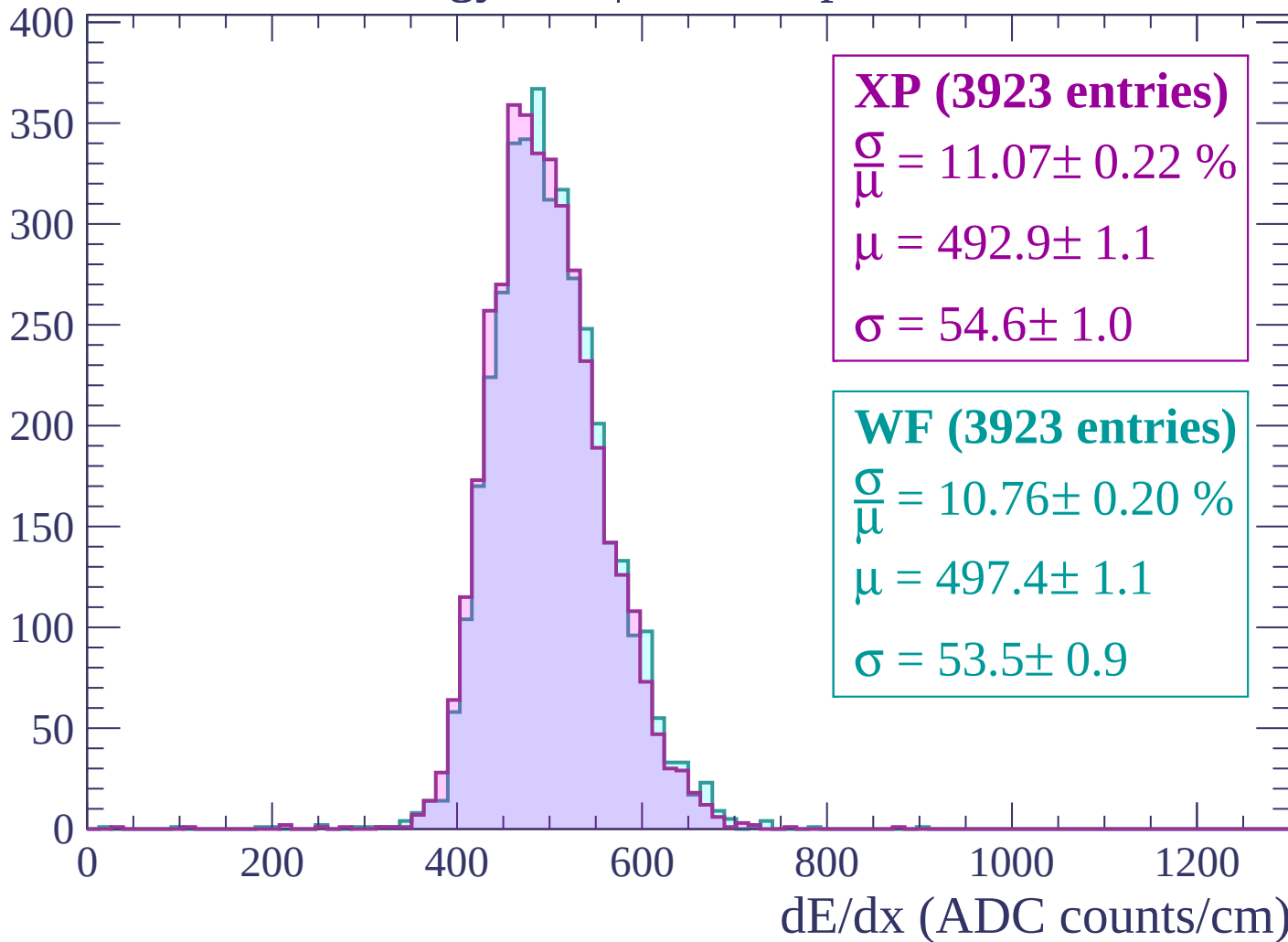
Energy loss | -1860 < p < -1800

Count



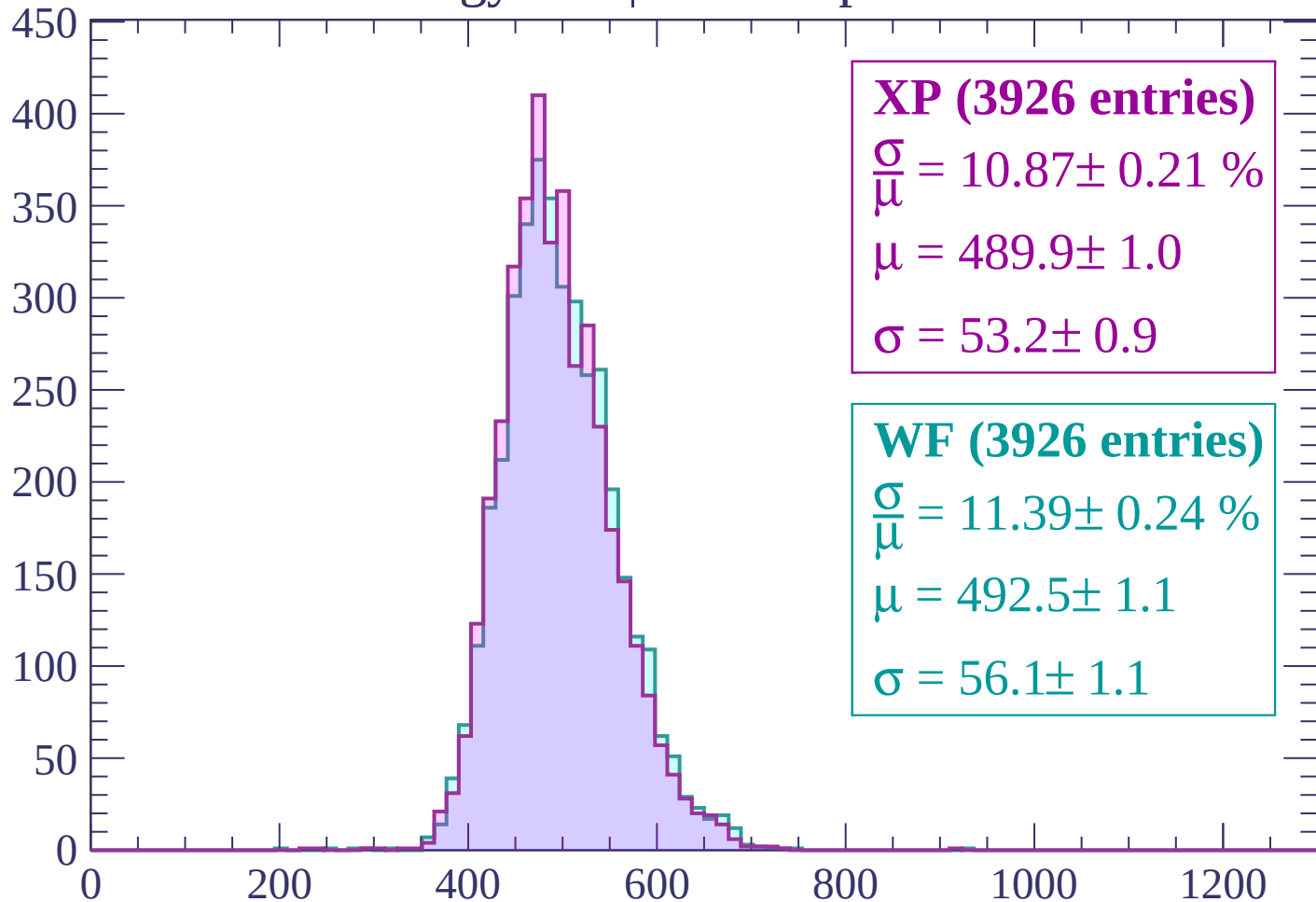
Energy loss | -1800 < p < -1740

Count



Energy loss | $-1740 < p < -1680$

Count



XP (3926 entries)

$\frac{\sigma}{\mu} = 10.87 \pm 0.21 \%$

$\mu = 489.9 \pm 1.0$

$\sigma = 53.2 \pm 0.9$

WF (3926 entries)

$\frac{\sigma}{\mu} = 11.39 \pm 0.24 \%$

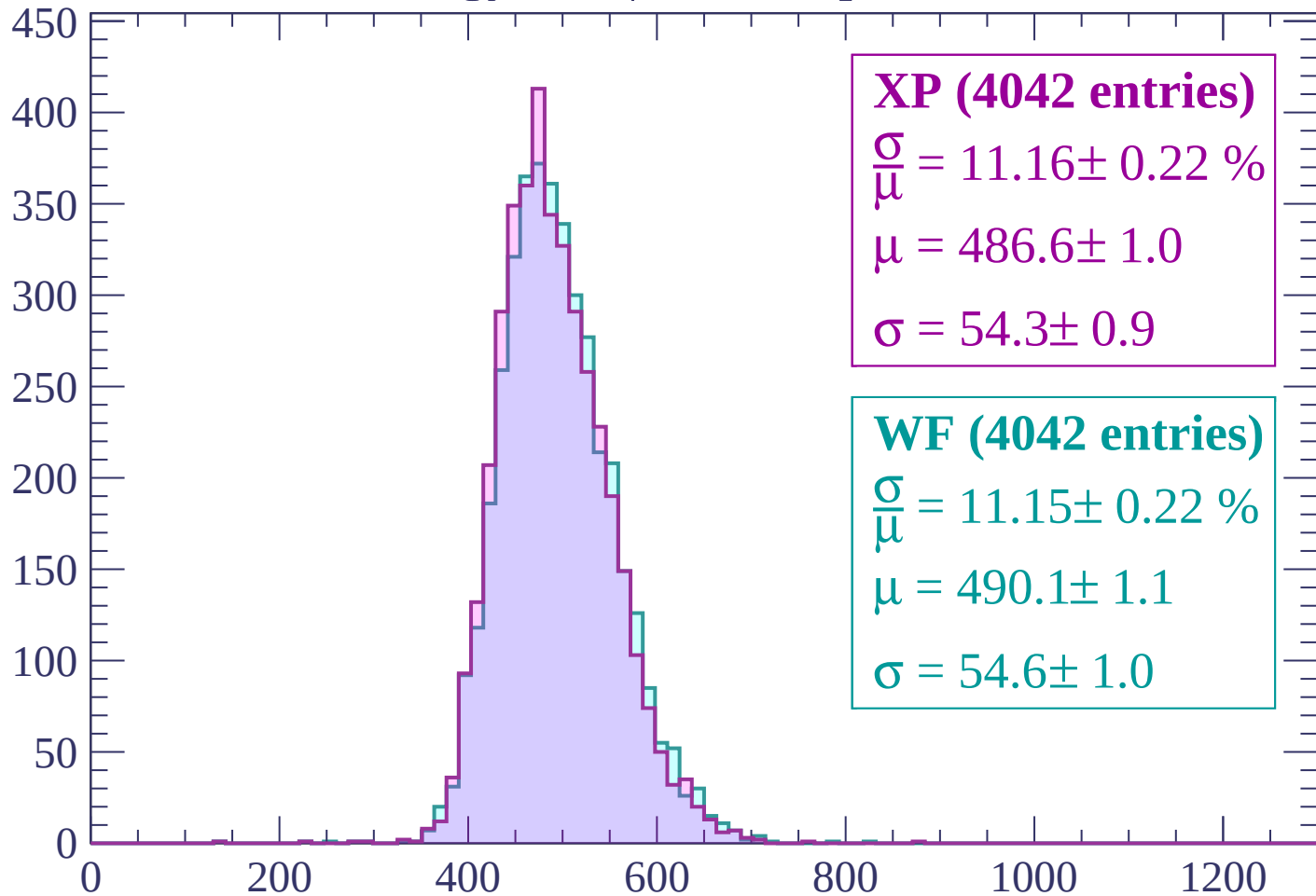
$\mu = 492.5 \pm 1.1$

$\sigma = 56.1 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss | -1680 < p < -1620

Count



XP (4042 entries)

$$\frac{\sigma}{\mu} = 11.16 \pm 0.22 \%$$

$$\mu = 486.6 \pm 1.0$$

$$\sigma = 54.3 \pm 0.9$$

WF (4042 entries)

$$\frac{\sigma}{\mu} = 11.15 \pm 0.22 \%$$

$$\mu = 490.1 \pm 1.1$$

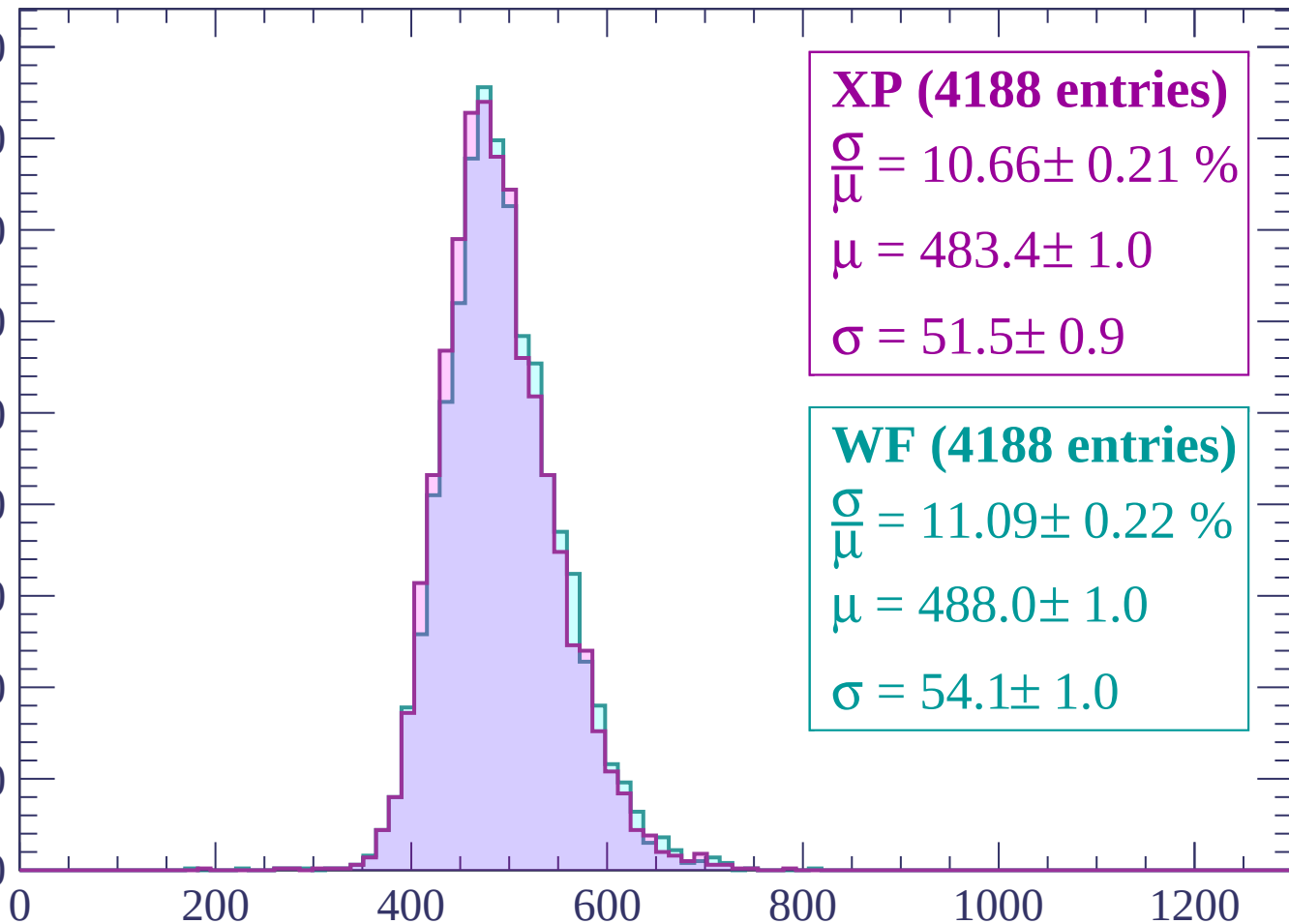
$$\sigma = 54.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1620 < p < -1560

Count

450
400
350
300
250
200
150
100
50
0



XP (4188 entries)

$\frac{\sigma}{\mu} = 10.66 \pm 0.21 \%$

$\mu = 483.4 \pm 1.0$

$\sigma = 51.5 \pm 0.9$

WF (4188 entries)

$\frac{\sigma}{\mu} = 11.09 \pm 0.22 \%$

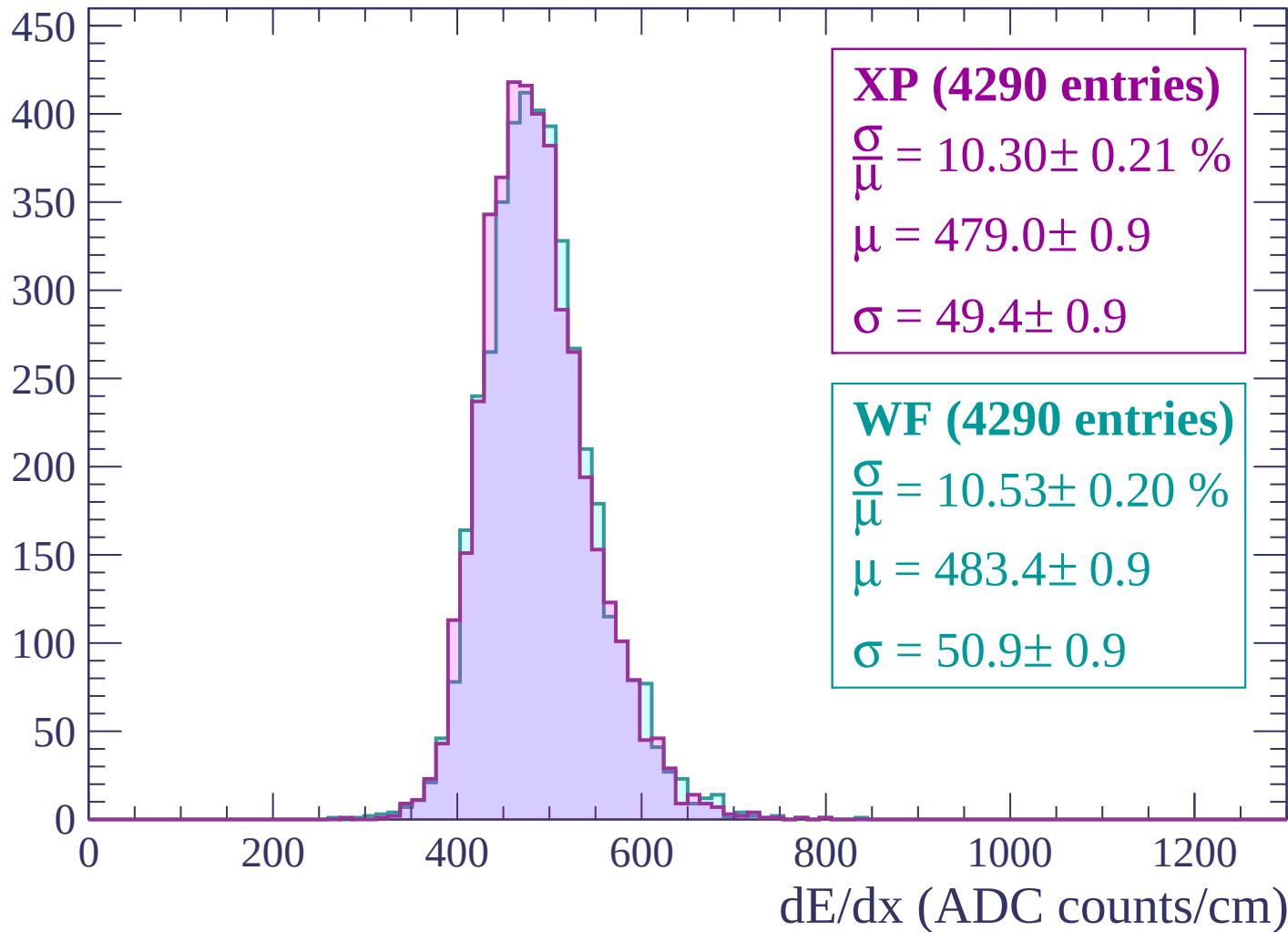
$\mu = 488.0 \pm 1.0$

$\sigma = 54.1 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1500

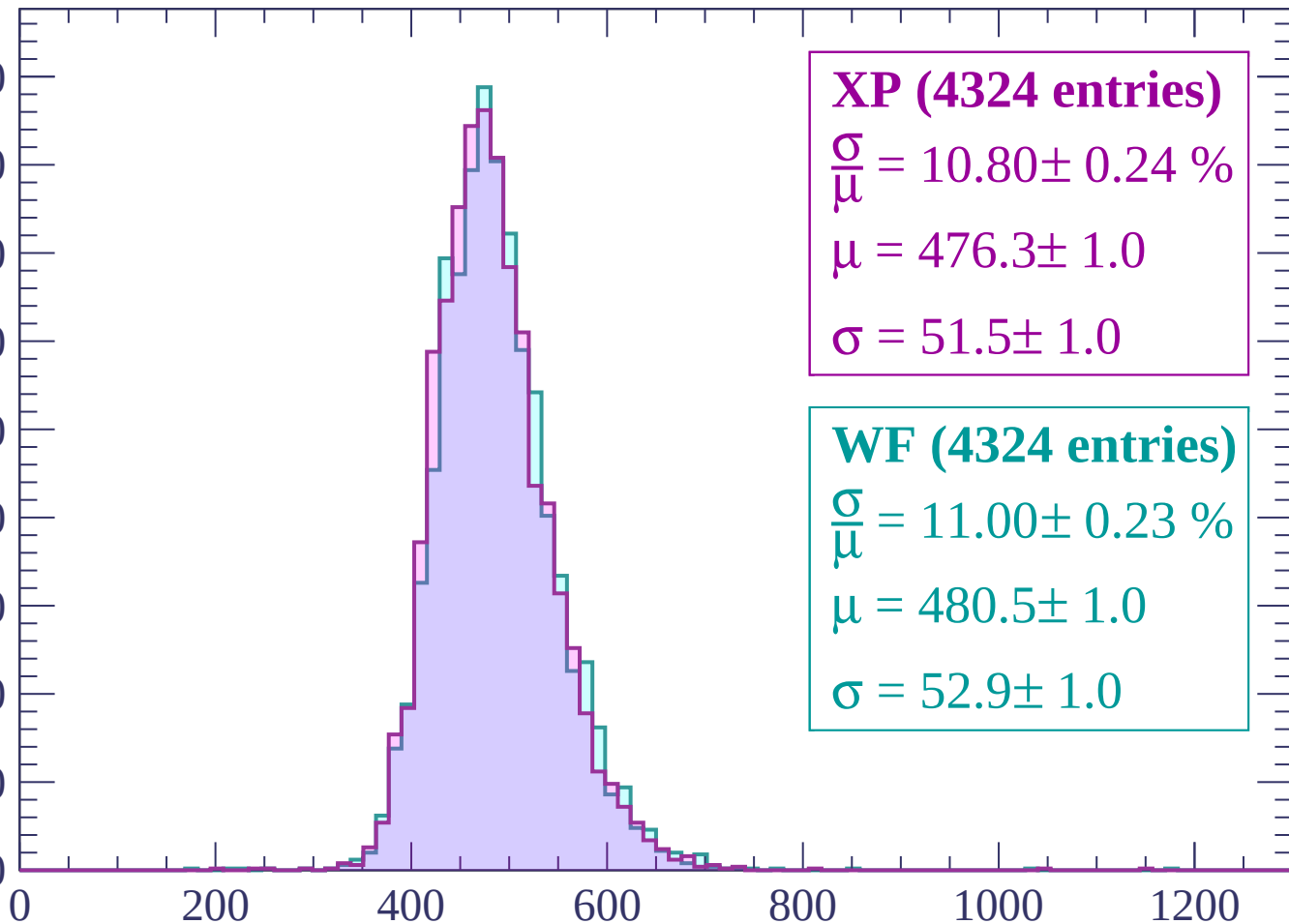
Count



Energy loss | -1500 < p < -1440

Count

450
400
350
300
250
200
150
100
50
0



XP (4324 entries)

$\frac{\sigma}{\mu} = 10.80 \pm 0.24 \%$

$\mu = 476.3 \pm 1.0$

$\sigma = 51.5 \pm 1.0$

WF (4324 entries)

$\frac{\sigma}{\mu} = 11.00 \pm 0.23 \%$

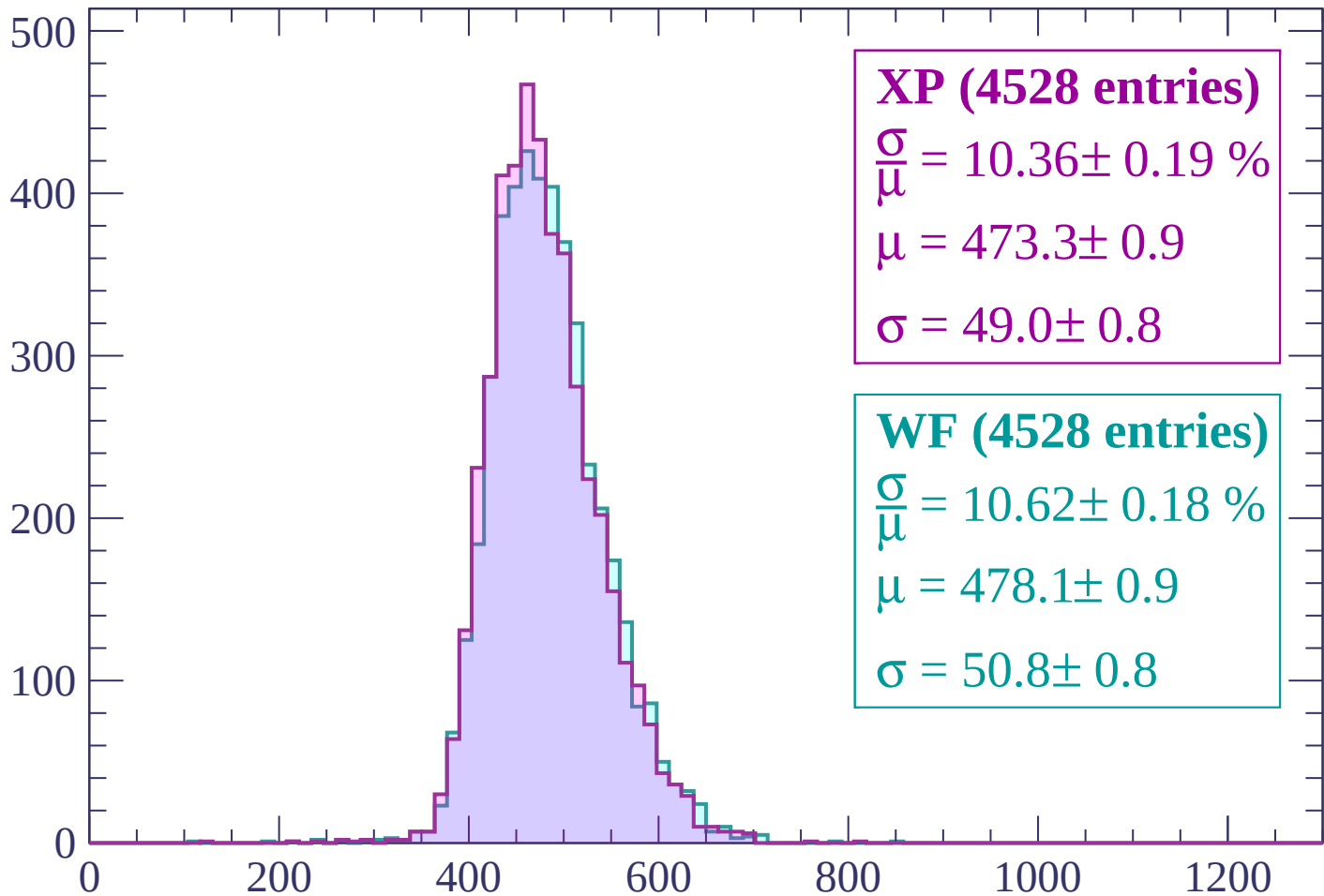
$\mu = 480.5 \pm 1.0$

$\sigma = 52.9 \pm 1.0$

dE/dx (ADC counts/cm)

Energy loss | -1440 < p < -1380

Count



XP (4528 entries)

$\frac{\sigma}{\mu} = 10.36 \pm 0.19 \%$

$\mu = 473.3 \pm 0.9$

$\sigma = 49.0 \pm 0.8$

WF (4528 entries)

$\frac{\sigma}{\mu} = 10.62 \pm 0.18 \%$

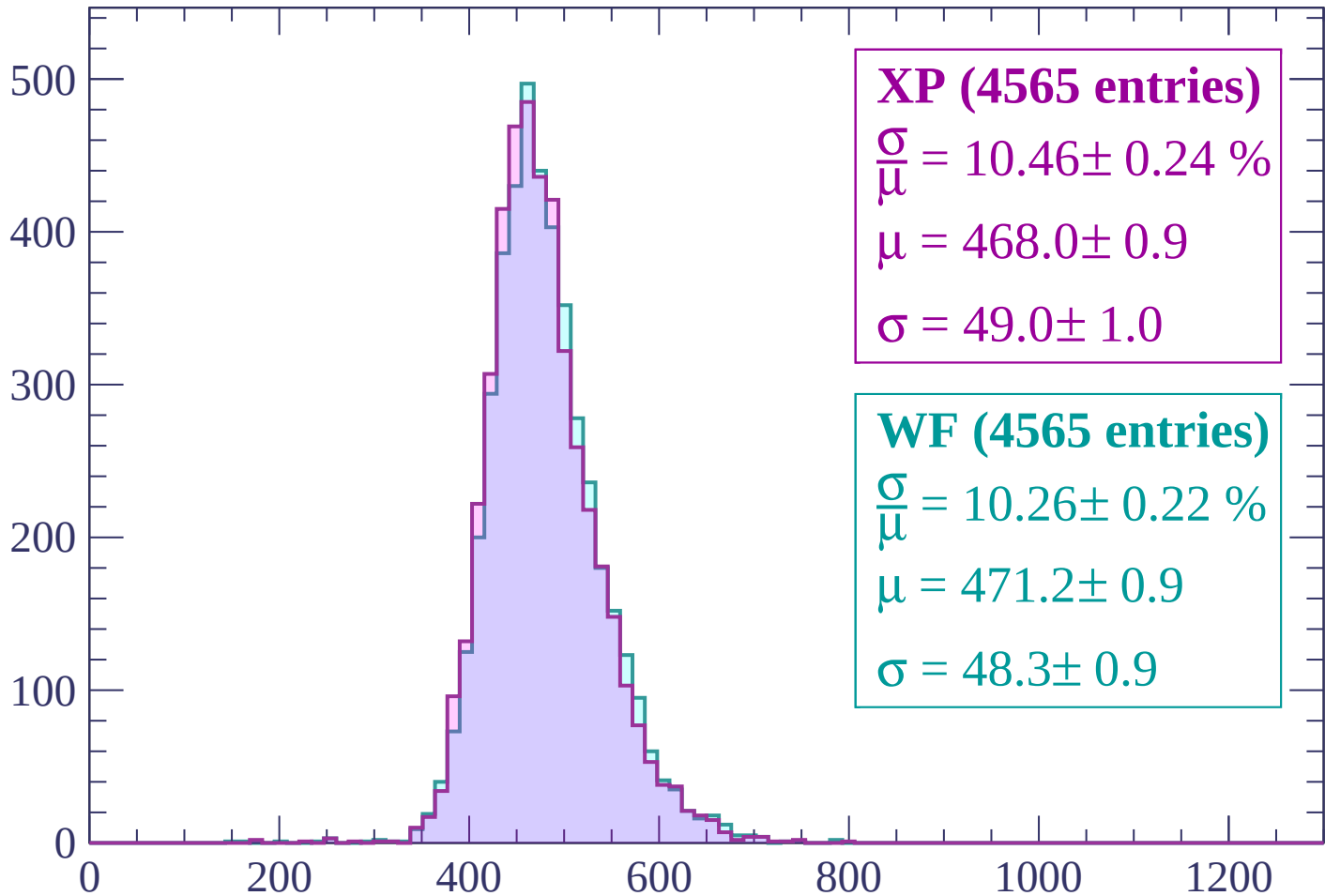
$\mu = 478.1 \pm 0.9$

$\sigma = 50.8 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | -1380 < p < -1320

Count



XP (4565 entries)

$\frac{\sigma}{\mu} = 10.46 \pm 0.24 \%$

$\mu = 468.0 \pm 0.9$

$\sigma = 49.0 \pm 1.0$

WF (4565 entries)

$\frac{\sigma}{\mu} = 10.26 \pm 0.22 \%$

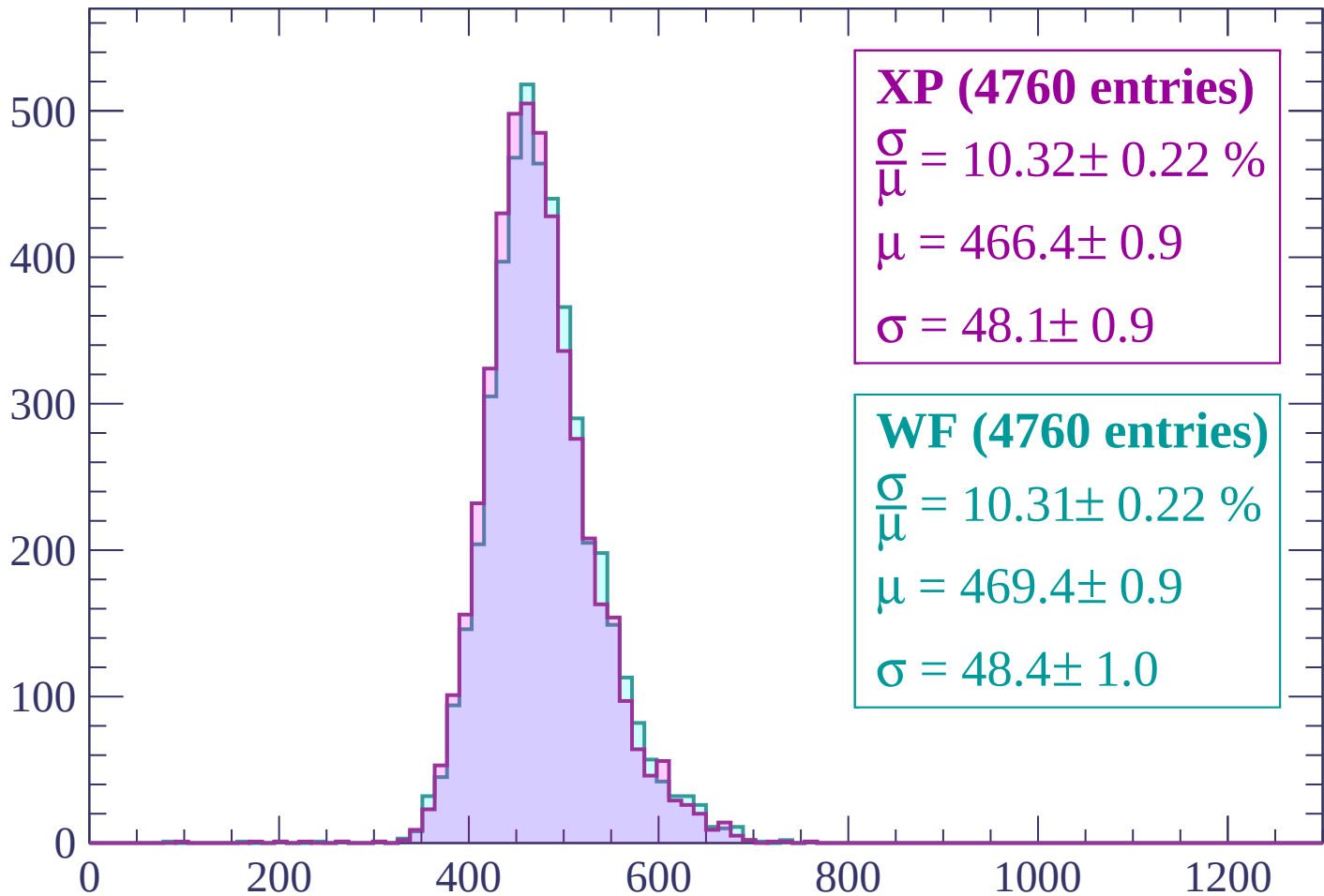
$\mu = 471.2 \pm 0.9$

$\sigma = 48.3 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | -1320 < p < -1260

Count



XP (4760 entries)

$$\frac{\sigma}{\mu} = 10.32 \pm 0.22 \%$$

$$\mu = 466.4 \pm 0.9$$

$$\sigma = 48.1 \pm 0.9$$

WF (4760 entries)

$$\frac{\sigma}{\mu} = 10.31 \pm 0.22 \%$$

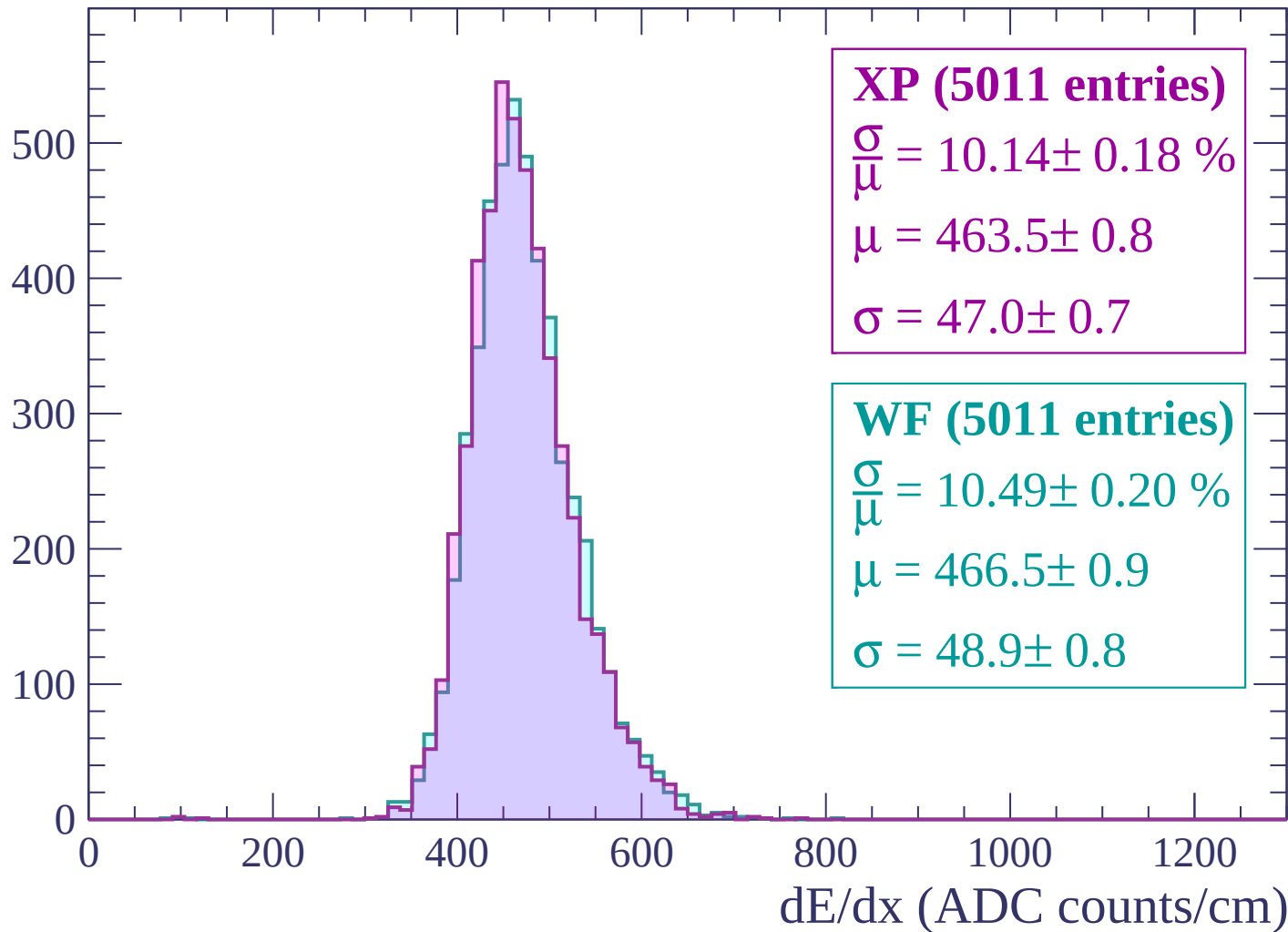
$$\mu = 469.4 \pm 0.9$$

$$\sigma = 48.4 \pm 1.0$$

dE/dx (ADC counts/cm)

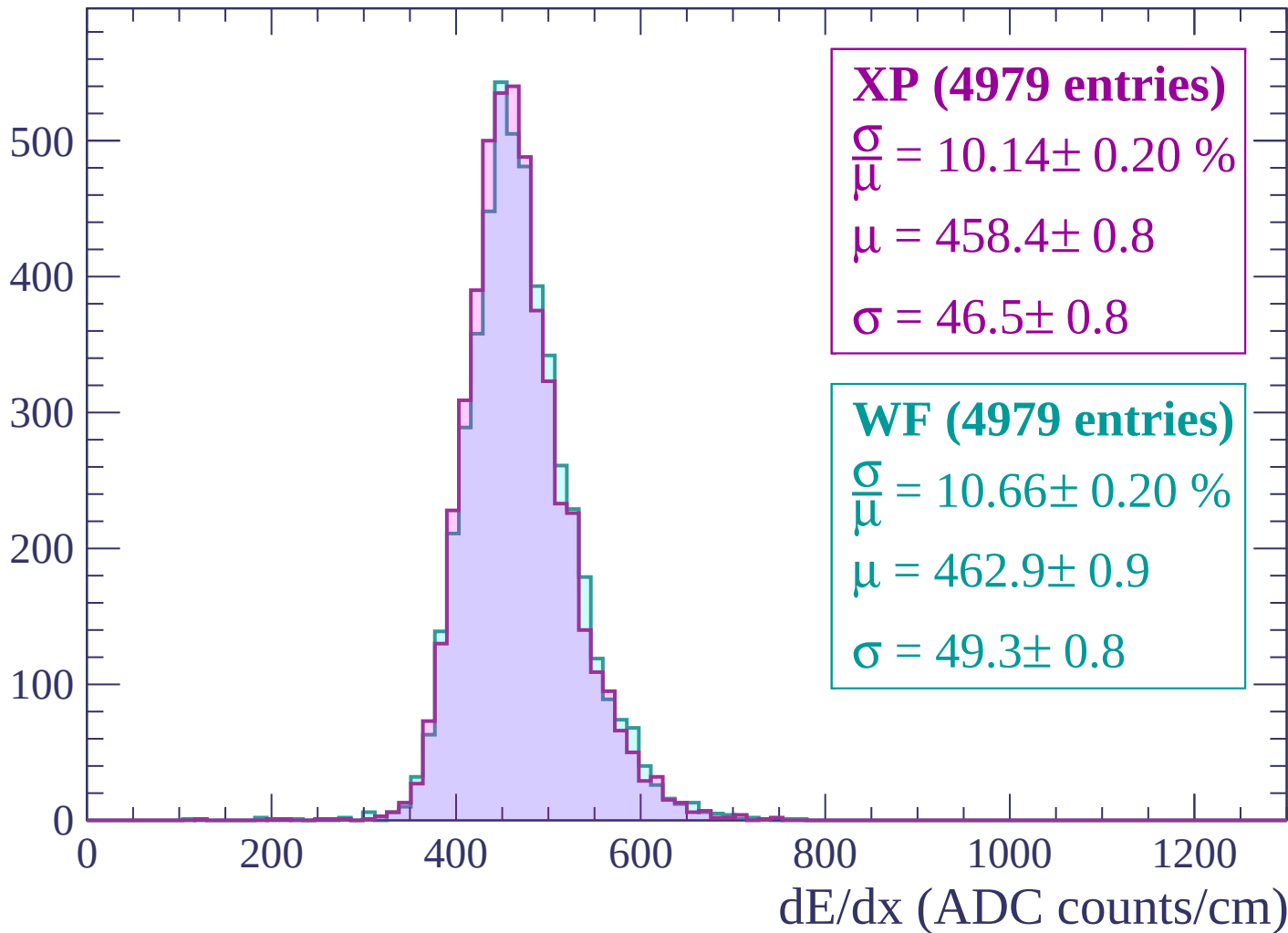
Energy loss | $-1260 < p < -1200$

Count



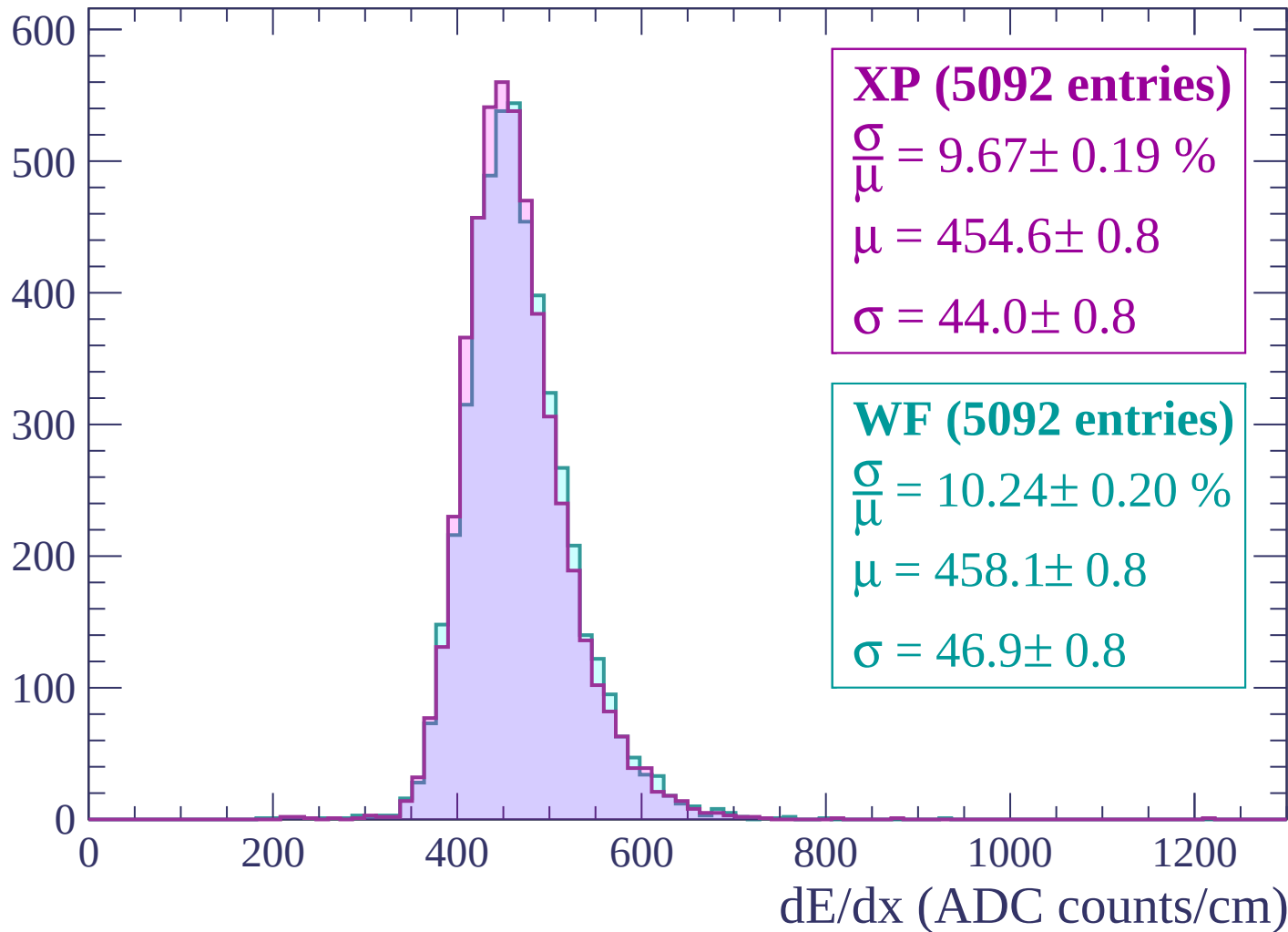
Energy loss | $-1200 < p < -1140$

Count



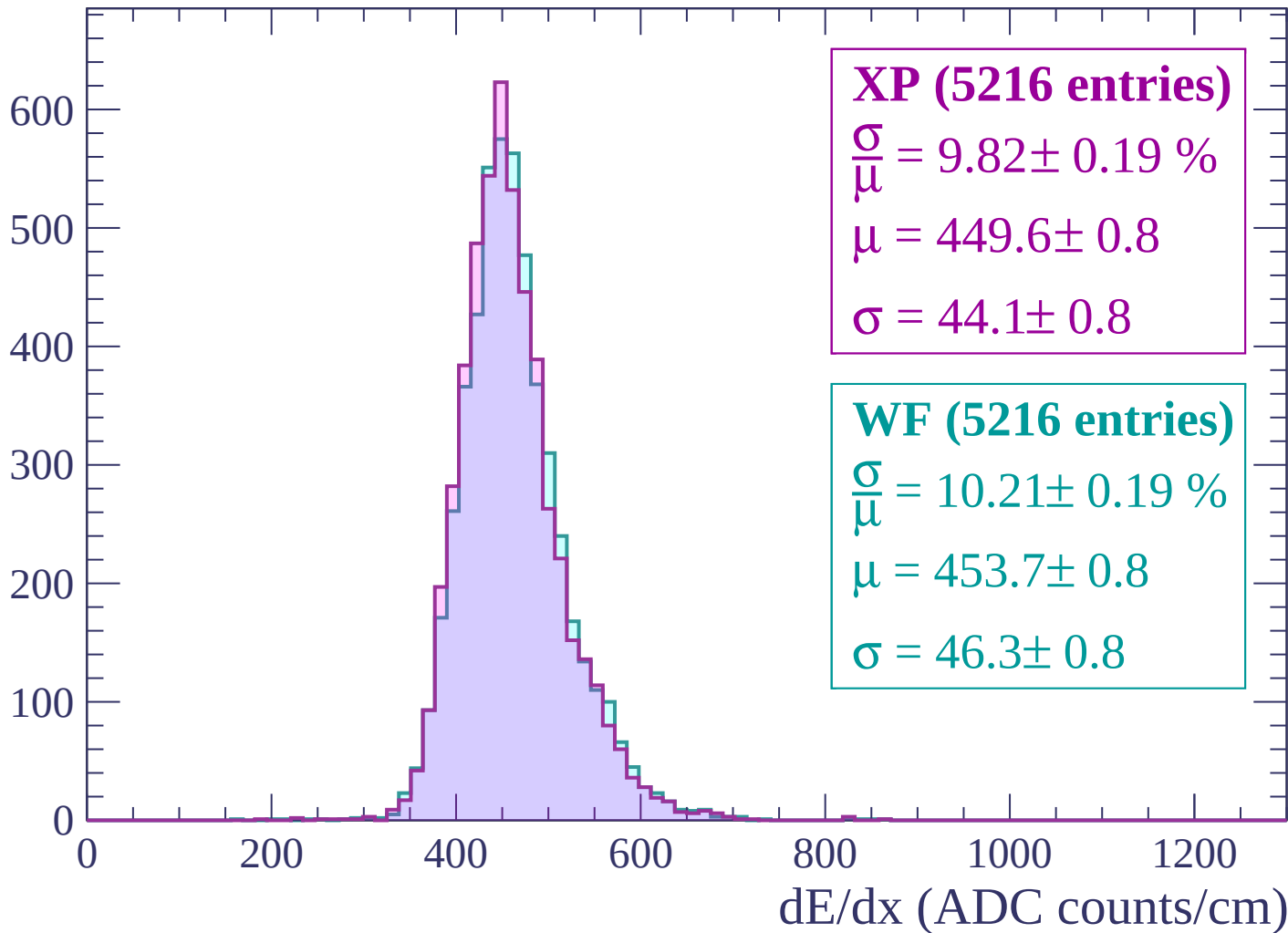
Energy loss | $-1140 < p < -1080$

Count



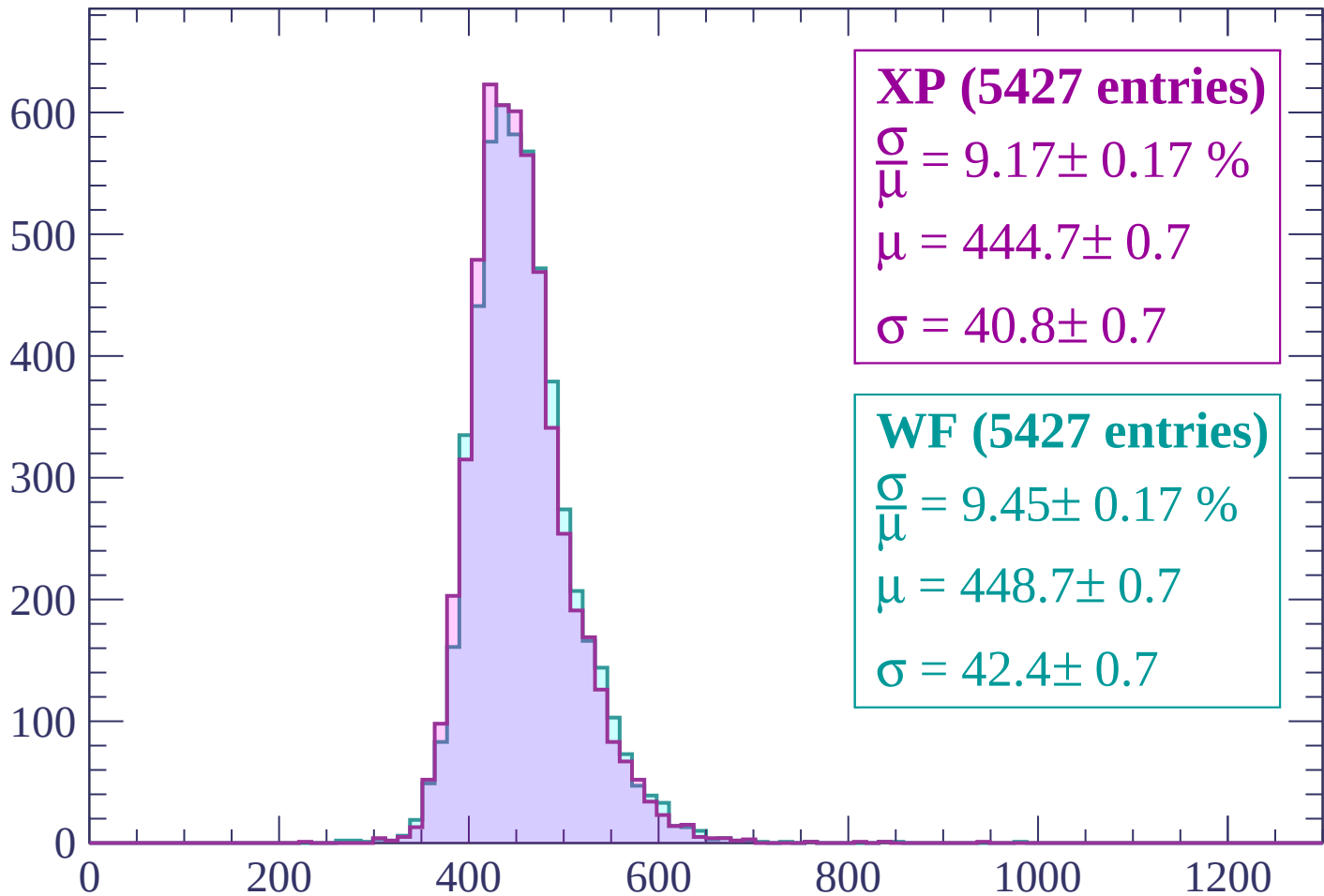
Energy loss | $-1080 < p < -1020$

Count



Energy loss | $-1020 < p < -960$

Count



XP (5427 entries)

$$\frac{\sigma}{\mu} = 9.17 \pm 0.17 \%$$

$$\mu = 444.7 \pm 0.7$$

$$\sigma = 40.8 \pm 0.7$$

WF (5427 entries)

$$\frac{\sigma}{\mu} = 9.45 \pm 0.17 \%$$

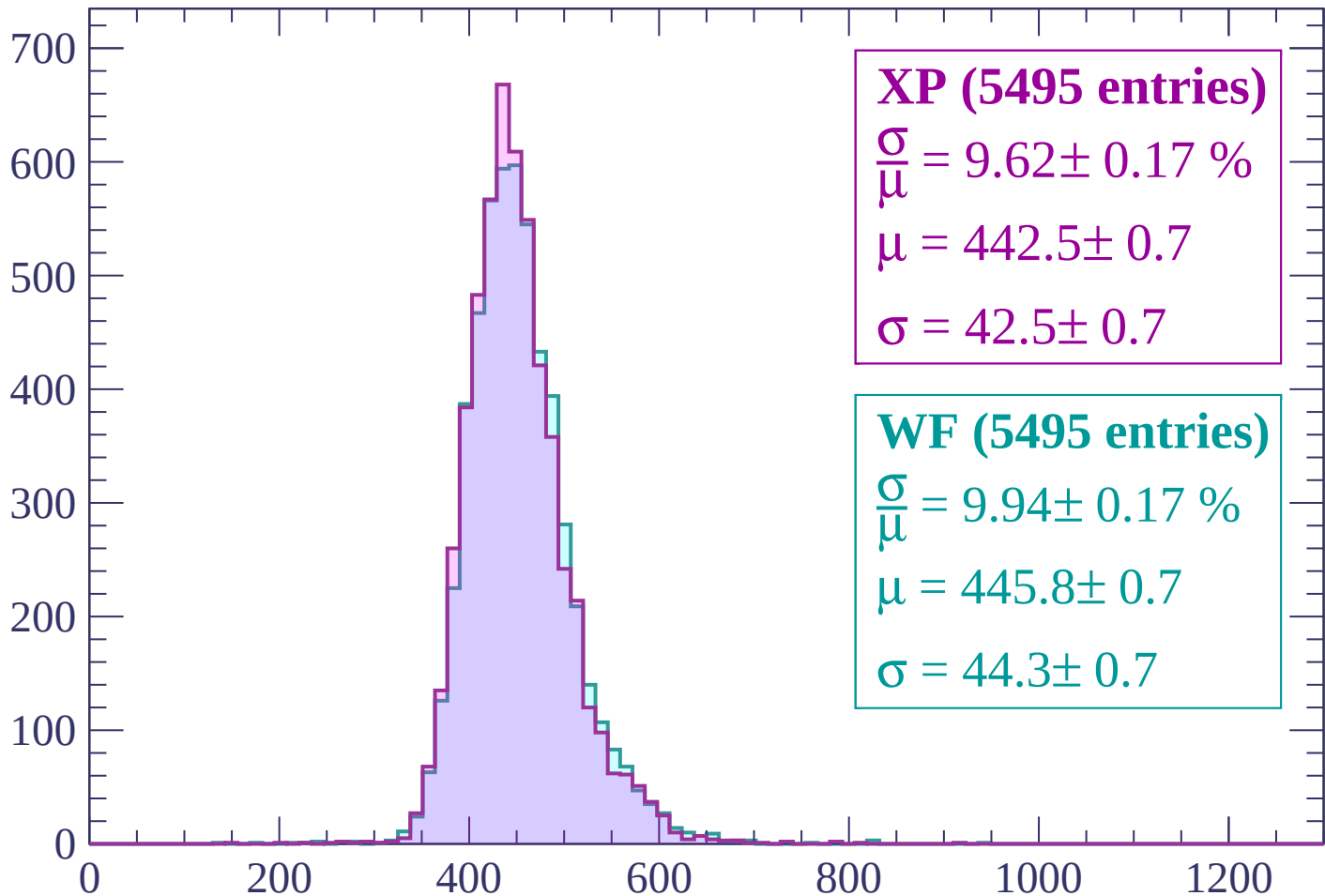
$$\mu = 448.7 \pm 0.7$$

$$\sigma = 42.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -900$

Count



XP (5495 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.17 \%$$

$$\mu = 442.5 \pm 0.7$$

$$\sigma = 42.5 \pm 0.7$$

WF (5495 entries)

$$\frac{\sigma}{\mu} = 9.94 \pm 0.17 \%$$

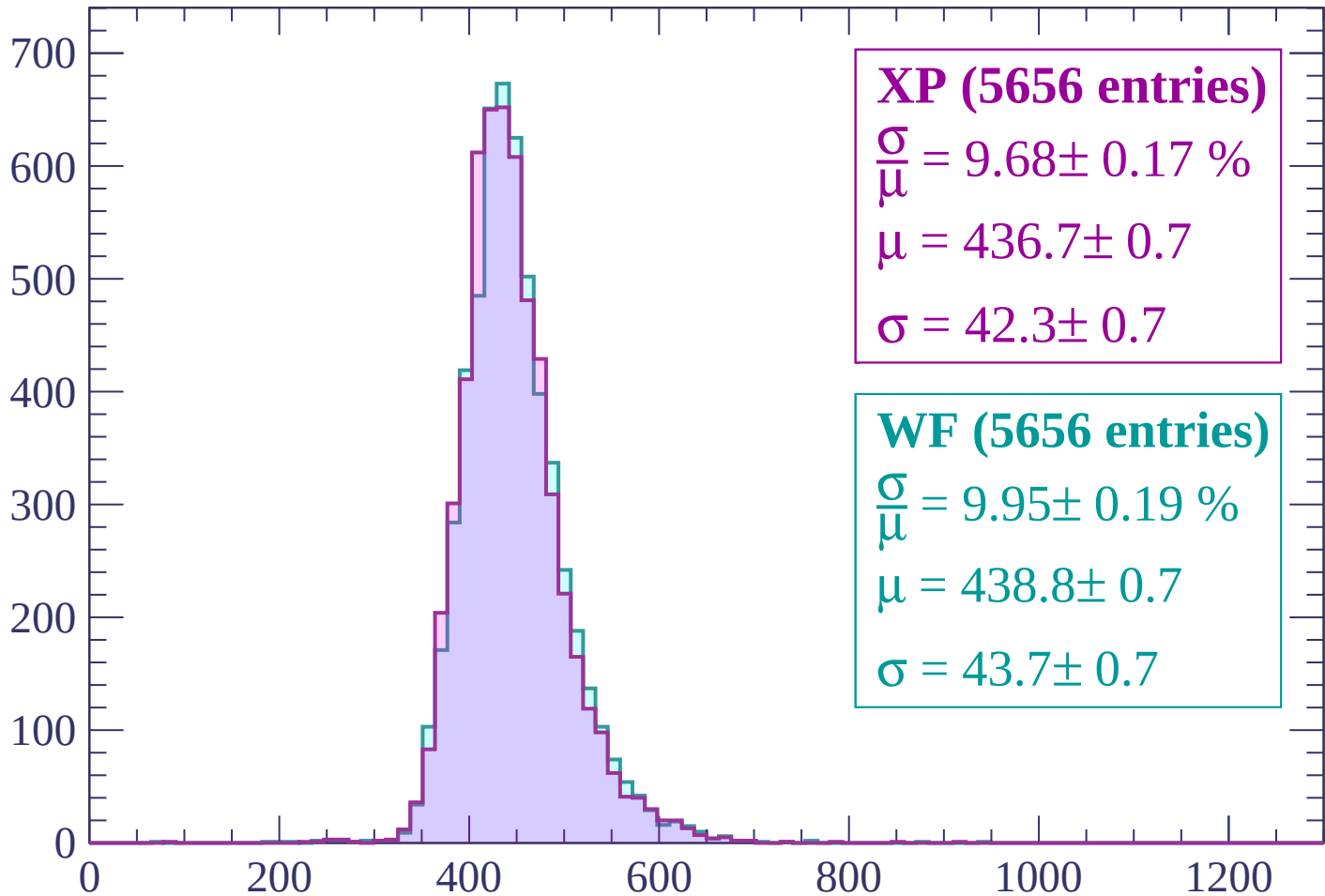
$$\mu = 445.8 \pm 0.7$$

$$\sigma = 44.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-900 < p < -840$

Count



XP (5656 entries)

$$\frac{\sigma}{\mu} = 9.68 \pm 0.17 \%$$

$$\mu = 436.7 \pm 0.7$$

$$\sigma = 42.3 \pm 0.7$$

WF (5656 entries)

$$\frac{\sigma}{\mu} = 9.95 \pm 0.19 \%$$

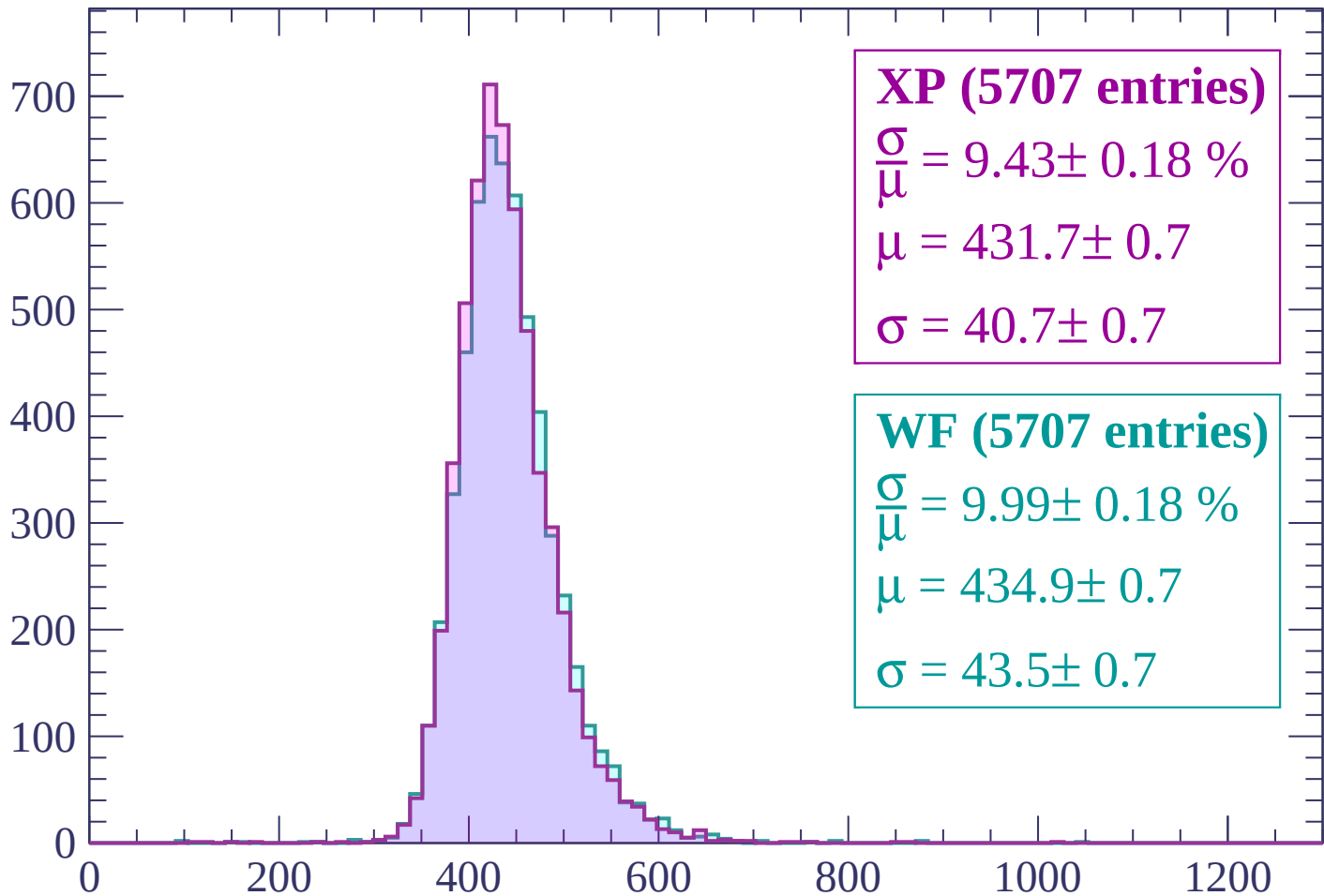
$$\mu = 438.8 \pm 0.7$$

$$\sigma = 43.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -780$

Count



XP (5707 entries)

$$\frac{\sigma}{\mu} = 9.43 \pm 0.18 \%$$

$$\mu = 431.7 \pm 0.7$$

$$\sigma = 40.7 \pm 0.7$$

WF (5707 entries)

$$\frac{\sigma}{\mu} = 9.99 \pm 0.18 \%$$

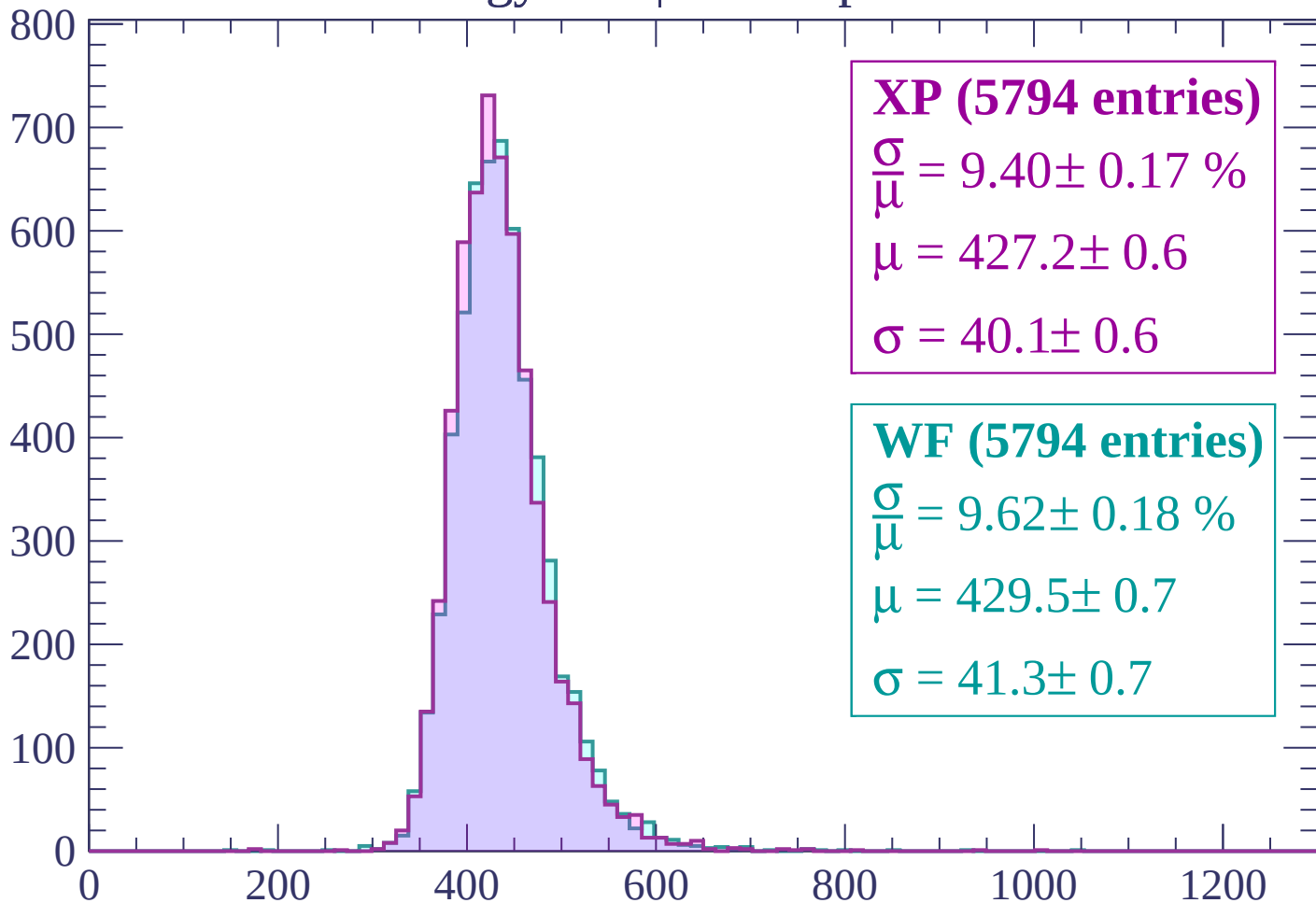
$$\mu = 434.9 \pm 0.7$$

$$\sigma = 43.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-780 < p < -720$

Count



XP (5794 entries)

$$\frac{\sigma}{\mu} = 9.40 \pm 0.17 \%$$

$$\mu = 427.2 \pm 0.6$$

$$\sigma = 40.1 \pm 0.6$$

WF (5794 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.18 \%$$

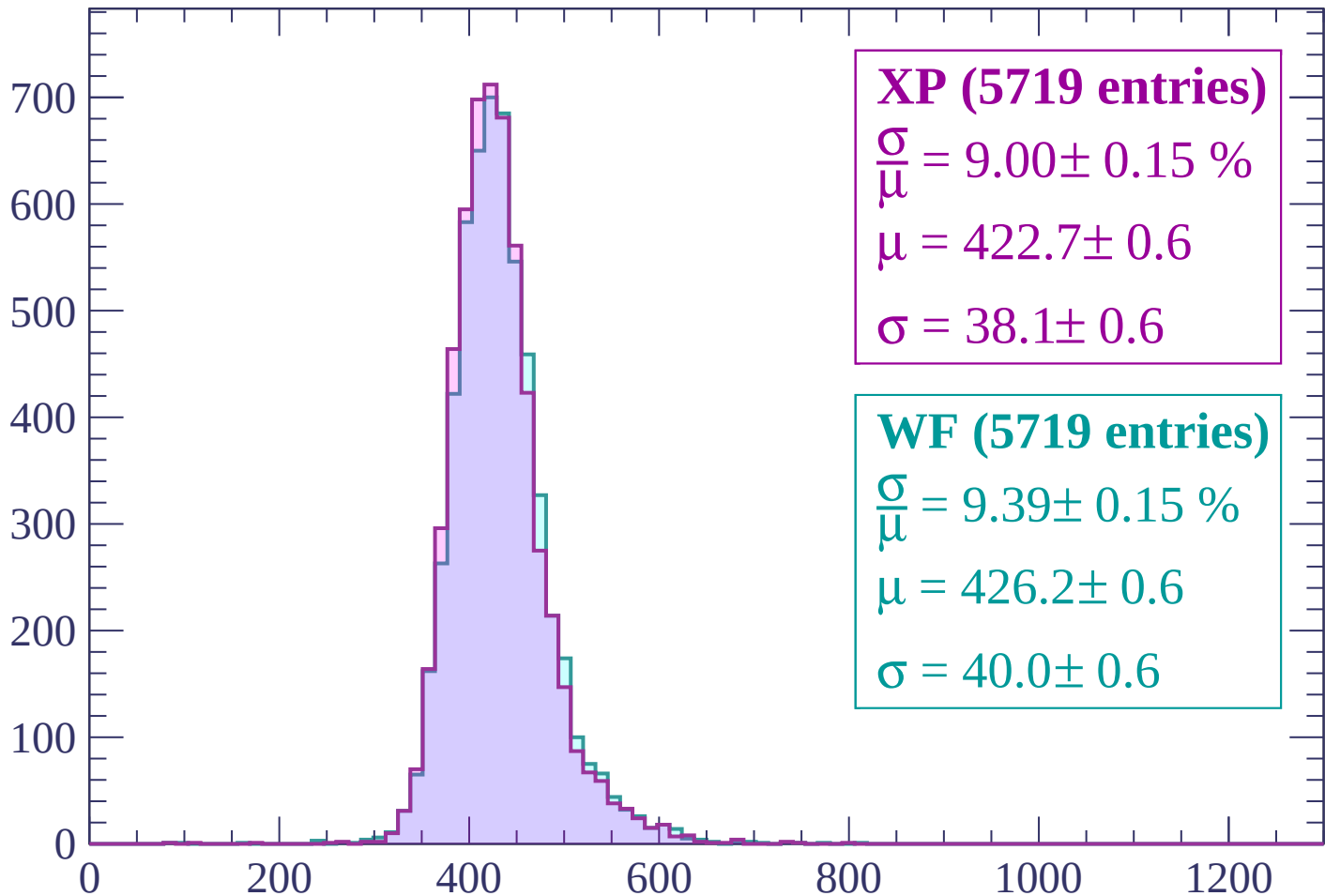
$$\mu = 429.5 \pm 0.7$$

$$\sigma = 41.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -660$

Count



XP (5719 entries)

$$\frac{\sigma}{\mu} = 9.00 \pm 0.15 \%$$

$$\mu = 422.7 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

WF (5719 entries)

$$\frac{\sigma}{\mu} = 9.39 \pm 0.15 \%$$

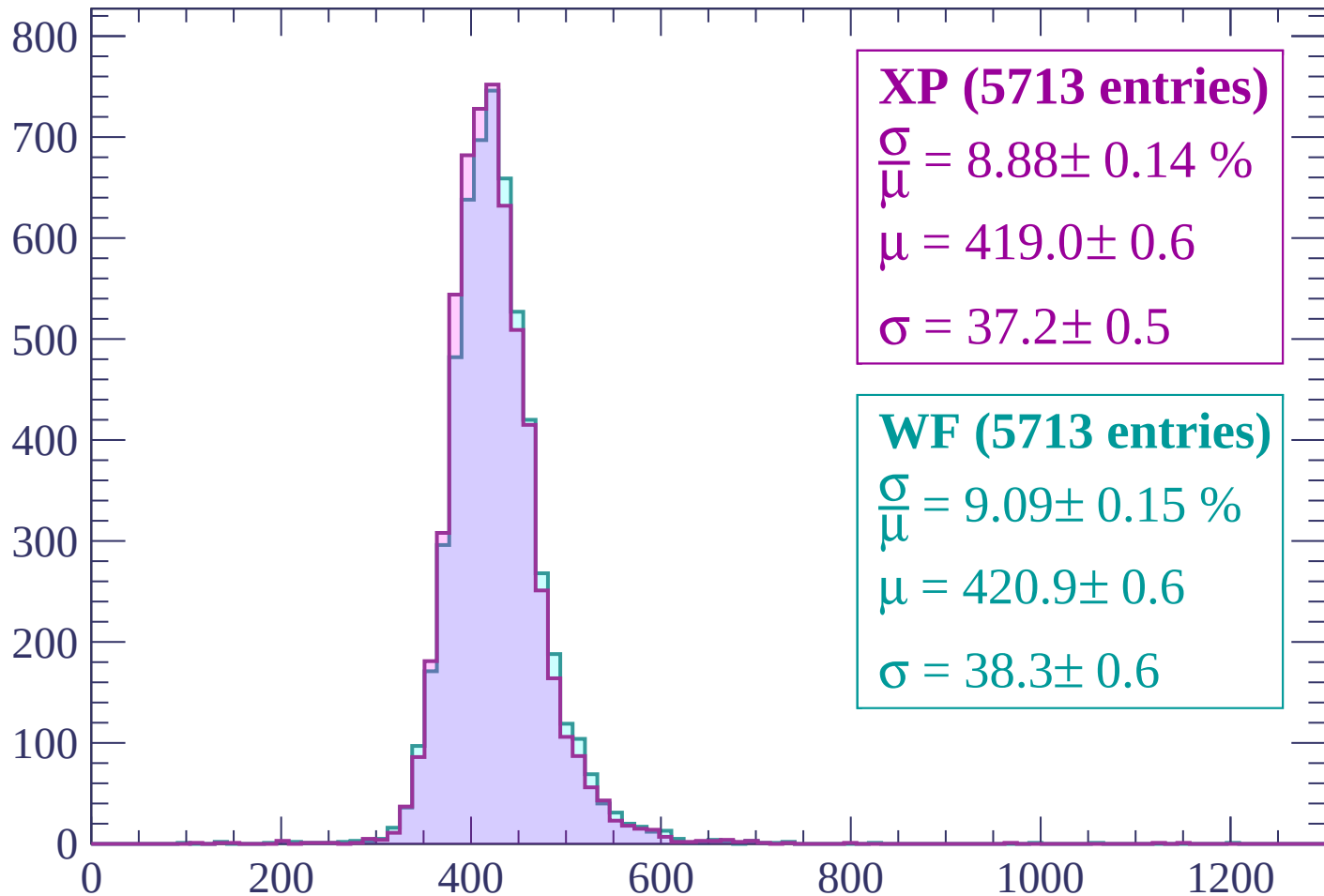
$$\mu = 426.2 \pm 0.6$$

$$\sigma = 40.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-660 < p < -600$

Count



XP (5713 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.14 \%$$

$$\mu = 419.0 \pm 0.6$$

$$\sigma = 37.2 \pm 0.5$$

WF (5713 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.15 \%$$

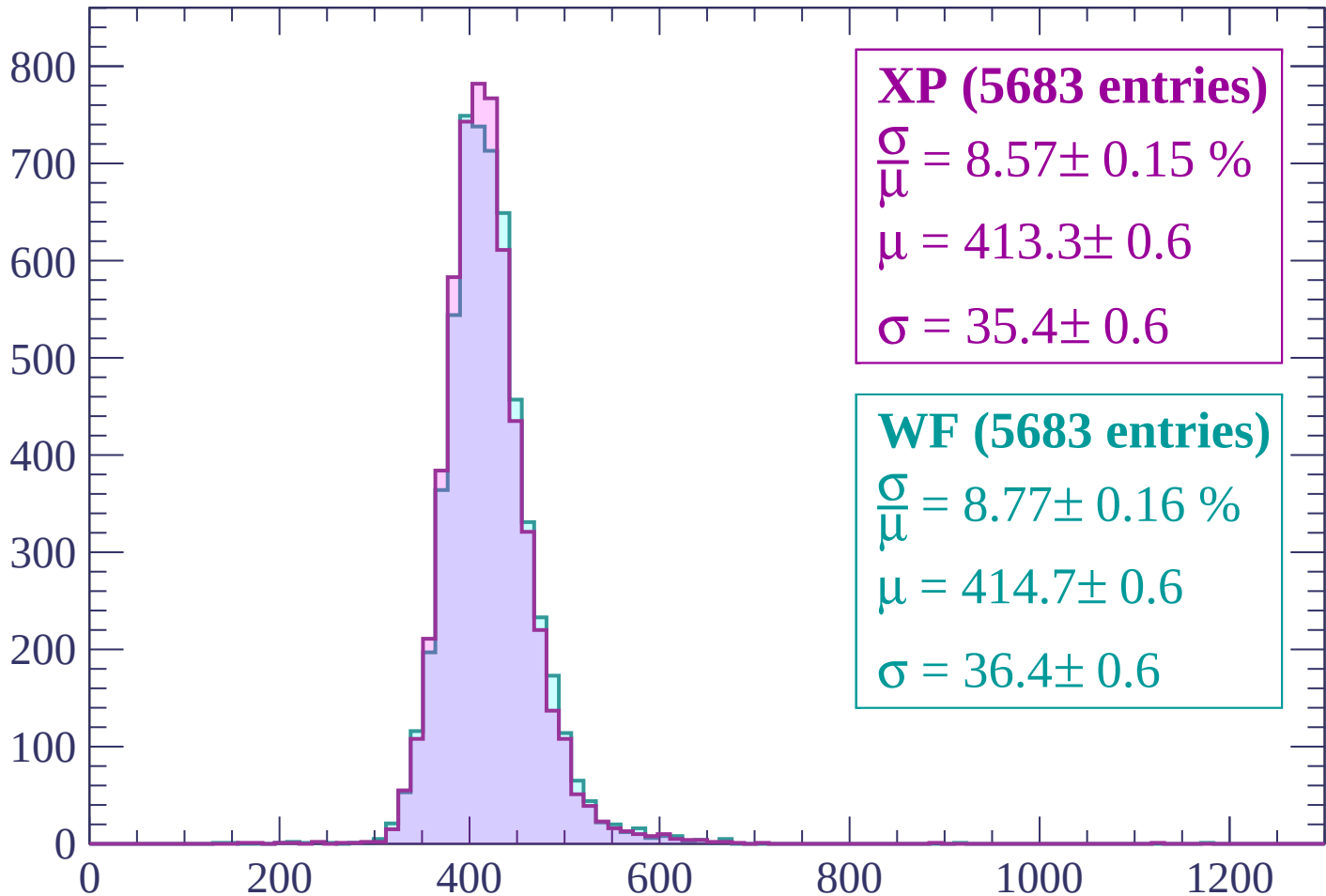
$$\mu = 420.9 \pm 0.6$$

$$\sigma = 38.3 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -540$

Count



XP (5683 entries)

$$\frac{\sigma}{\mu} = 8.57 \pm 0.15 \%$$

$$\mu = 413.3 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

WF (5683 entries)

$$\frac{\sigma}{\mu} = 8.77 \pm 0.16 \%$$

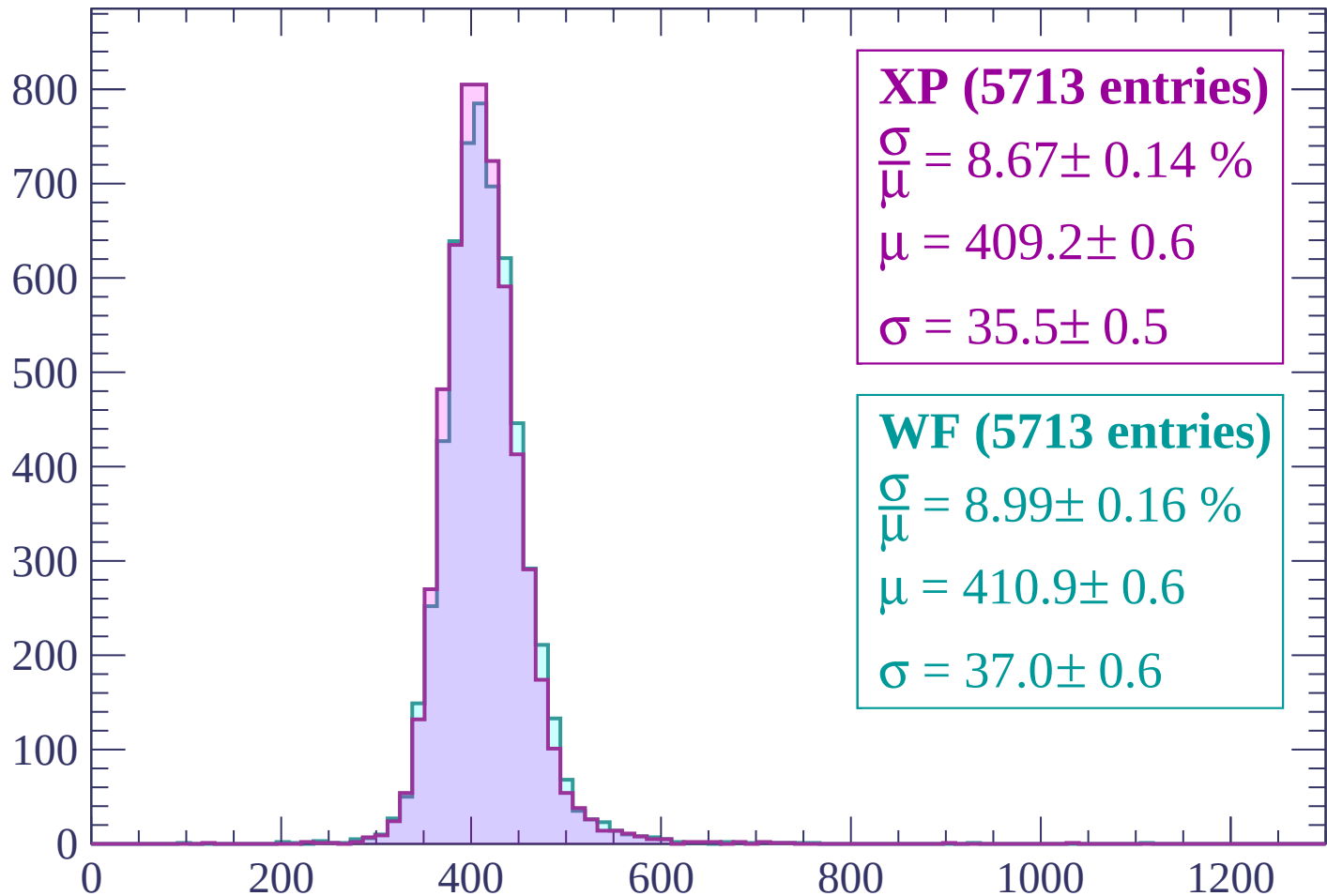
$$\mu = 414.7 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-540 < p < -480$

Count



XP (5713 entries)

$$\frac{\sigma}{\mu} = 8.67 \pm 0.14 \%$$

$$\mu = 409.2 \pm 0.6$$

$$\sigma = 35.5 \pm 0.5$$

WF (5713 entries)

$$\frac{\sigma}{\mu} = 8.99 \pm 0.16 \%$$

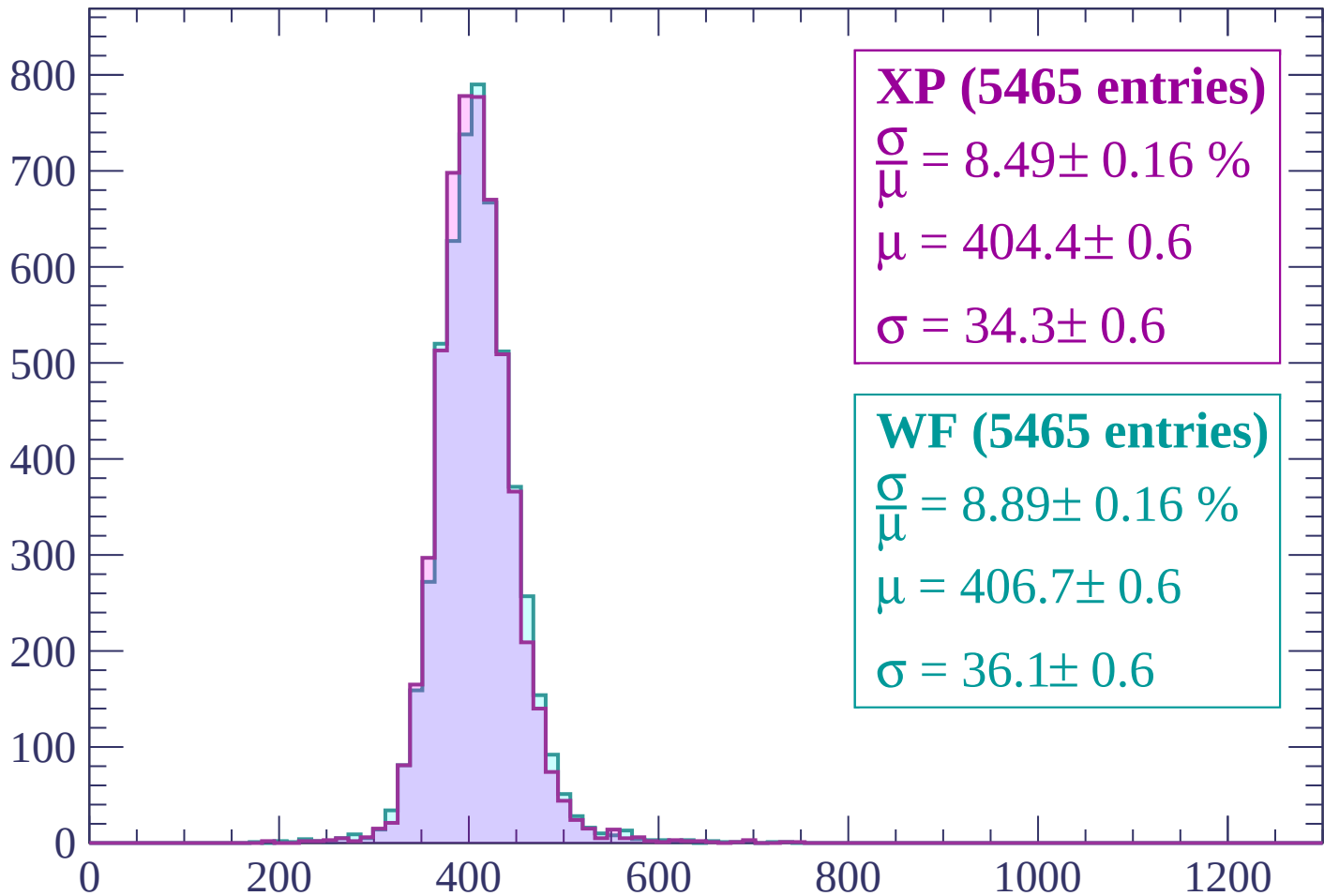
$$\mu = 410.9 \pm 0.6$$

$$\sigma = 37.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -420$

Count



XP (5465 entries)

$$\frac{\sigma}{\mu} = 8.49 \pm 0.16 \%$$

$$\mu = 404.4 \pm 0.6$$

$$\sigma = 34.3 \pm 0.6$$

WF (5465 entries)

$$\frac{\sigma}{\mu} = 8.89 \pm 0.16 \%$$

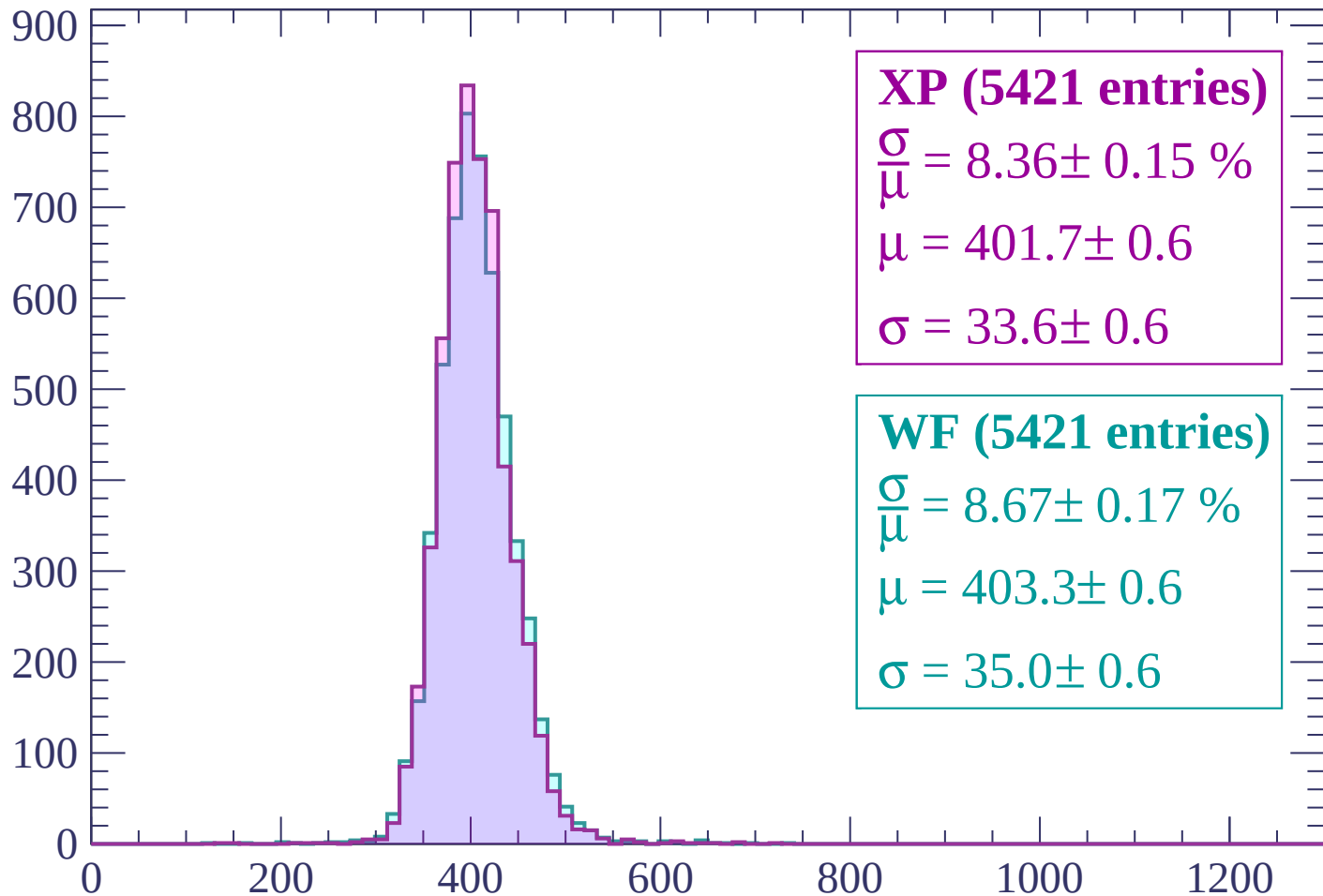
$$\mu = 406.7 \pm 0.6$$

$$\sigma = 36.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-420 < p < -360$

Count



XP (5421 entries)

$$\frac{\sigma}{\mu} = 8.36 \pm 0.15 \%$$

$$\mu = 401.7 \pm 0.6$$

$$\sigma = 33.6 \pm 0.6$$

WF (5421 entries)

$$\frac{\sigma}{\mu} = 8.67 \pm 0.17 \%$$

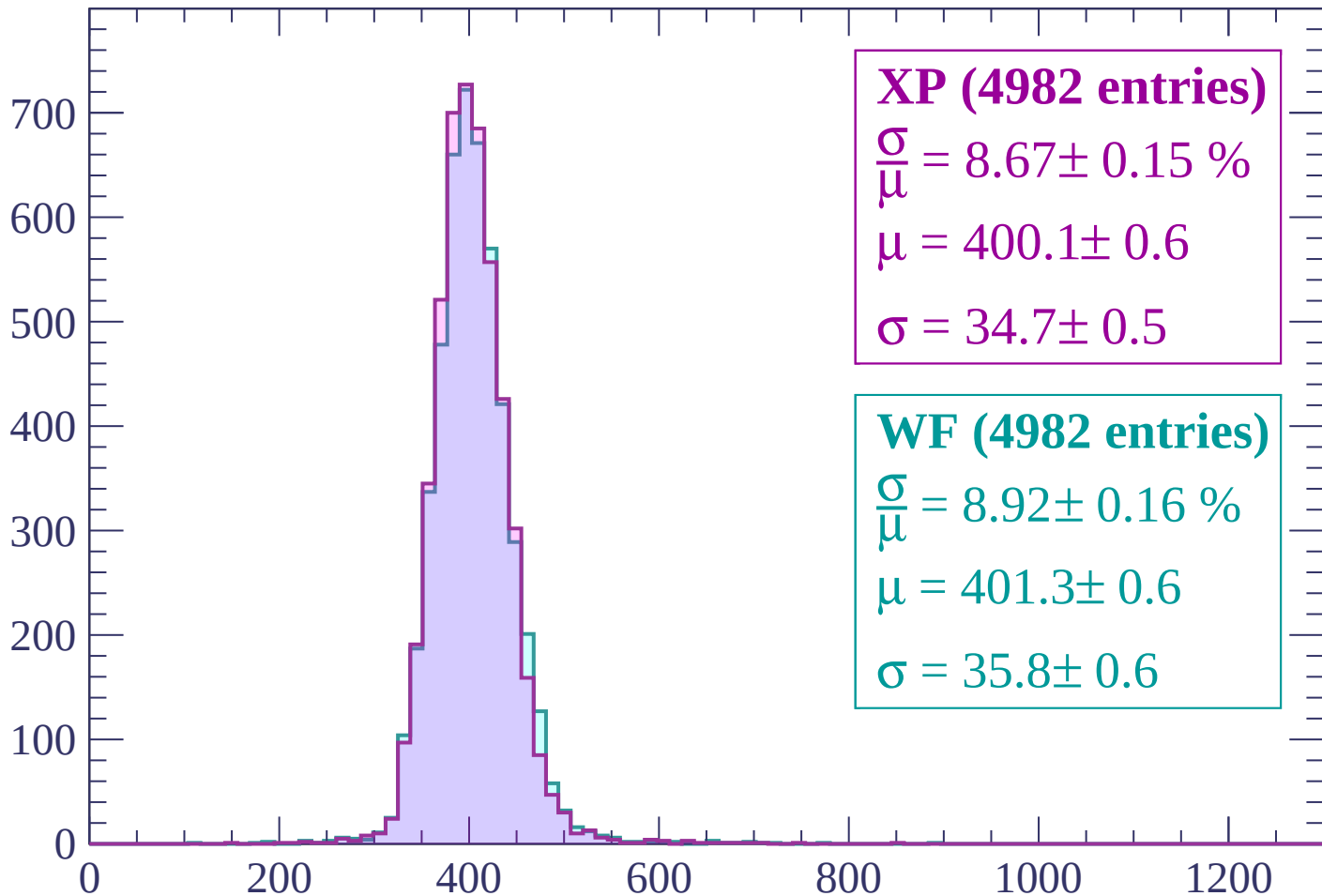
$$\mu = 403.3 \pm 0.6$$

$$\sigma = 35.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -300$

Count



XP (4982 entries)

$$\frac{\sigma}{\mu} = 8.67 \pm 0.15 \%$$

$$\mu = 400.1 \pm 0.6$$

$$\sigma = 34.7 \pm 0.5$$

WF (4982 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.16 \%$$

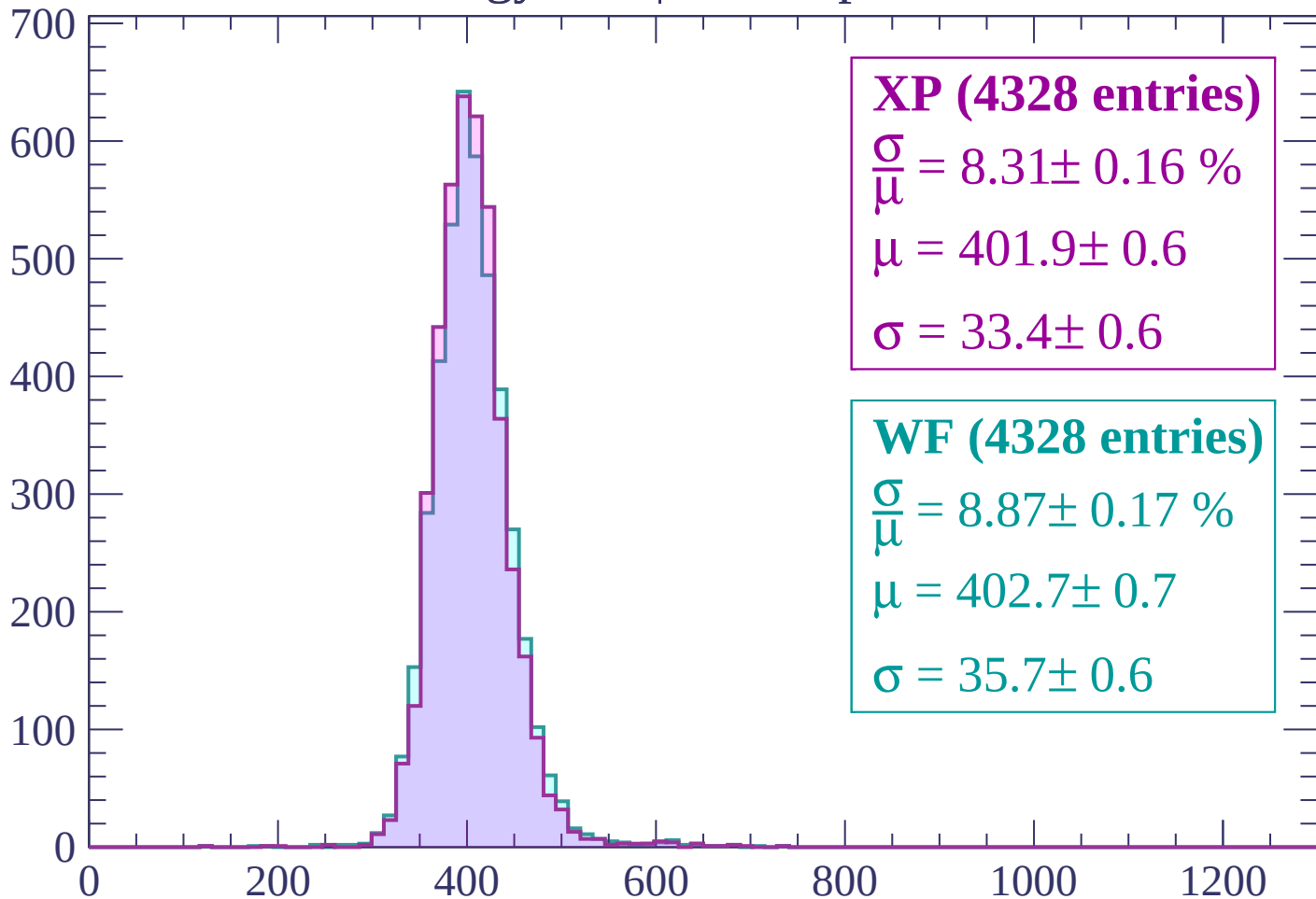
$$\mu = 401.3 \pm 0.6$$

$$\sigma = 35.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-300 < p < -240$

Count



XP (4328 entries)

$$\frac{\sigma}{\mu} = 8.31 \pm 0.16 \%$$

$$\mu = 401.9 \pm 0.6$$

$$\sigma = 33.4 \pm 0.6$$

WF (4328 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.17 \%$$

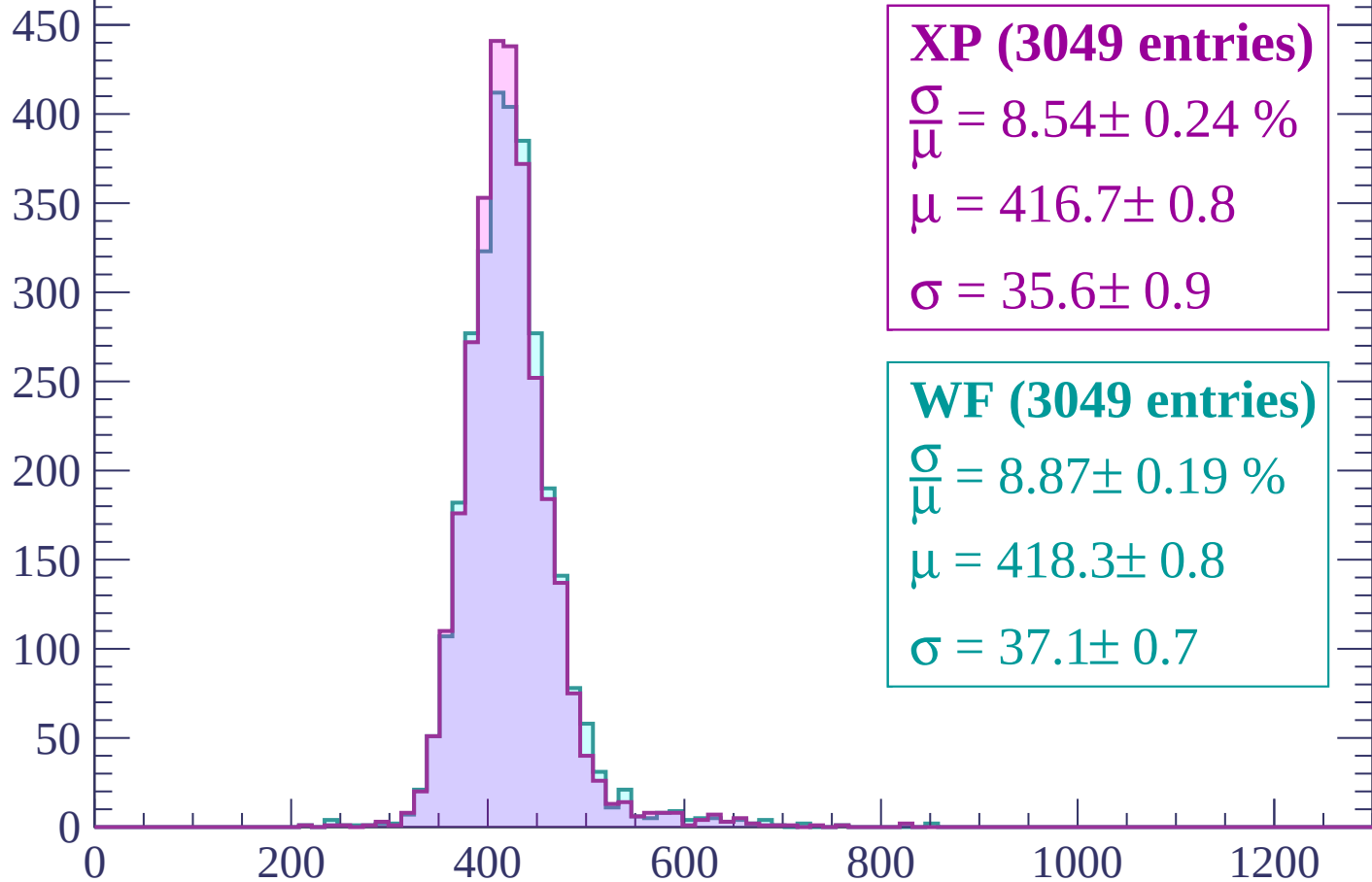
$$\mu = 402.7 \pm 0.7$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -180$

Count



XP (3049 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.24 \%$$

$$\mu = 416.7 \pm 0.8$$

$$\sigma = 35.6 \pm 0.9$$

WF (3049 entries)

$$\frac{\sigma}{\mu} = 8.87 \pm 0.19 \%$$

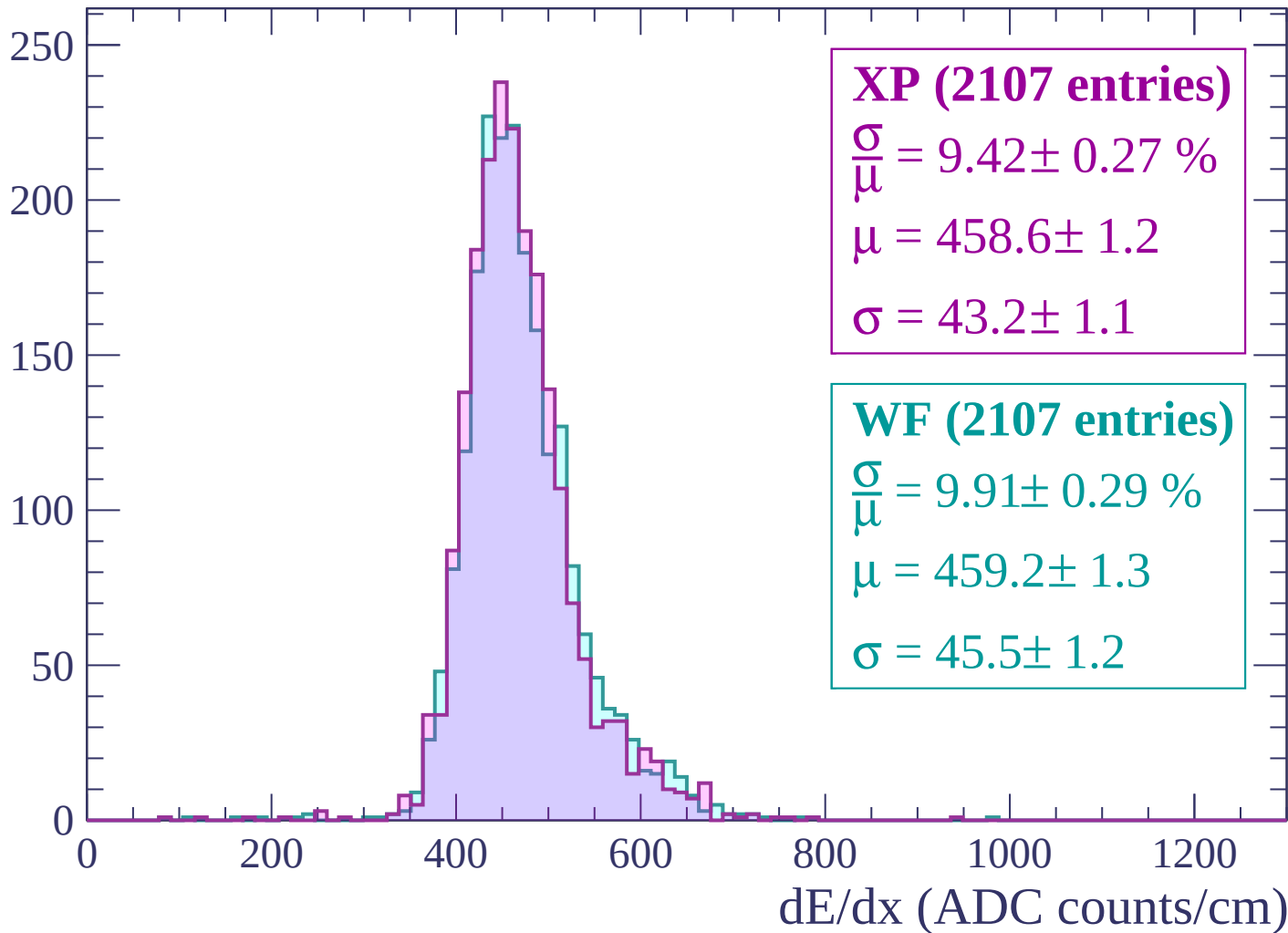
$$\mu = 418.3 \pm 0.8$$

$$\sigma = 37.1 \pm 0.7$$

dE/dx (ADC counts/cm)

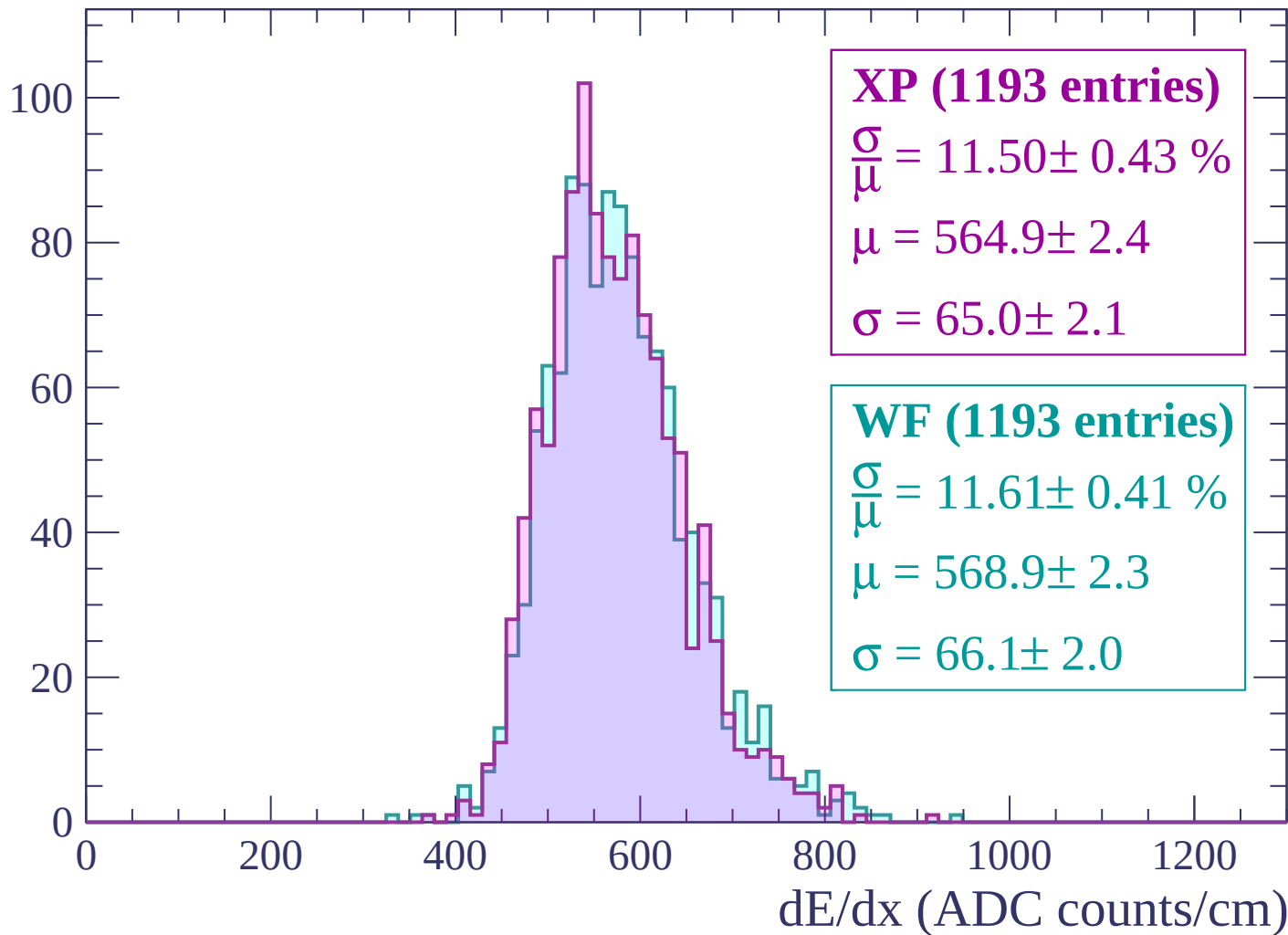
Energy loss | $-180 < p < -120$

Count



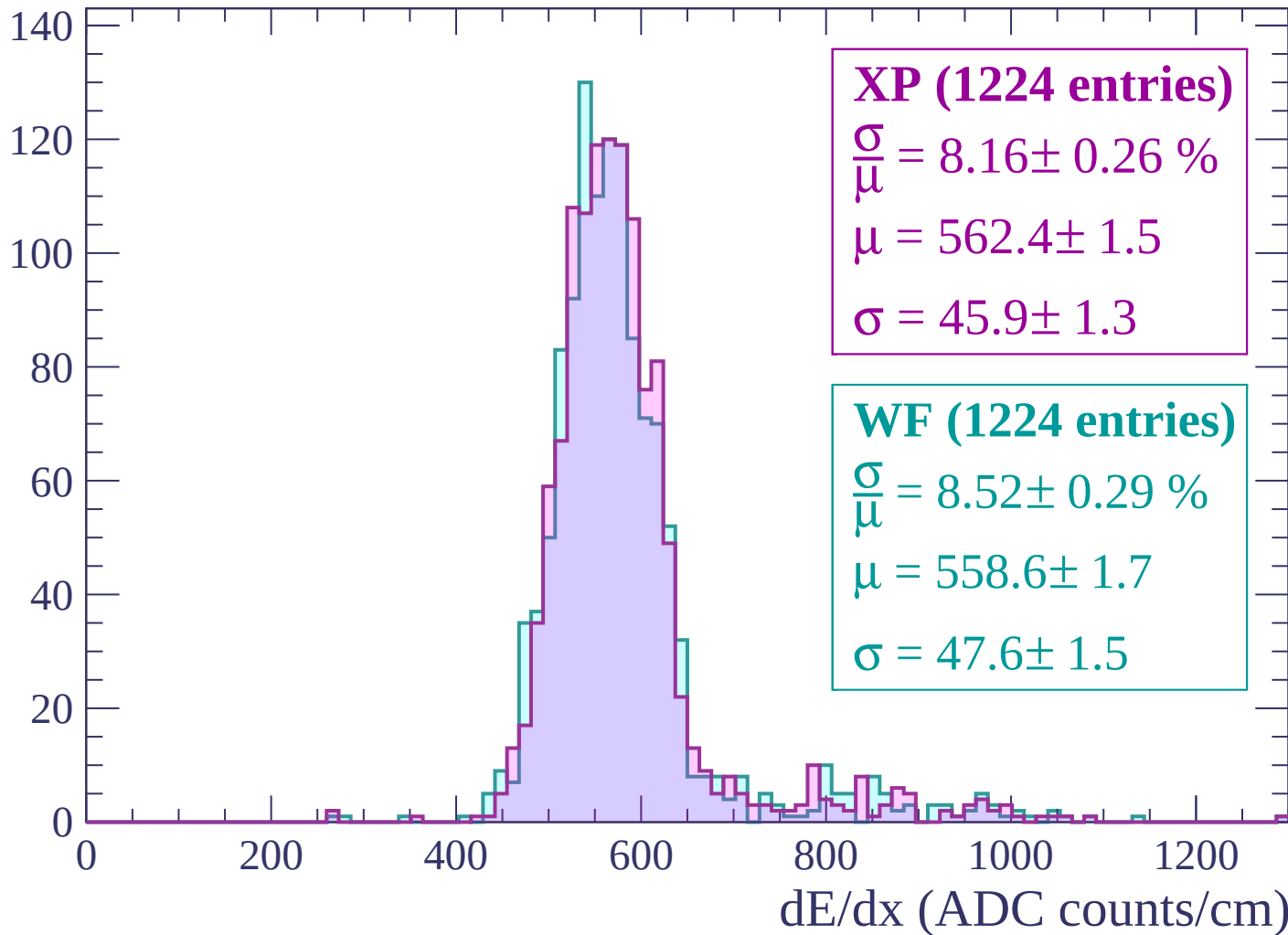
Energy loss | $-120 < p < -60$

Count



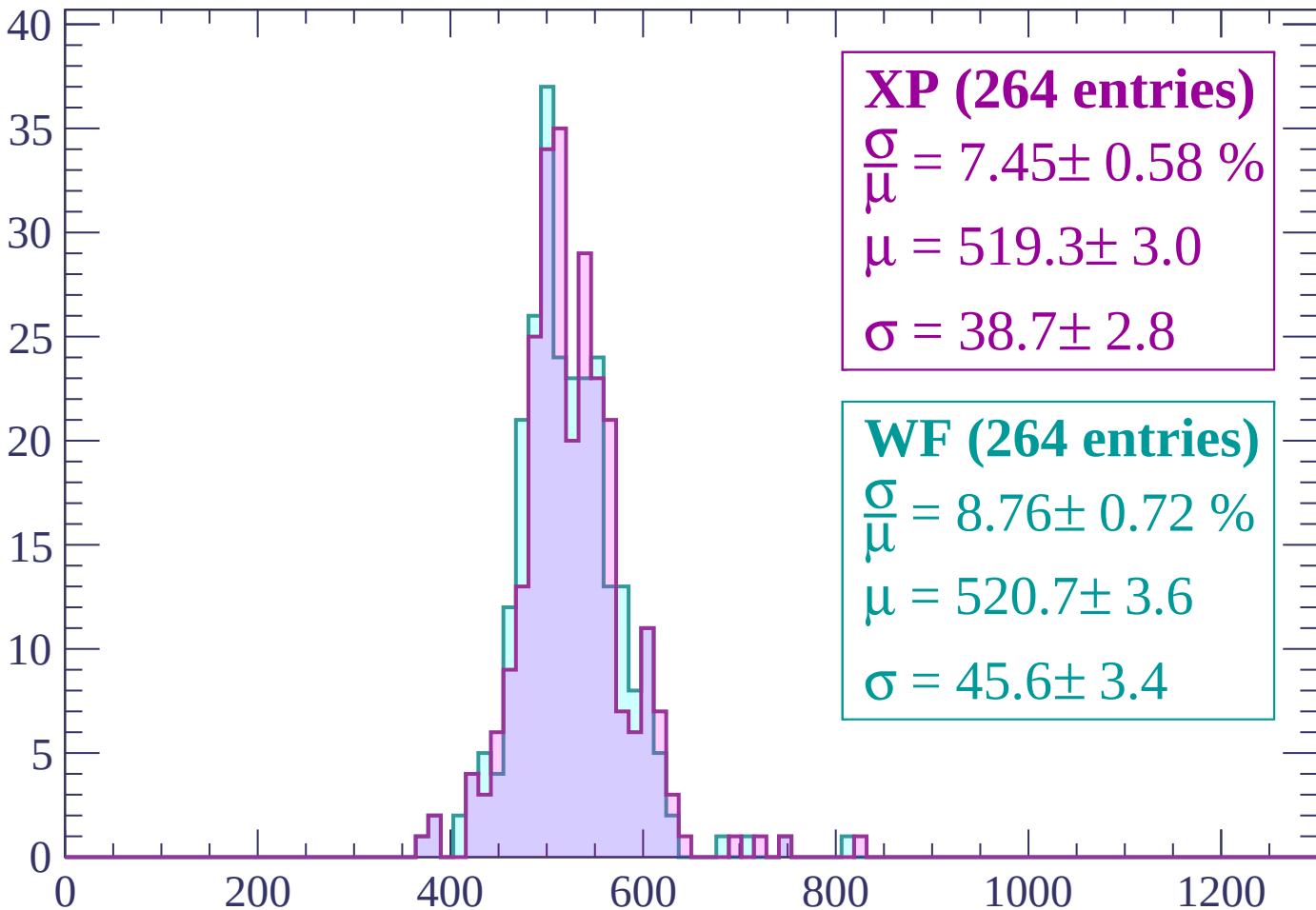
Energy loss | $-60 < p < 0$

Count



Energy loss | $0 < p < 60$

Count



XP (264 entries)

$\frac{\sigma}{\mu} = 7.45 \pm 0.58 \%$

$\mu = 519.3 \pm 3.0$

$\sigma = 38.7 \pm 2.8$

WF (264 entries)

$\frac{\sigma}{\mu} = 8.76 \pm 0.72 \%$

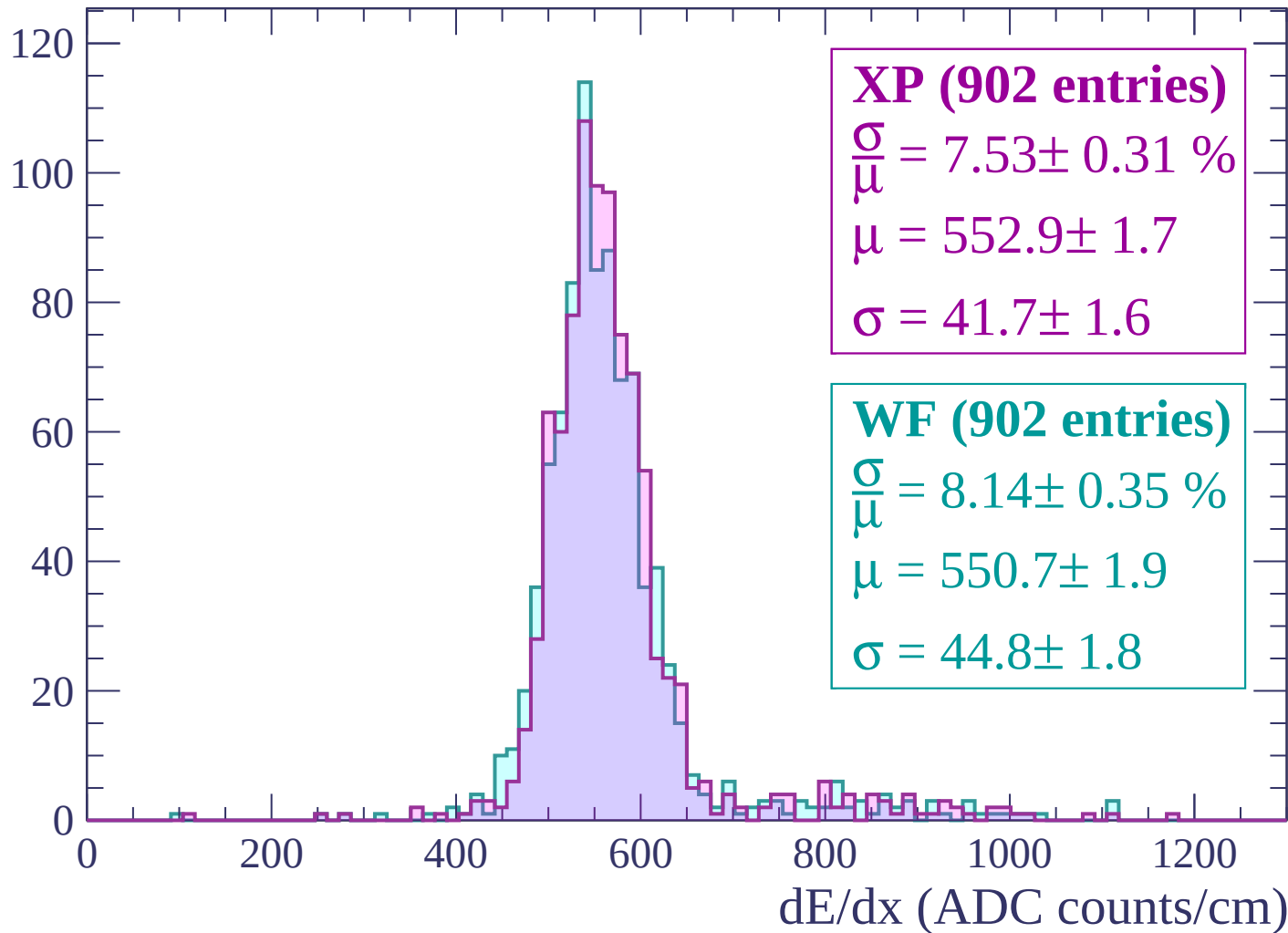
$\mu = 520.7 \pm 3.6$

$\sigma = 45.6 \pm 3.4$

dE/dx (ADC counts/cm)

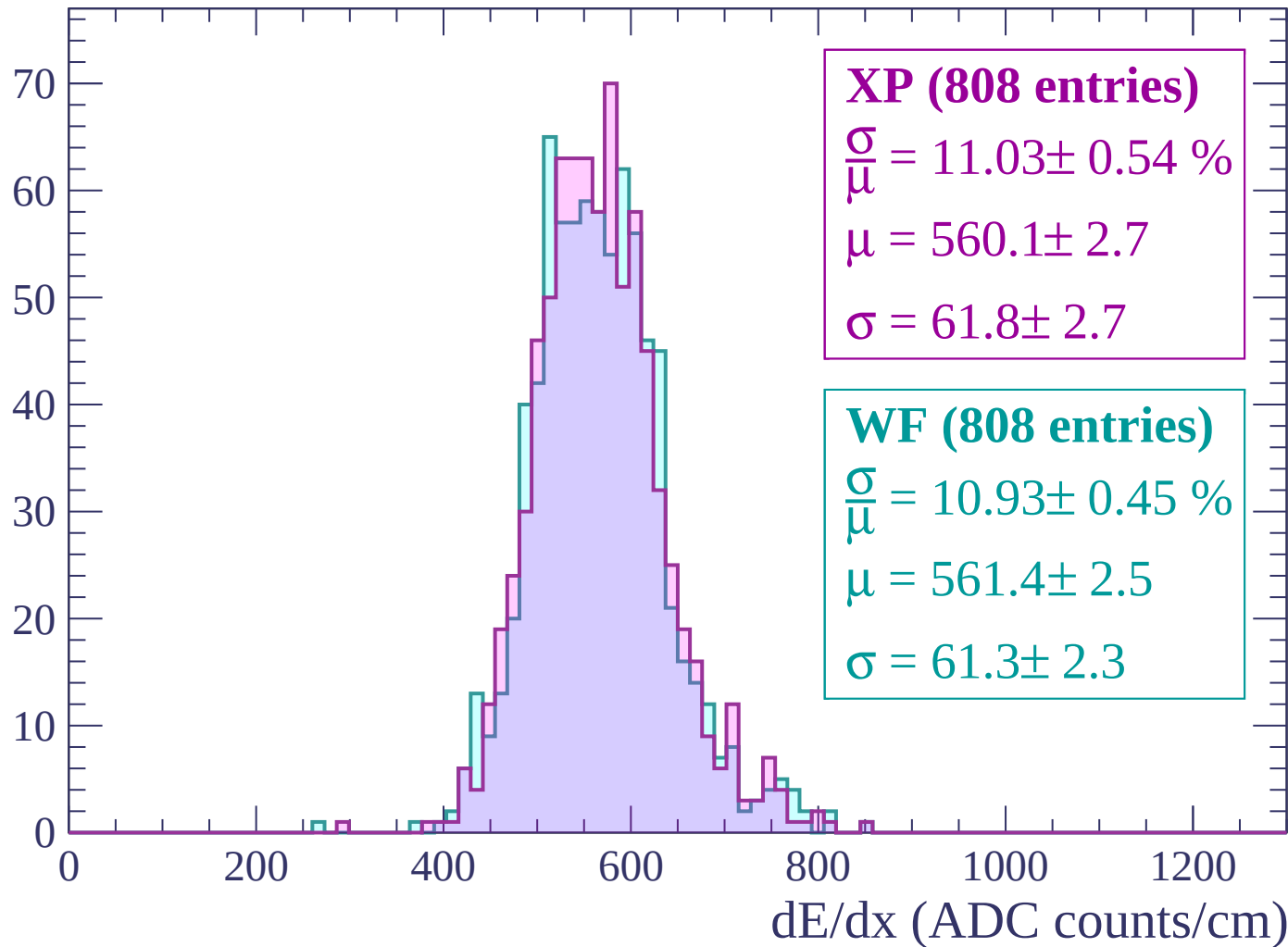
Energy loss | $60 < p < 120$

Count



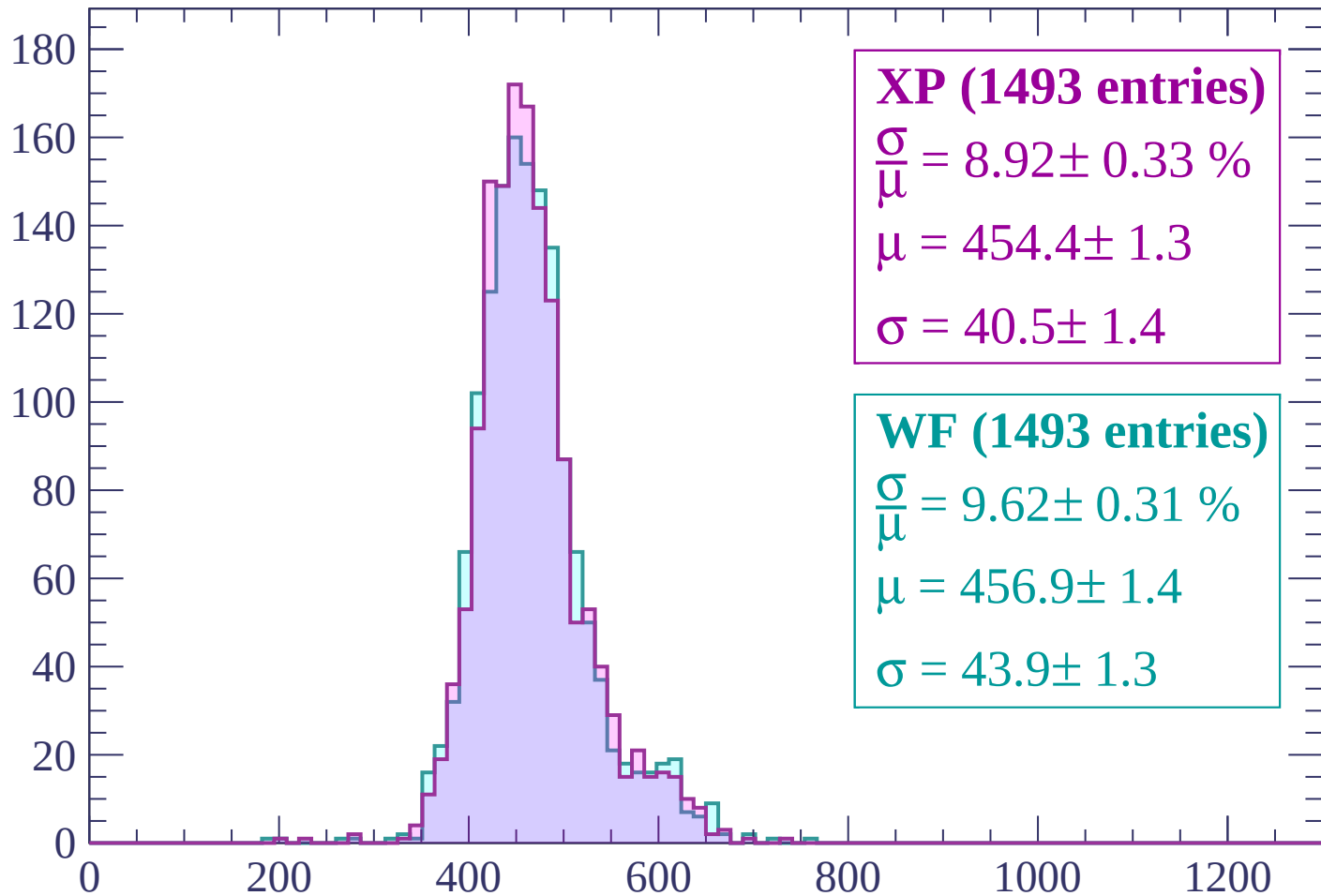
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP (1493 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.33 \%$$

$$\mu = 454.4 \pm 1.3$$

$$\sigma = 40.5 \pm 1.4$$

WF (1493 entries)

$$\frac{\sigma}{\mu} = 9.62 \pm 0.31 \%$$

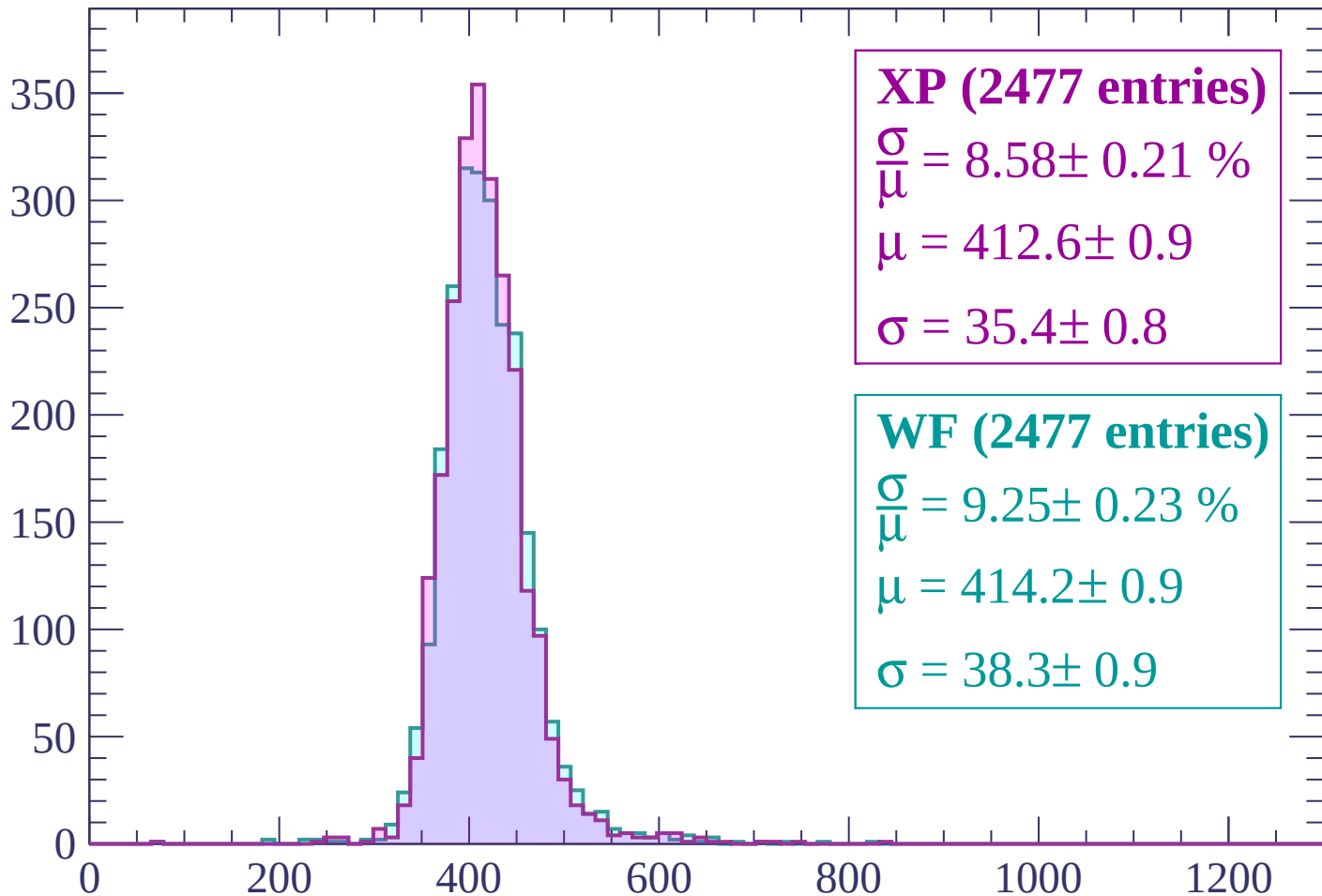
$$\mu = 456.9 \pm 1.4$$

$$\sigma = 43.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 300$

Count



XP (2477 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.21 \%$

$\mu = 412.6 \pm 0.9$

$\sigma = 35.4 \pm 0.8$

WF (2477 entries)

$\frac{\sigma}{\mu} = 9.25 \pm 0.23 \%$

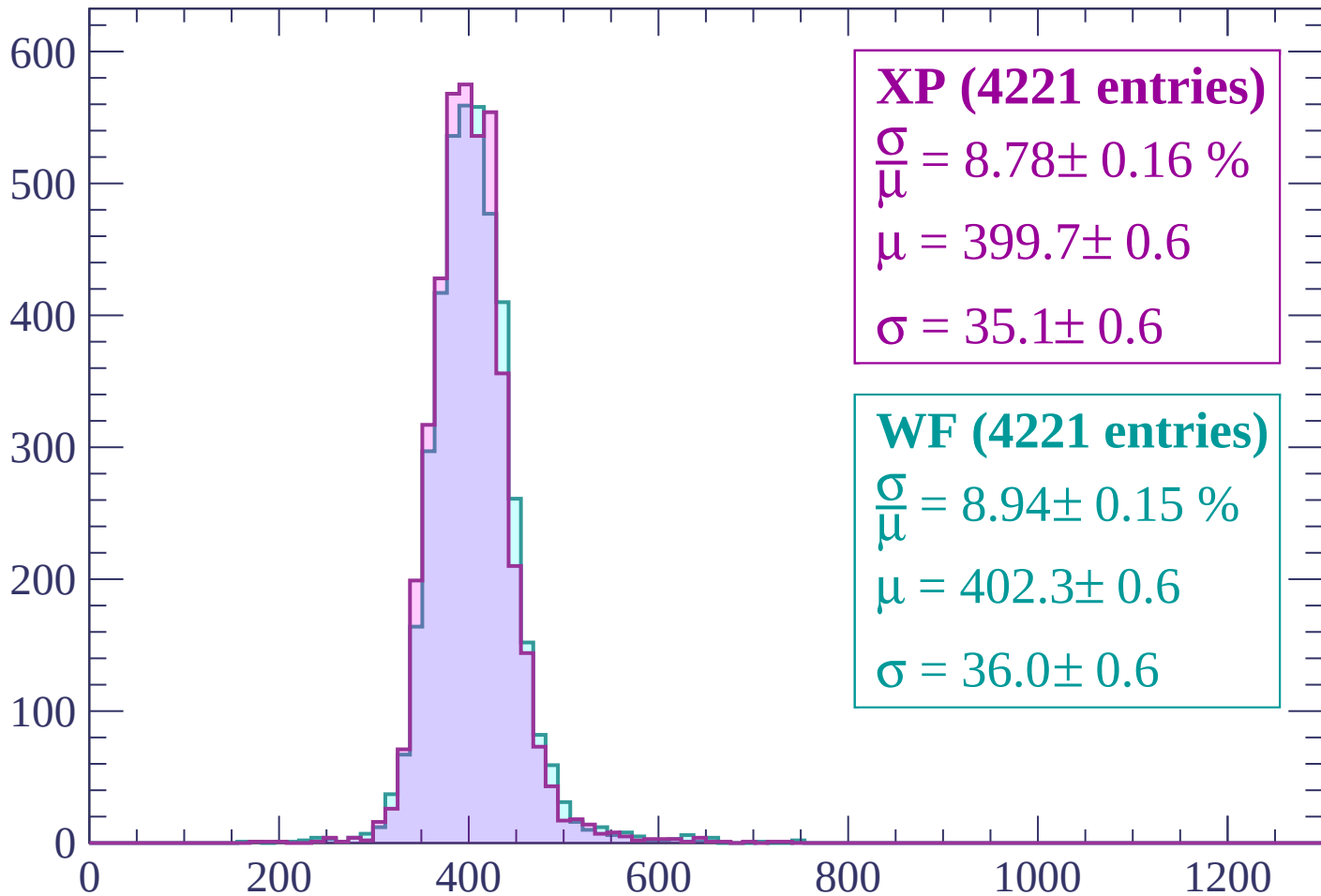
$\mu = 414.2 \pm 0.9$

$\sigma = 38.3 \pm 0.9$

dE/dx (ADC counts/cm)

Energy loss | $300 < p < 360$

Count



XP (4221 entries)

$$\frac{\sigma}{\mu} = 8.78 \pm 0.16 \%$$

$$\mu = 399.7 \pm 0.6$$

$$\sigma = 35.1 \pm 0.6$$

WF (4221 entries)

$$\frac{\sigma}{\mu} = 8.94 \pm 0.15 \%$$

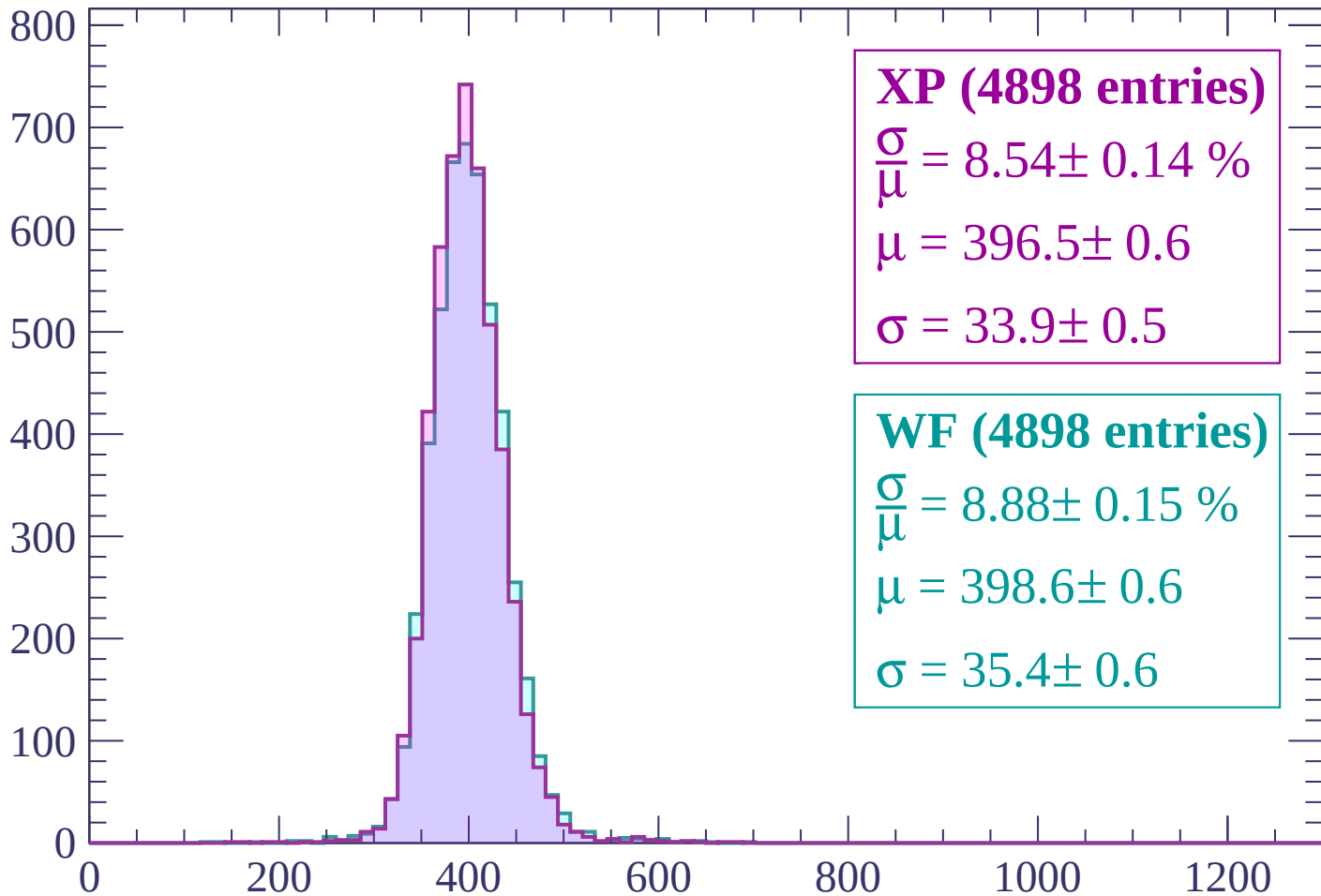
$$\mu = 402.3 \pm 0.6$$

$$\sigma = 36.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 420$

Count



XP (4898 entries)

$$\frac{\sigma}{\mu} = 8.54 \pm 0.14 \%$$

$$\mu = 396.5 \pm 0.6$$

$$\sigma = 33.9 \pm 0.5$$

WF (4898 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.15 \%$$

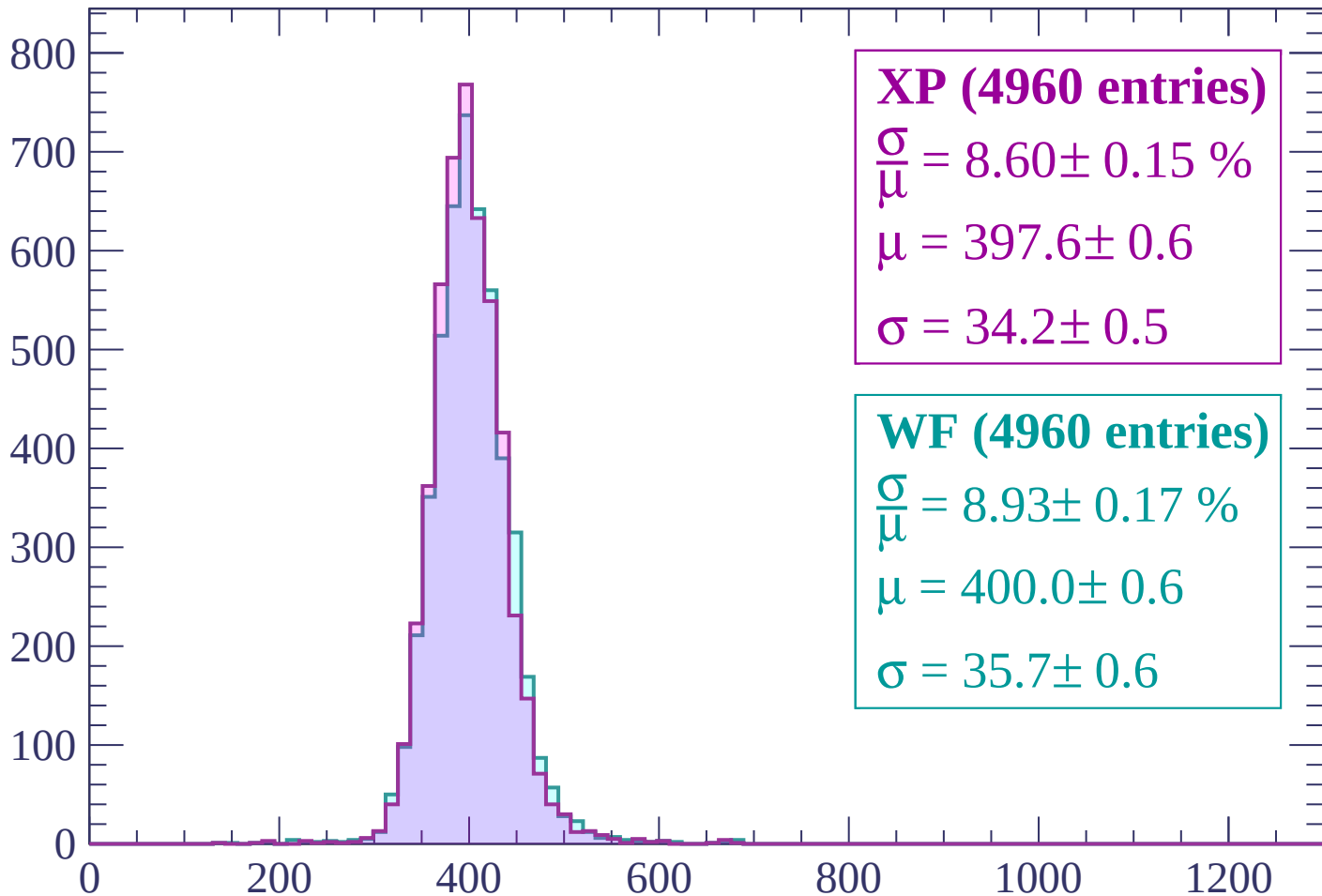
$$\mu = 398.6 \pm 0.6$$

$$\sigma = 35.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $420 < p < 480$

Count



XP (4960 entries)

$$\frac{\sigma}{\mu} = 8.60 \pm 0.15 \%$$

$$\mu = 397.6 \pm 0.6$$

$$\sigma = 34.2 \pm 0.5$$

WF (4960 entries)

$$\frac{\sigma}{\mu} = 8.93 \pm 0.17 \%$$

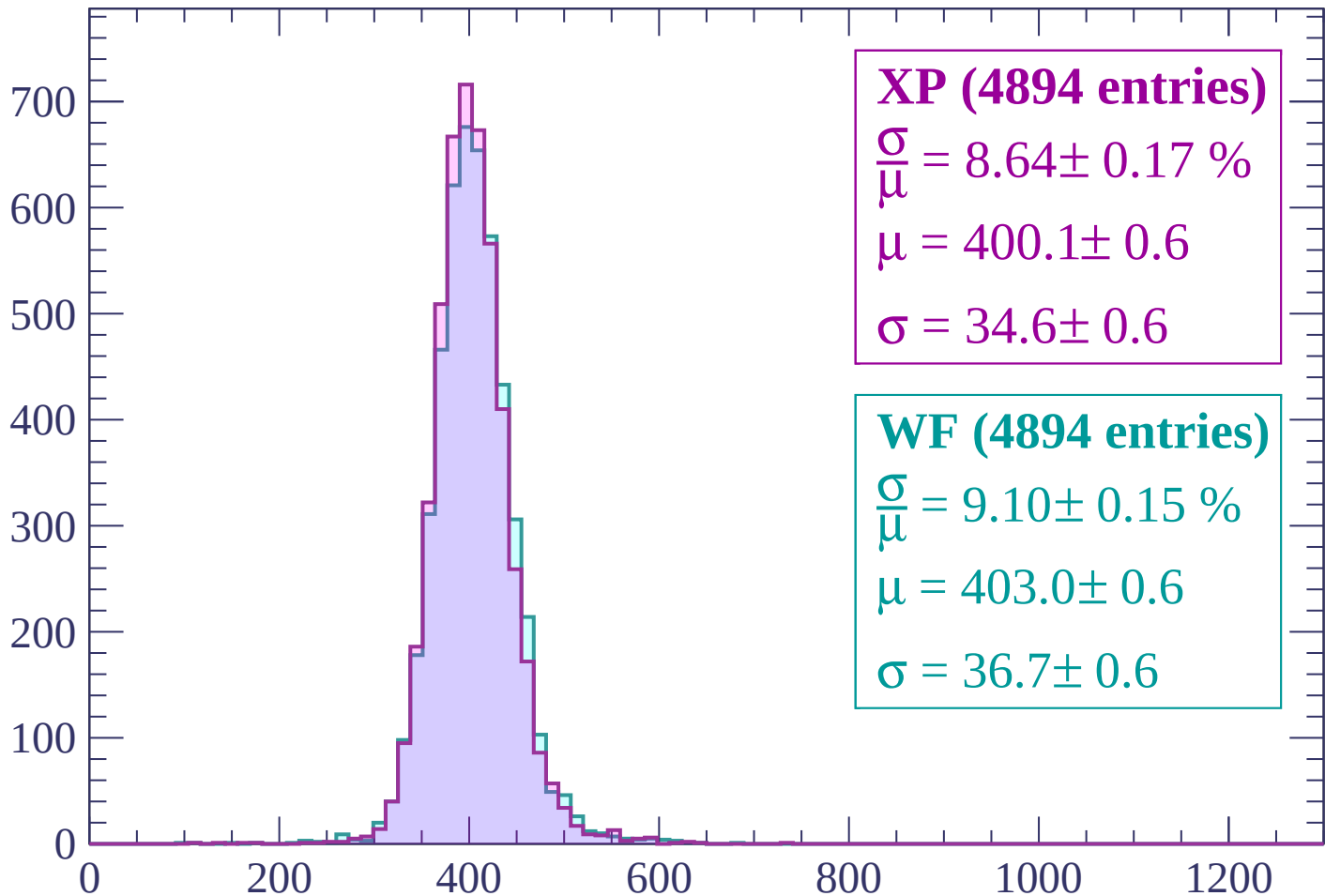
$$\mu = 400.0 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 540$

Count



XP (4894 entries)

$$\frac{\sigma}{\mu} = 8.64 \pm 0.17 \%$$

$$\mu = 400.1 \pm 0.6$$

$$\sigma = 34.6 \pm 0.6$$

WF (4894 entries)

$$\frac{\sigma}{\mu} = 9.10 \pm 0.15 \%$$

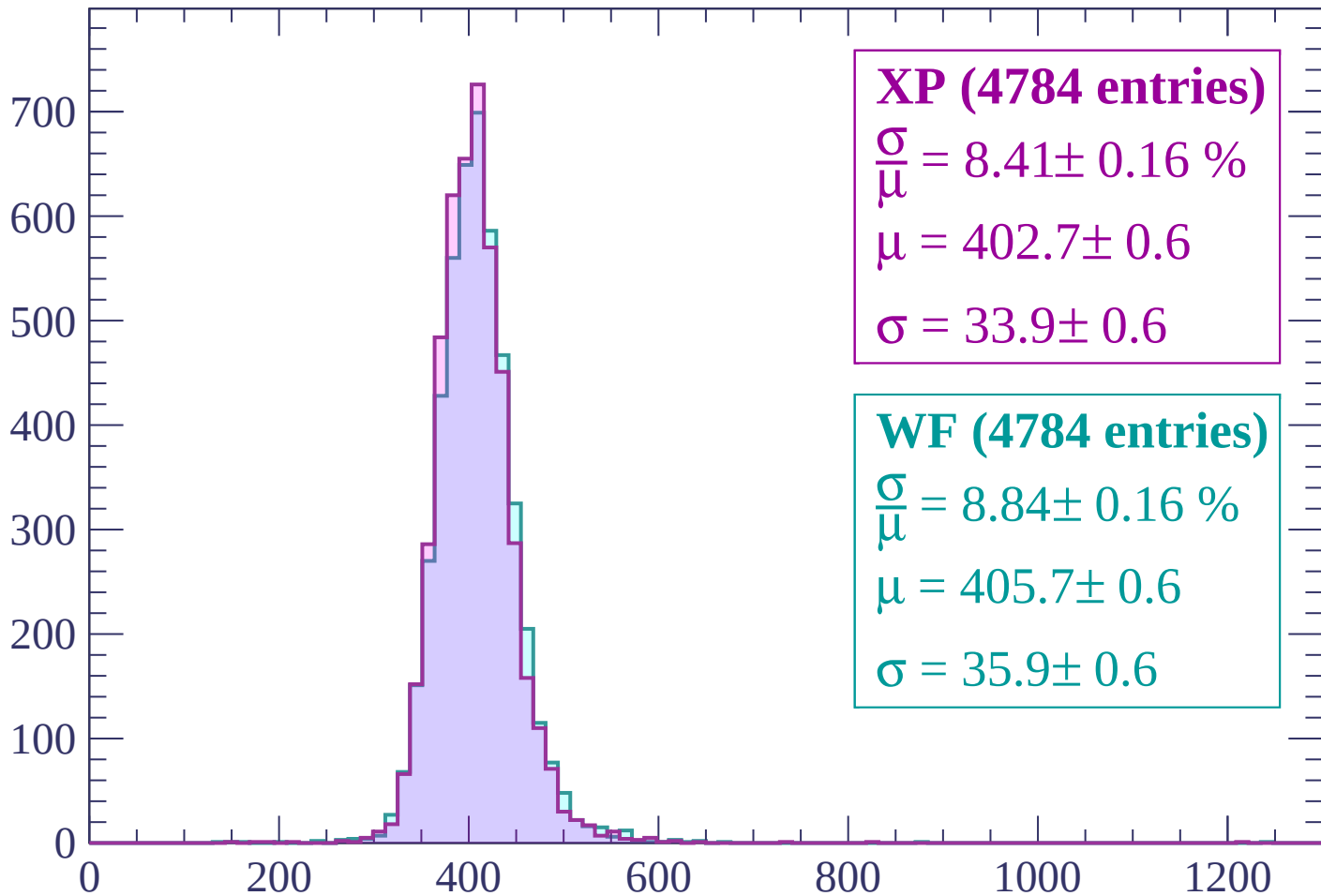
$$\mu = 403.0 \pm 0.6$$

$$\sigma = 36.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $540 < p < 600$

Count



XP (4784 entries)

$$\frac{\sigma}{\mu} = 8.41 \pm 0.16 \%$$

$$\mu = 402.7 \pm 0.6$$

$$\sigma = 33.9 \pm 0.6$$

WF (4784 entries)

$$\frac{\sigma}{\mu} = 8.84 \pm 0.16 \%$$

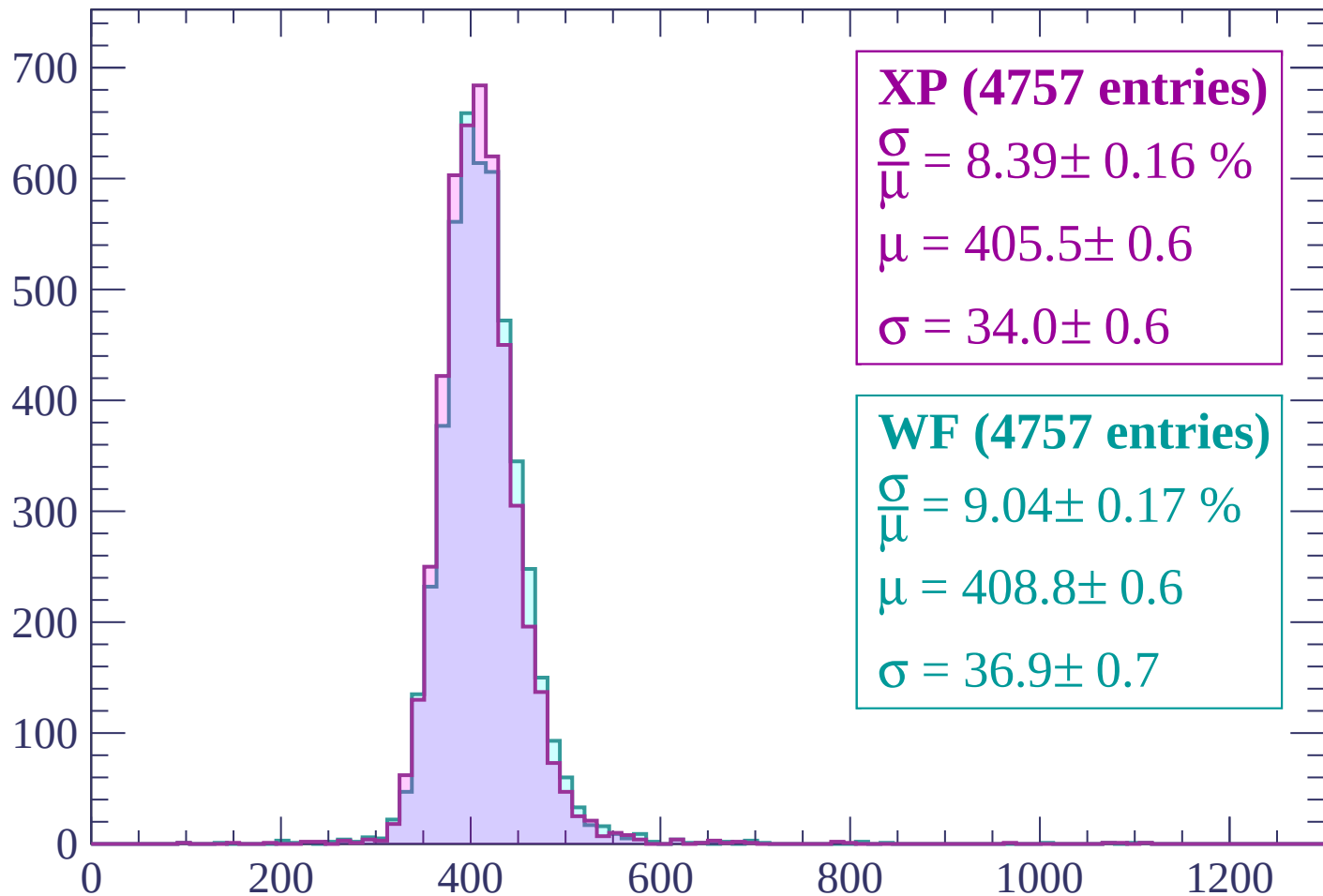
$$\mu = 405.7 \pm 0.6$$

$$\sigma = 35.9 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 660$

Count



XP (4757 entries)

$$\frac{\sigma}{\mu} = 8.39 \pm 0.16 \%$$

$$\mu = 405.5 \pm 0.6$$

$$\sigma = 34.0 \pm 0.6$$

WF (4757 entries)

$$\frac{\sigma}{\mu} = 9.04 \pm 0.17 \%$$

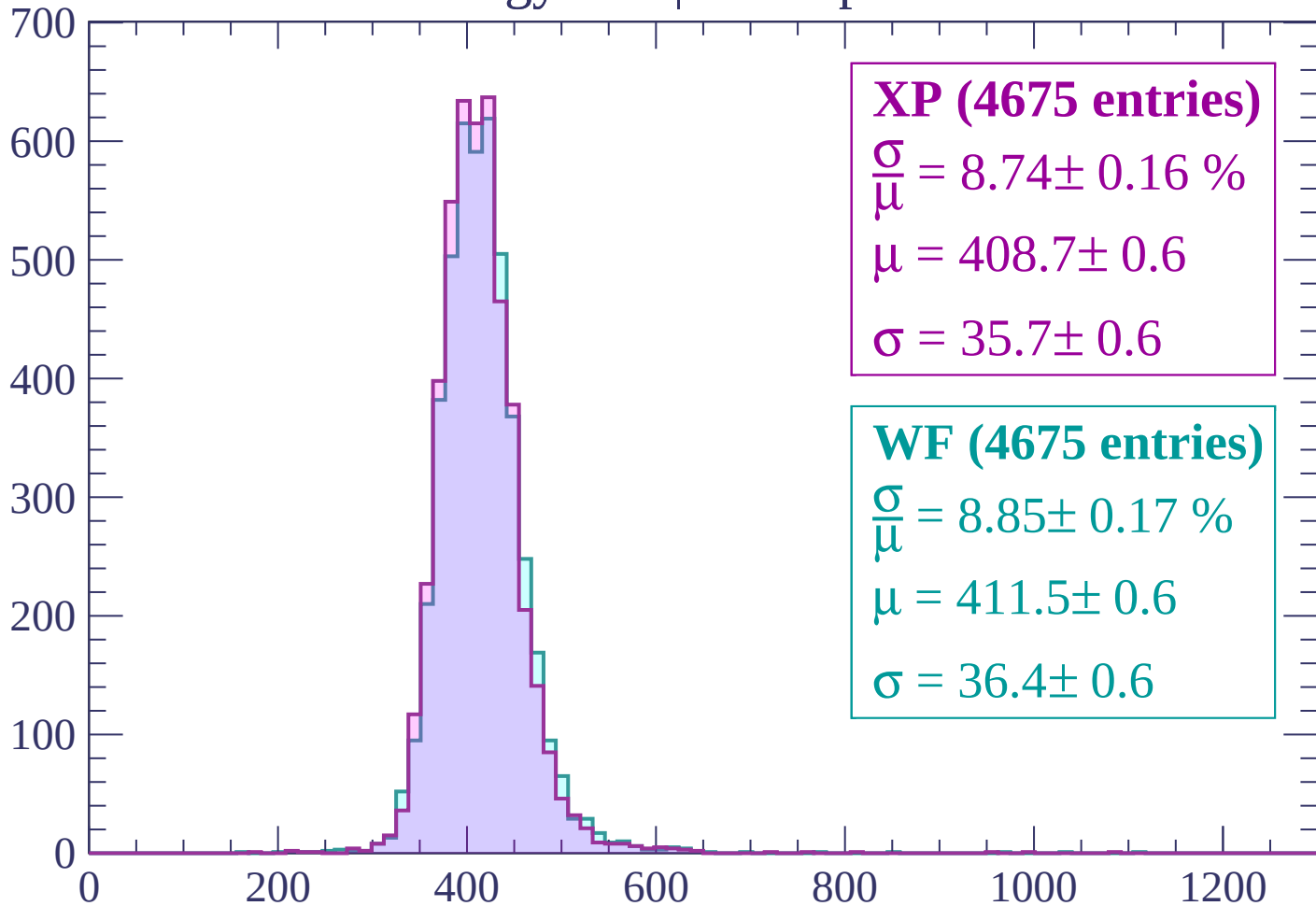
$$\mu = 408.8 \pm 0.6$$

$$\sigma = 36.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $660 < p < 720$

Count



XP (4675 entries)

$$\frac{\sigma}{\mu} = 8.74 \pm 0.16 \%$$

$$\mu = 408.7 \pm 0.6$$

$$\sigma = 35.7 \pm 0.6$$

WF (4675 entries)

$$\frac{\sigma}{\mu} = 8.85 \pm 0.17 \%$$

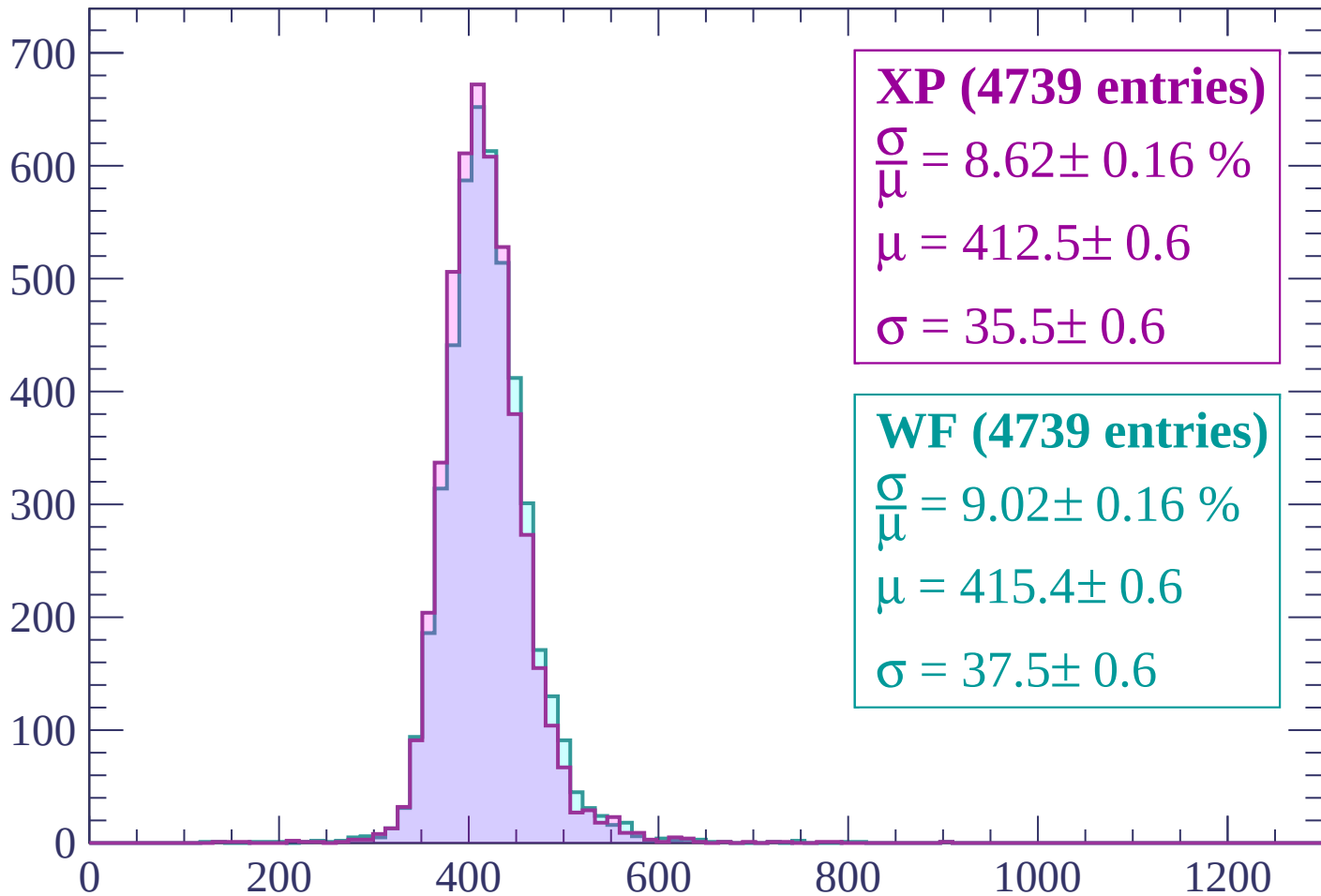
$$\mu = 411.5 \pm 0.6$$

$$\sigma = 36.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 780$

Count



XP (4739 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.16 \%$$

$$\mu = 412.5 \pm 0.6$$

$$\sigma = 35.5 \pm 0.6$$

WF (4739 entries)

$$\frac{\sigma}{\mu} = 9.02 \pm 0.16 \%$$

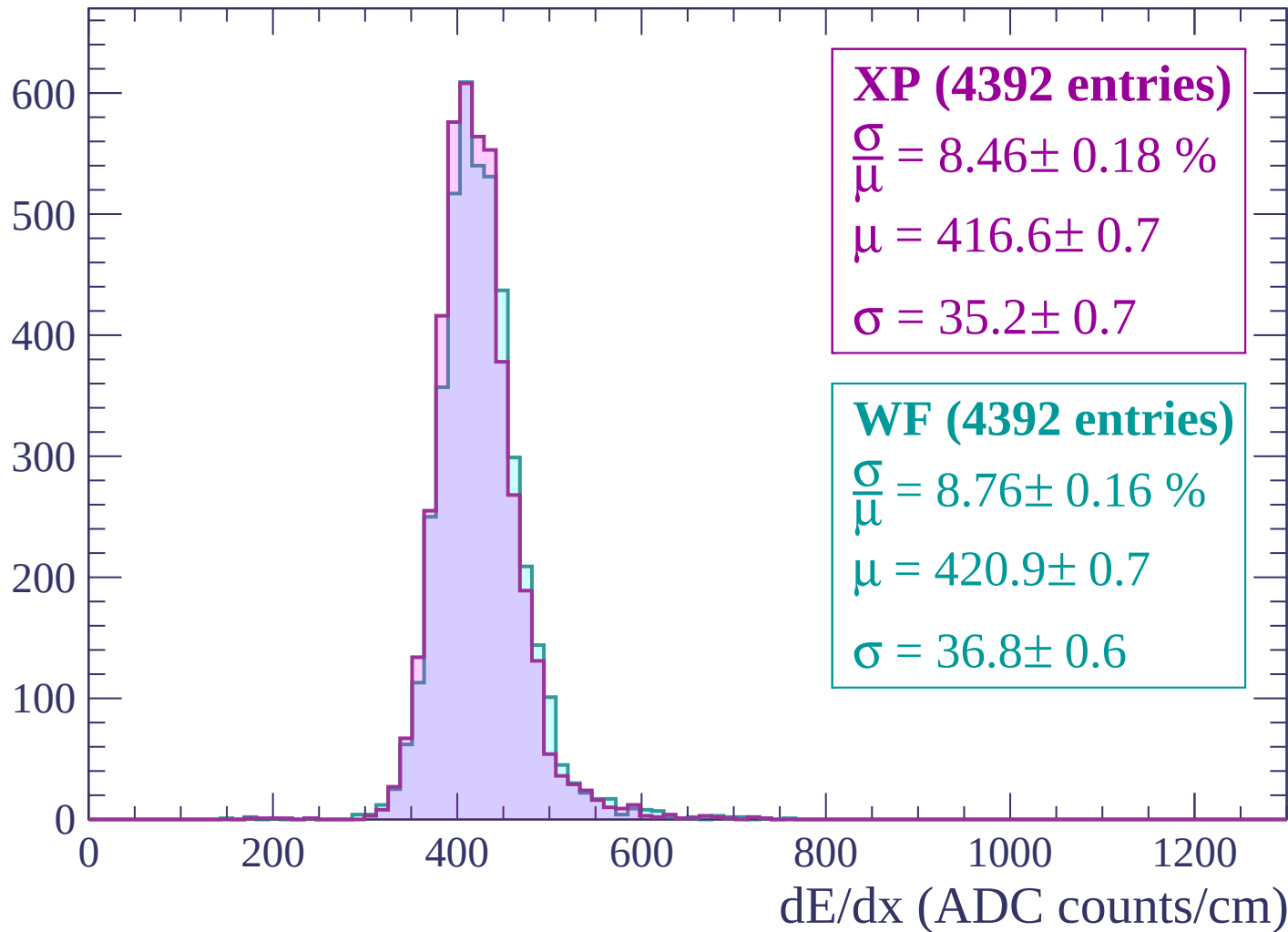
$$\mu = 415.4 \pm 0.6$$

$$\sigma = 37.5 \pm 0.6$$

dE/dx (ADC counts/cm)

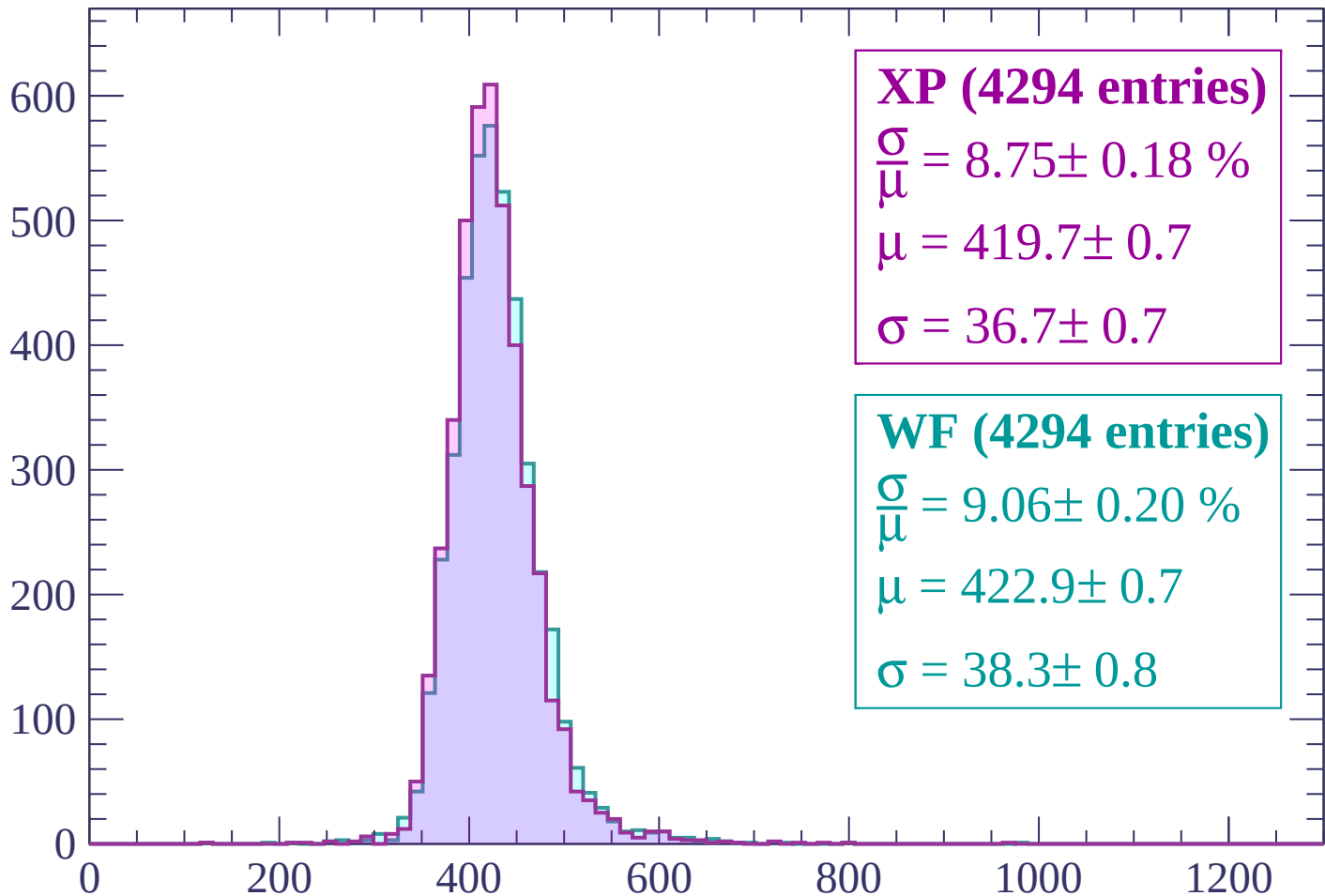
Energy loss | $780 < p < 840$

Count



Energy loss | $840 < p < 900$

Count



XP (4294 entries)

$$\frac{\sigma}{\mu} = 8.75 \pm 0.18 \%$$

$$\mu = 419.7 \pm 0.7$$

$$\sigma = 36.7 \pm 0.7$$

WF (4294 entries)

$$\frac{\sigma}{\mu} = 9.06 \pm 0.20 \%$$

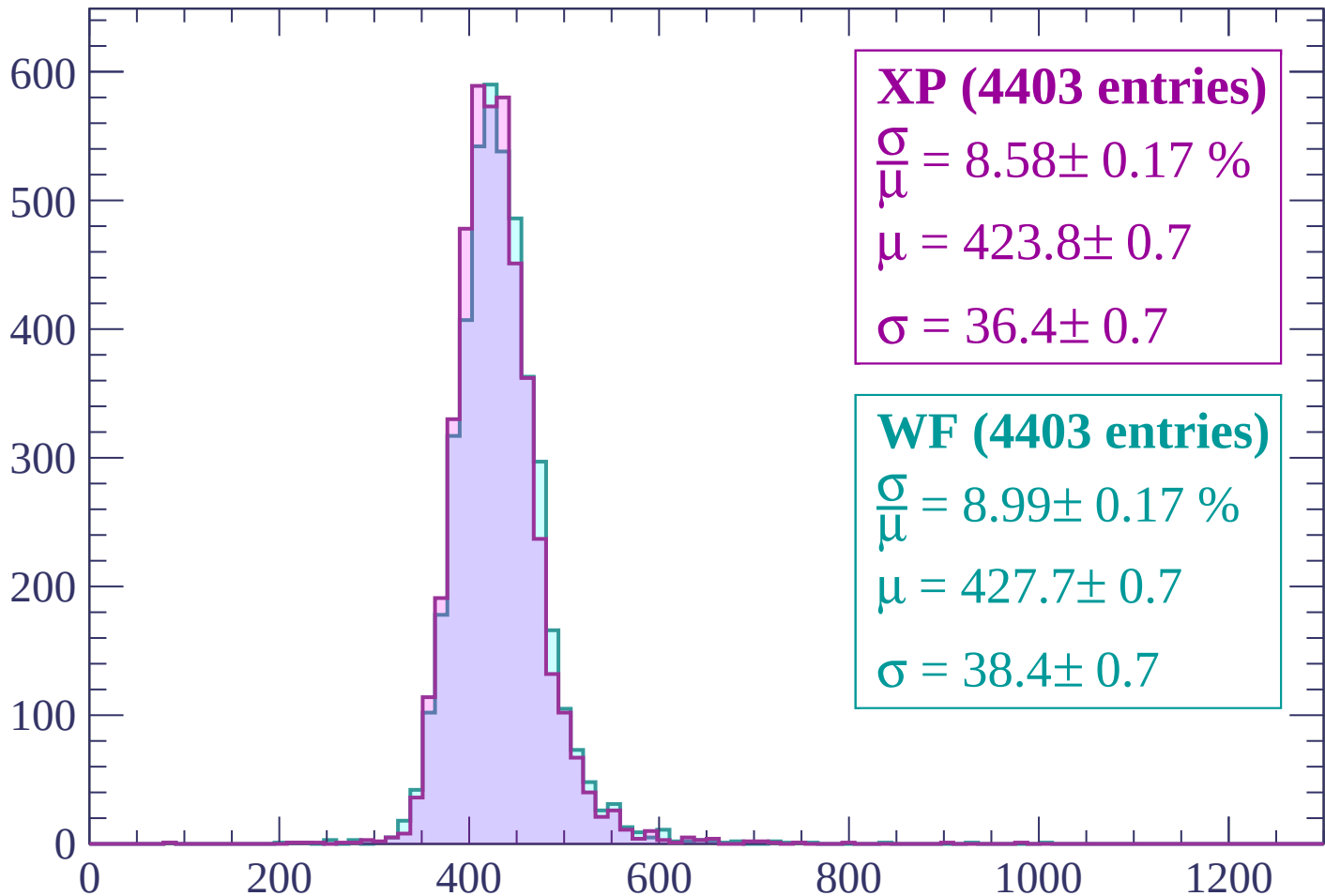
$$\mu = 422.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $900 < p < 960$

Count



XP (4403 entries)

$\frac{\sigma}{\mu} = 8.58 \pm 0.17 \%$

$\mu = 423.8 \pm 0.7$

$\sigma = 36.4 \pm 0.7$

WF (4403 entries)

$\frac{\sigma}{\mu} = 8.99 \pm 0.17 \%$

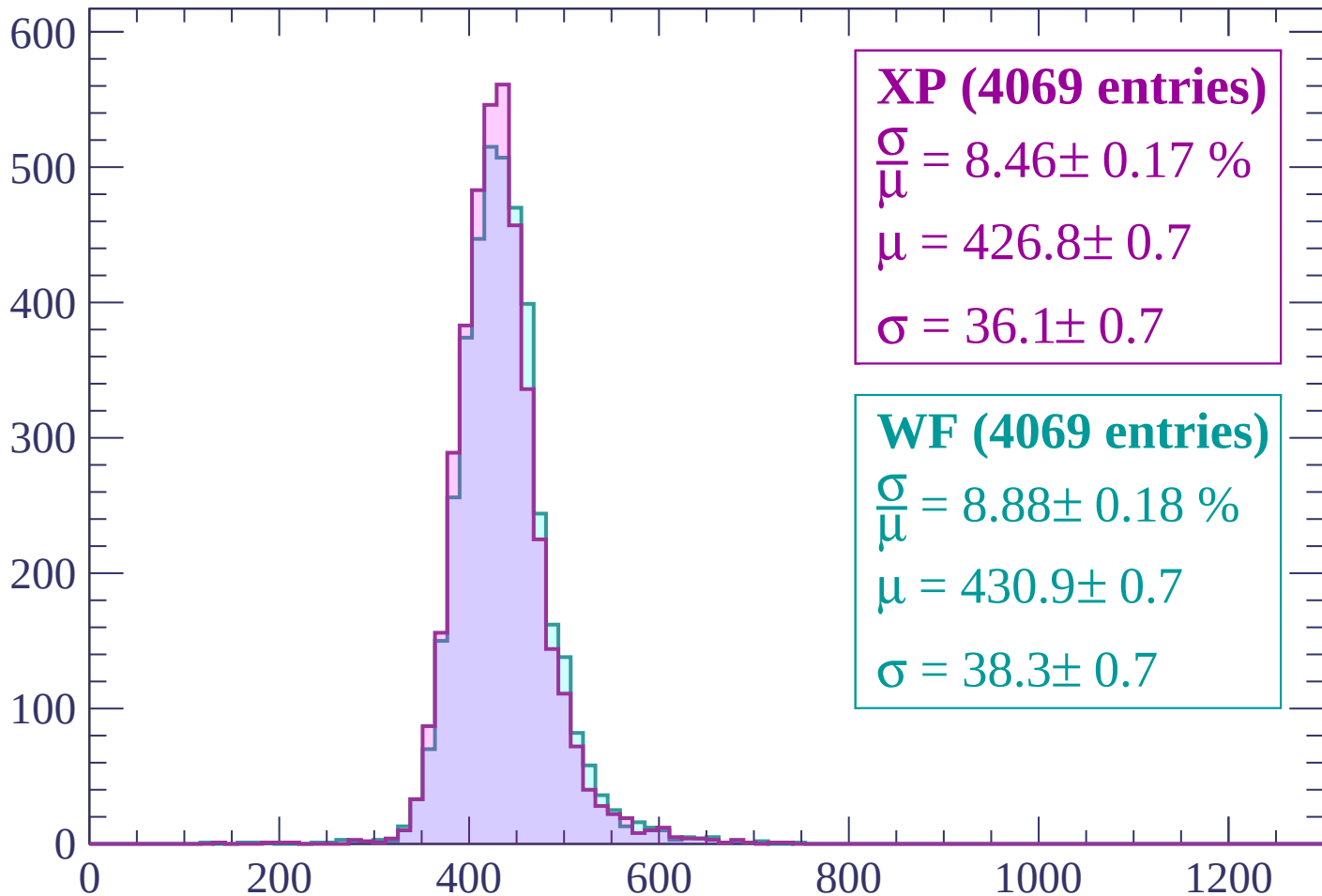
$\mu = 427.7 \pm 0.7$

$\sigma = 38.4 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1020$

Count



XP (4069 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.17 \%$$

$$\mu = 426.8 \pm 0.7$$

$$\sigma = 36.1 \pm 0.7$$

WF (4069 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.18 \%$$

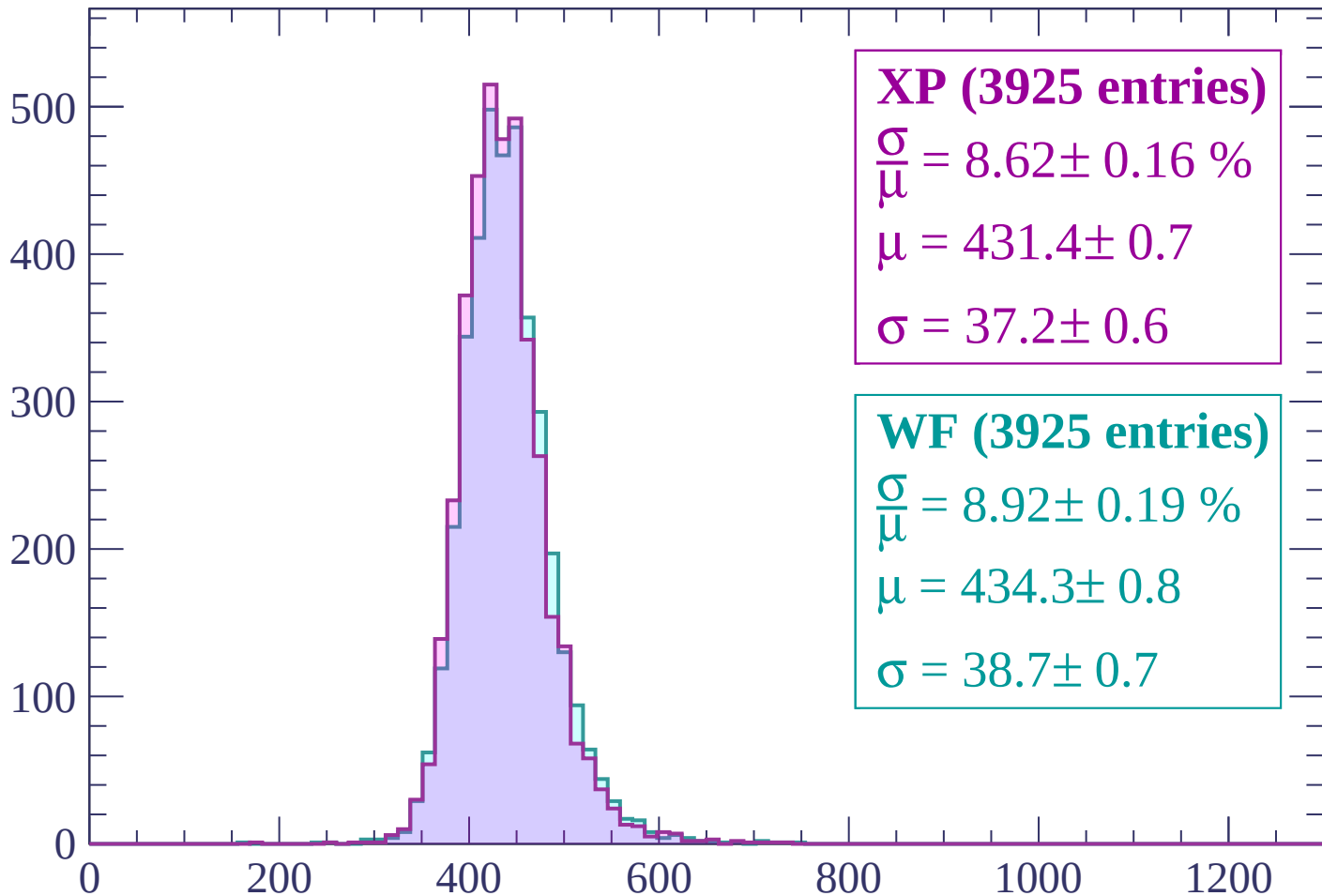
$$\mu = 430.9 \pm 0.7$$

$$\sigma = 38.3 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1020 < p < 1080$

Count



XP (3925 entries)

$$\frac{\sigma}{\mu} = 8.62 \pm 0.16 \%$$

$$\mu = 431.4 \pm 0.7$$

$$\sigma = 37.2 \pm 0.6$$

WF (3925 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.19 \%$$

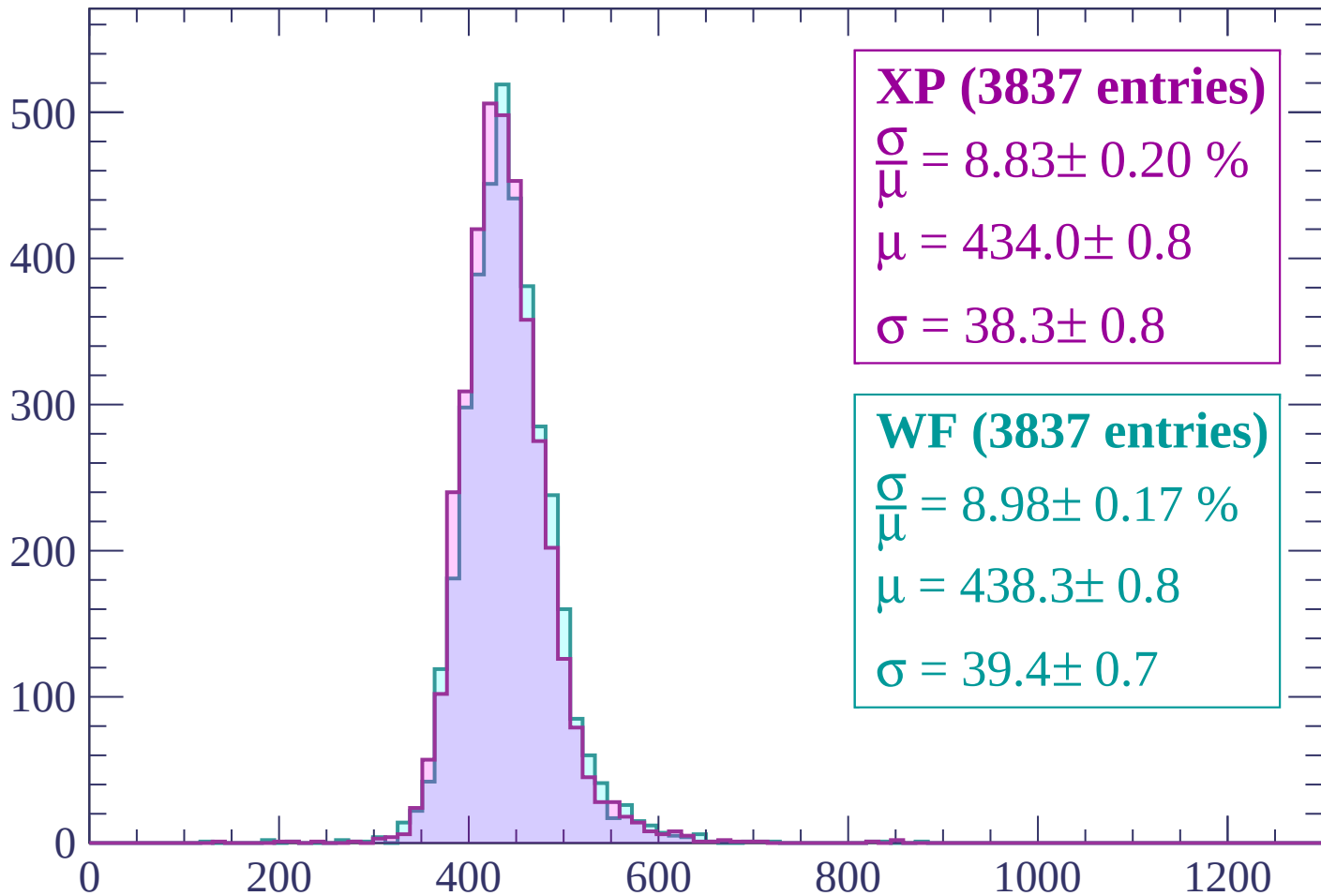
$$\mu = 434.3 \pm 0.8$$

$$\sigma = 38.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1140$

Count



XP (3837 entries)

$$\frac{\sigma}{\mu} = 8.83 \pm 0.20 \%$$

$$\mu = 434.0 \pm 0.8$$

$$\sigma = 38.3 \pm 0.8$$

WF (3837 entries)

$$\frac{\sigma}{\mu} = 8.98 \pm 0.17 \%$$

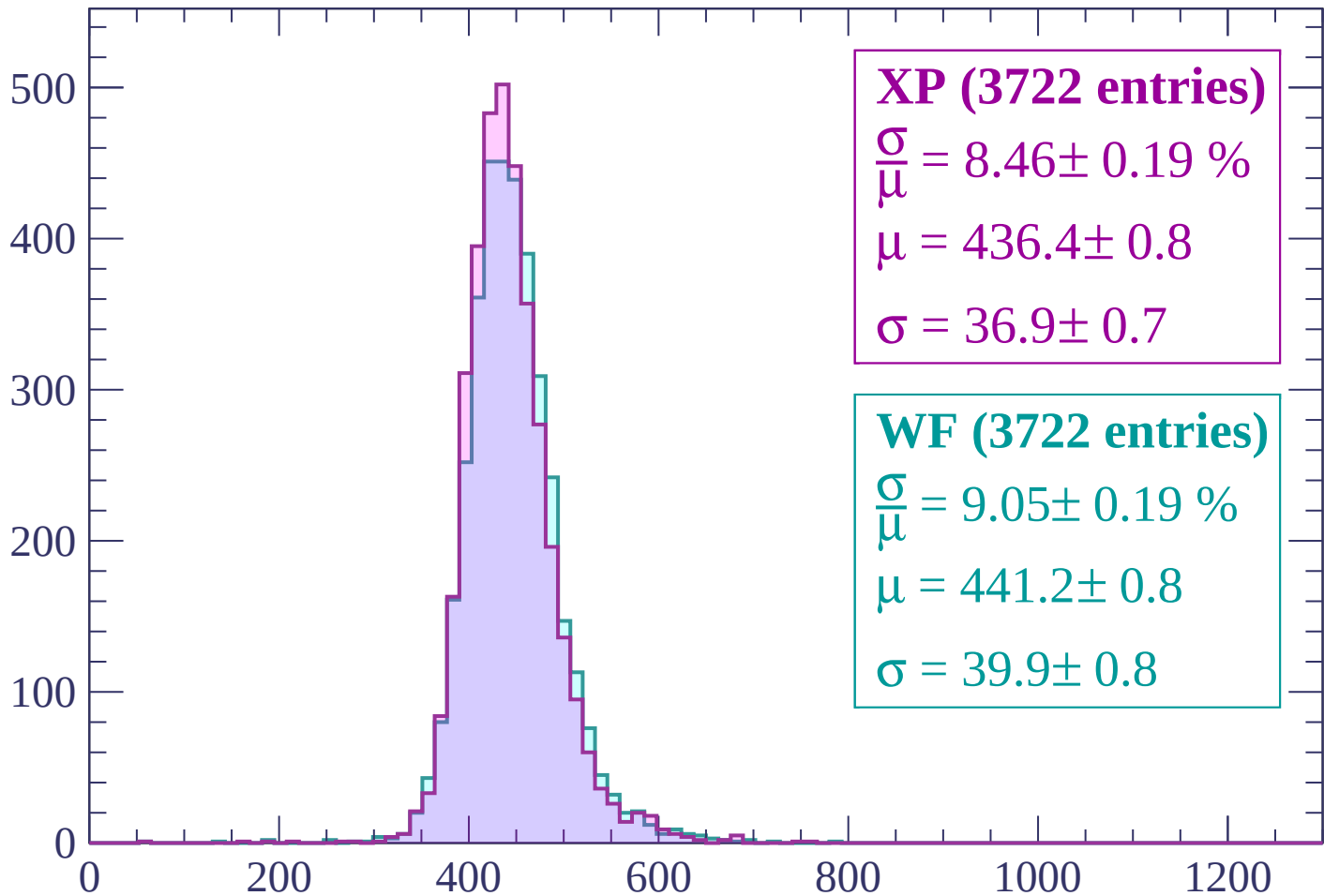
$$\mu = 438.3 \pm 0.8$$

$$\sigma = 39.4 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $1140 < p < 1200$

Count



XP (3722 entries)

$$\frac{\sigma}{\mu} = 8.46 \pm 0.19 \%$$

$$\mu = 436.4 \pm 0.8$$

$$\sigma = 36.9 \pm 0.7$$

WF (3722 entries)

$$\frac{\sigma}{\mu} = 9.05 \pm 0.19 \%$$

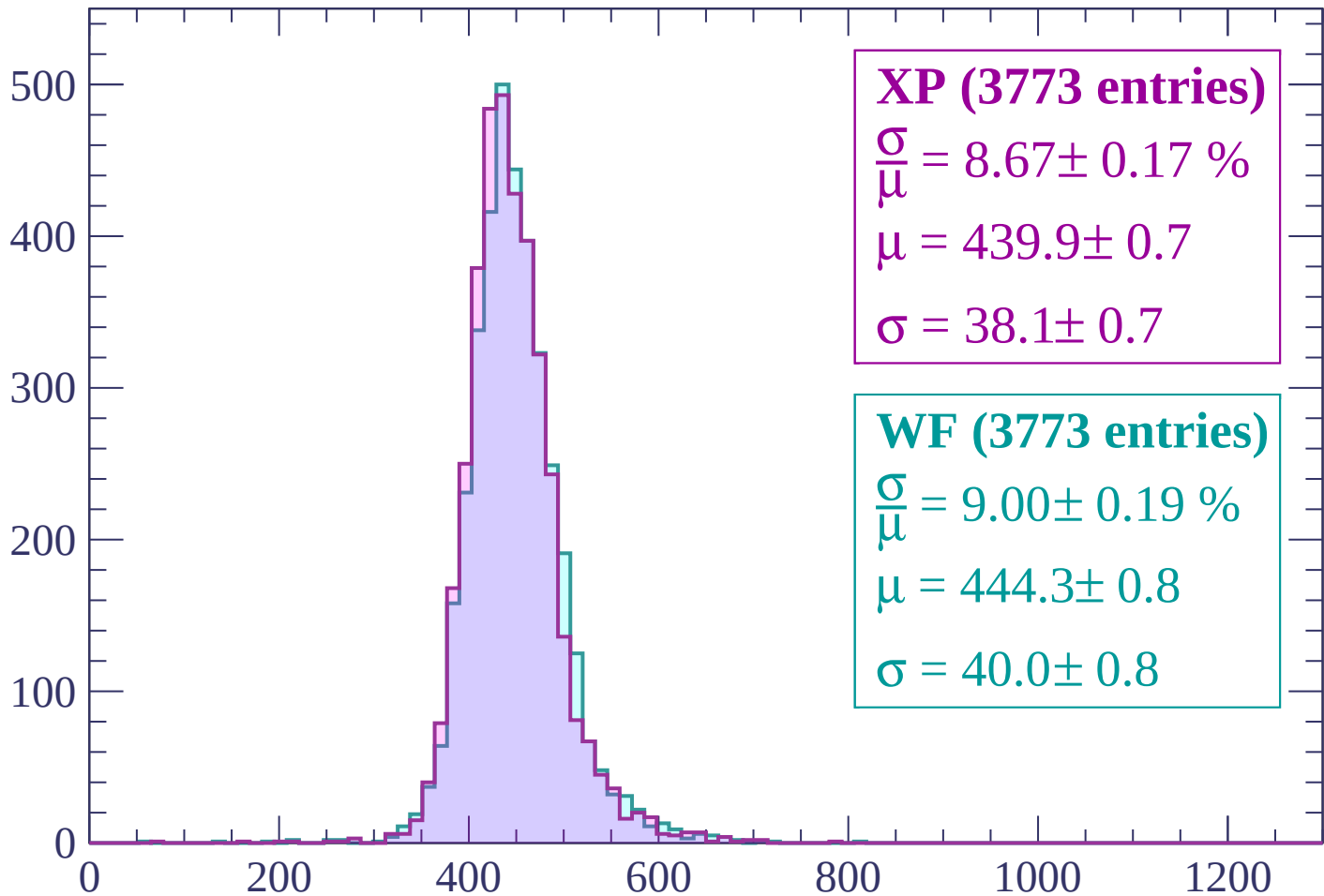
$$\mu = 441.2 \pm 0.8$$

$$\sigma = 39.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1260$

Count



XP (3773 entries)

$\frac{\sigma}{\mu} = 8.67 \pm 0.17 \%$

$\mu = 439.9 \pm 0.7$

$\sigma = 38.1 \pm 0.7$

WF (3773 entries)

$\frac{\sigma}{\mu} = 9.00 \pm 0.19 \%$

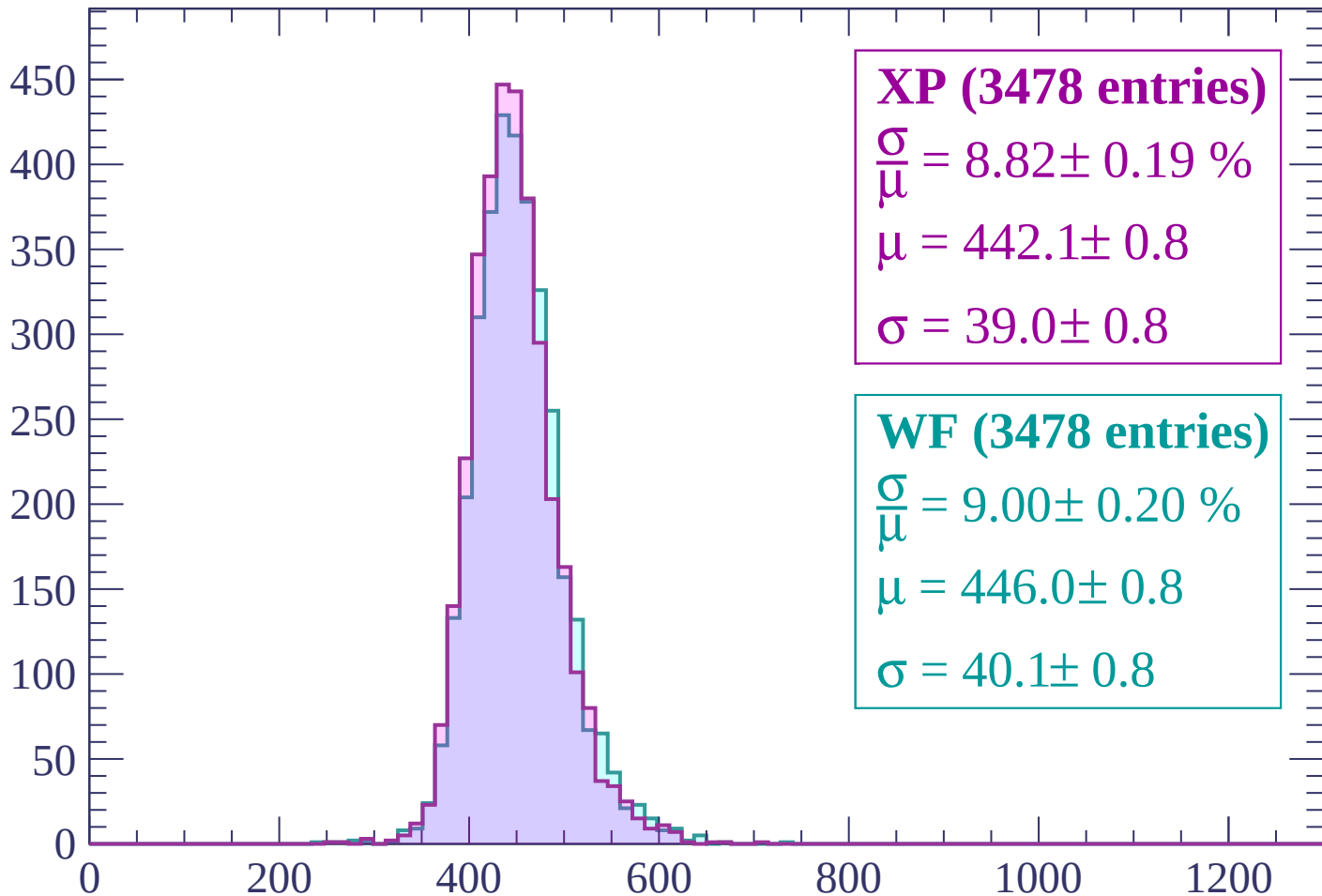
$\mu = 444.3 \pm 0.8$

$\sigma = 40.0 \pm 0.8$

dE/dx (ADC counts/cm)

Energy loss | $1260 < p < 1320$

Count



XP (3478 entries)

$$\frac{\sigma}{\mu} = 8.82 \pm 0.19 \%$$

$$\mu = 442.1 \pm 0.8$$

$$\sigma = 39.0 \pm 0.8$$

WF (3478 entries)

$$\frac{\sigma}{\mu} = 9.00 \pm 0.20 \%$$

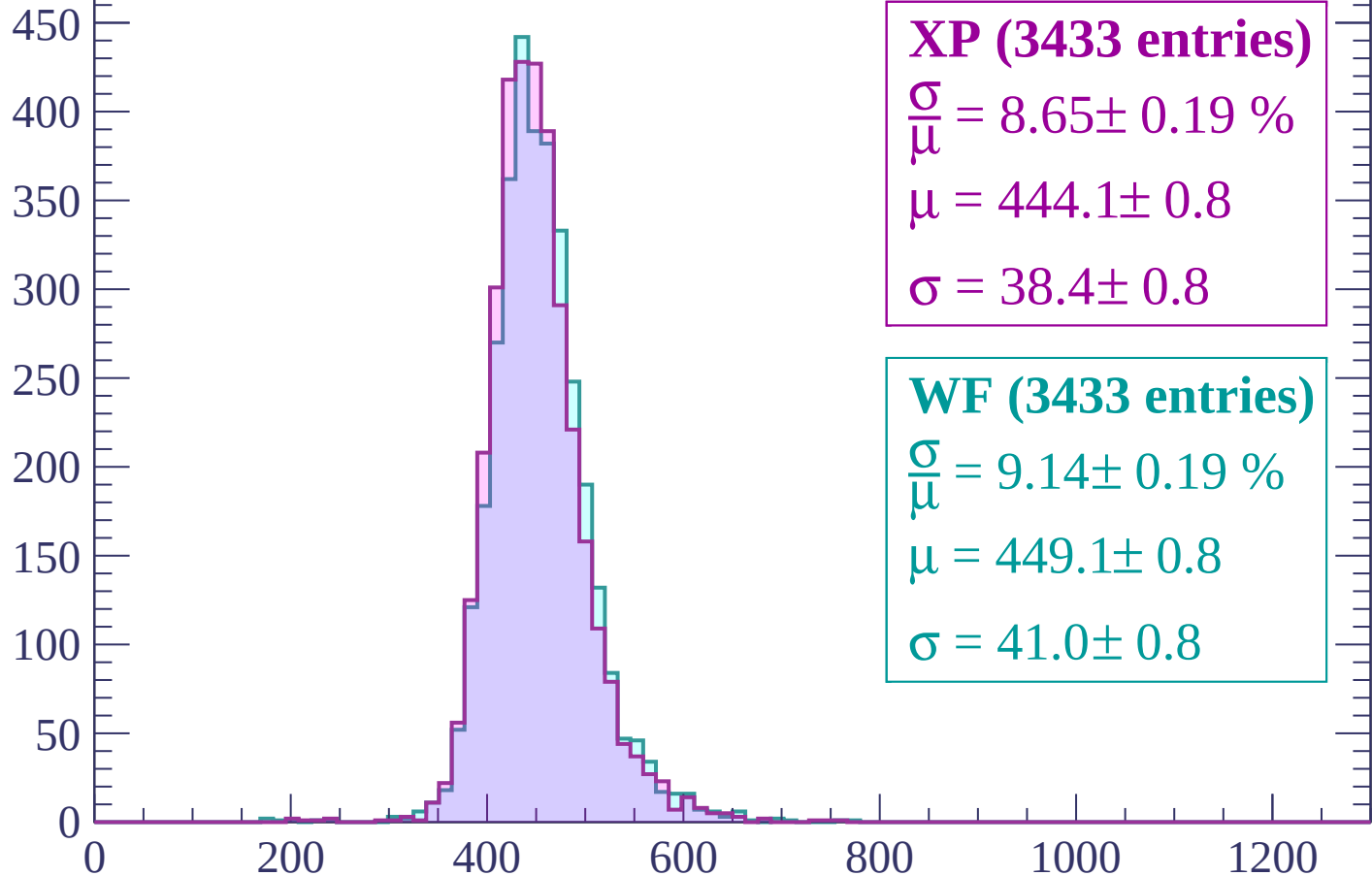
$$\mu = 446.0 \pm 0.8$$

$$\sigma = 40.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1380$

Count



XP (3433 entries)

$$\frac{\sigma}{\mu} = 8.65 \pm 0.19 \%$$

$$\mu = 444.1 \pm 0.8$$

$$\sigma = 38.4 \pm 0.8$$

WF (3433 entries)

$$\frac{\sigma}{\mu} = 9.14 \pm 0.19 \%$$

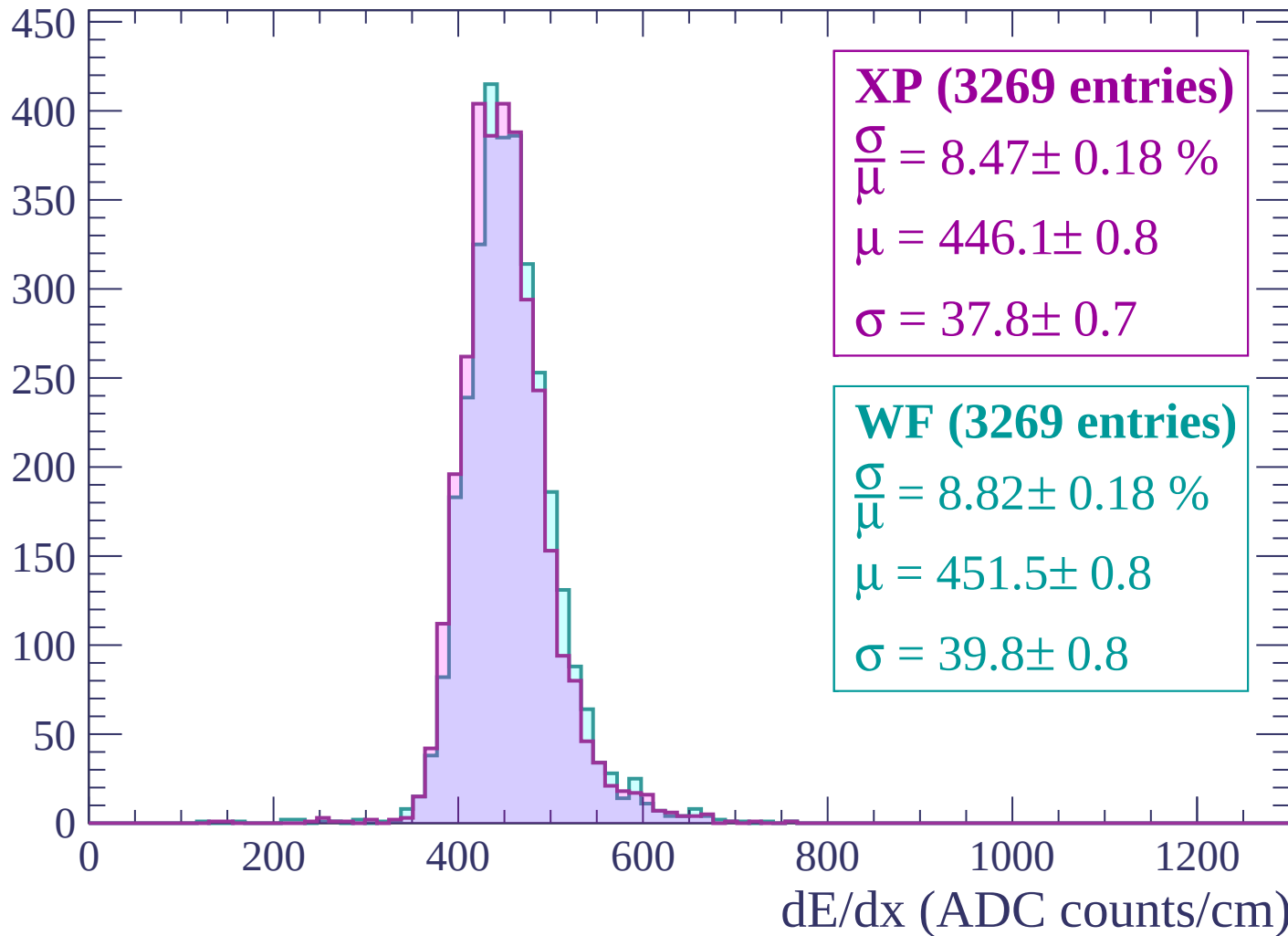
$$\mu = 449.1 \pm 0.8$$

$$\sigma = 41.0 \pm 0.8$$

dE/dx (ADC counts/cm)

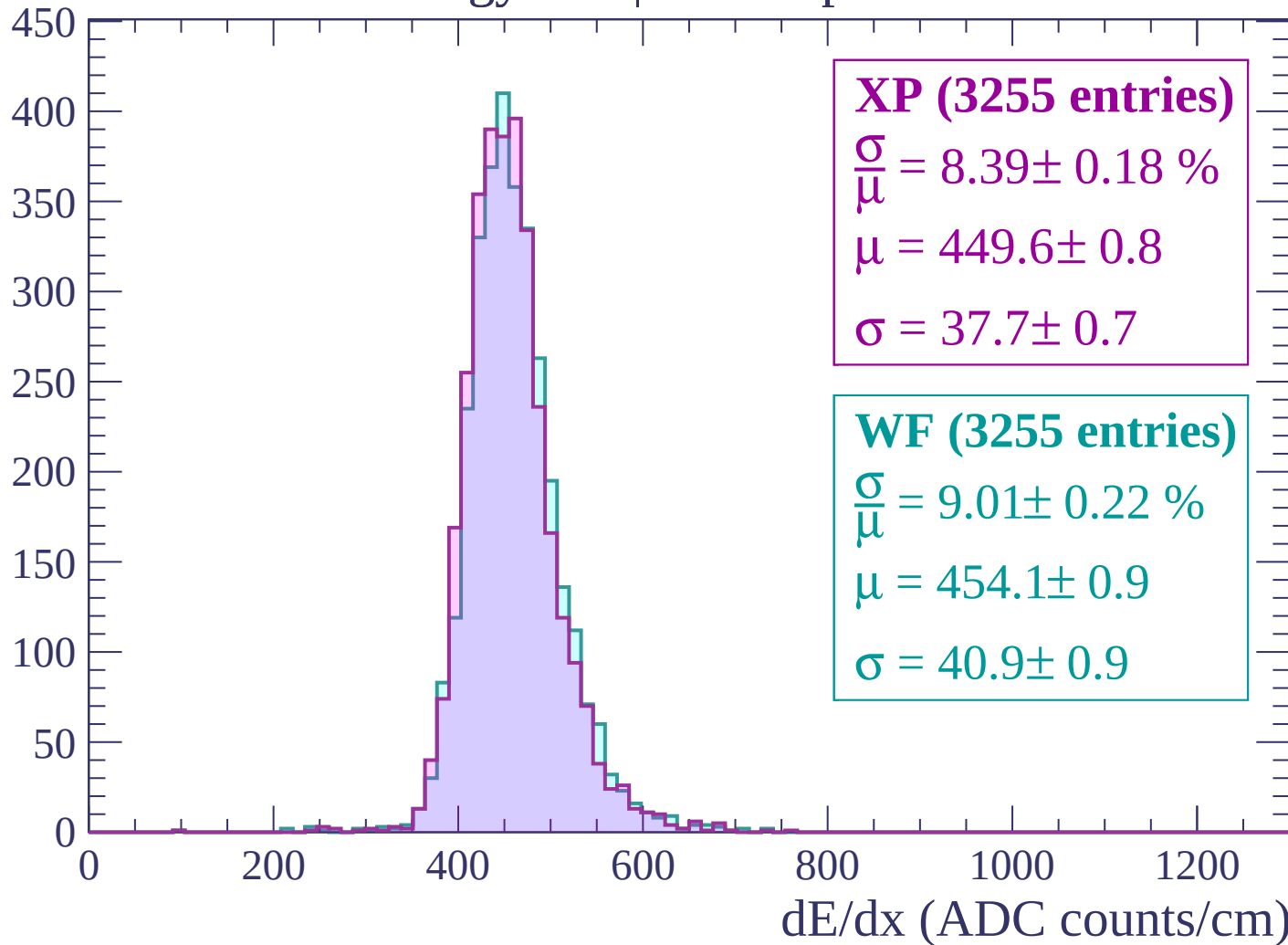
Energy loss | $1380 < p < 1440$

Count



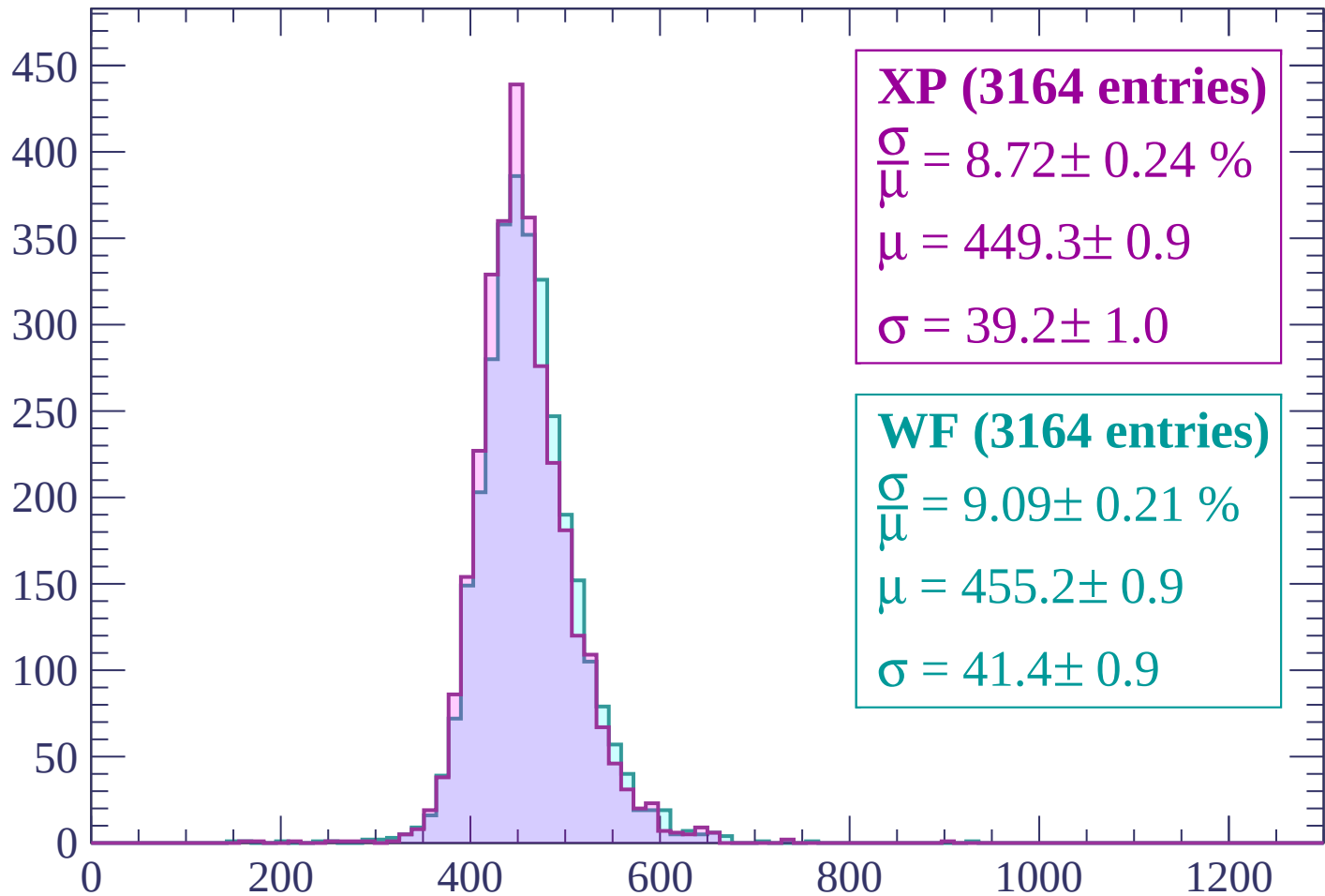
Energy loss | $1440 < p < 1500$

Count



Energy loss | $1500 < p < 1560$

Count



XP (3164 entries)

$$\frac{\sigma}{\mu} = 8.72 \pm 0.24 \%$$

$$\mu = 449.3 \pm 0.9$$

$$\sigma = 39.2 \pm 1.0$$

WF (3164 entries)

$$\frac{\sigma}{\mu} = 9.09 \pm 0.21 \%$$

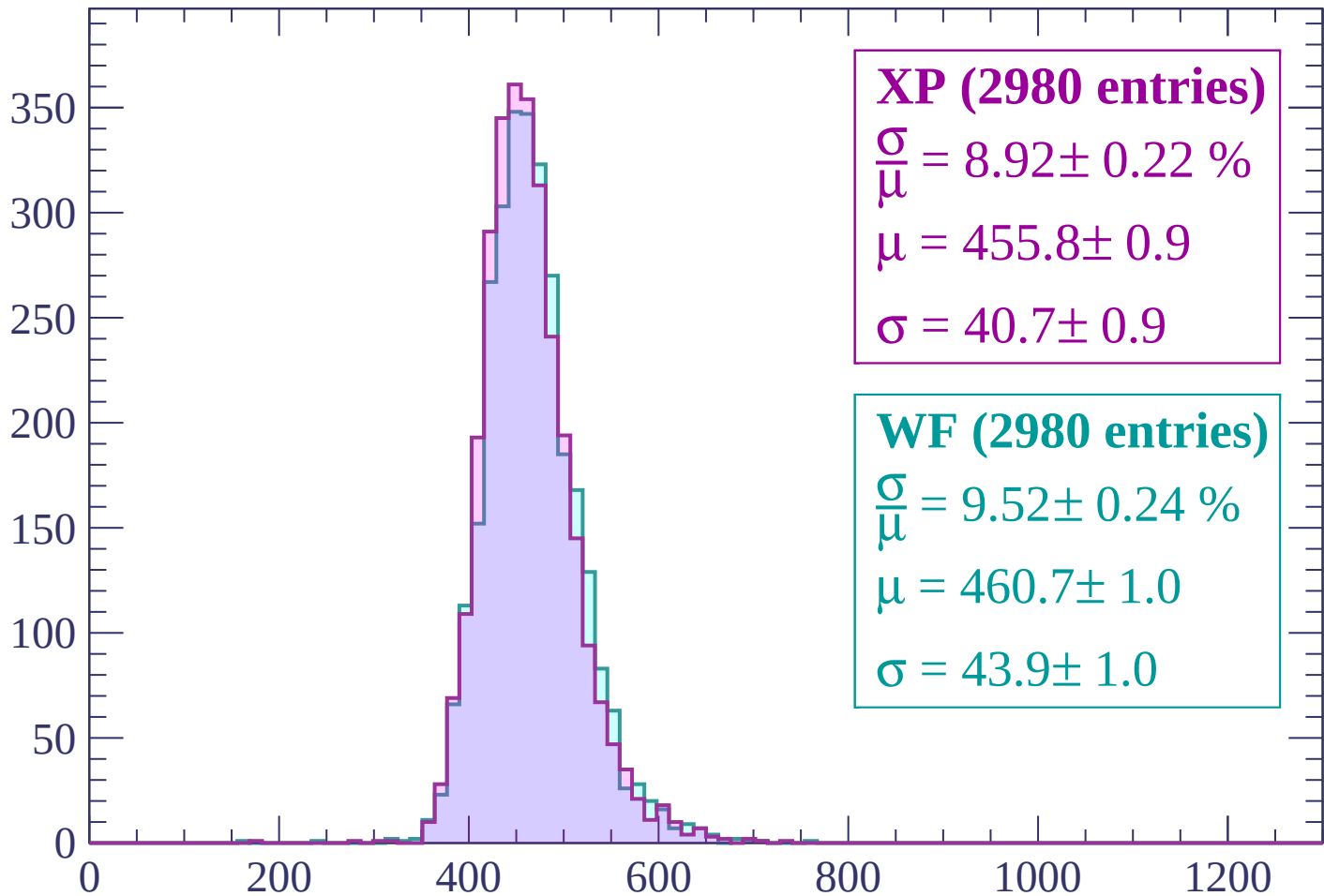
$$\mu = 455.2 \pm 0.9$$

$$\sigma = 41.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1560 < p < 1620$

Count



XP (2980 entries)

$$\frac{\sigma}{\mu} = 8.92 \pm 0.22 \%$$

$$\mu = 455.8 \pm 0.9$$

$$\sigma = 40.7 \pm 0.9$$

WF (2980 entries)

$$\frac{\sigma}{\mu} = 9.52 \pm 0.24 \%$$

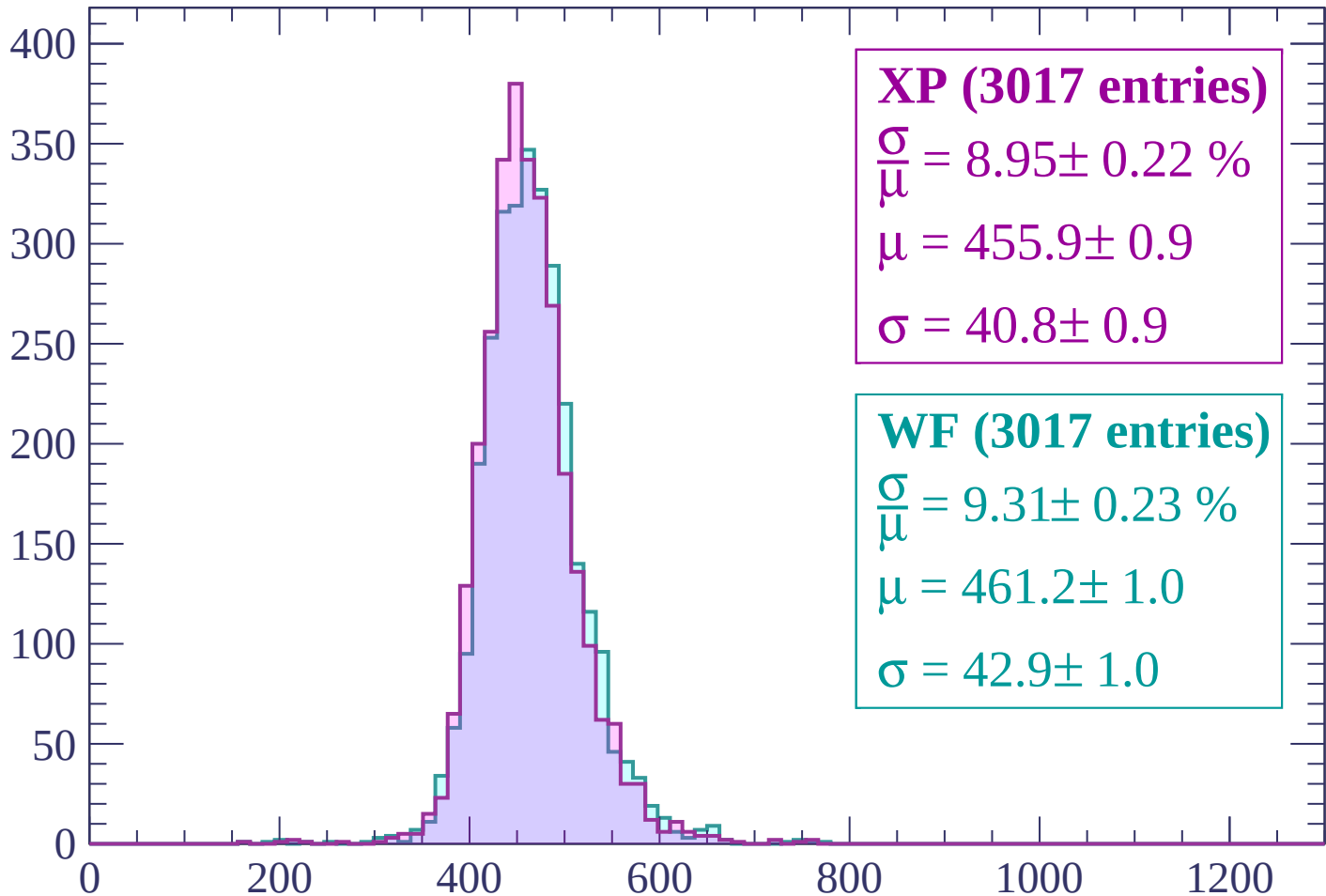
$$\mu = 460.7 \pm 1.0$$

$$\sigma = 43.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1620 < p < 1680$

Count



XP (3017 entries)

$$\frac{\sigma}{\mu} = 8.95 \pm 0.22 \%$$

$$\mu = 455.9 \pm 0.9$$

$$\sigma = 40.8 \pm 0.9$$

WF (3017 entries)

$$\frac{\sigma}{\mu} = 9.31 \pm 0.23 \%$$

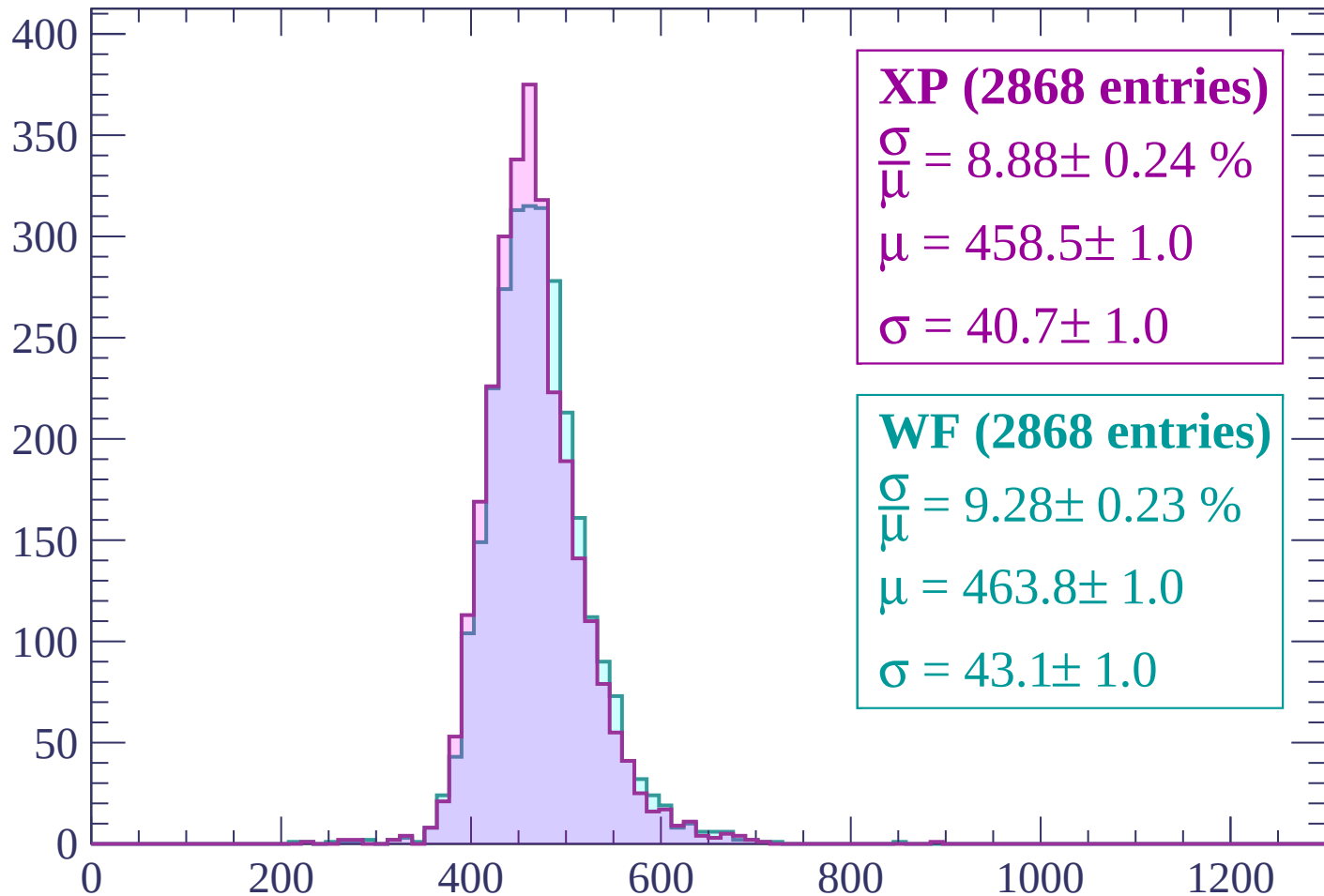
$$\mu = 461.2 \pm 1.0$$

$$\sigma = 42.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1740$

Count



XP (2868 entries)

$$\frac{\sigma}{\mu} = 8.88 \pm 0.24 \%$$

$$\mu = 458.5 \pm 1.0$$

$$\sigma = 40.7 \pm 1.0$$

WF (2868 entries)

$$\frac{\sigma}{\mu} = 9.28 \pm 0.23 \%$$

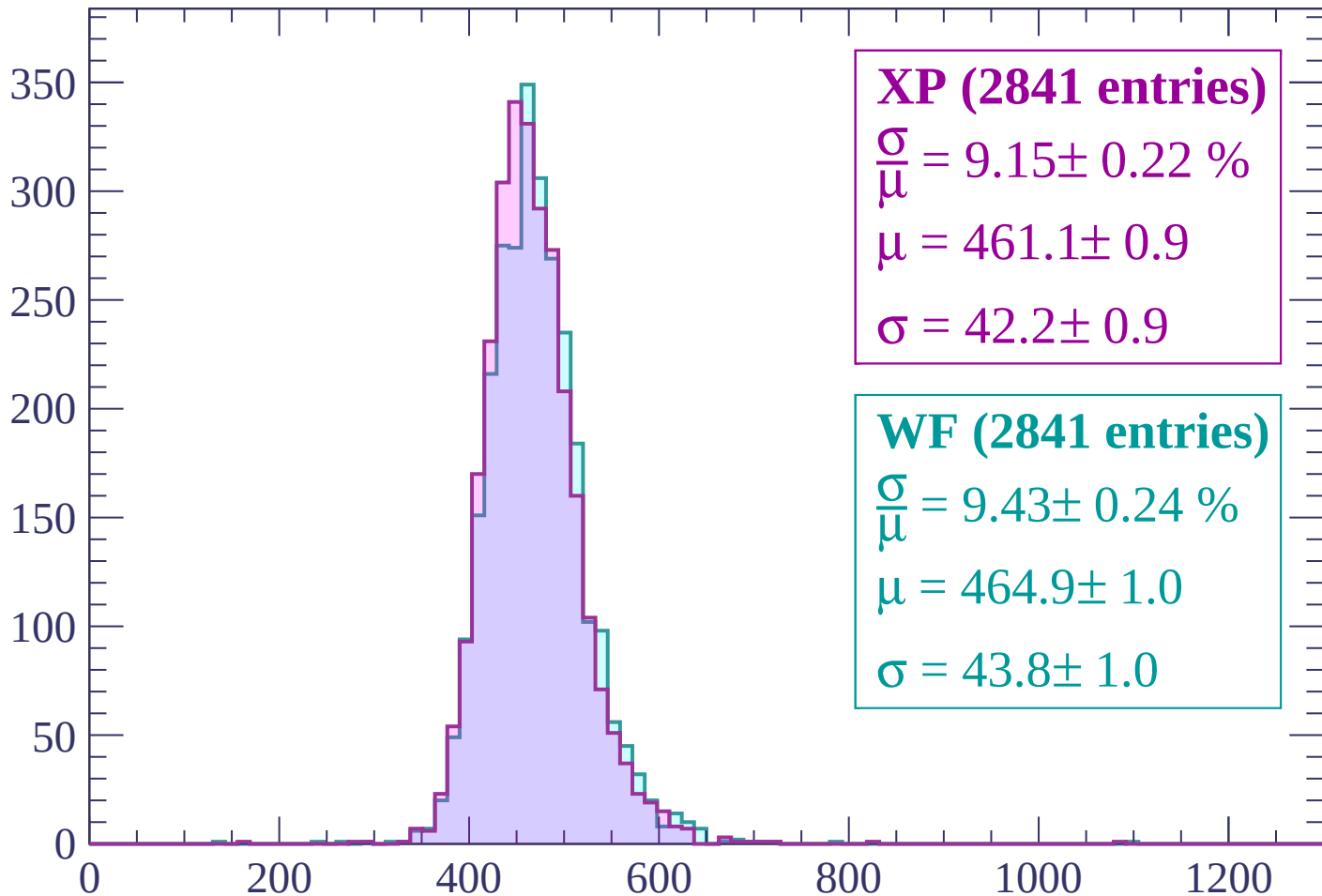
$$\mu = 463.8 \pm 1.0$$

$$\sigma = 43.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1740 < p < 1800$

Count



XP (2841 entries)

$\frac{\sigma}{\mu} = 9.15 \pm 0.22 \%$

$\mu = 461.1 \pm 0.9$

$\sigma = 42.2 \pm 0.9$

WF (2841 entries)

$\frac{\sigma}{\mu} = 9.43 \pm 0.24 \%$

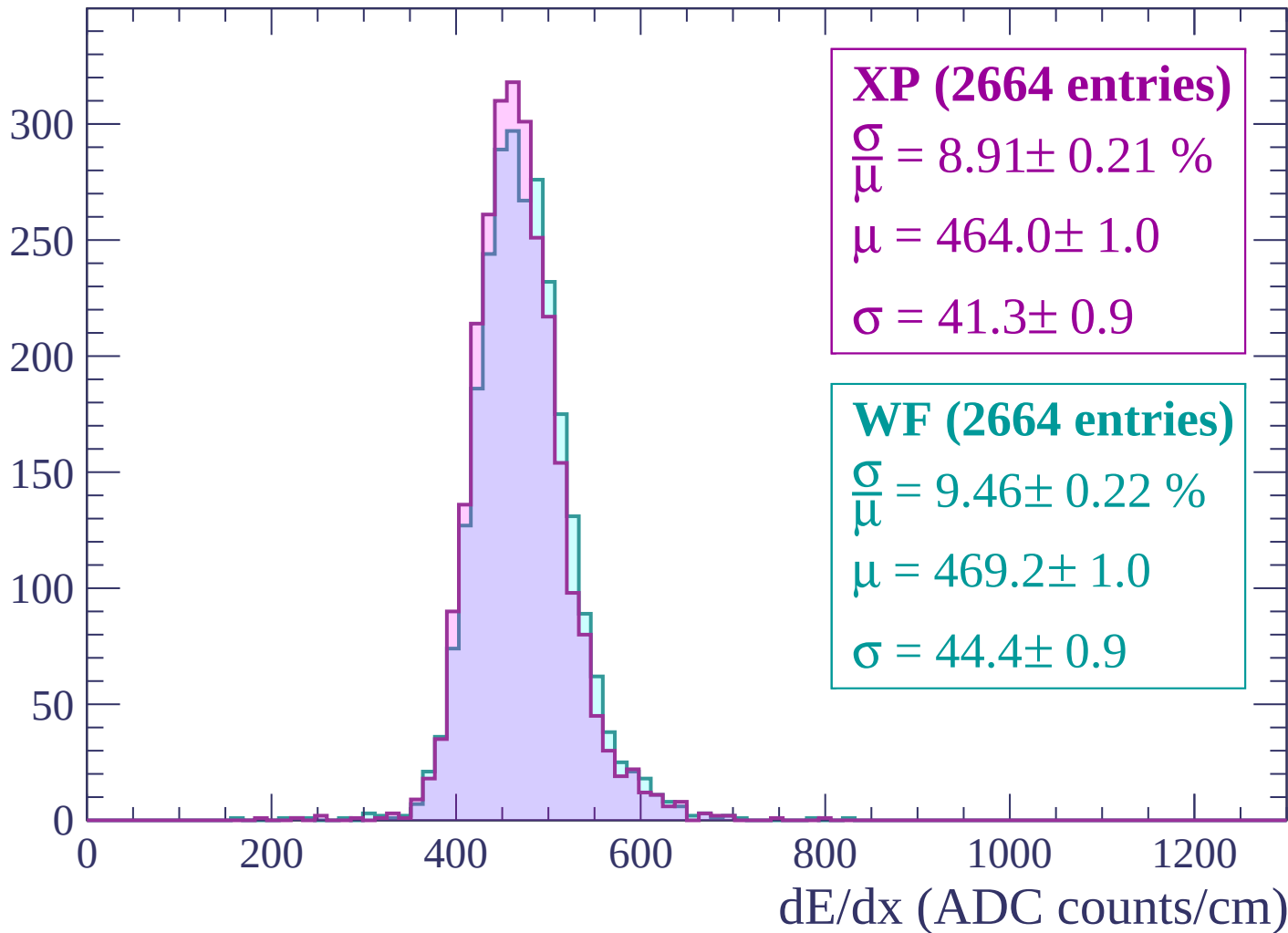
$\mu = 464.9 \pm 1.0$

$\sigma = 43.8 \pm 1.0$

dE/dx (ADC counts/cm)

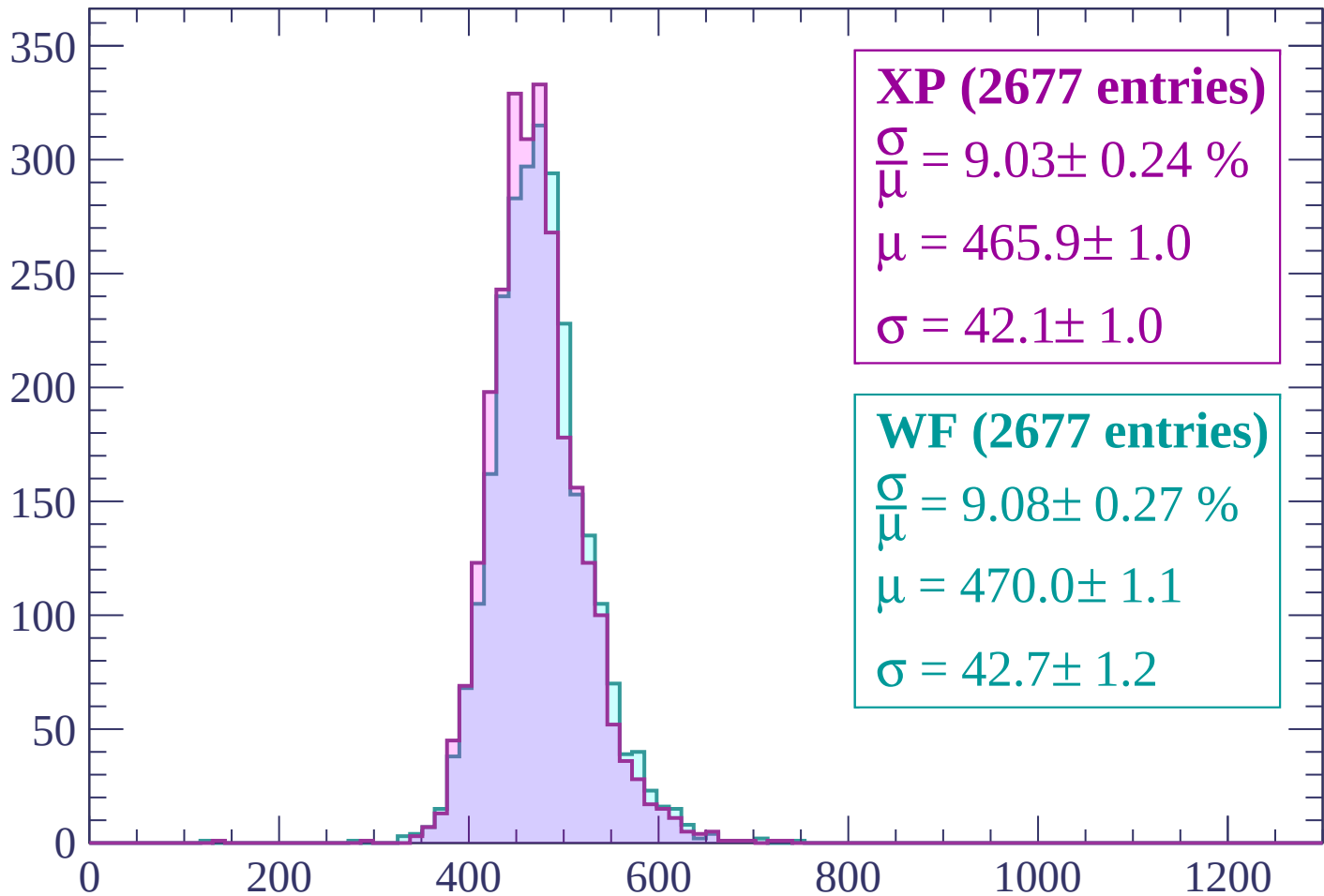
Energy loss | $1800 < p < 1860$

Count



Energy loss | $1860 < p < 1920$

Count



XP (2677 entries)

$$\frac{\sigma}{\mu} = 9.03 \pm 0.24 \%$$

$$\mu = 465.9 \pm 1.0$$

$$\sigma = 42.1 \pm 1.0$$

WF (2677 entries)

$$\frac{\sigma}{\mu} = 9.08 \pm 0.27 \%$$

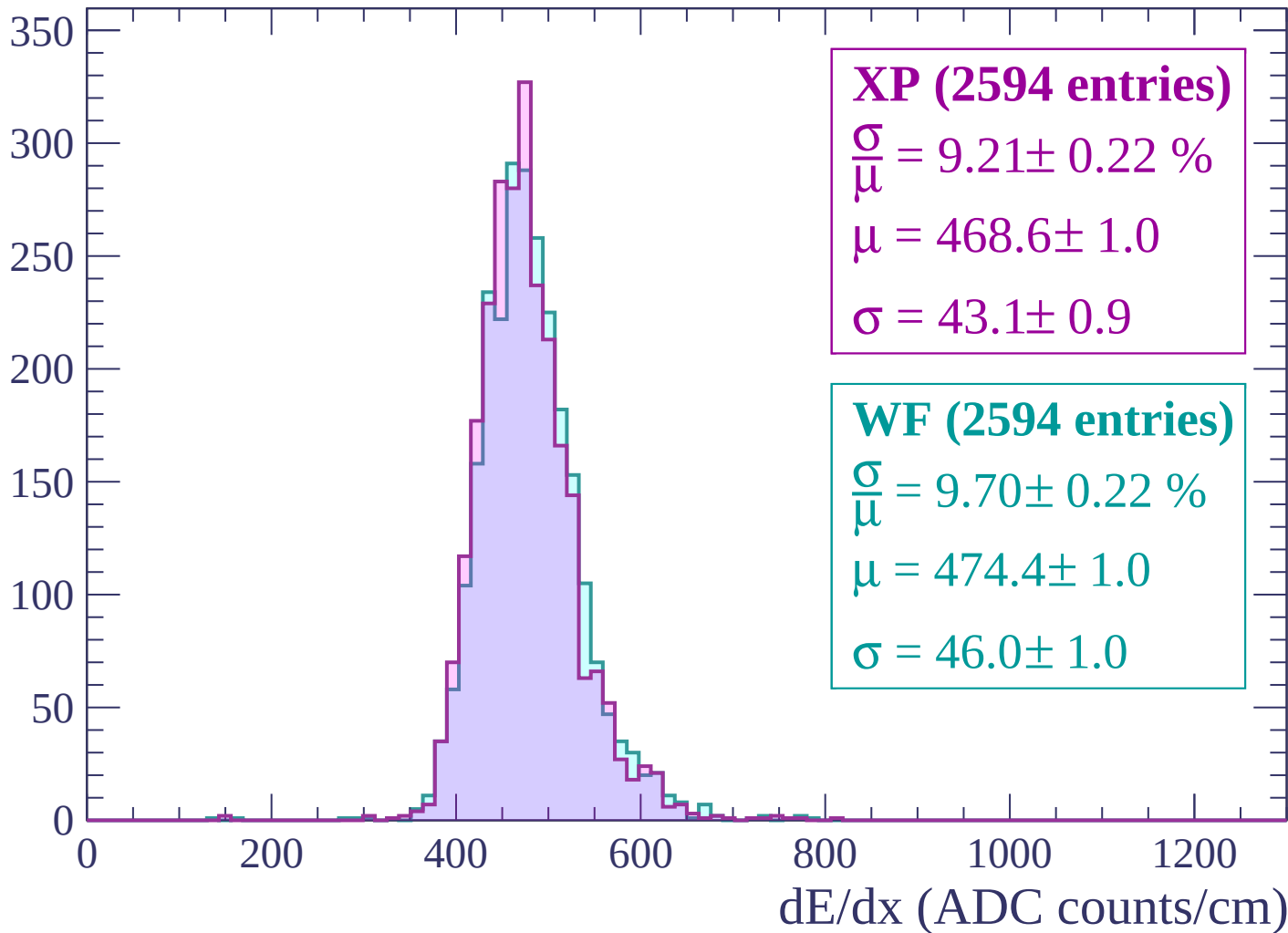
$$\mu = 470.0 \pm 1.1$$

$$\sigma = 42.7 \pm 1.2$$

dE/dx (ADC counts/cm)

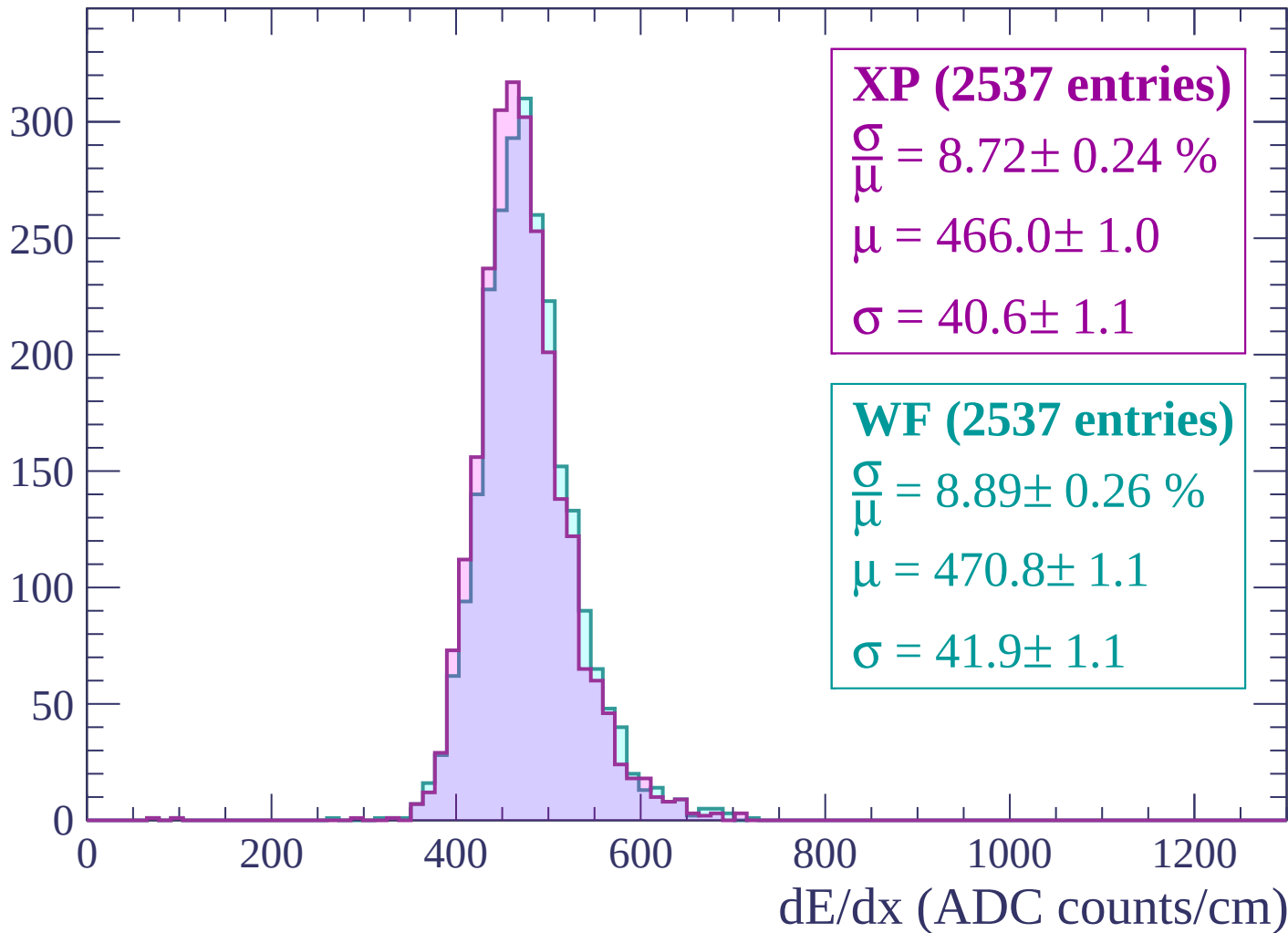
Energy loss | $1920 < p < 1980$

Count



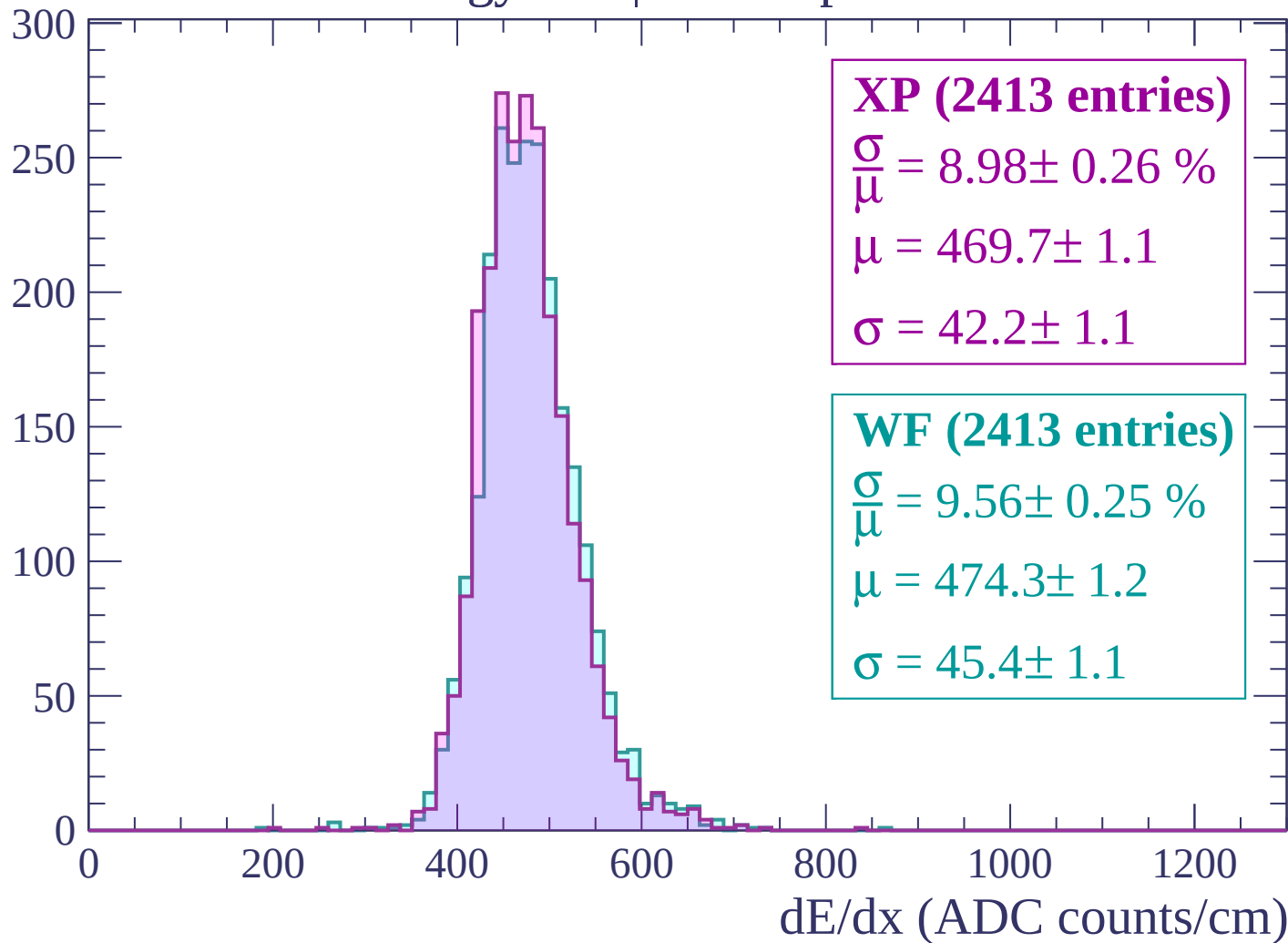
Energy loss | $1980 < p < 2040$

Count



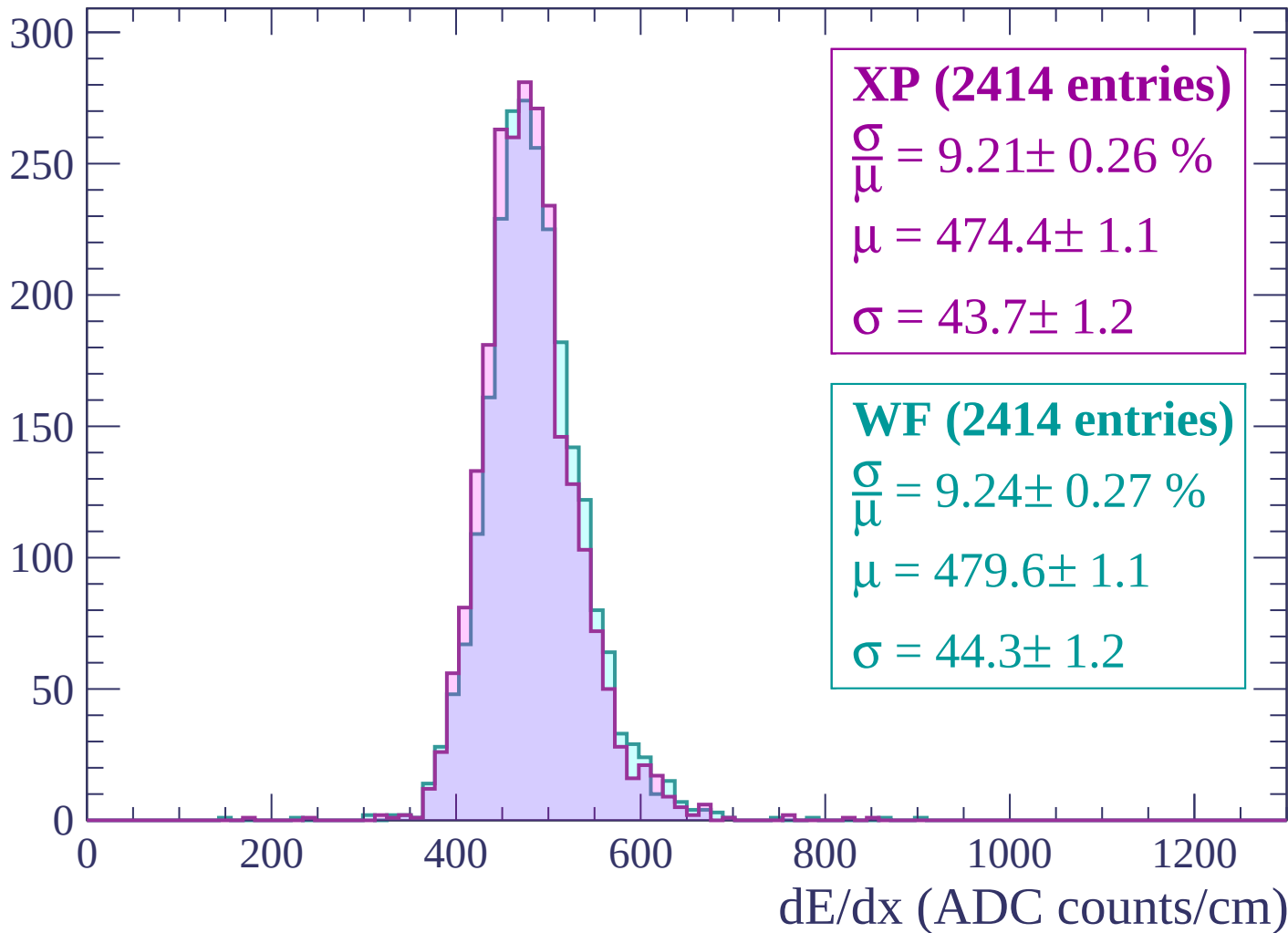
Energy loss | $2040 < p < 2100$

Count



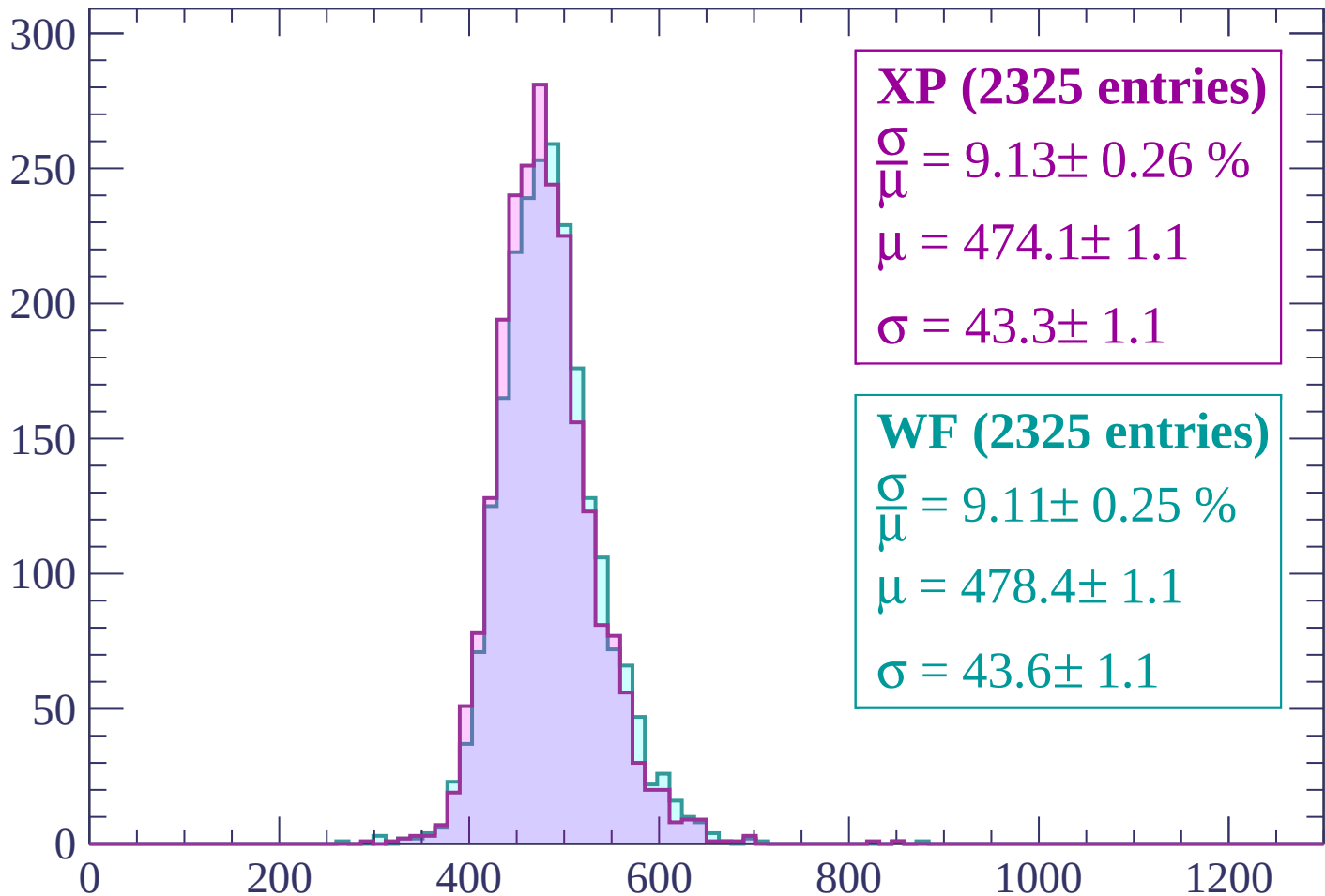
Energy loss | $2100 < p < 2160$

Count



Energy loss | $2160 < p < 2220$

Count



XP (2325 entries)

$$\frac{\sigma}{\mu} = 9.13 \pm 0.26 \%$$

$$\mu = 474.1 \pm 1.1$$

$$\sigma = 43.3 \pm 1.1$$

WF (2325 entries)

$$\frac{\sigma}{\mu} = 9.11 \pm 0.25 \%$$

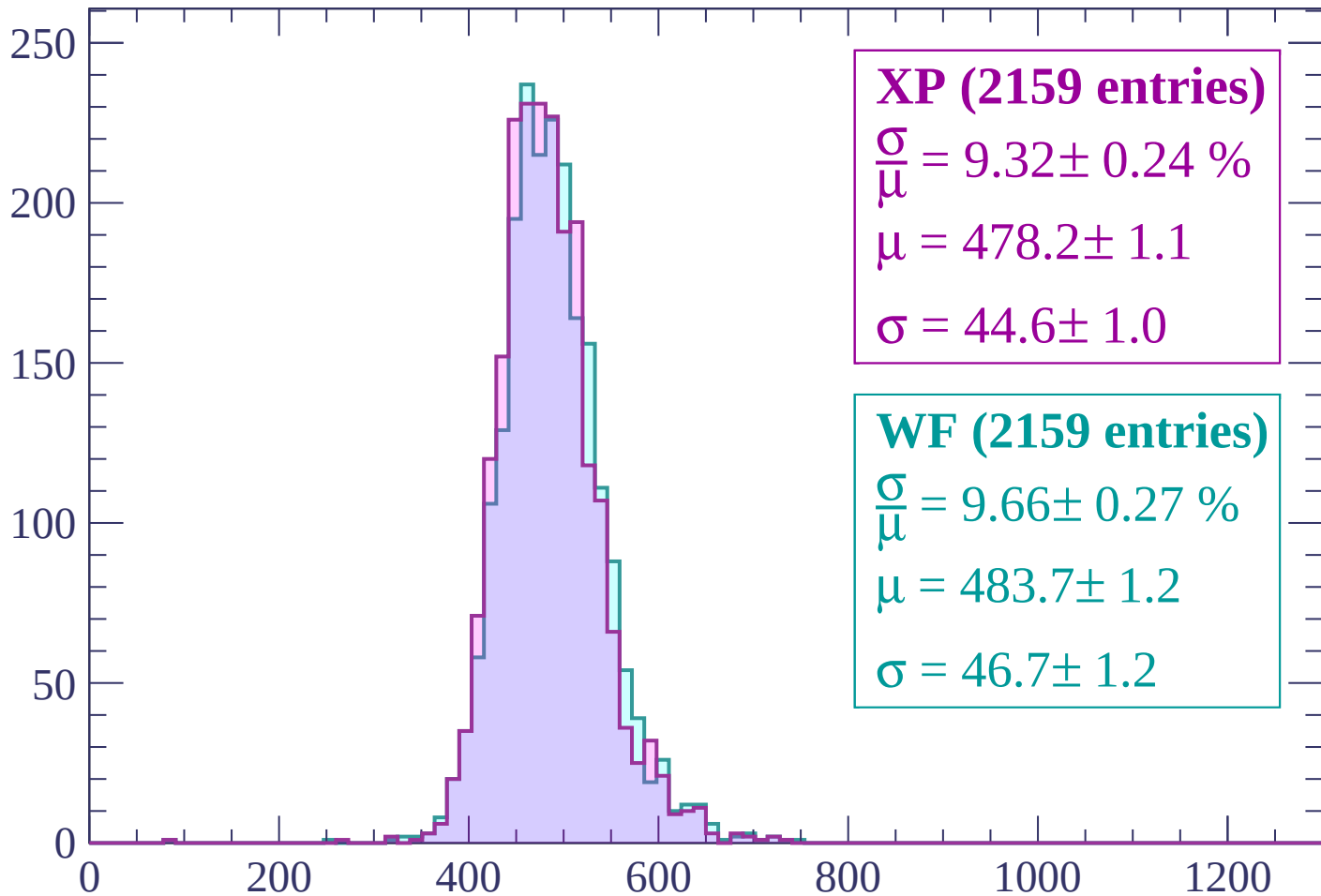
$$\mu = 478.4 \pm 1.1$$

$$\sigma = 43.6 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $2220 < p < 2280$

Count



XP (2159 entries)

$$\frac{\sigma}{\mu} = 9.32 \pm 0.24 \%$$

$$\mu = 478.2 \pm 1.1$$

$$\sigma = 44.6 \pm 1.0$$

WF (2159 entries)

$$\frac{\sigma}{\mu} = 9.66 \pm 0.27 \%$$

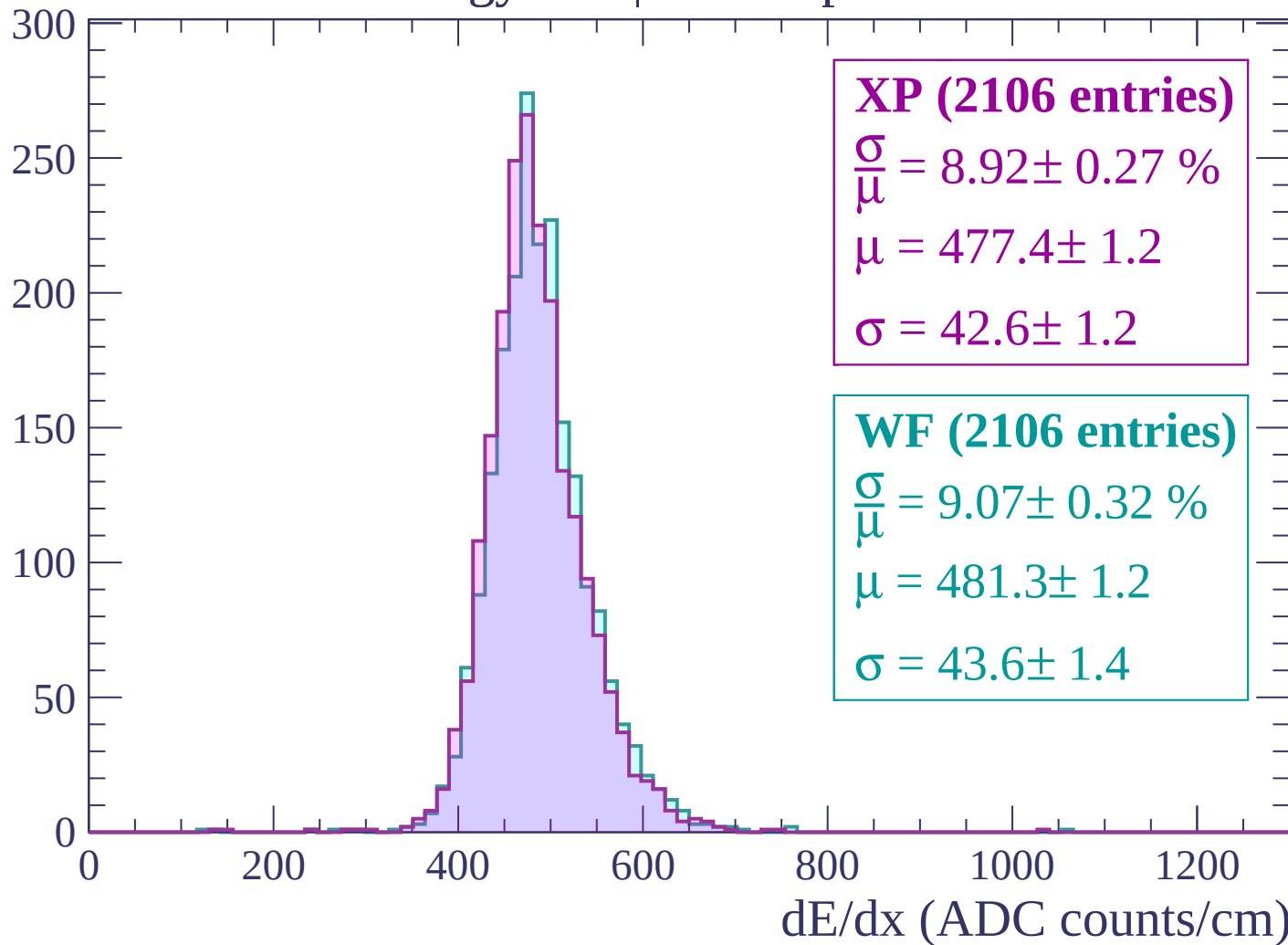
$$\mu = 483.7 \pm 1.2$$

$$\sigma = 46.7 \pm 1.2$$

dE/dx (ADC counts/cm)

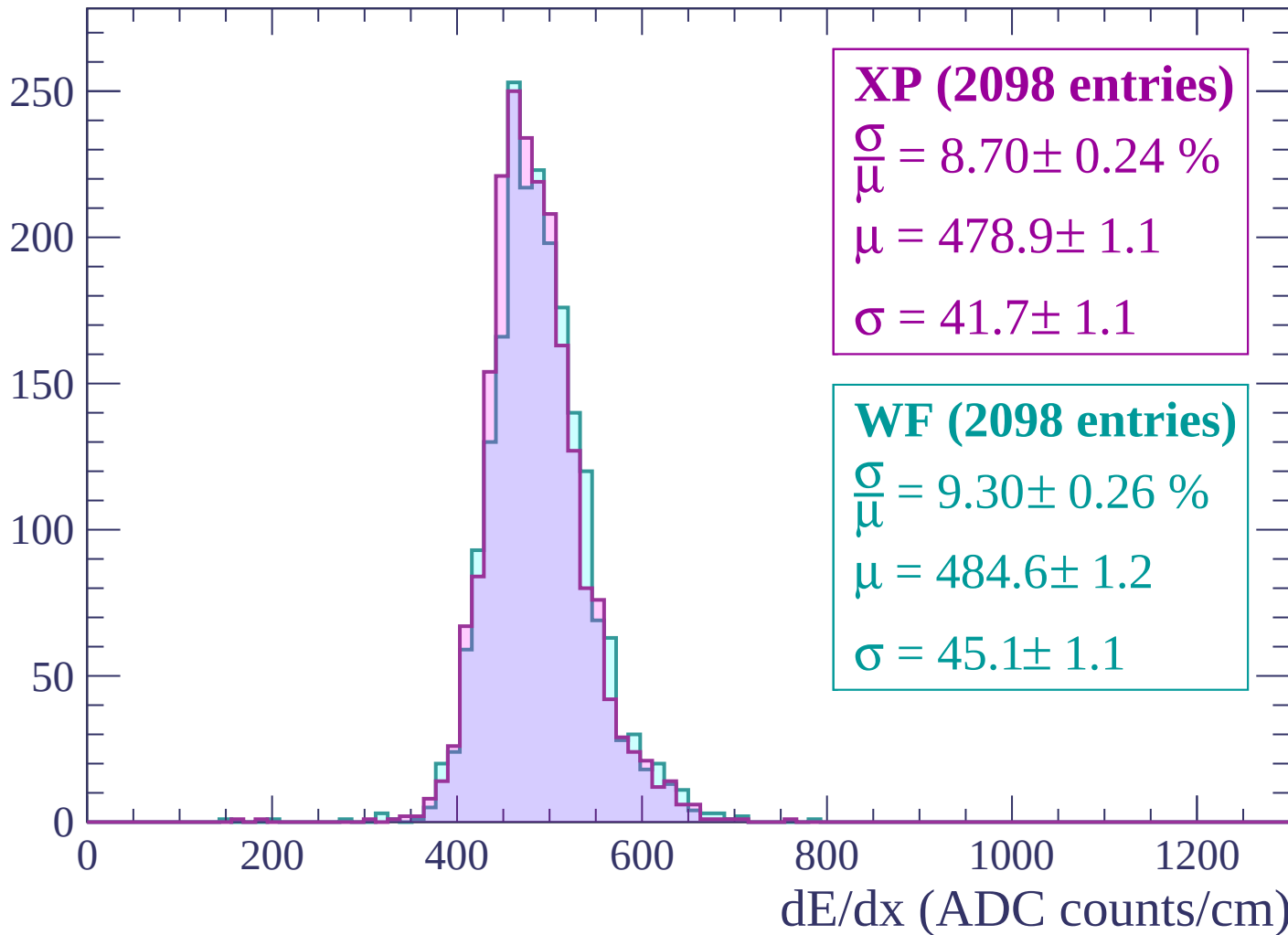
Energy loss | $2280 < p < 2340$

Count



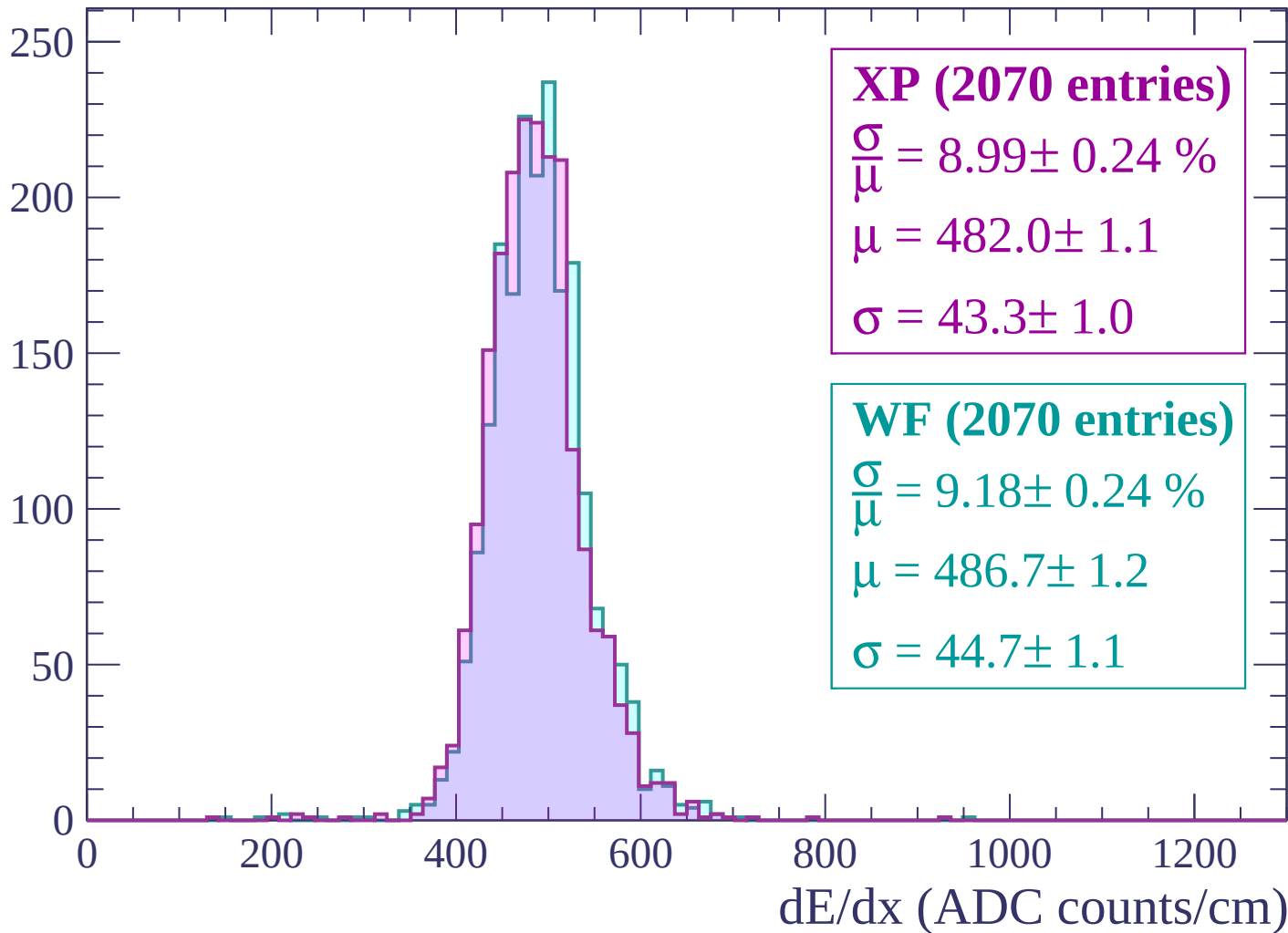
Energy loss | $2340 < p < 2400$

Count



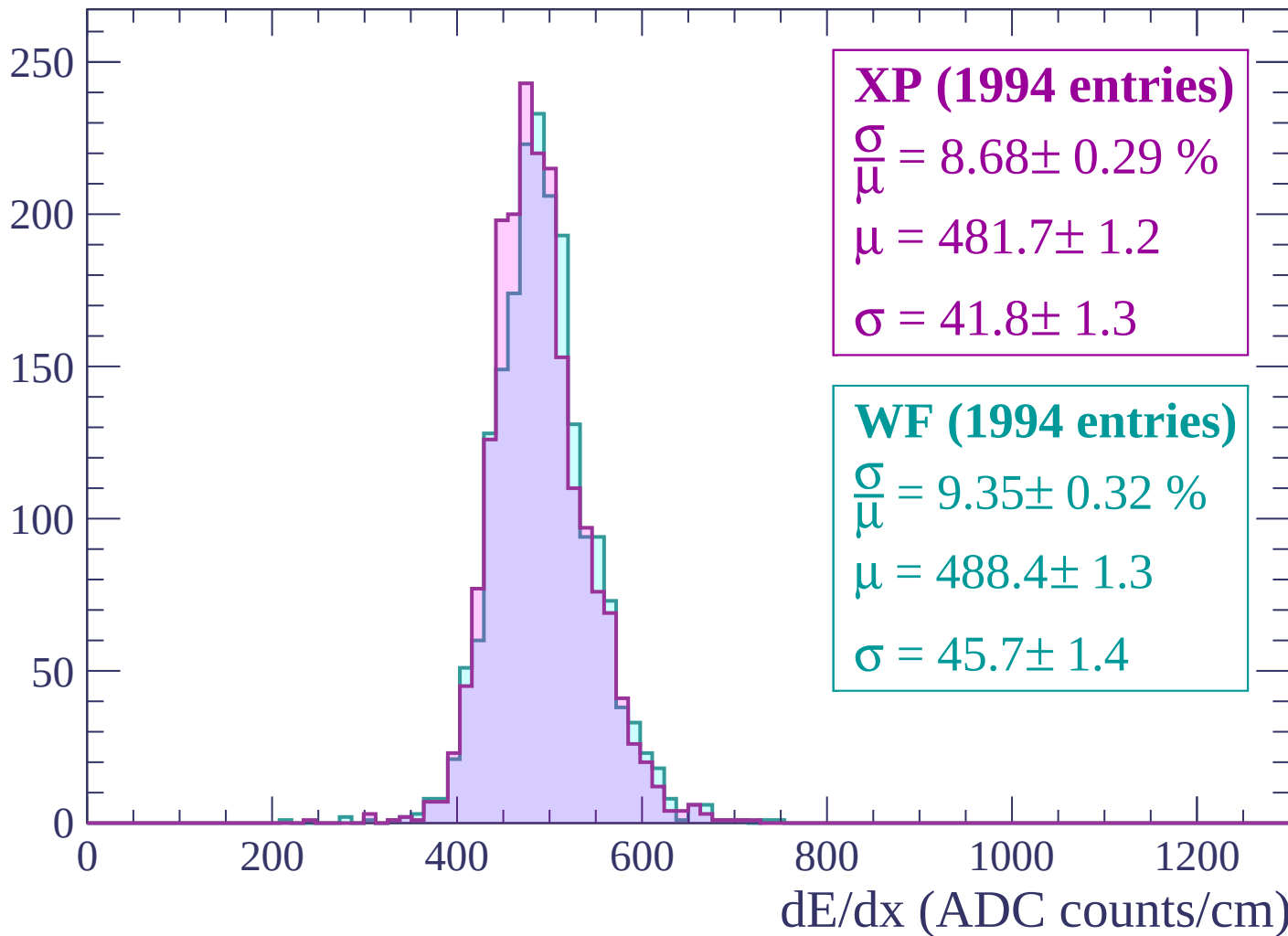
Energy loss | $2400 < p < 2460$

Count



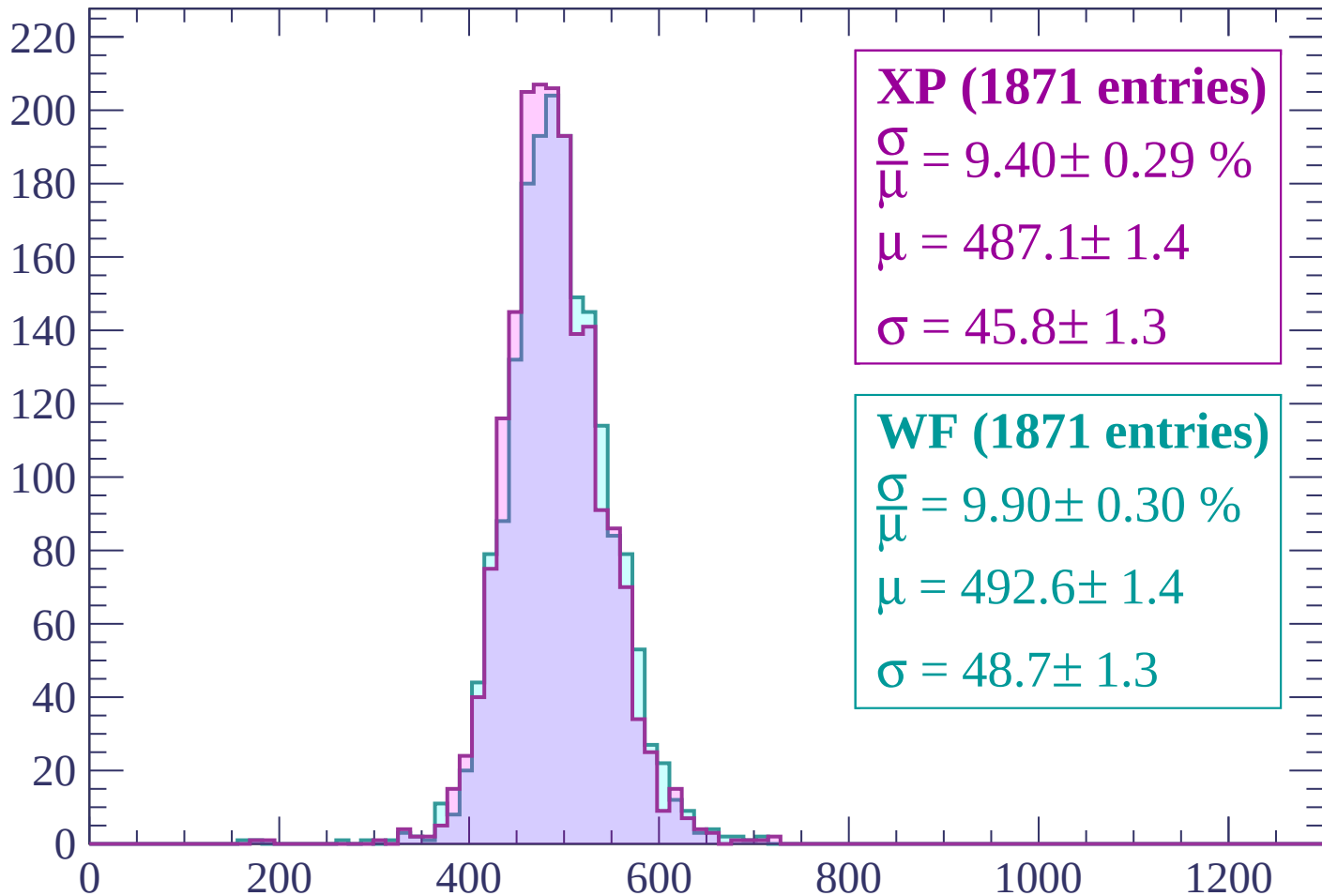
Energy loss | $2460 < p < 2520$

Count



Energy loss | $2520 < p < 2580$

Count



XP (1871 entries)

$$\frac{\sigma}{\mu} = 9.40 \pm 0.29 \%$$

$$\mu = 487.1 \pm 1.4$$

$$\sigma = 45.8 \pm 1.3$$

WF (1871 entries)

$$\frac{\sigma}{\mu} = 9.90 \pm 0.30 \%$$

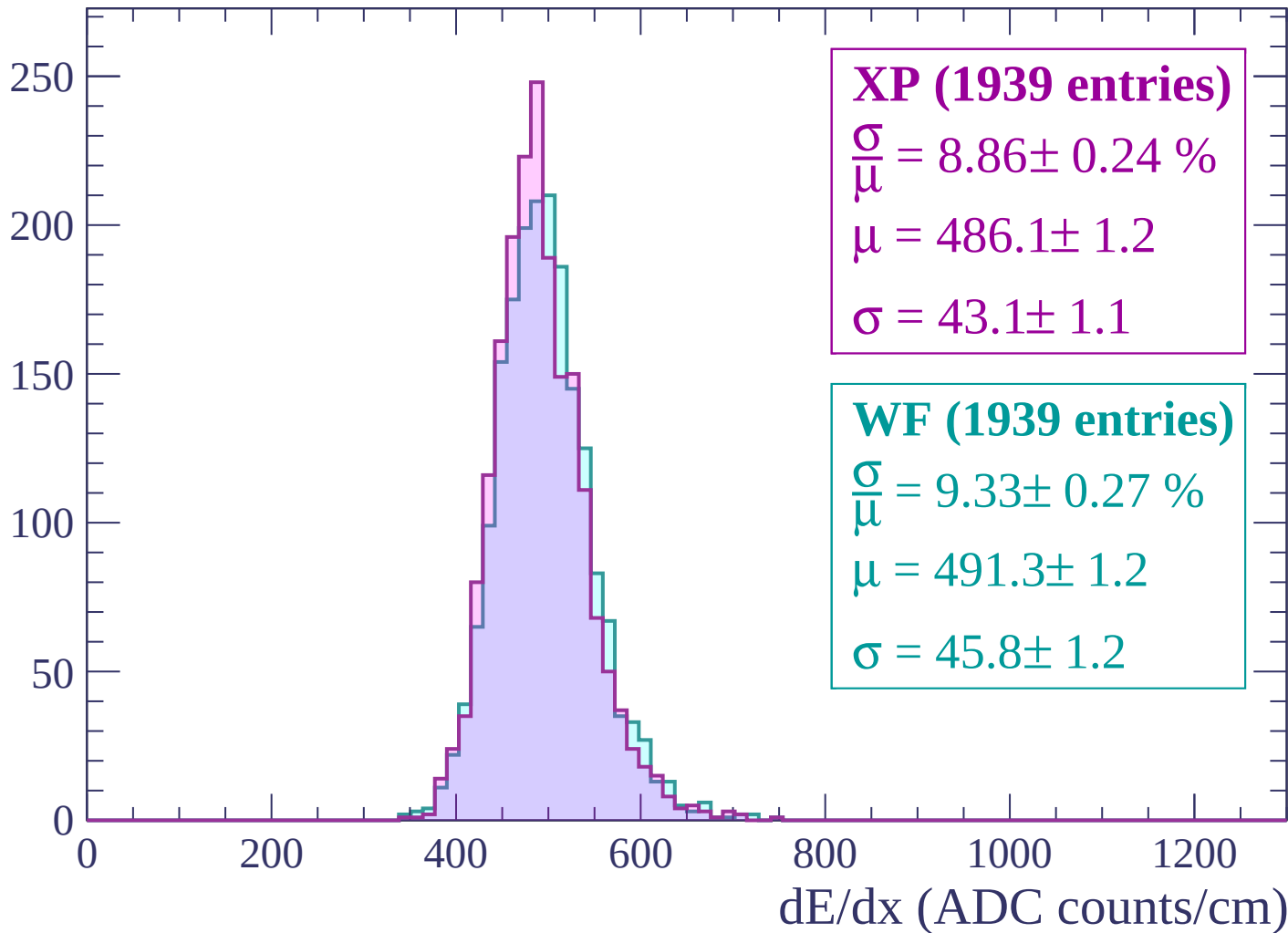
$$\mu = 492.6 \pm 1.4$$

$$\sigma = 48.7 \pm 1.3$$

dE/dx (ADC counts/cm)

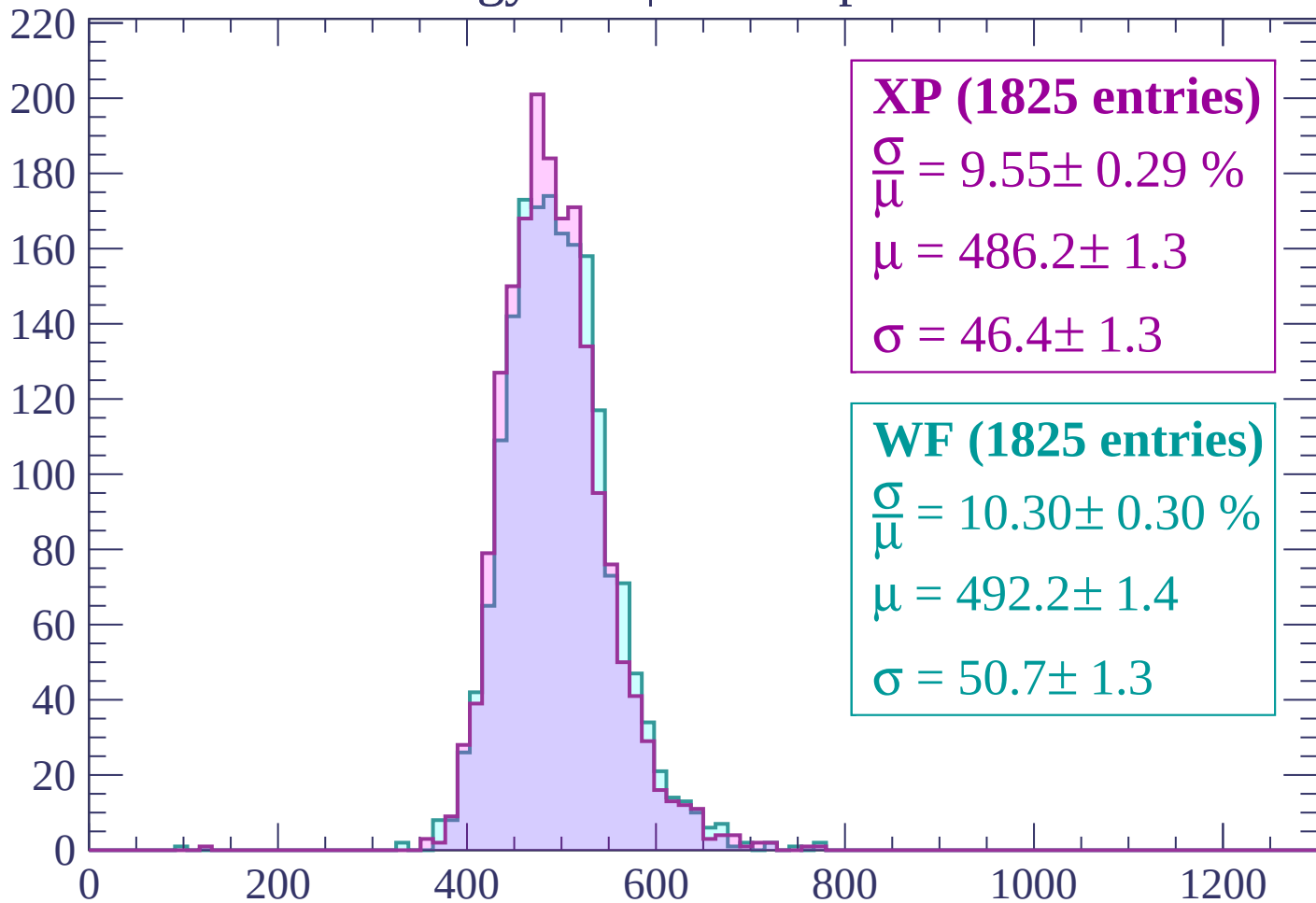
Energy loss | $2580 < p < 2640$

Count



Energy loss | $2640 < p < 2700$

Count



XP (1825 entries)

$$\frac{\sigma}{\mu} = 9.55 \pm 0.29 \%$$

$$\mu = 486.2 \pm 1.3$$

$$\sigma = 46.4 \pm 1.3$$

WF (1825 entries)

$$\frac{\sigma}{\mu} = 10.30 \pm 0.30 \%$$

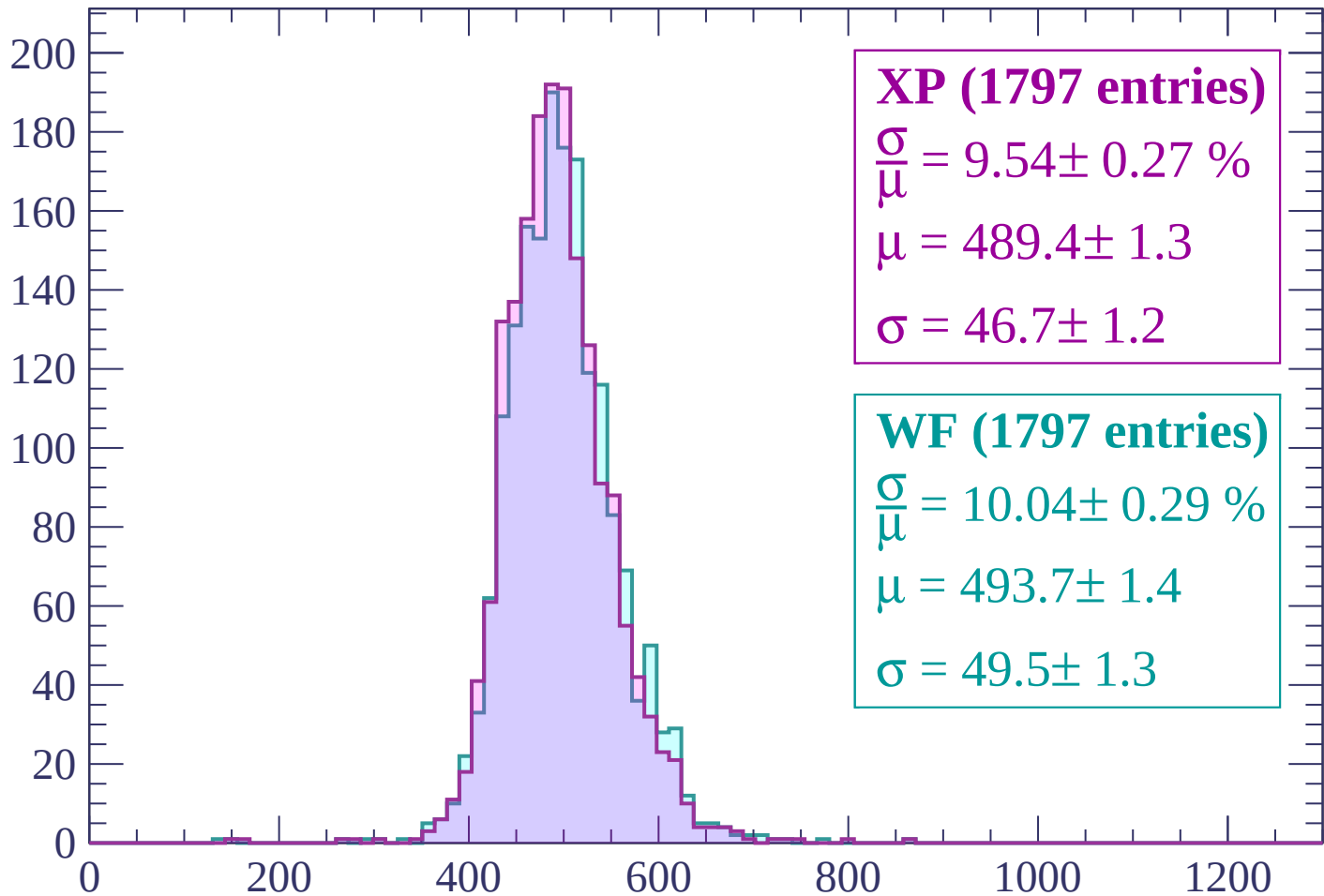
$$\mu = 492.2 \pm 1.4$$

$$\sigma = 50.7 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $2700 < p < 2760$

Count



XP (1797 entries)

$$\frac{\sigma}{\mu} = 9.54 \pm 0.27 \%$$

$$\mu = 489.4 \pm 1.3$$

$$\sigma = 46.7 \pm 1.2$$

WF (1797 entries)

$$\frac{\sigma}{\mu} = 10.04 \pm 0.29 \%$$

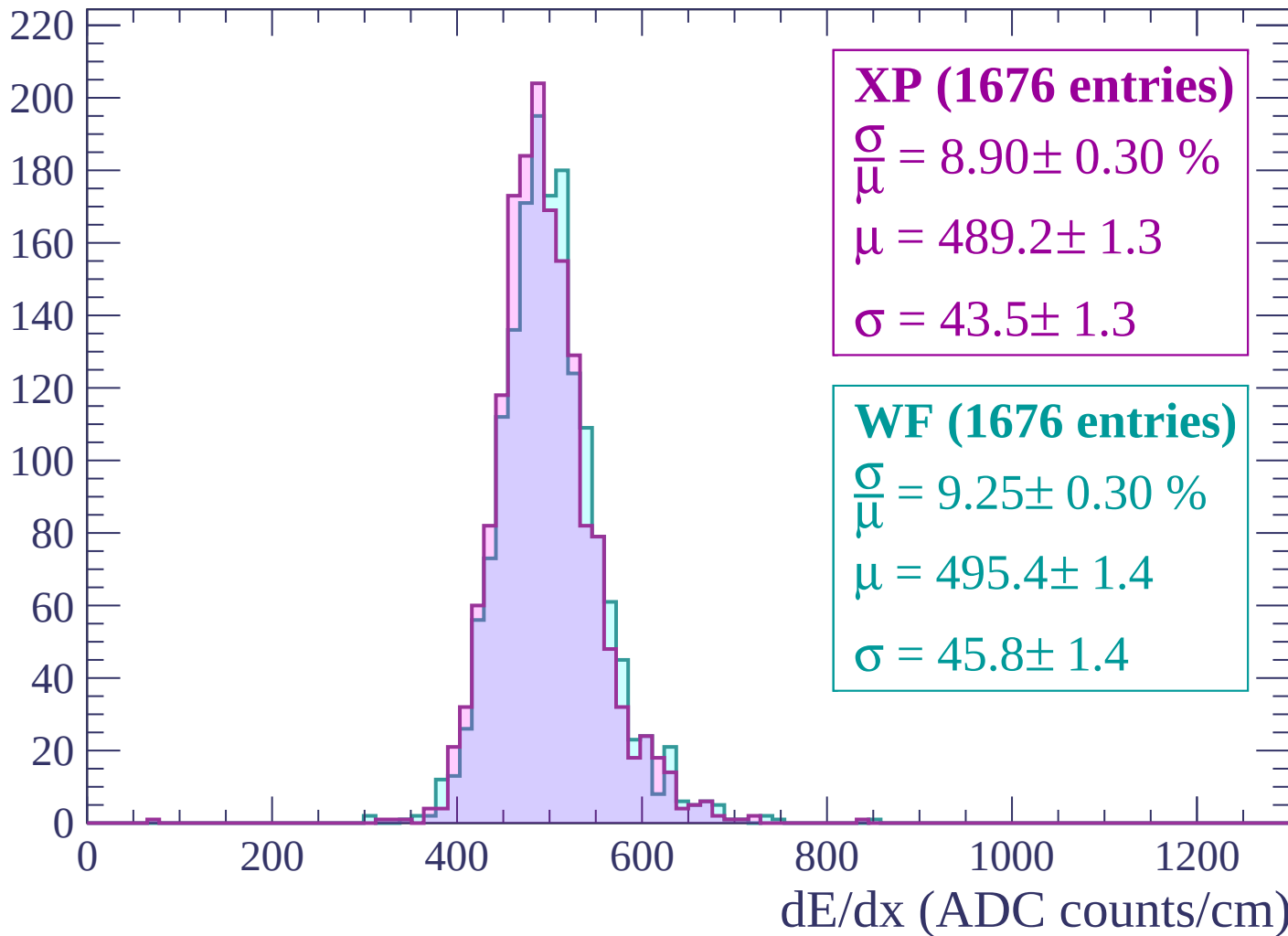
$$\mu = 493.7 \pm 1.4$$

$$\sigma = 49.5 \pm 1.3$$

dE/dx (ADC counts/cm)

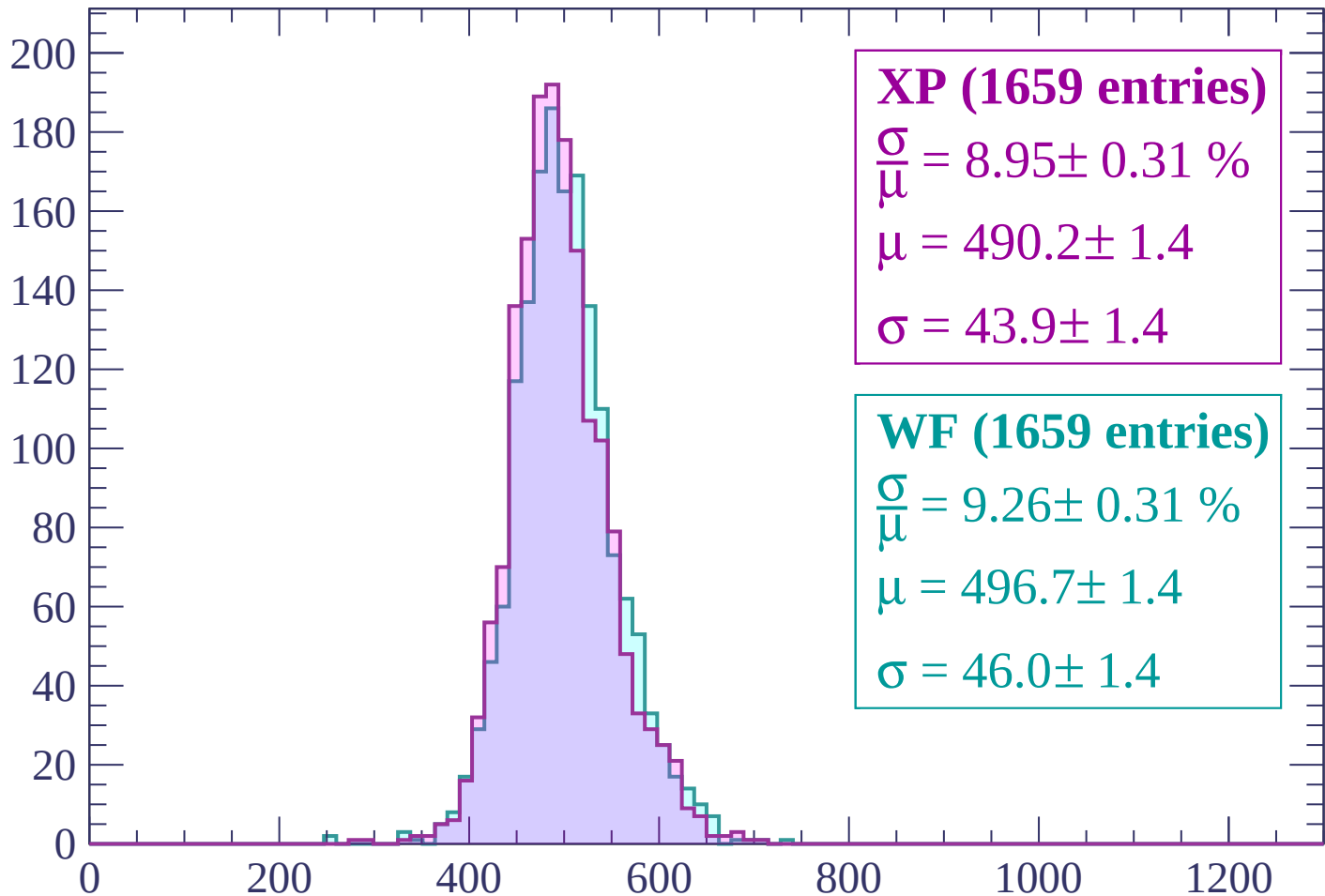
Energy loss | $2760 < p < 2820$

Count



Energy loss | $2820 < p < 2880$

Count



XP (1659 entries)

$$\frac{\sigma}{\mu} = 8.95 \pm 0.31 \%$$

$$\mu = 490.2 \pm 1.4$$

$$\sigma = 43.9 \pm 1.4$$

WF (1659 entries)

$$\frac{\sigma}{\mu} = 9.26 \pm 0.31 \%$$

$$\mu = 496.7 \pm 1.4$$

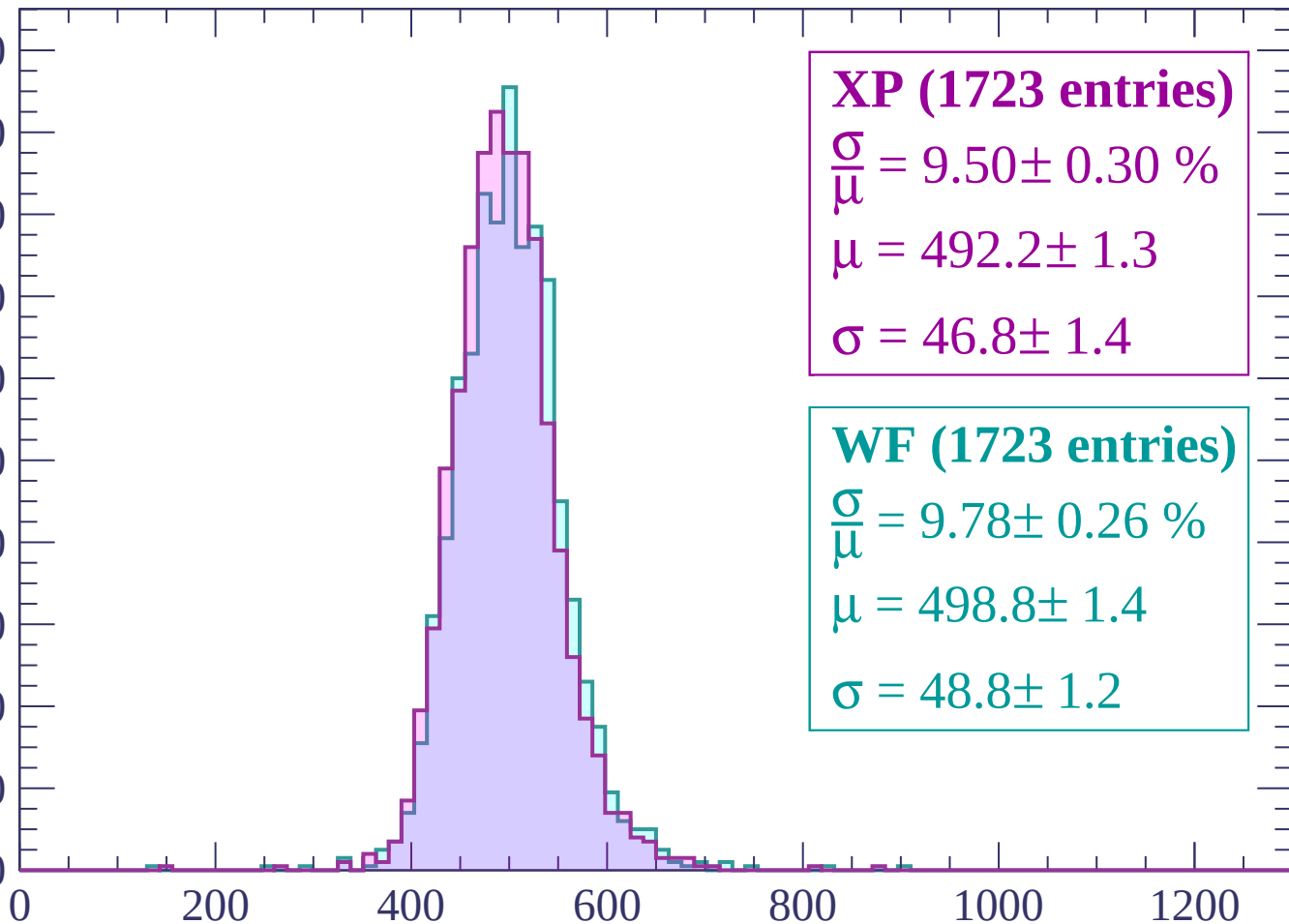
$$\sigma = 46.0 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $2880 < p < 2940$

Count

200
180
160
140
120
100
80
60
40
20
0



XP (1723 entries)

$\frac{\sigma}{\mu} = 9.50 \pm 0.30 \%$

$\mu = 492.2 \pm 1.3$

$\sigma = 46.8 \pm 1.4$

WF (1723 entries)

$\frac{\sigma}{\mu} = 9.78 \pm 0.26 \%$

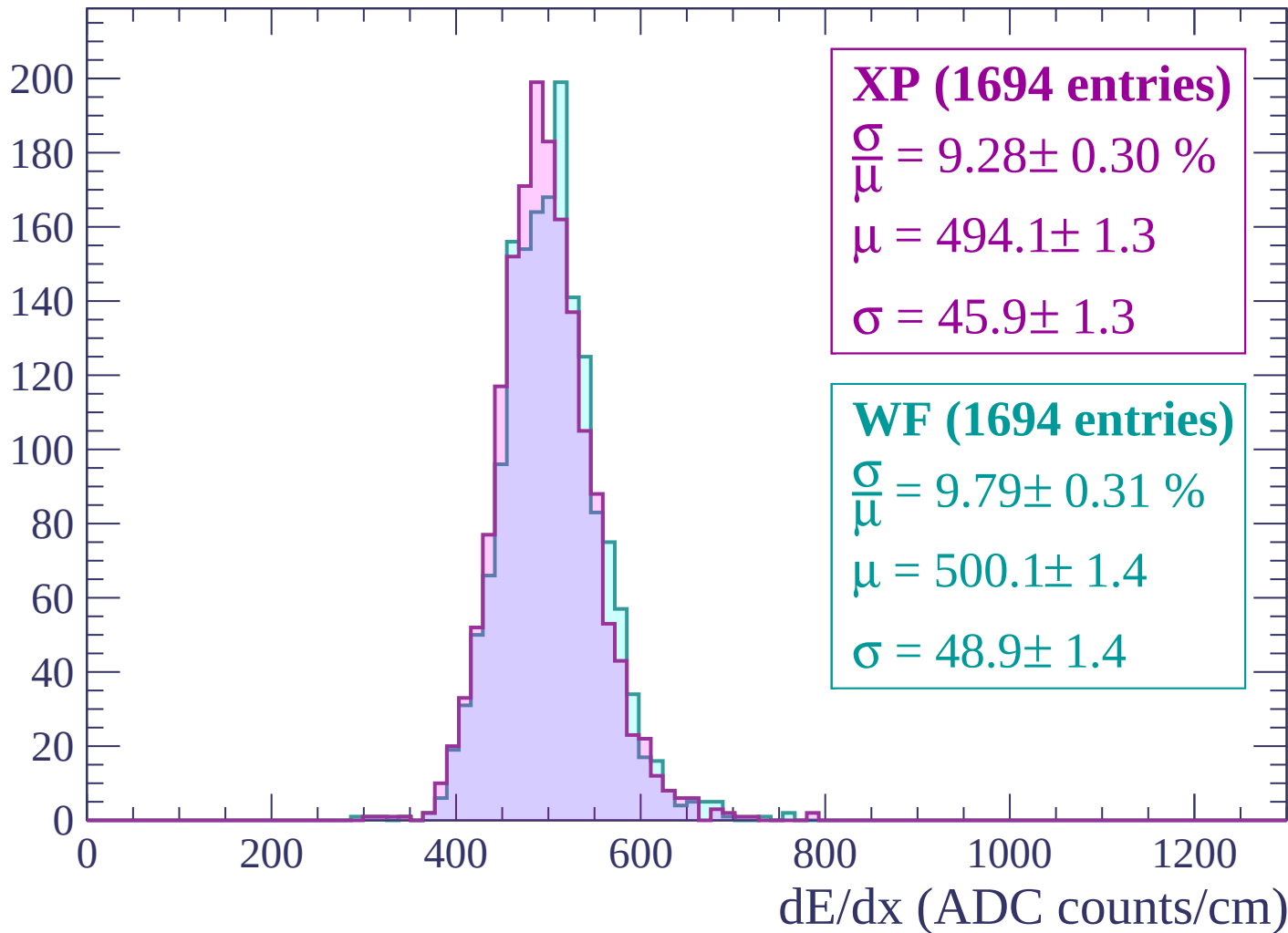
$\mu = 498.8 \pm 1.4$

$\sigma = 48.8 \pm 1.2$

dE/dx (ADC counts/cm)

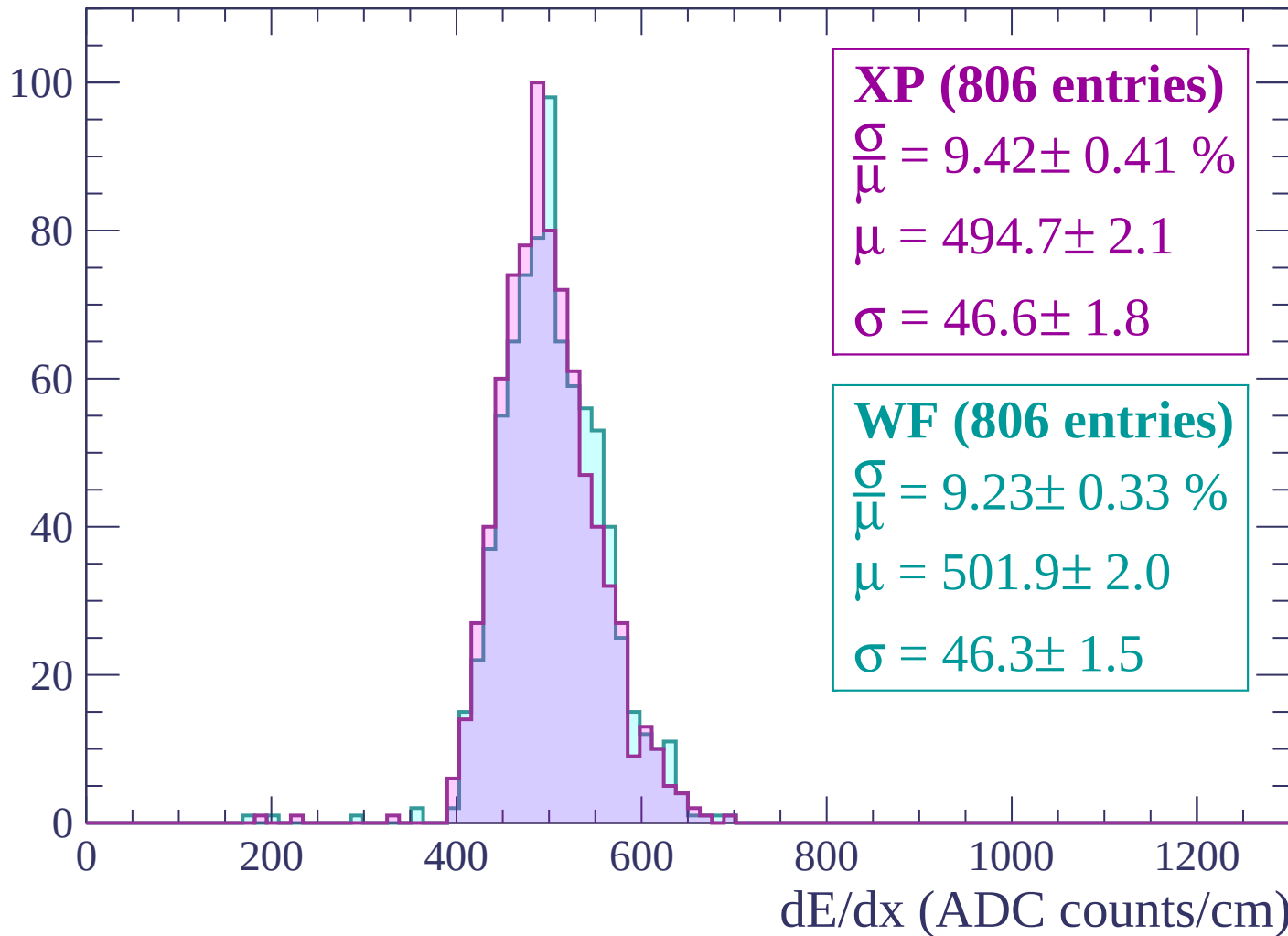
Energy loss | $2940 < p < 3000$

Count



Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-88 < \theta < -86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-86 < \theta < -84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-84 < \theta < -82$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-82 < \theta < -80$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

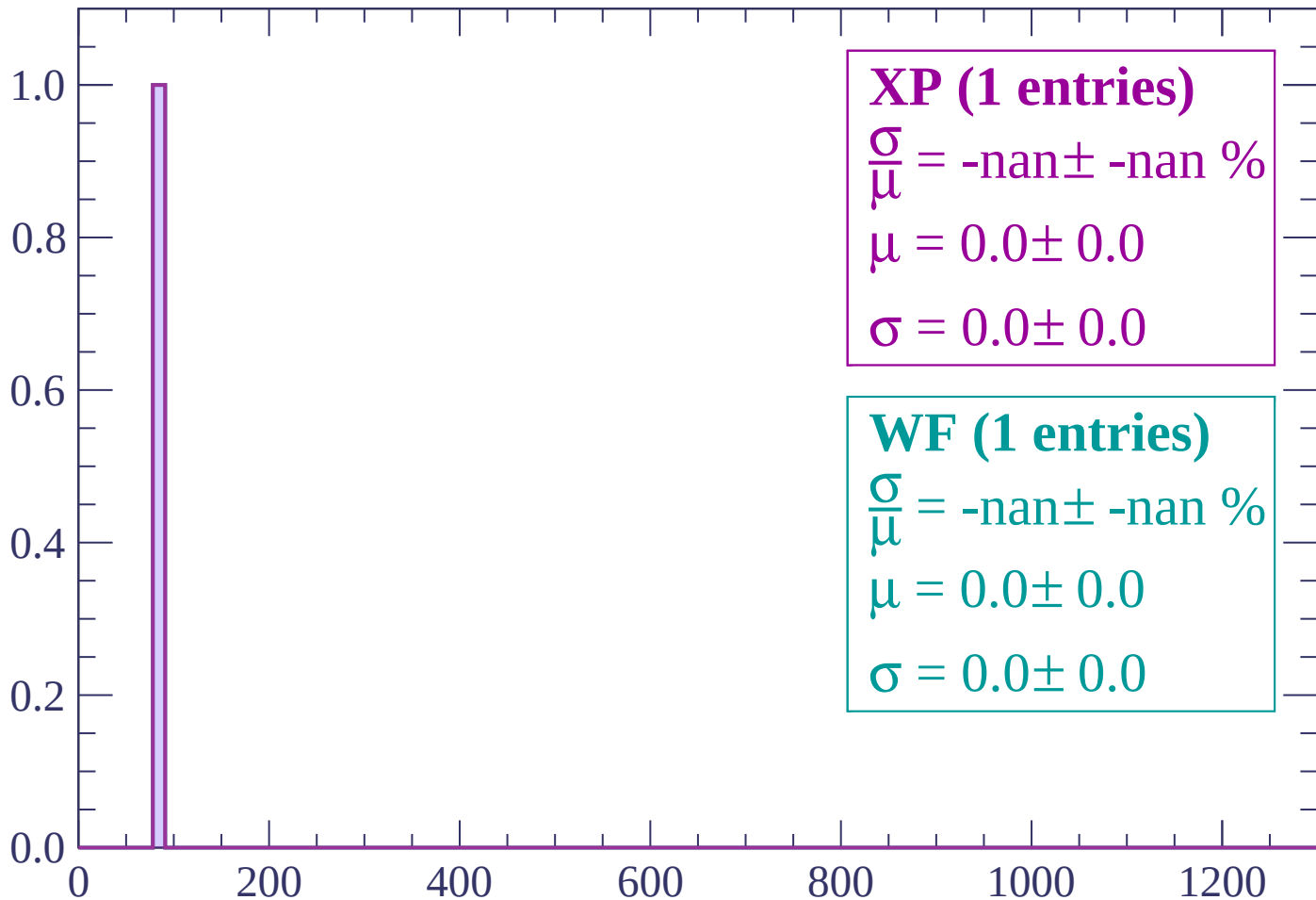
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

Count



XP (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (1 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

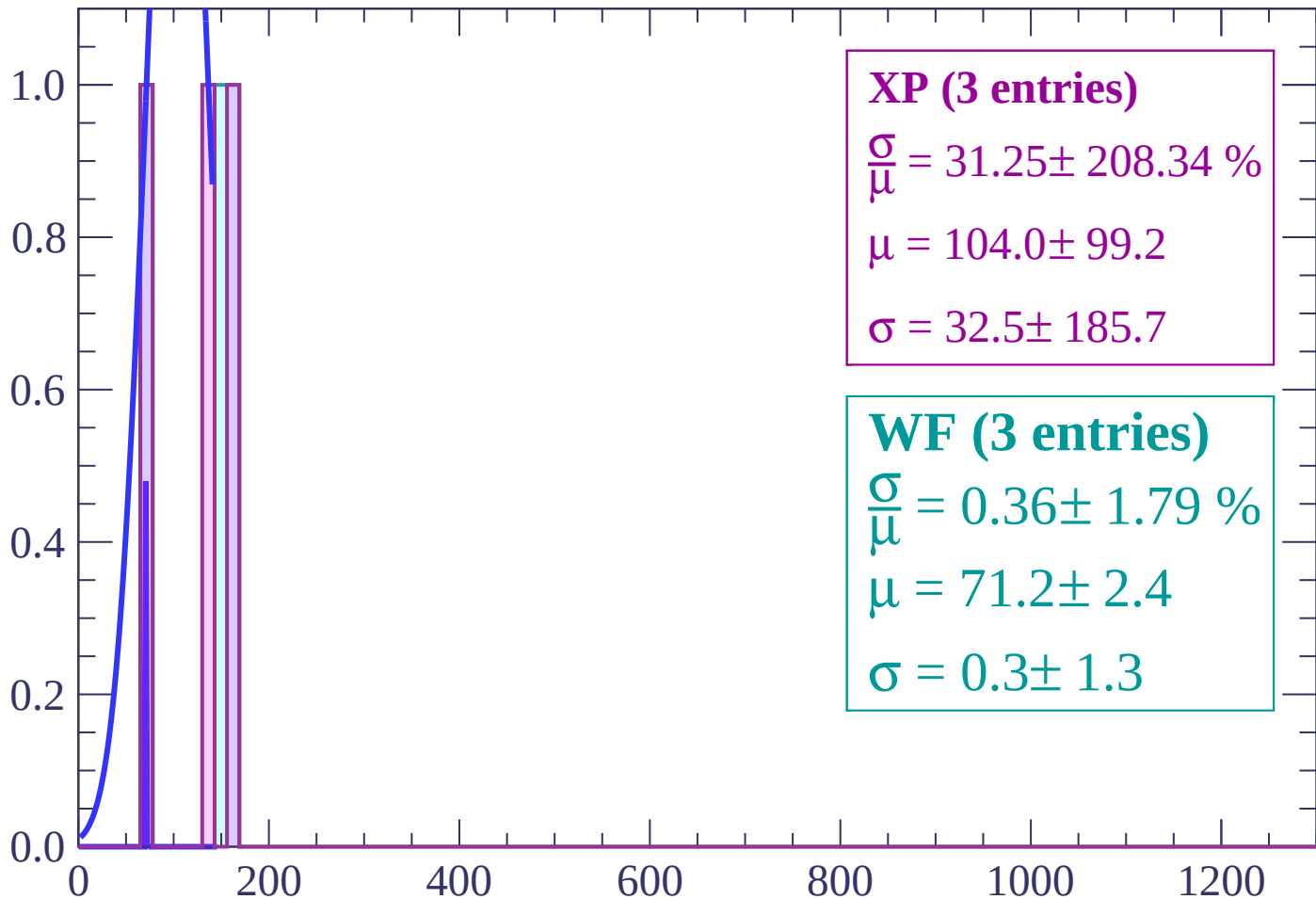
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

Count



XP (3 entries)

$$\frac{\sigma}{\mu} = 31.25 \pm 208.34 \%$$

$$\mu = 104.0 \pm 99.2$$

$$\sigma = 32.5 \pm 185.7$$

WF (3 entries)

$$\frac{\sigma}{\mu} = 0.36 \pm 1.79 \%$$

$$\mu = 71.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

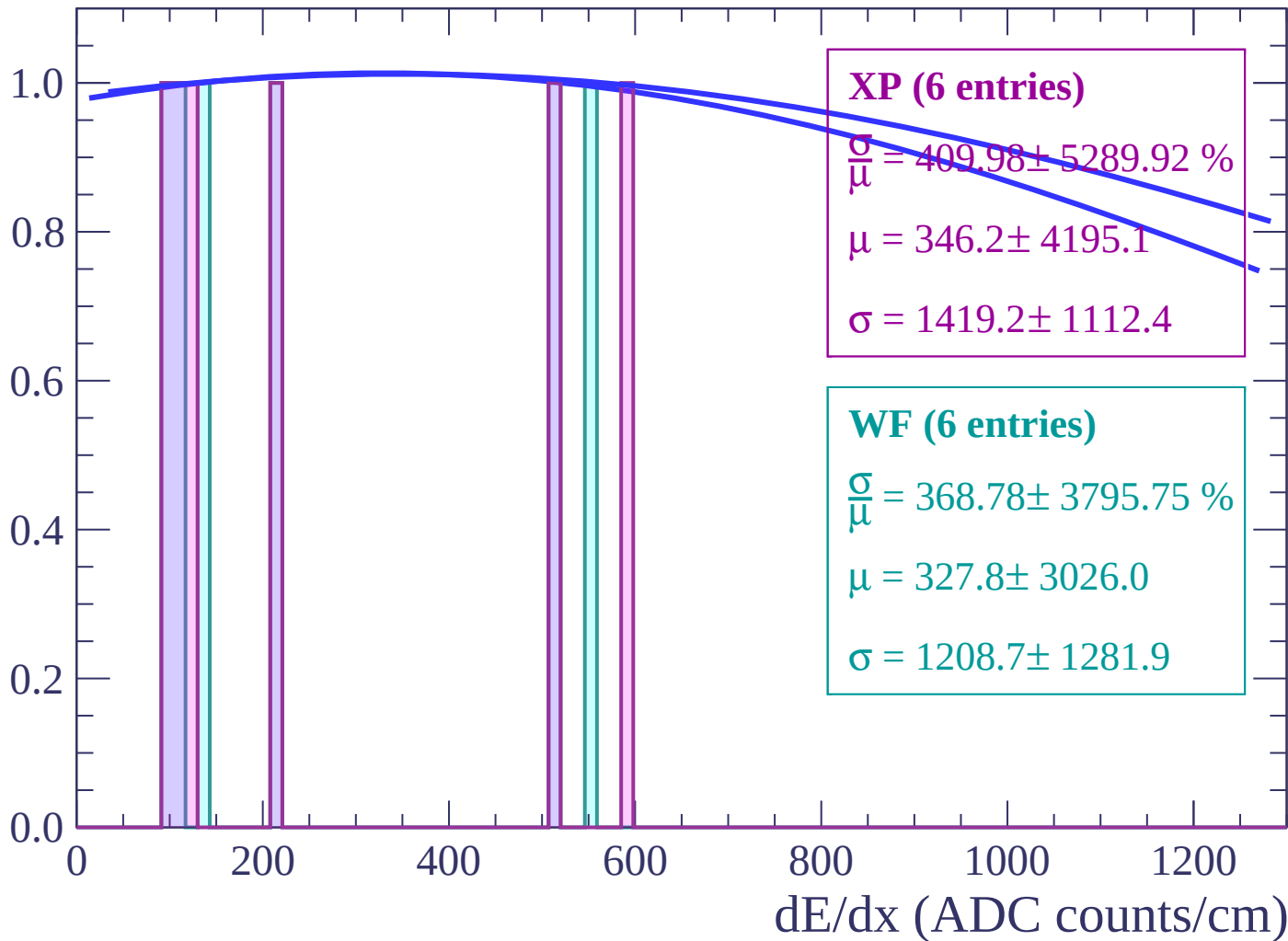
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

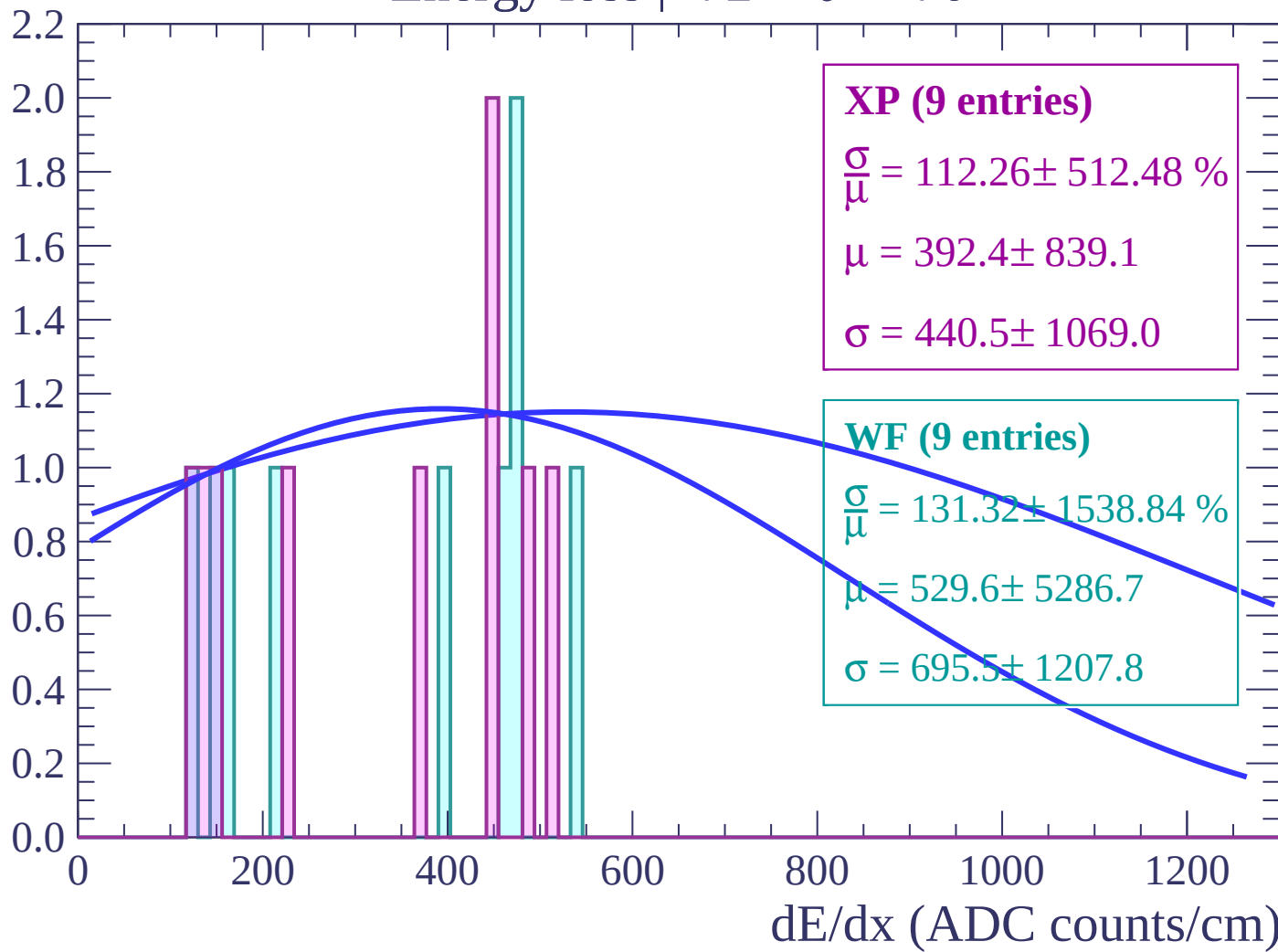
Energy loss | $-74 < \theta < -72$

Count



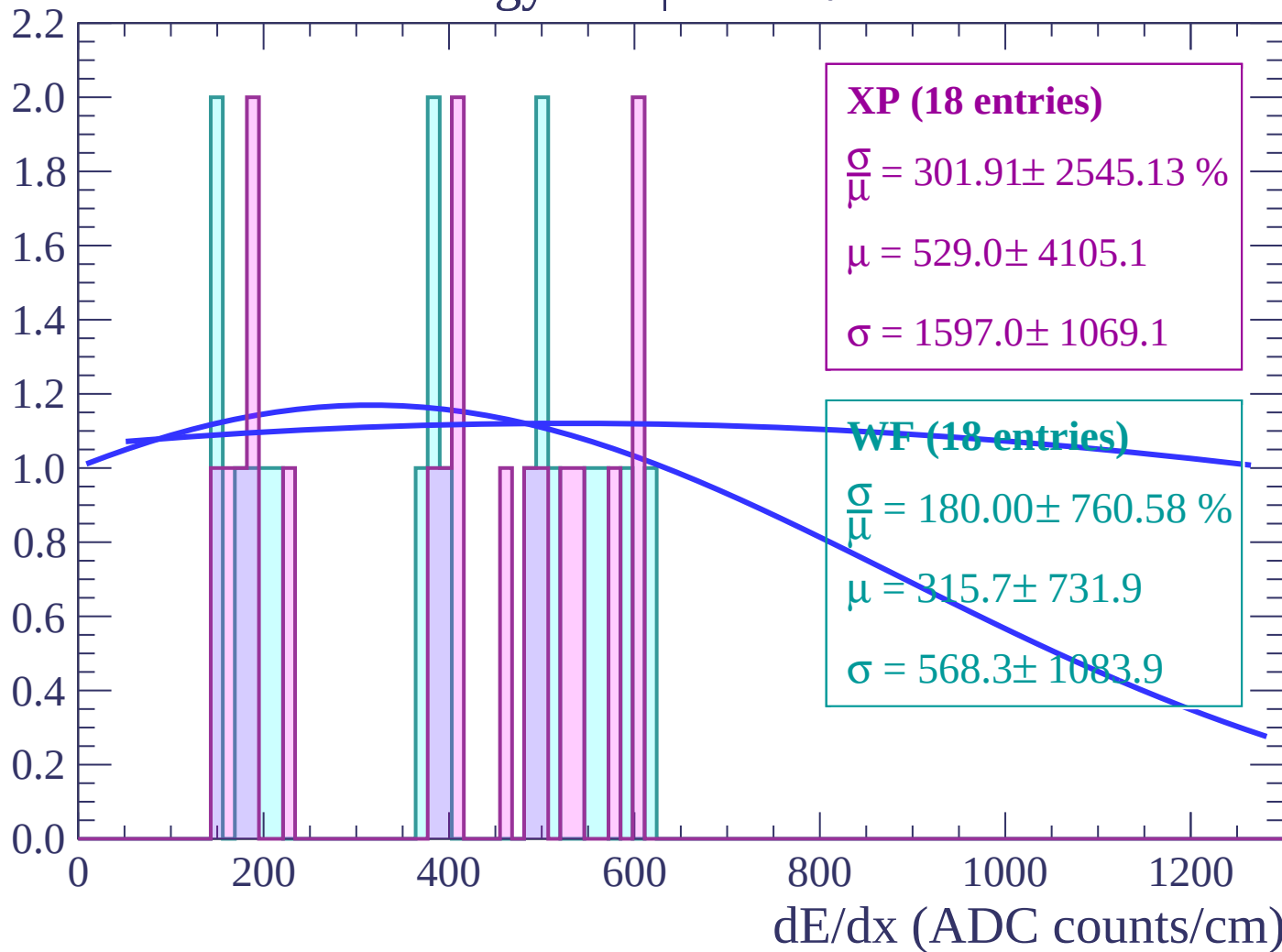
Energy loss | $-72 < \theta < -70$

Count



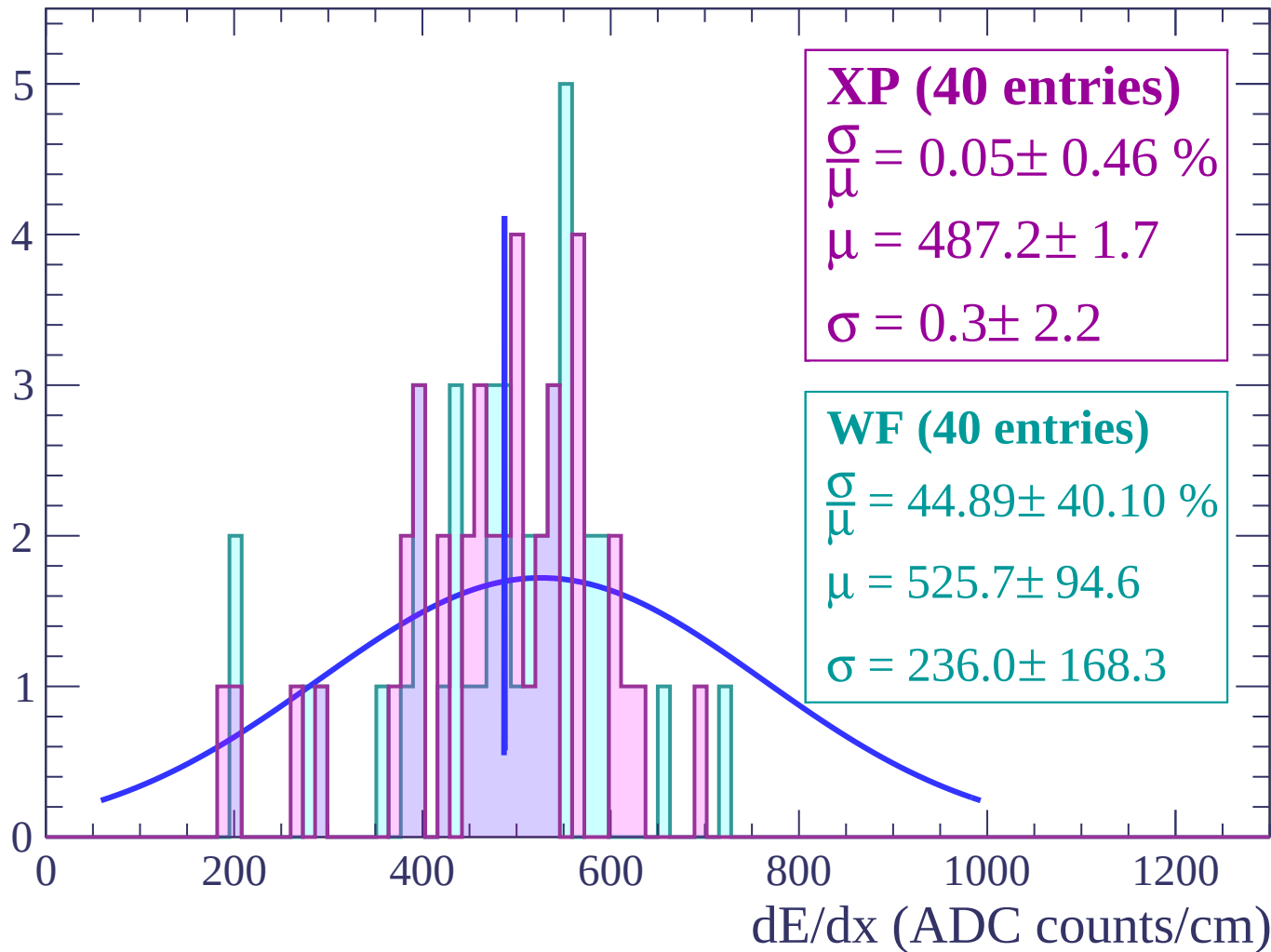
Energy loss | $-70 < \theta < -68$

Count



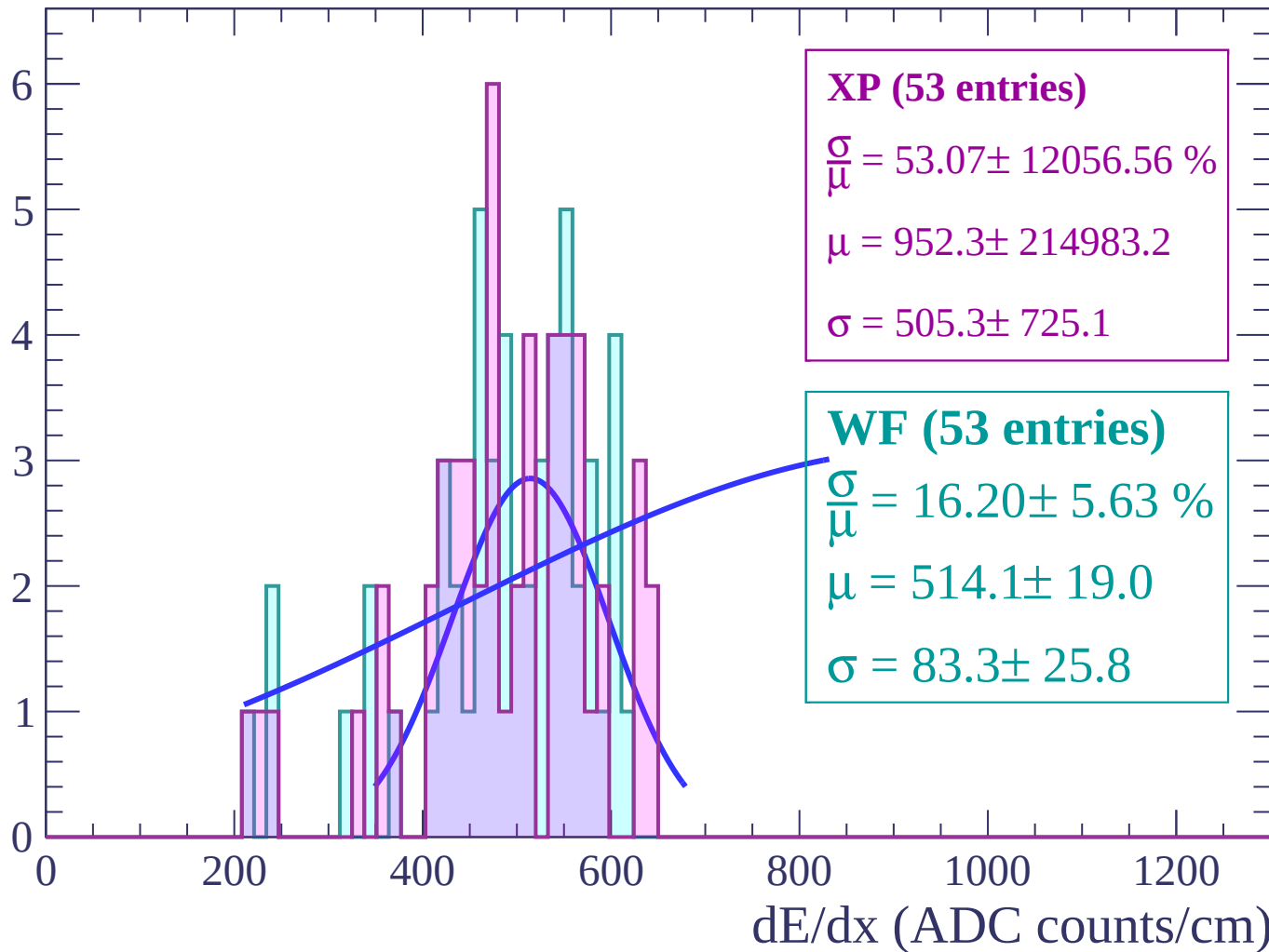
Energy loss | $-68 < \theta < -66$

Count



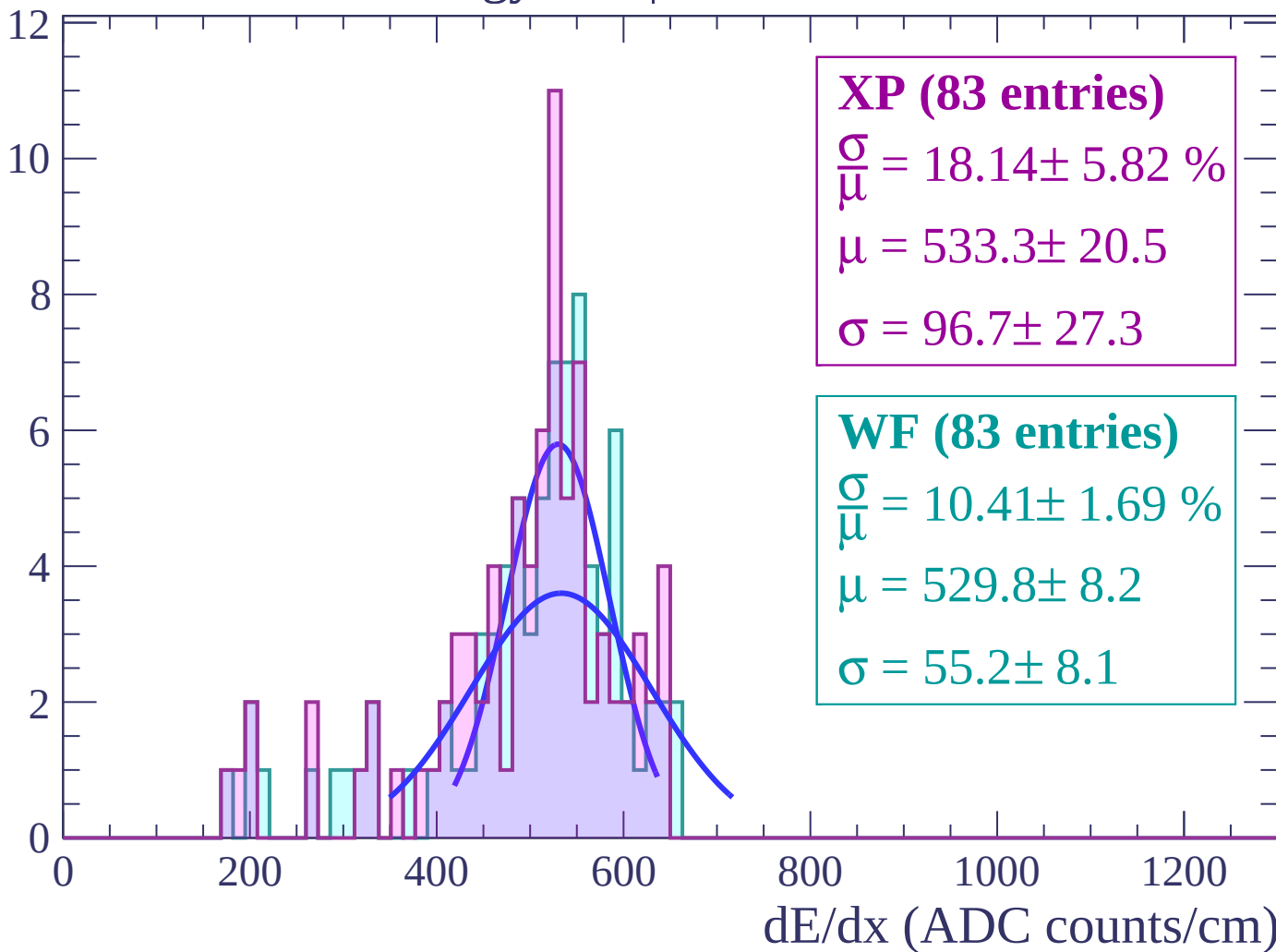
Energy loss | $-66 < \theta < -64$

Count



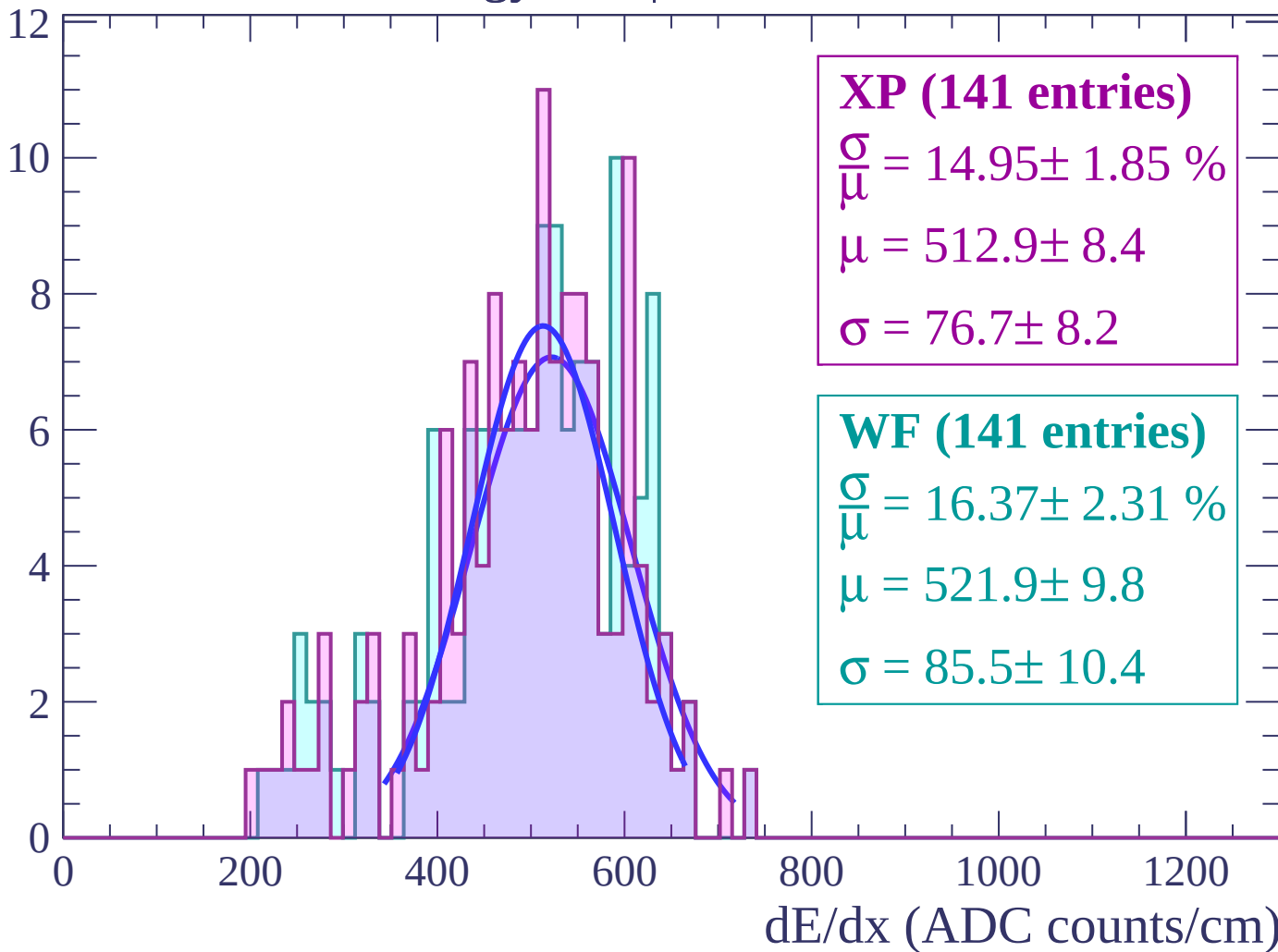
Energy loss | $-64 < \theta < -62$

Count



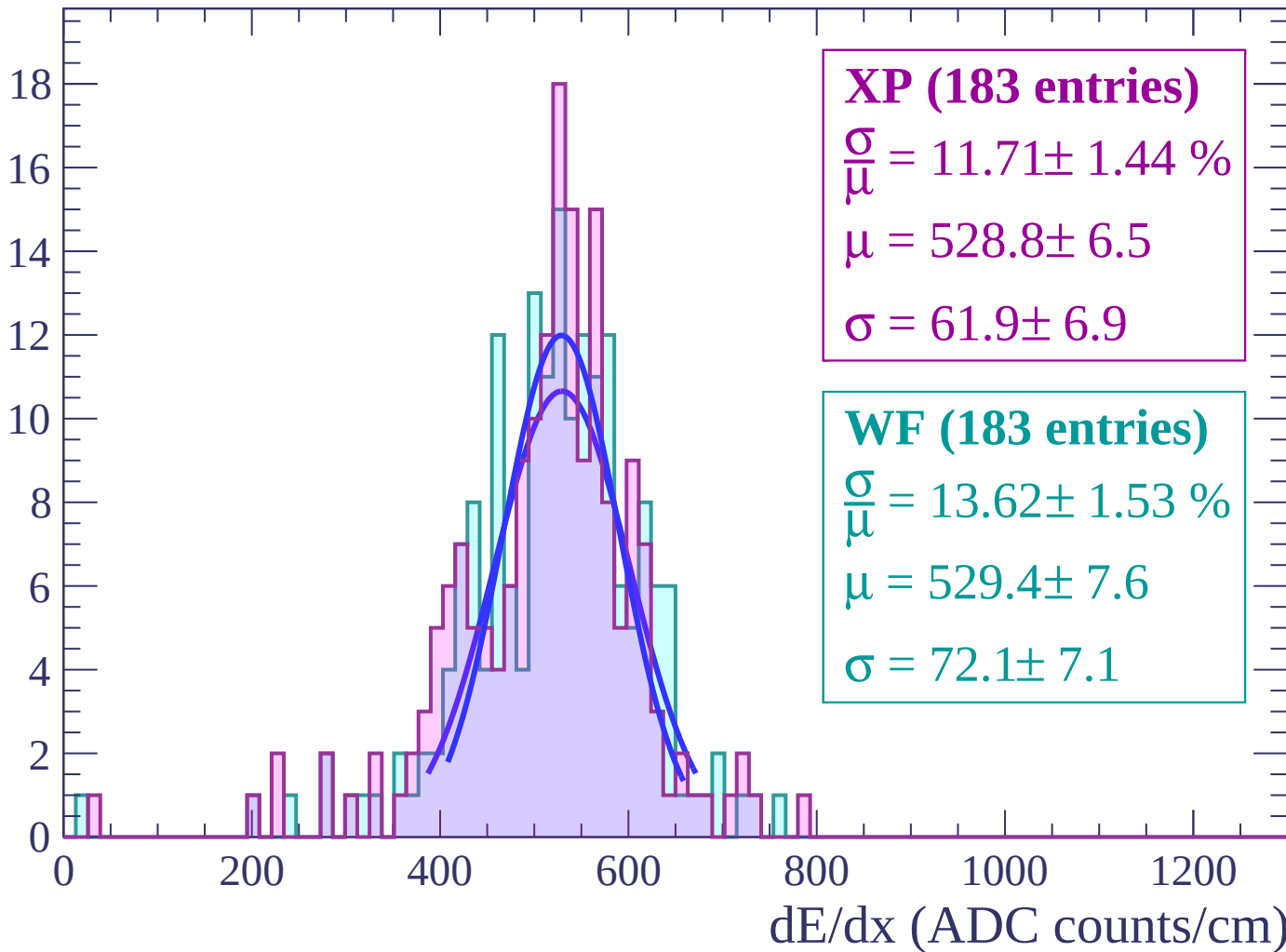
Energy loss | $-62 < \theta < -60$

Count



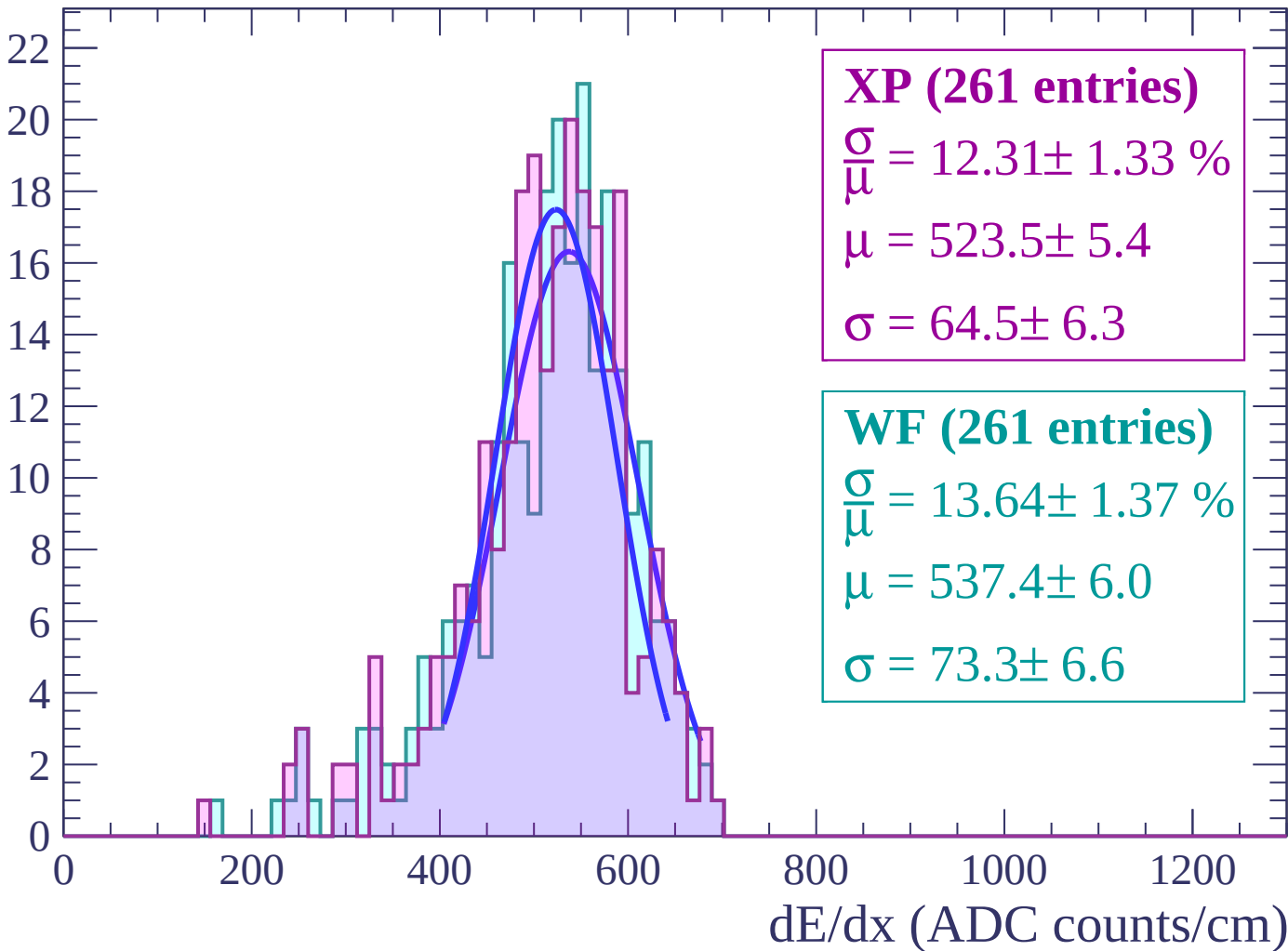
Energy loss | $-60 < \theta < -58$

Count



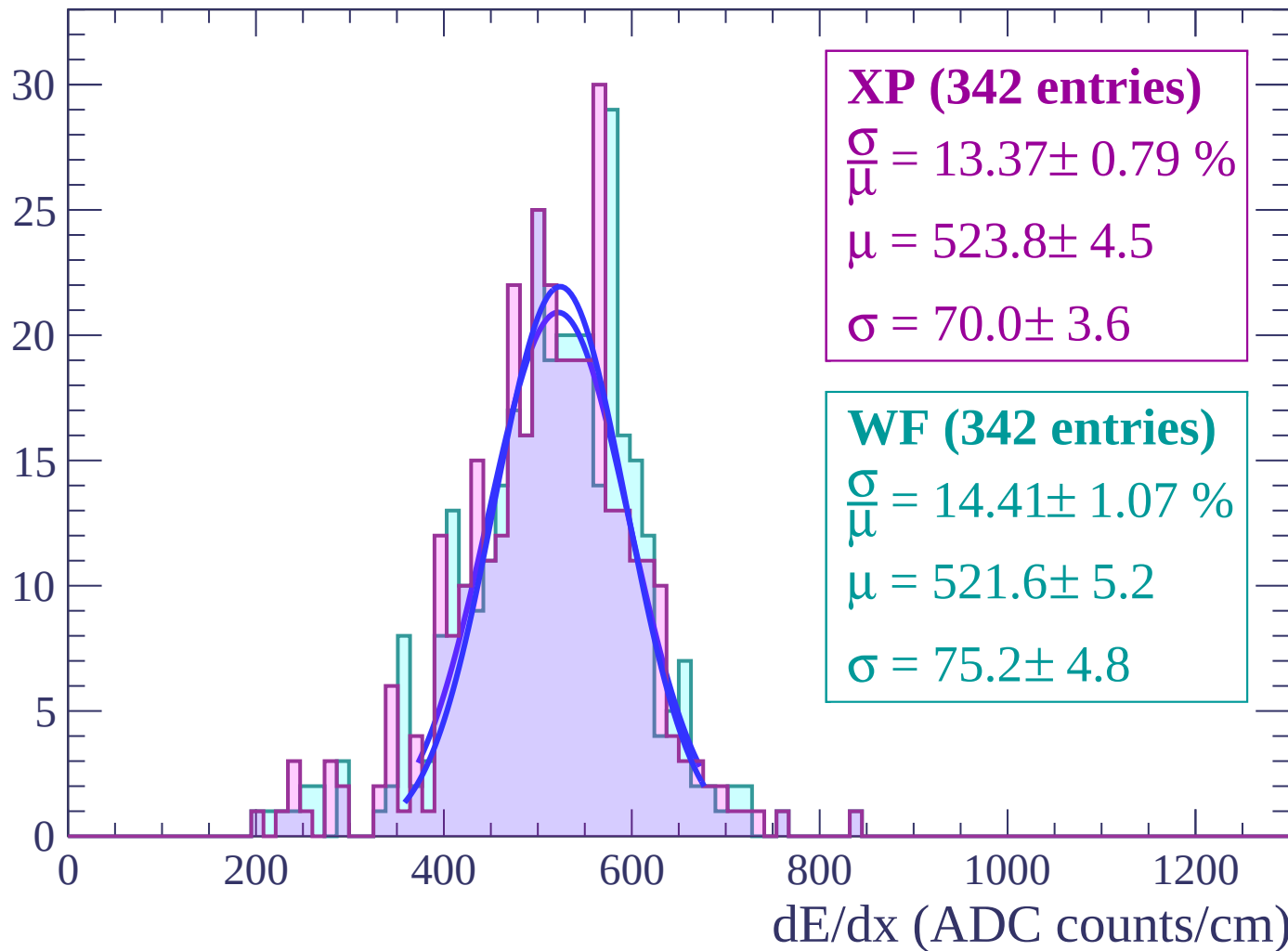
Energy loss | $-58 < \theta < -56$

Count



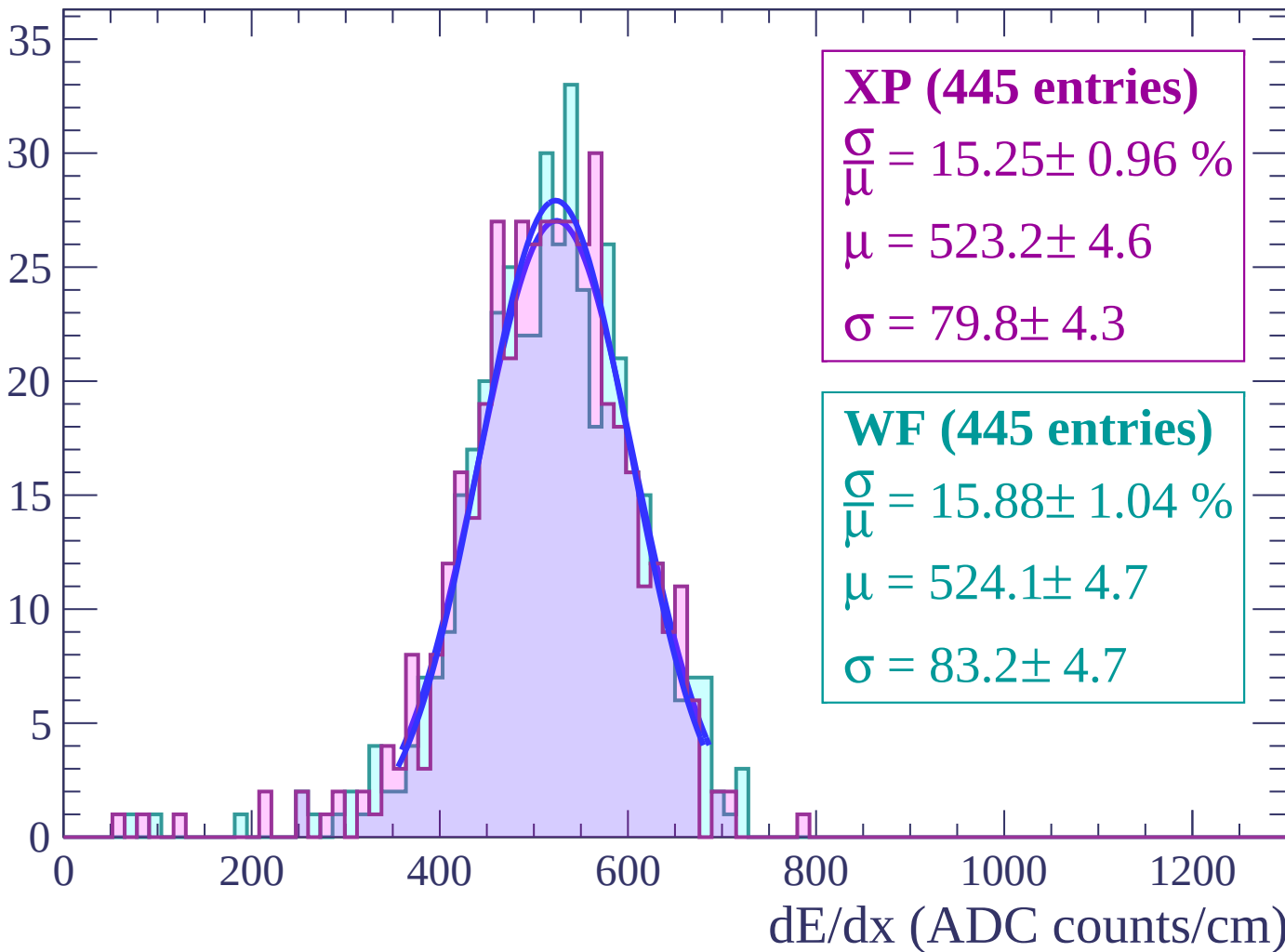
Energy loss | $-56 < \theta < -54$

Count



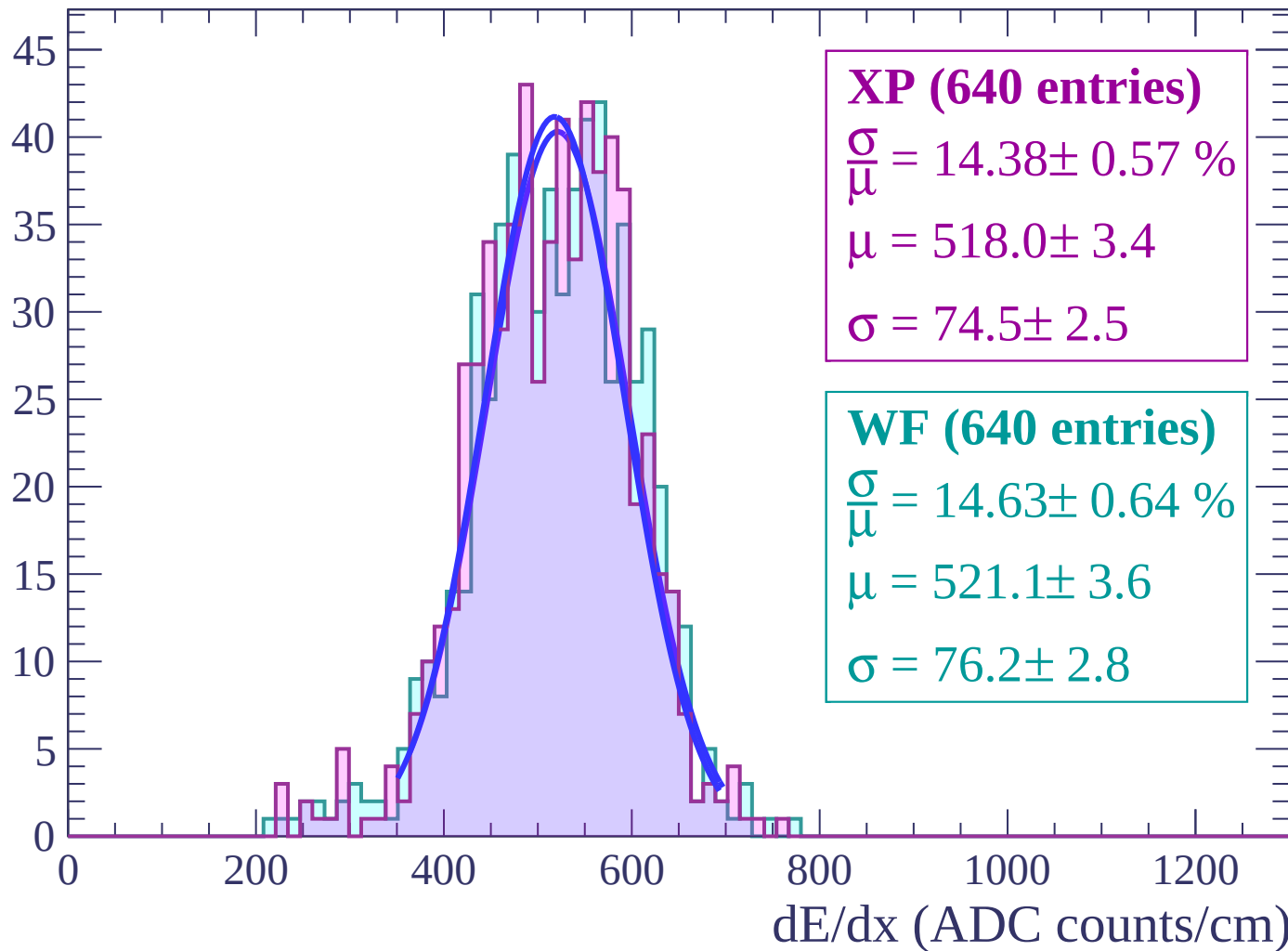
Energy loss | $-54 < \theta < -52$

Count



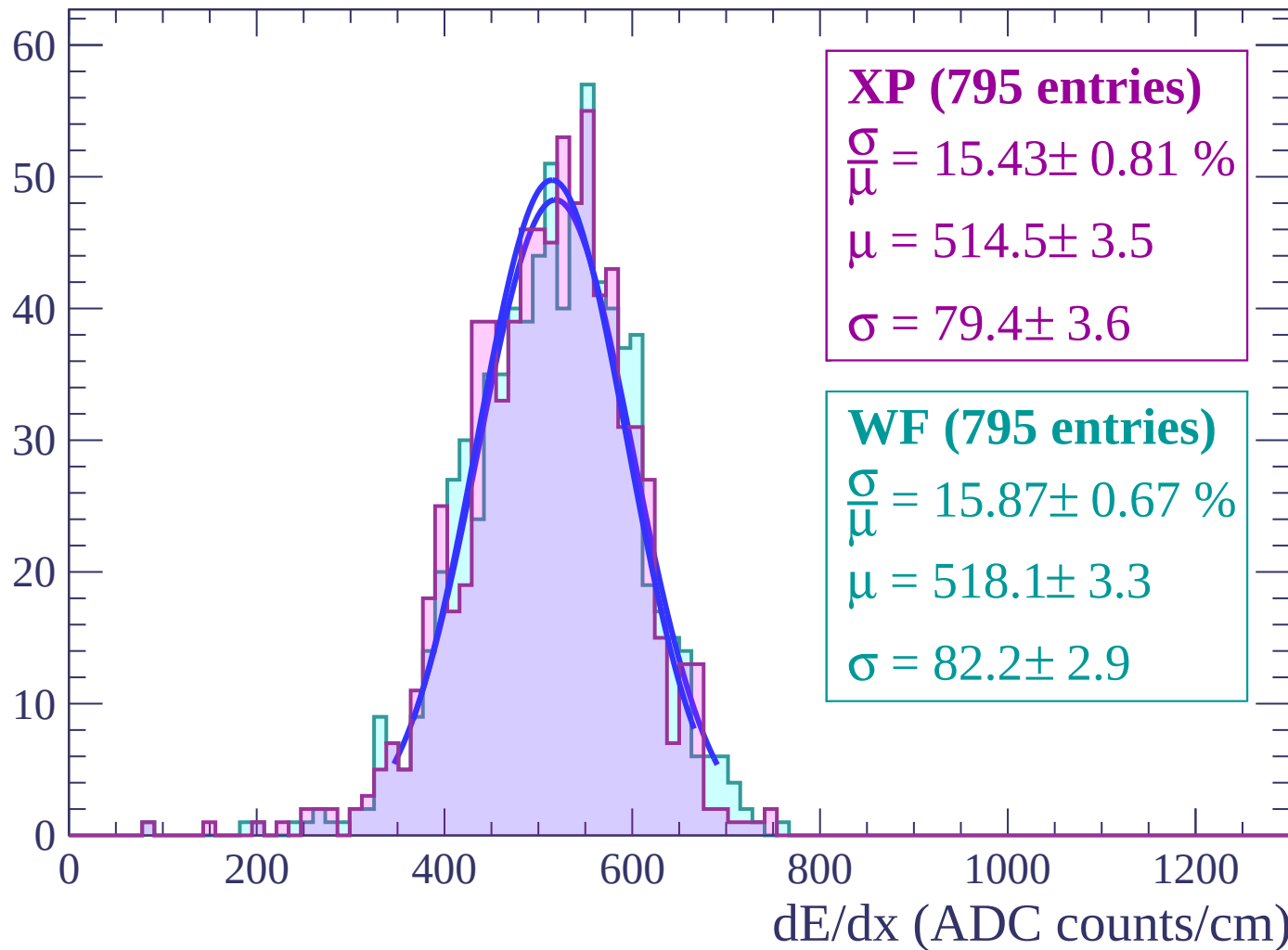
Energy loss | $-52 < \theta < -50$

Count



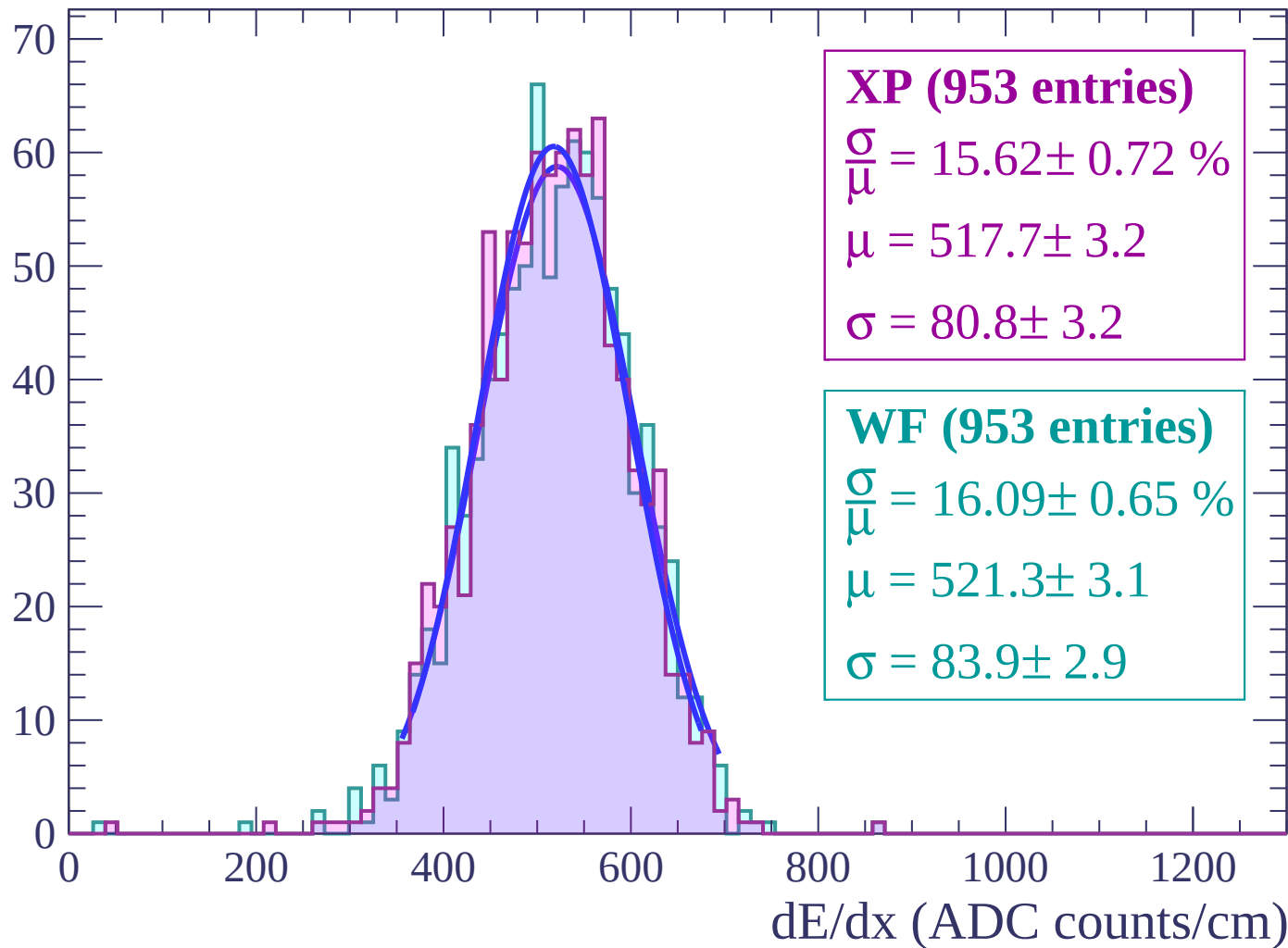
Energy loss | $-50 < \theta < -48$

Count



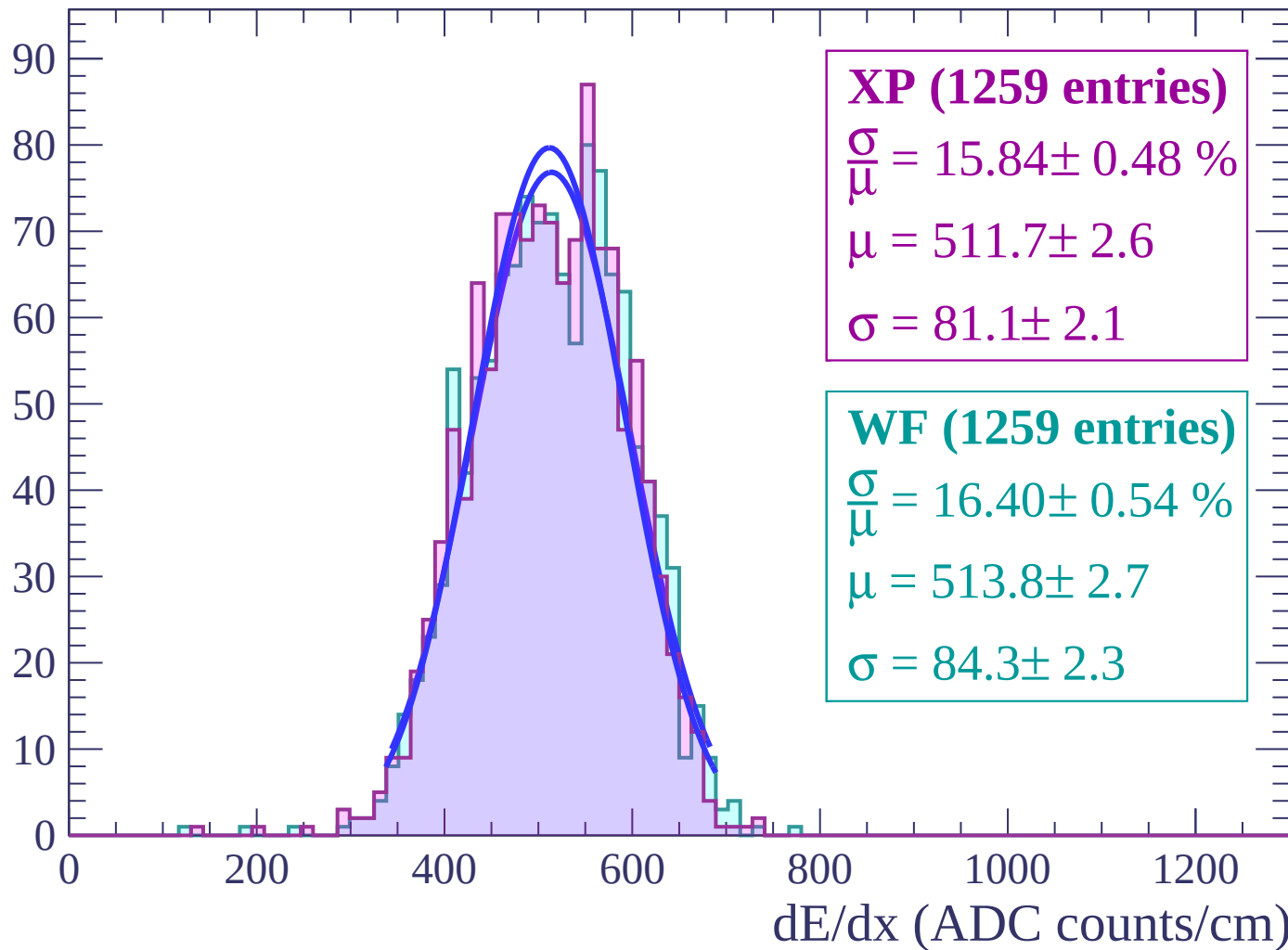
Energy loss | $-48 < \theta < -46$

Count



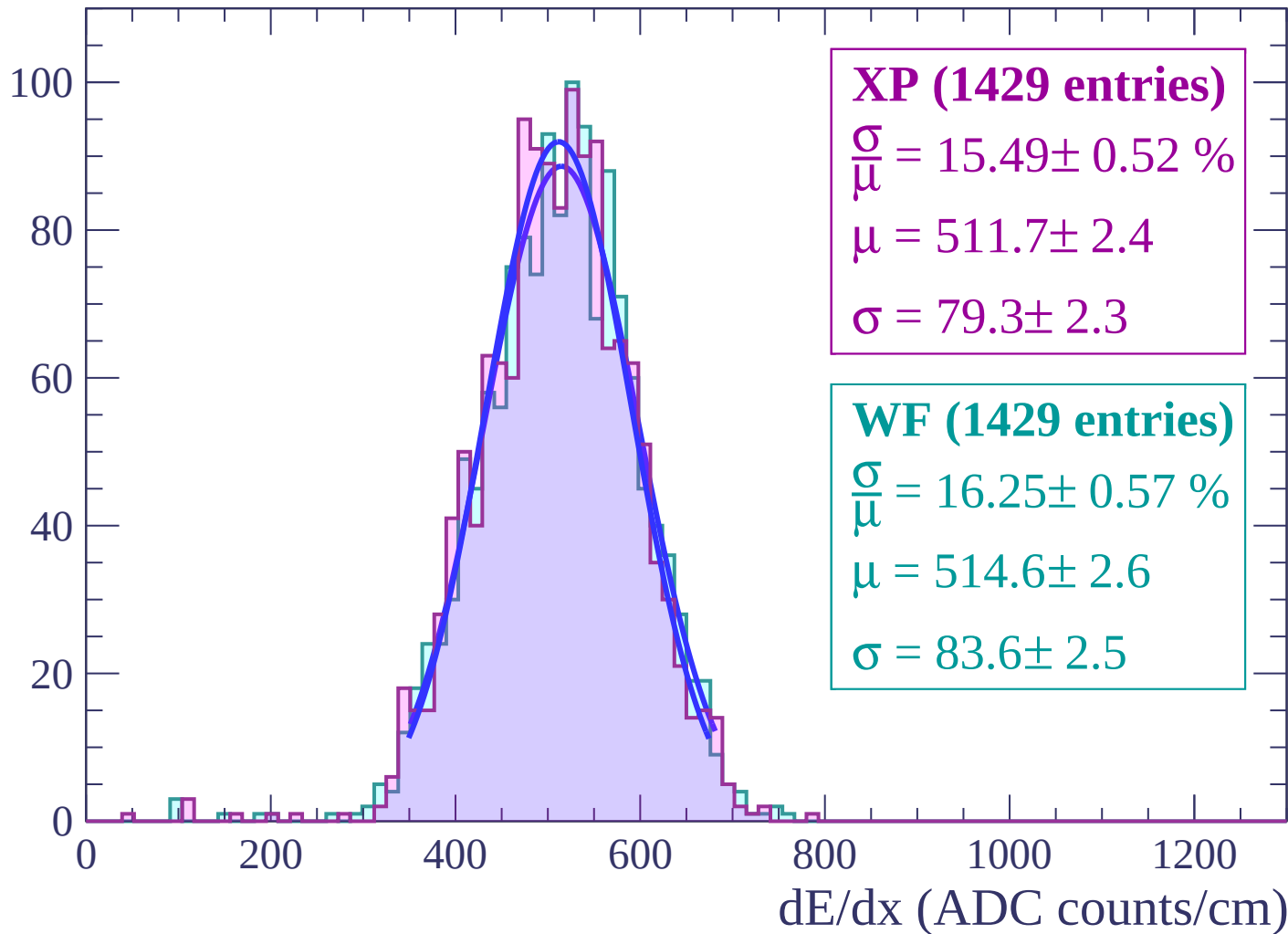
Energy loss | $-46 < \theta < -44$

Count



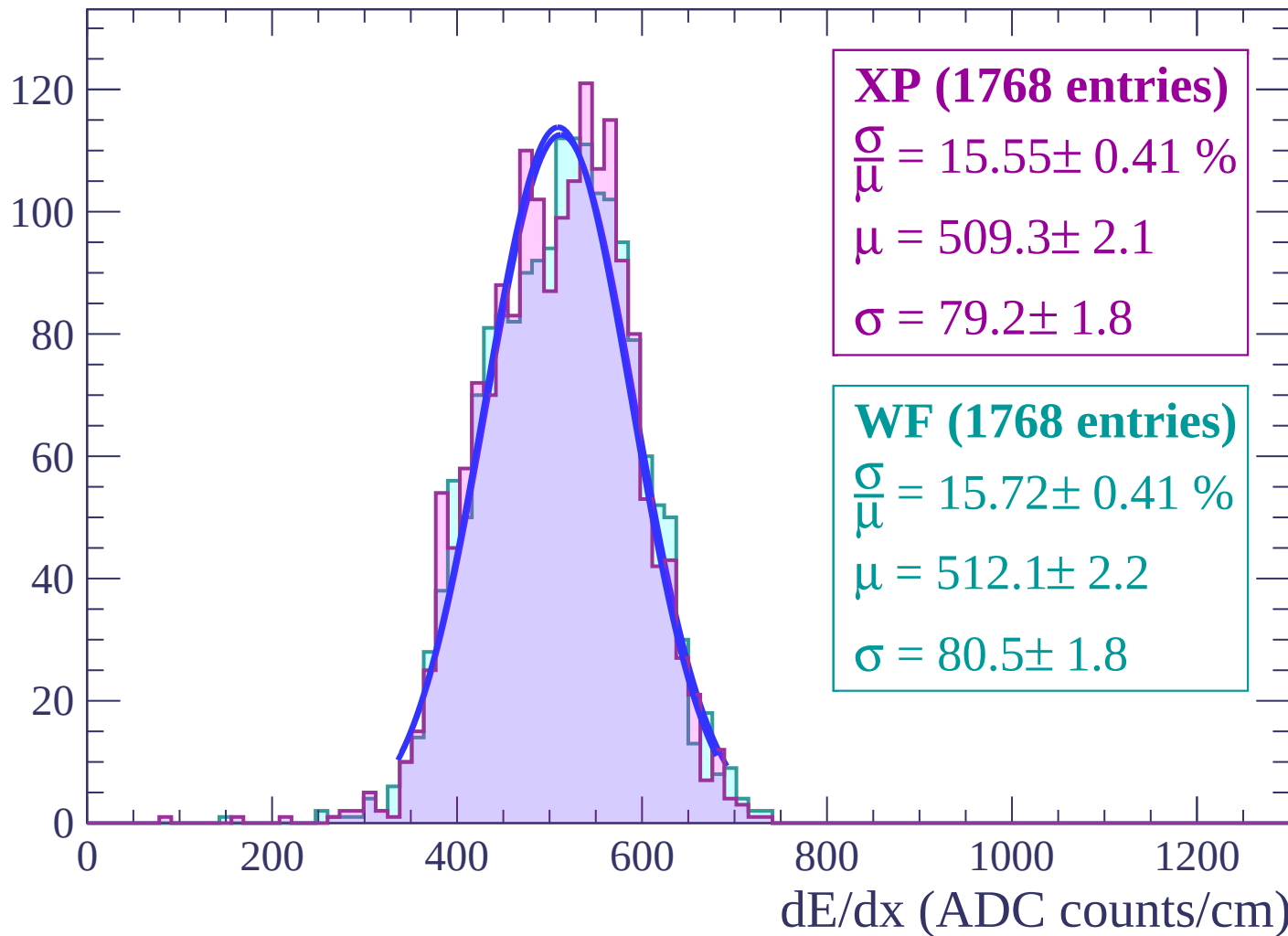
Energy loss | $-44 < \theta < -42$

Count

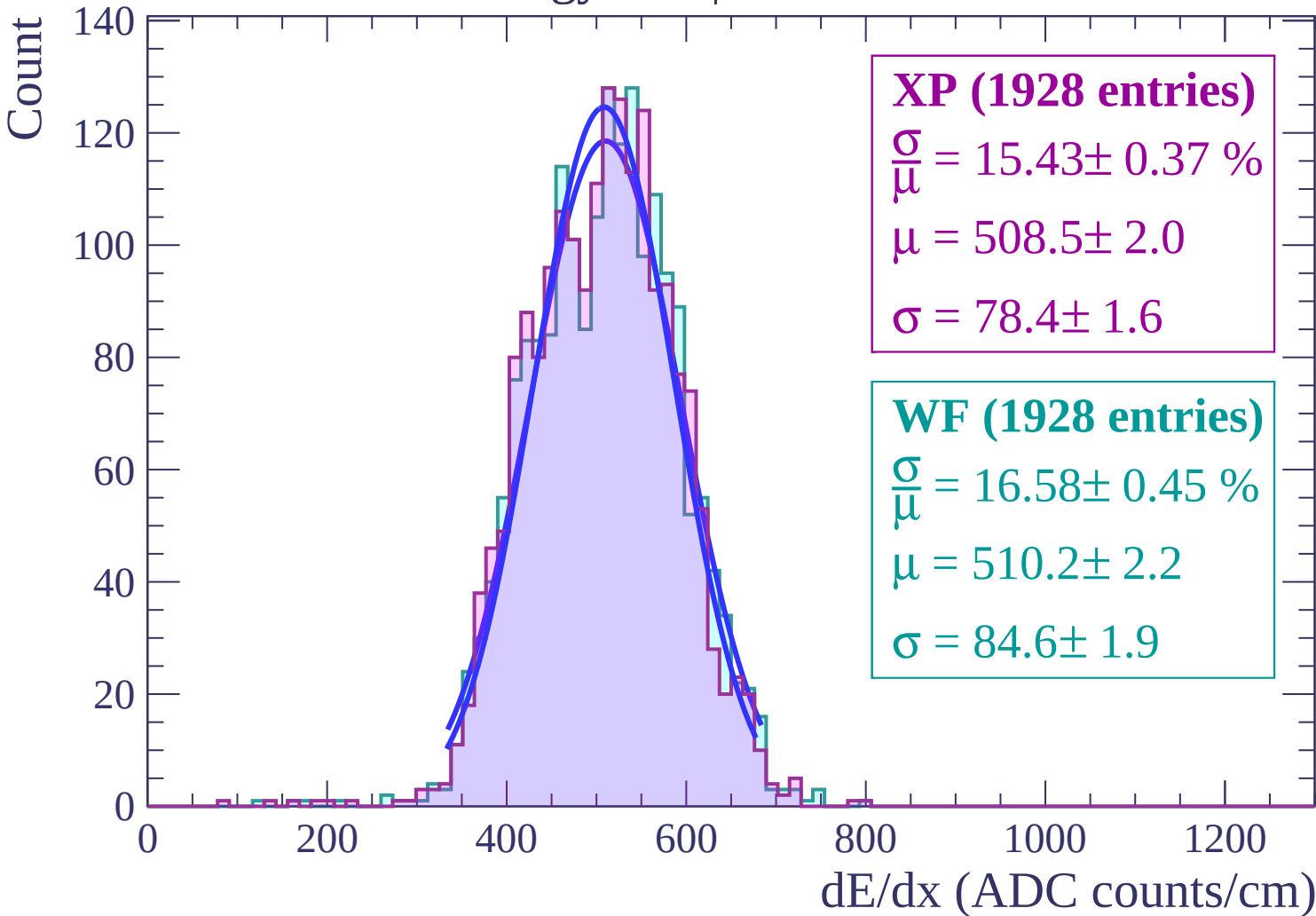


Energy loss | $-42 < \theta < -40$

Count

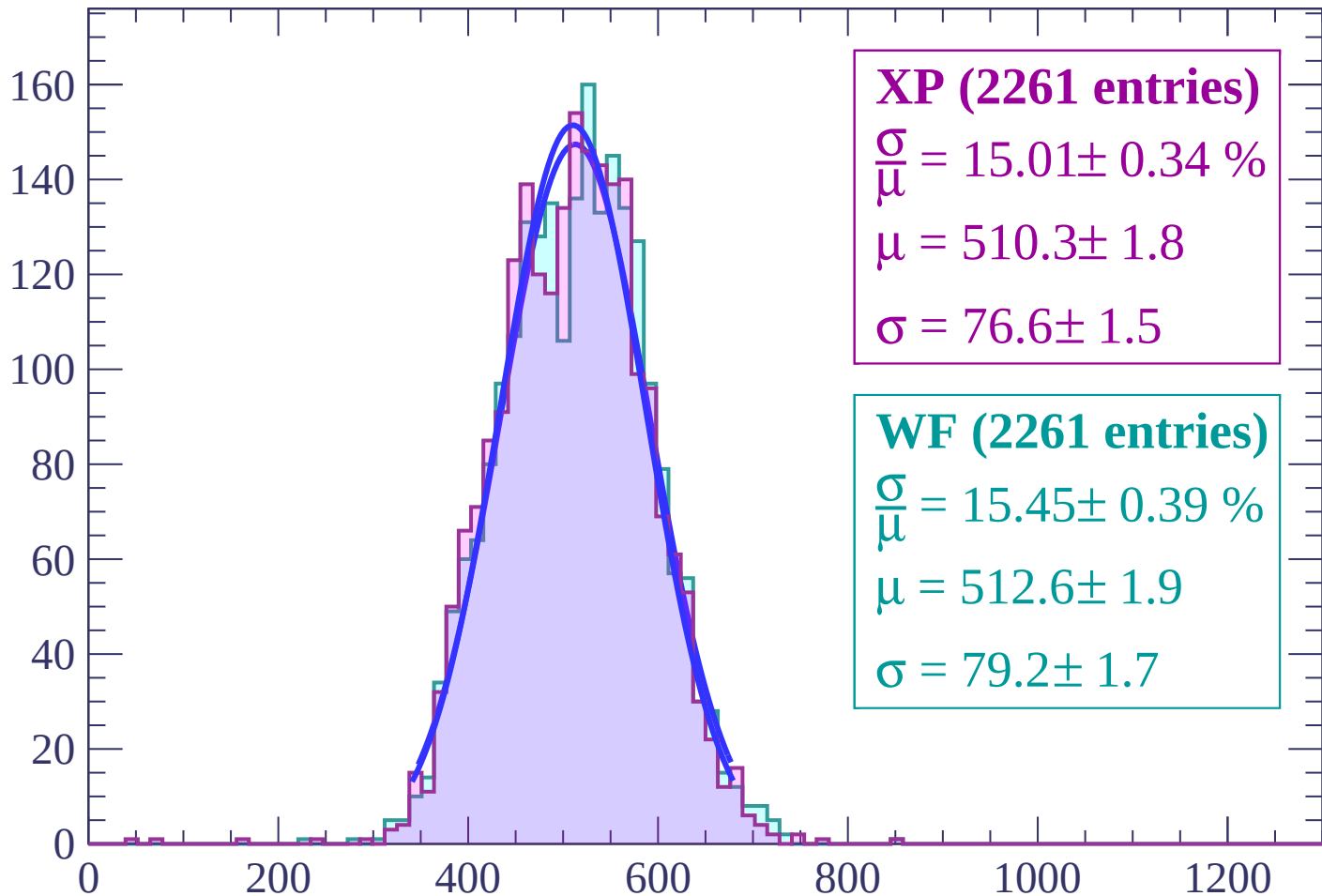


Energy loss | $-40 < \theta < -38$



Energy loss | $-38 < \theta < -36$

Count



XP (2261 entries)

$\frac{\sigma}{\mu} = 15.01 \pm 0.34 \%$

$\mu = 510.3 \pm 1.8$

$\sigma = 76.6 \pm 1.5$

WF (2261 entries)

$\frac{\sigma}{\mu} = 15.45 \pm 0.39 \%$

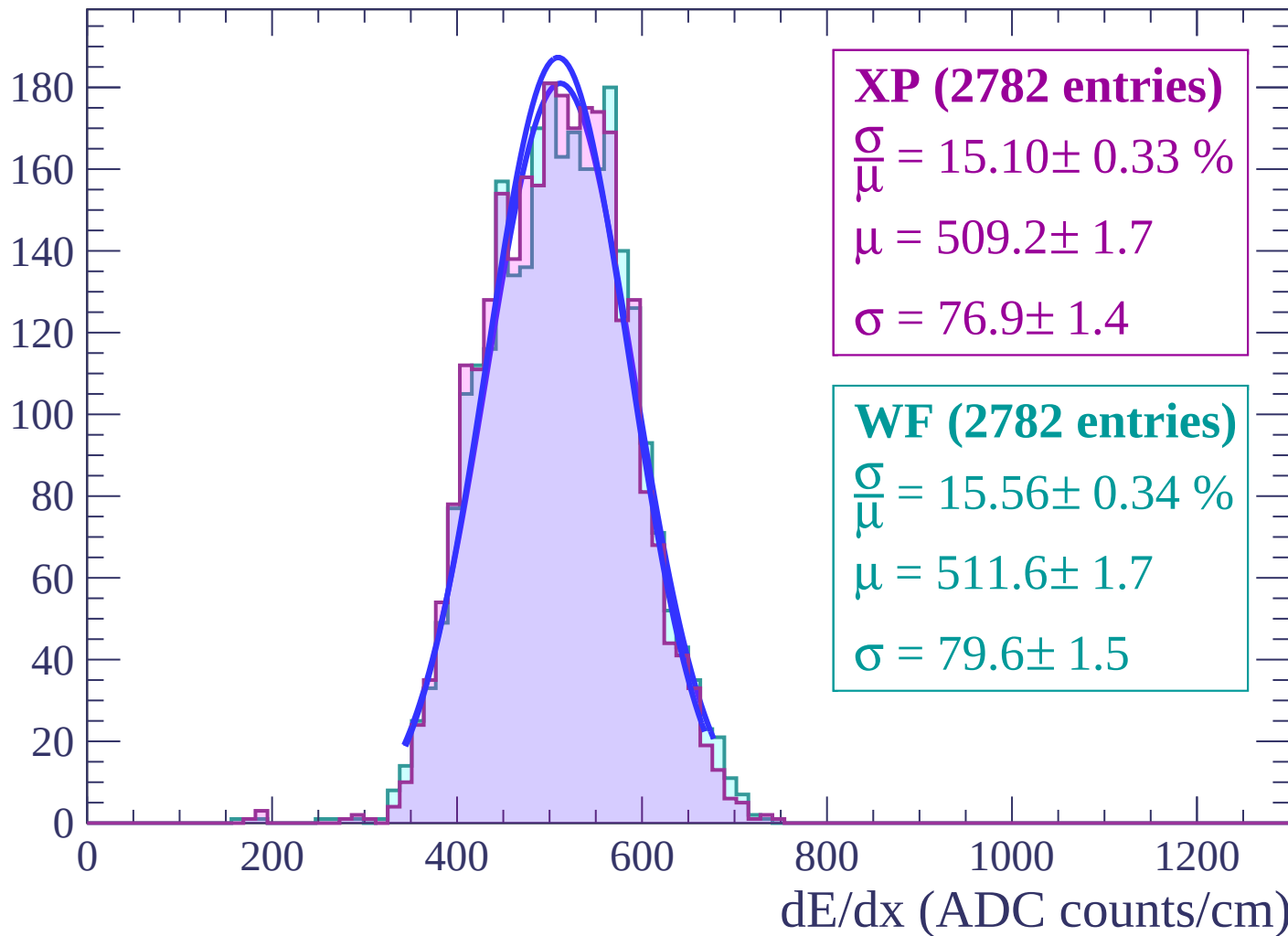
$\mu = 512.6 \pm 1.9$

$\sigma = 79.2 \pm 1.7$

dE/dx (ADC counts/cm)

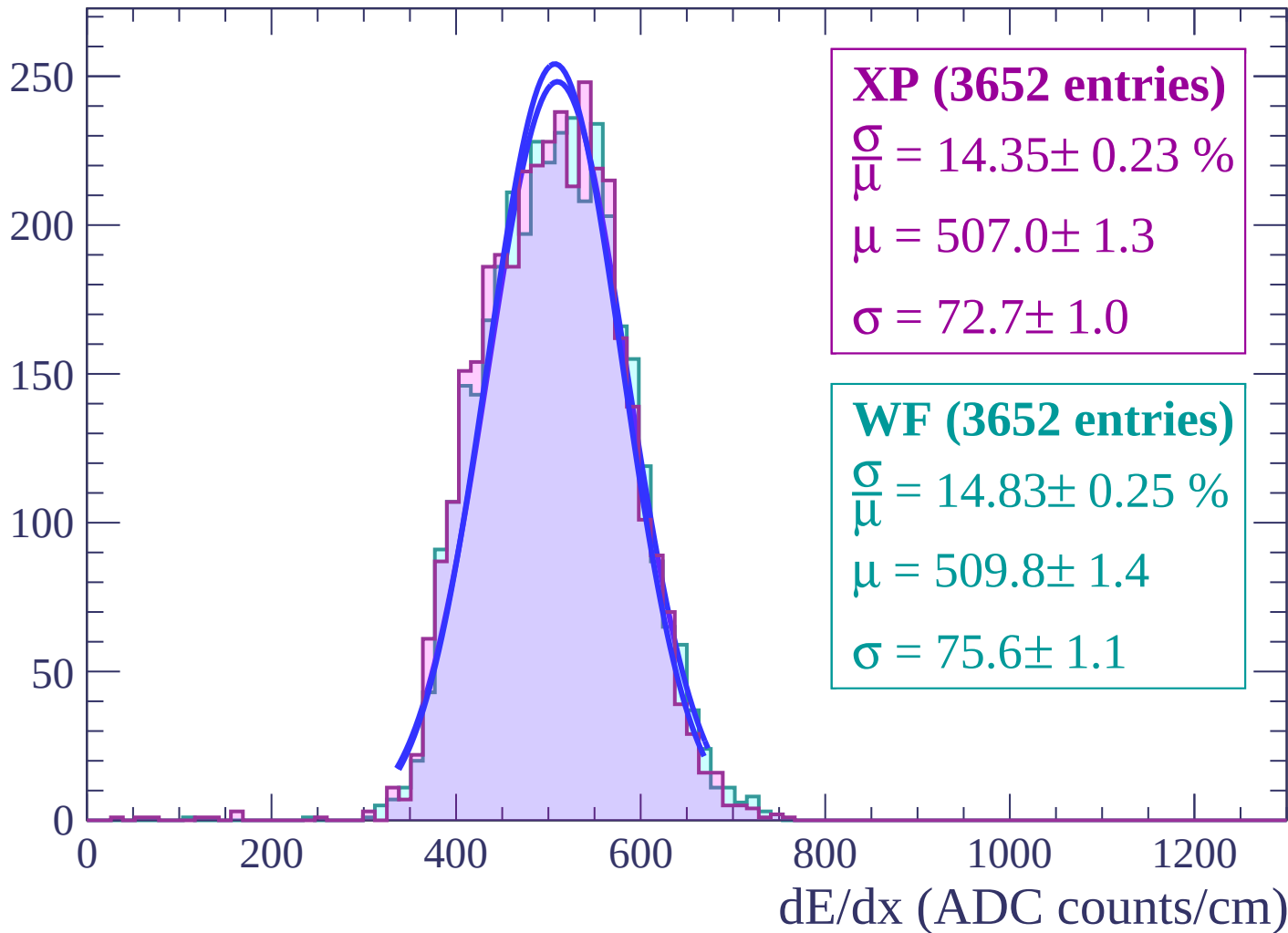
Energy loss | $-36 < \theta < -34$

Count



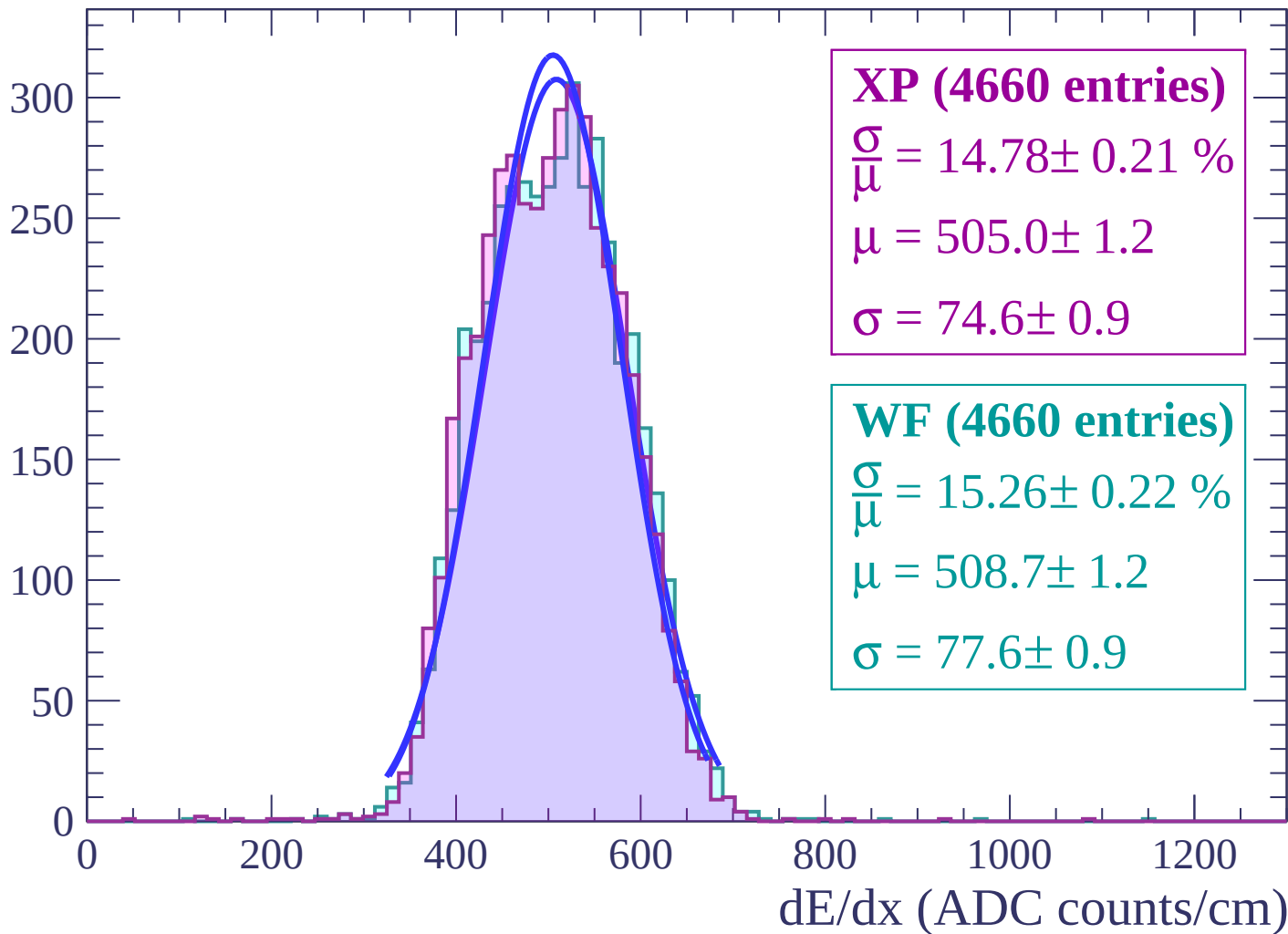
Energy loss | $-34 < \theta < -32$

Count



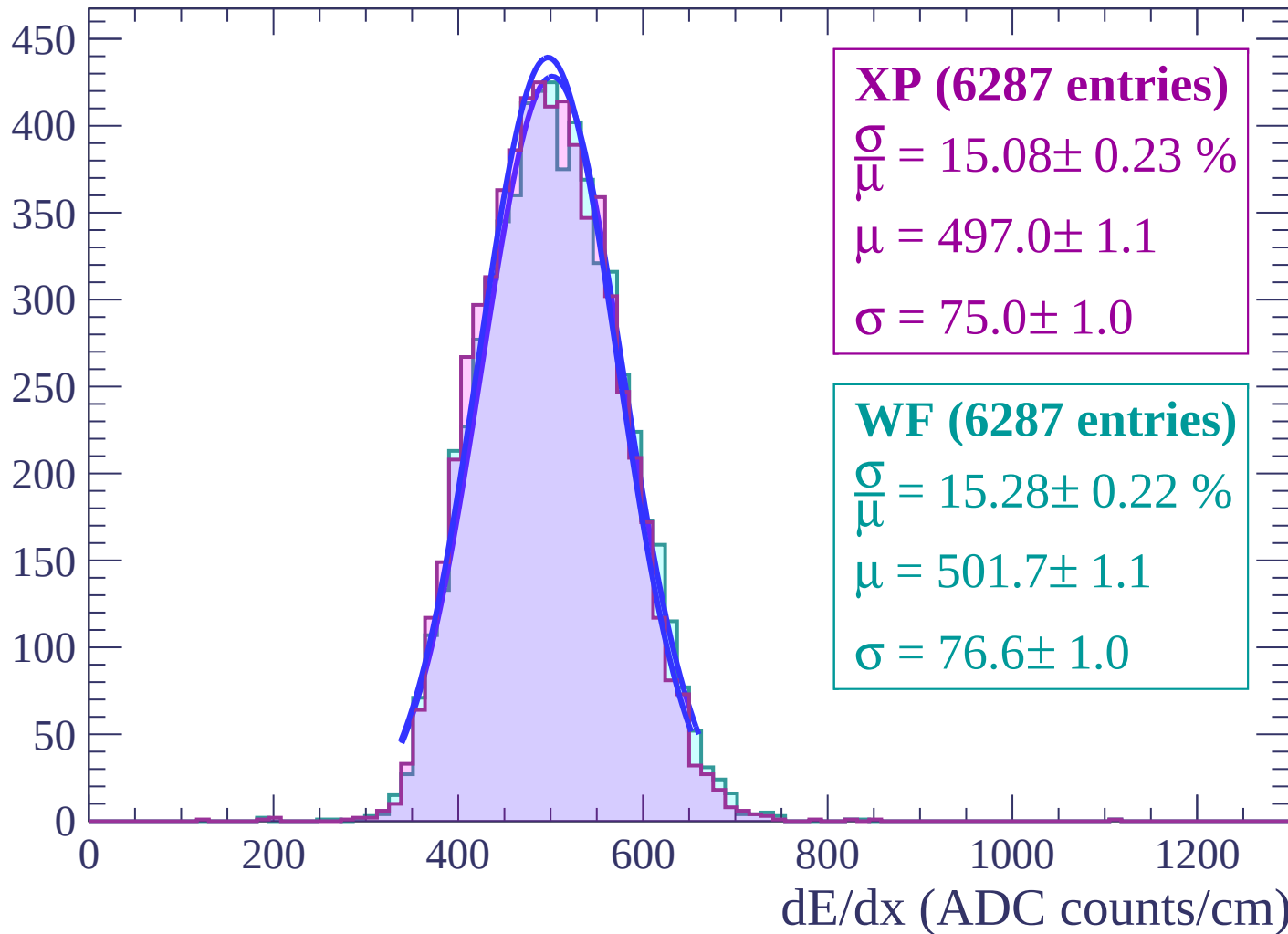
Energy loss | $-32 < \theta < -30$

Count

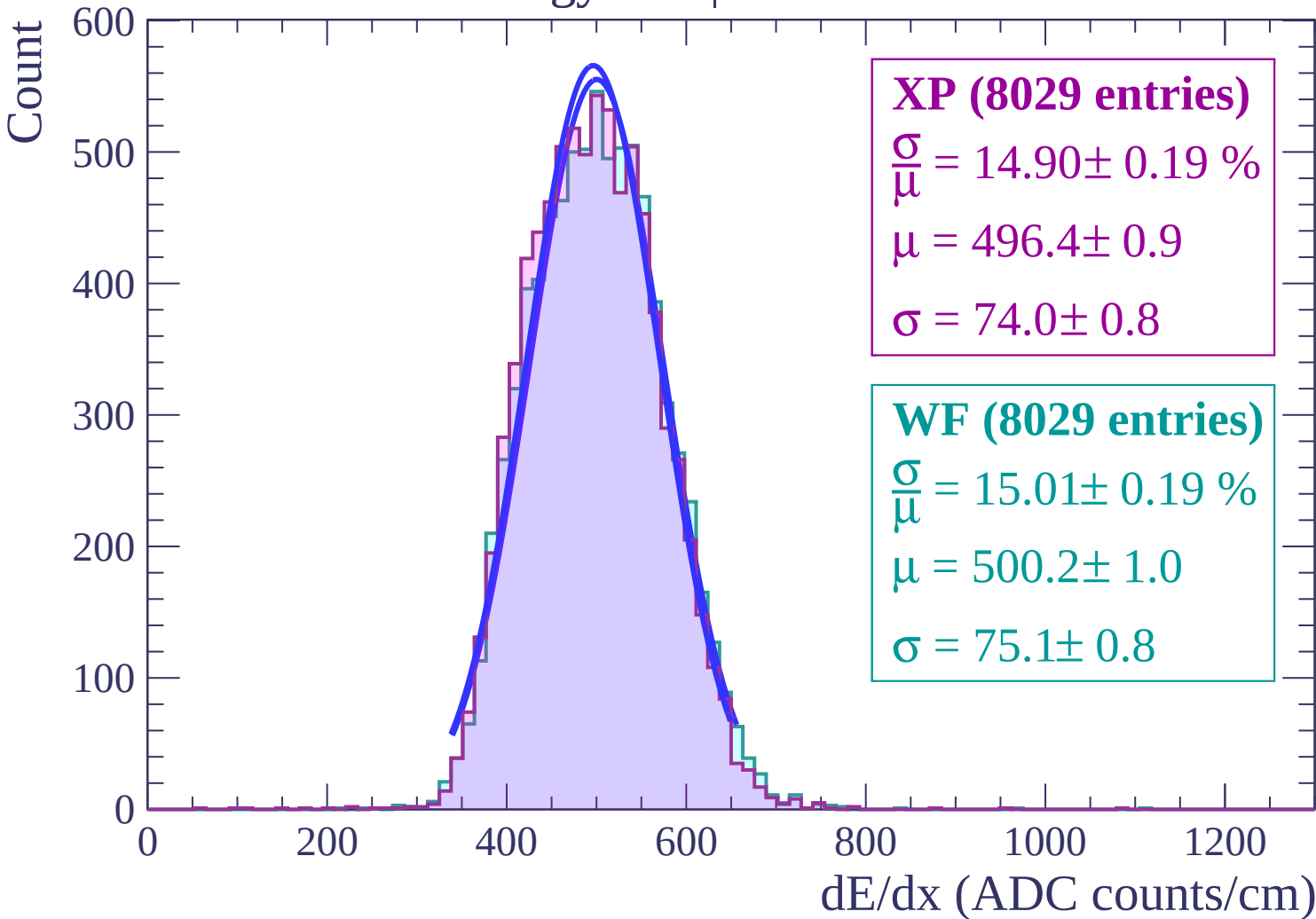


Energy loss | $-30 < \theta < -28$

Count

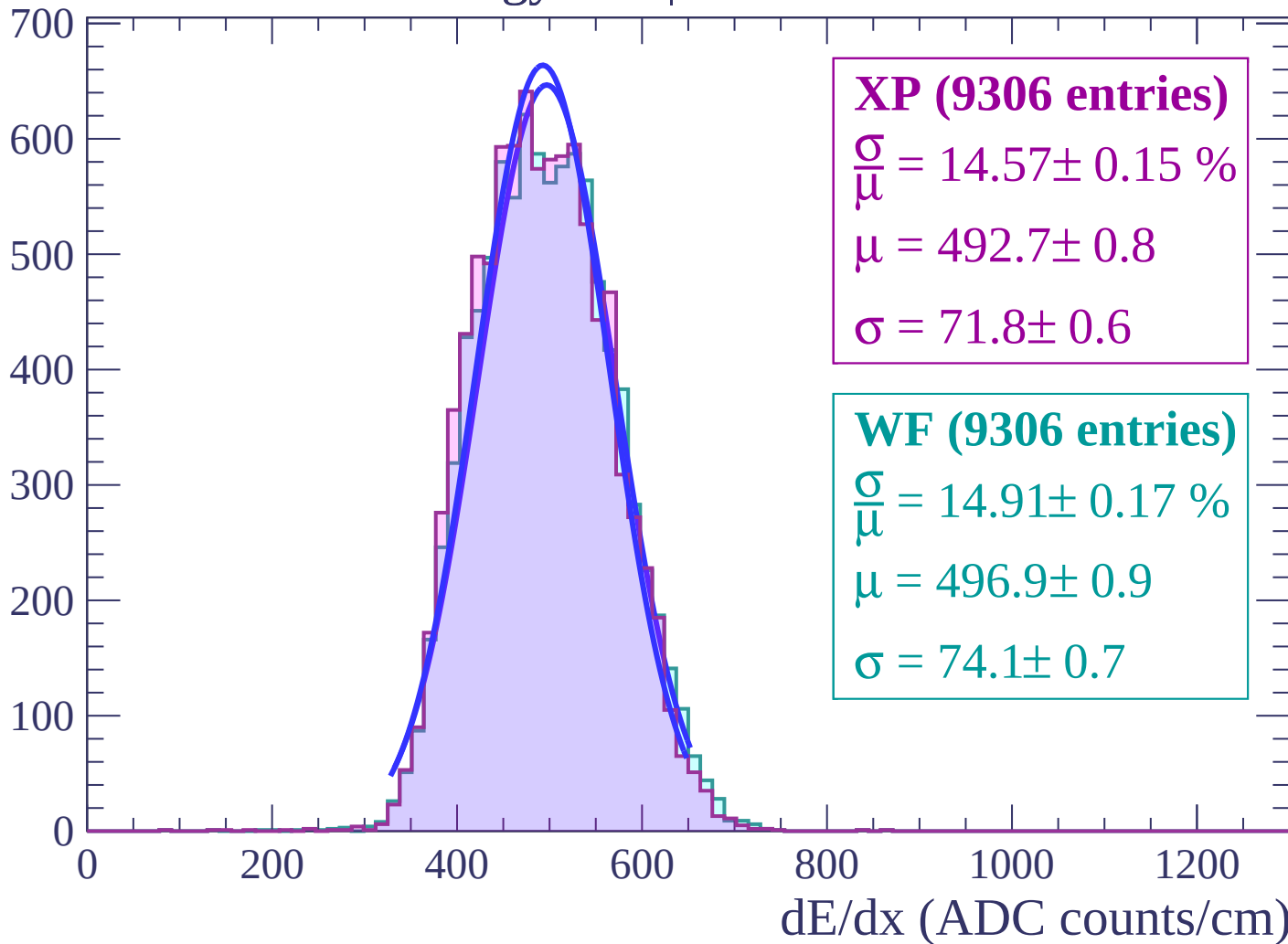


Energy loss | $-28 < \theta < -26$



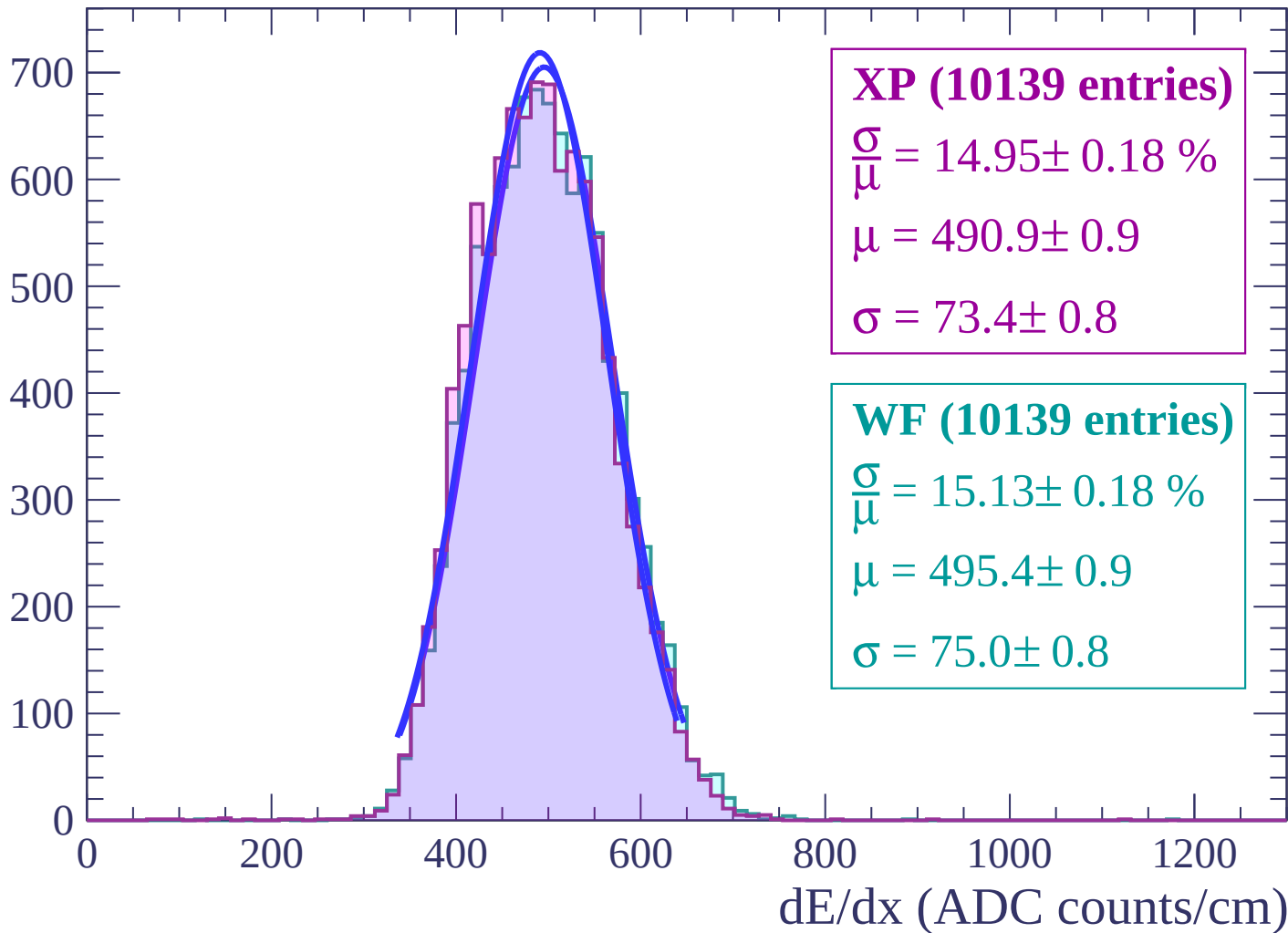
Energy loss | $-26 < \theta < -24$

Count



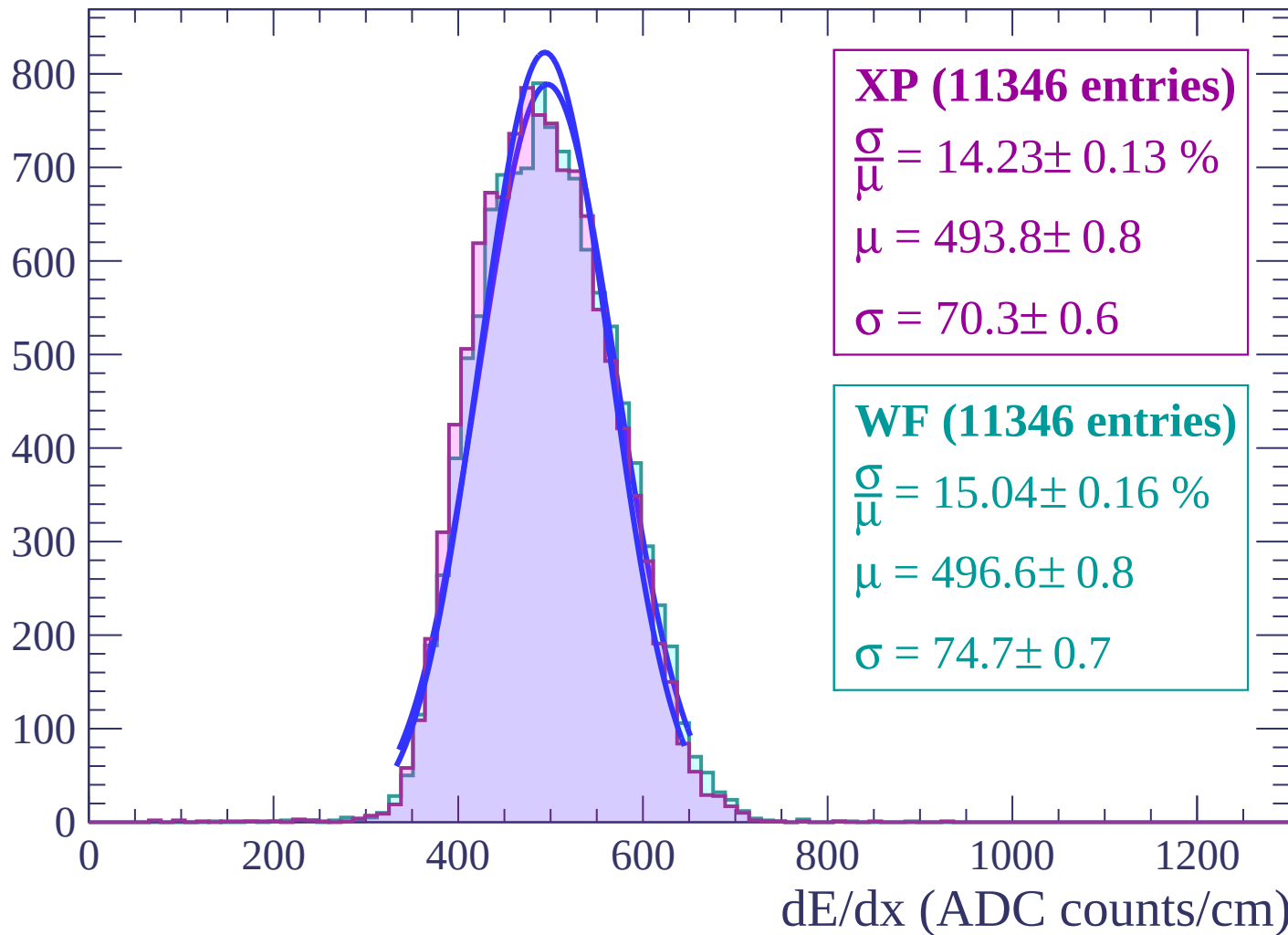
Energy loss | $-24 < \theta < -22$

Count



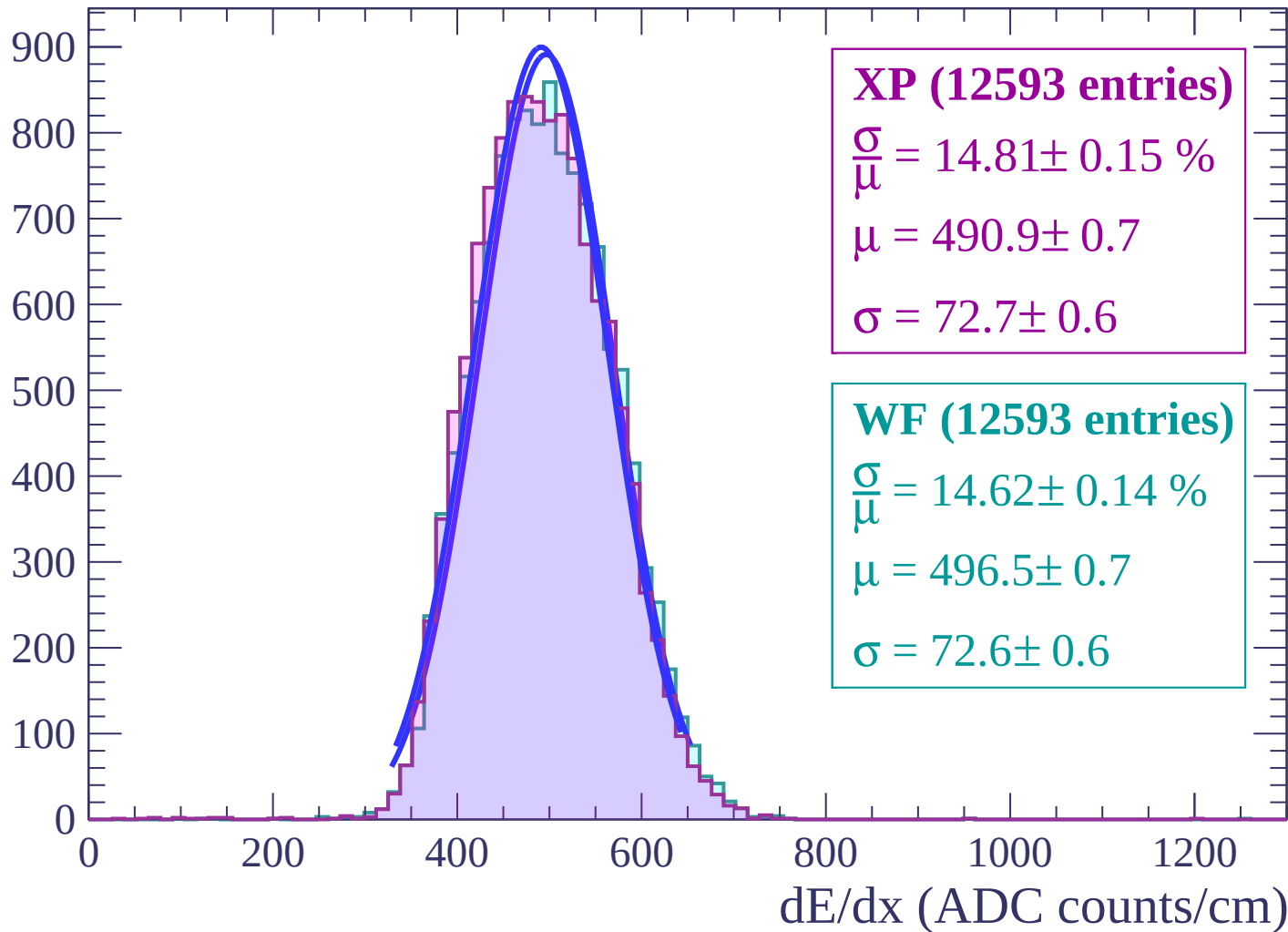
Energy loss | $-22 < \theta < -20$

Count

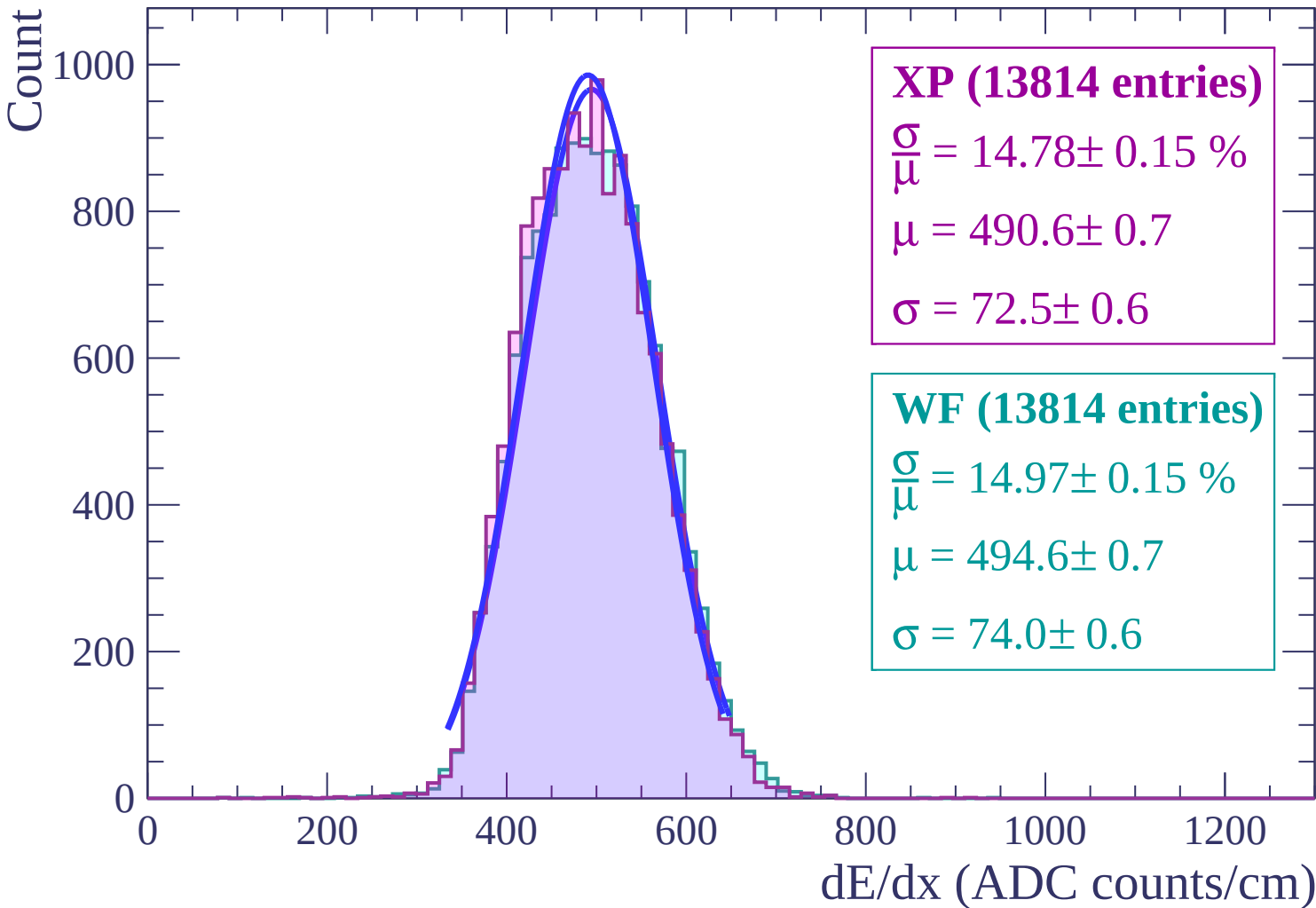


Energy loss | $-20 < \theta < -18$

Count



Energy loss | $-18 < \theta < -16$



Energy loss | $-16 < \theta < -14$

Count

1000

800

600

400

200

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (14900 entries)

$$\frac{\sigma}{\mu} = 14.72 \pm 0.14 \%$$

$$\mu = 490.3 \pm 0.7$$

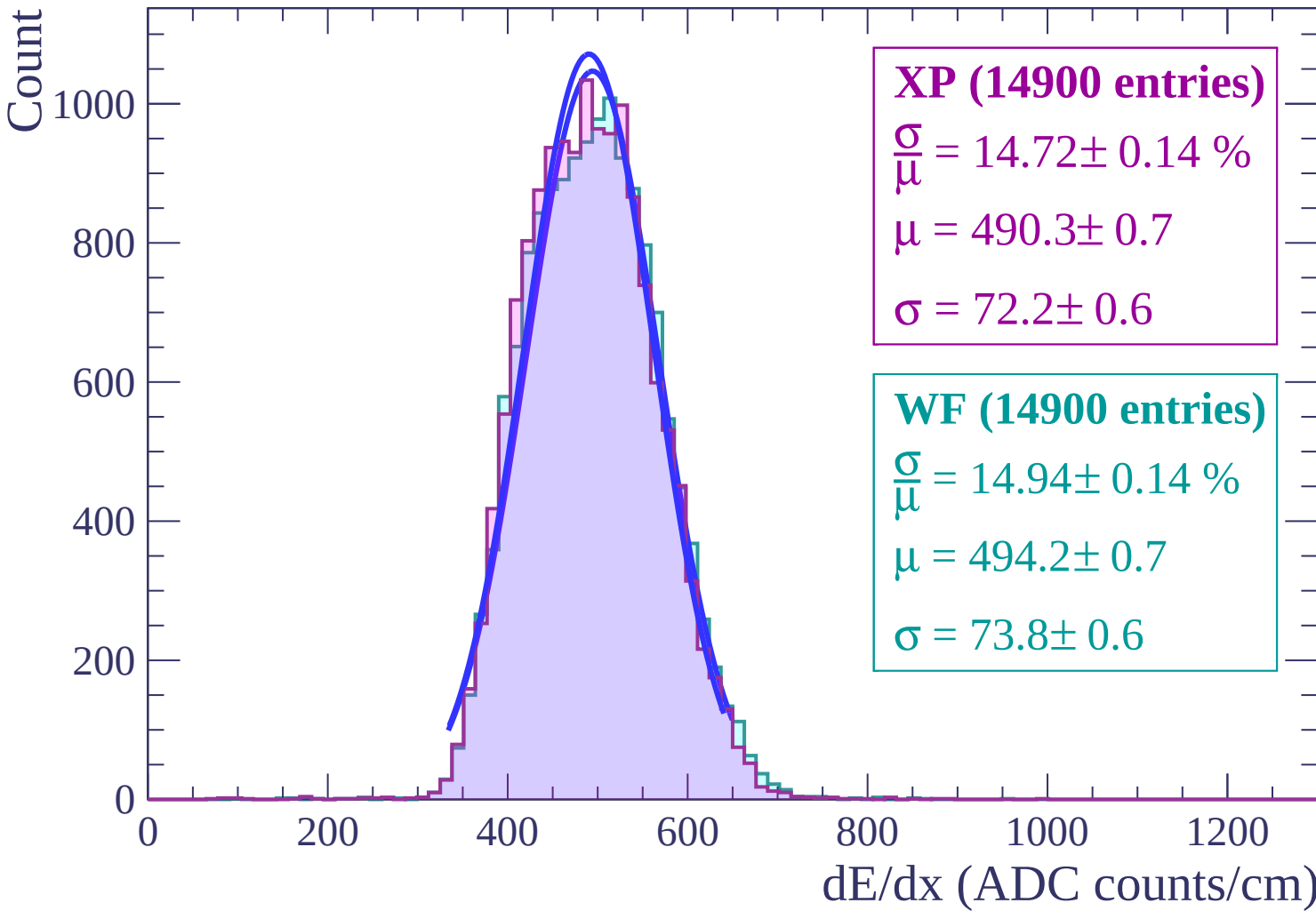
$$\sigma = 72.2 \pm 0.6$$

WF (14900 entries)

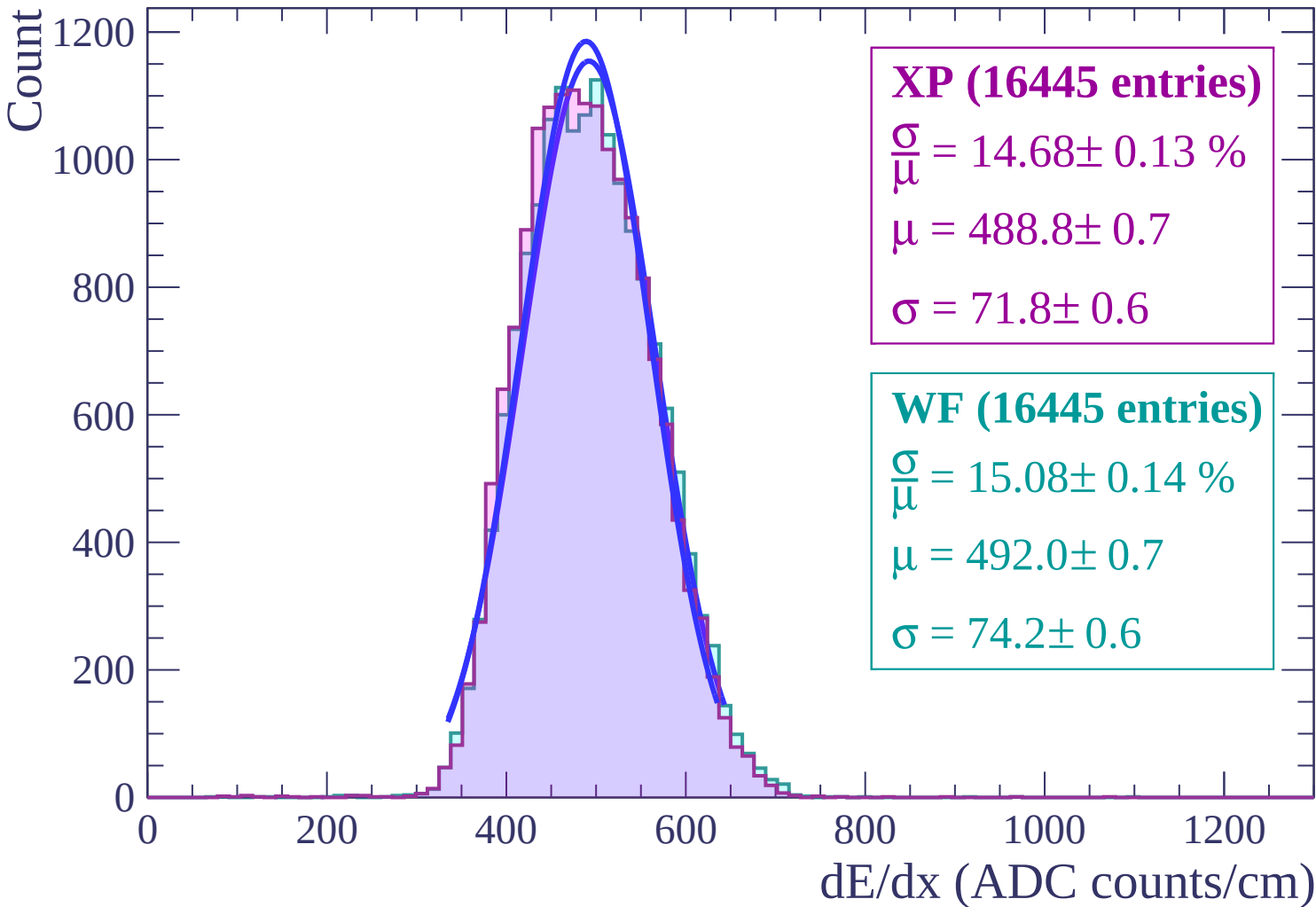
$$\frac{\sigma}{\mu} = 14.94 \pm 0.14 \%$$

$$\mu = 494.2 \pm 0.7$$

$$\sigma = 73.8 \pm 0.6$$

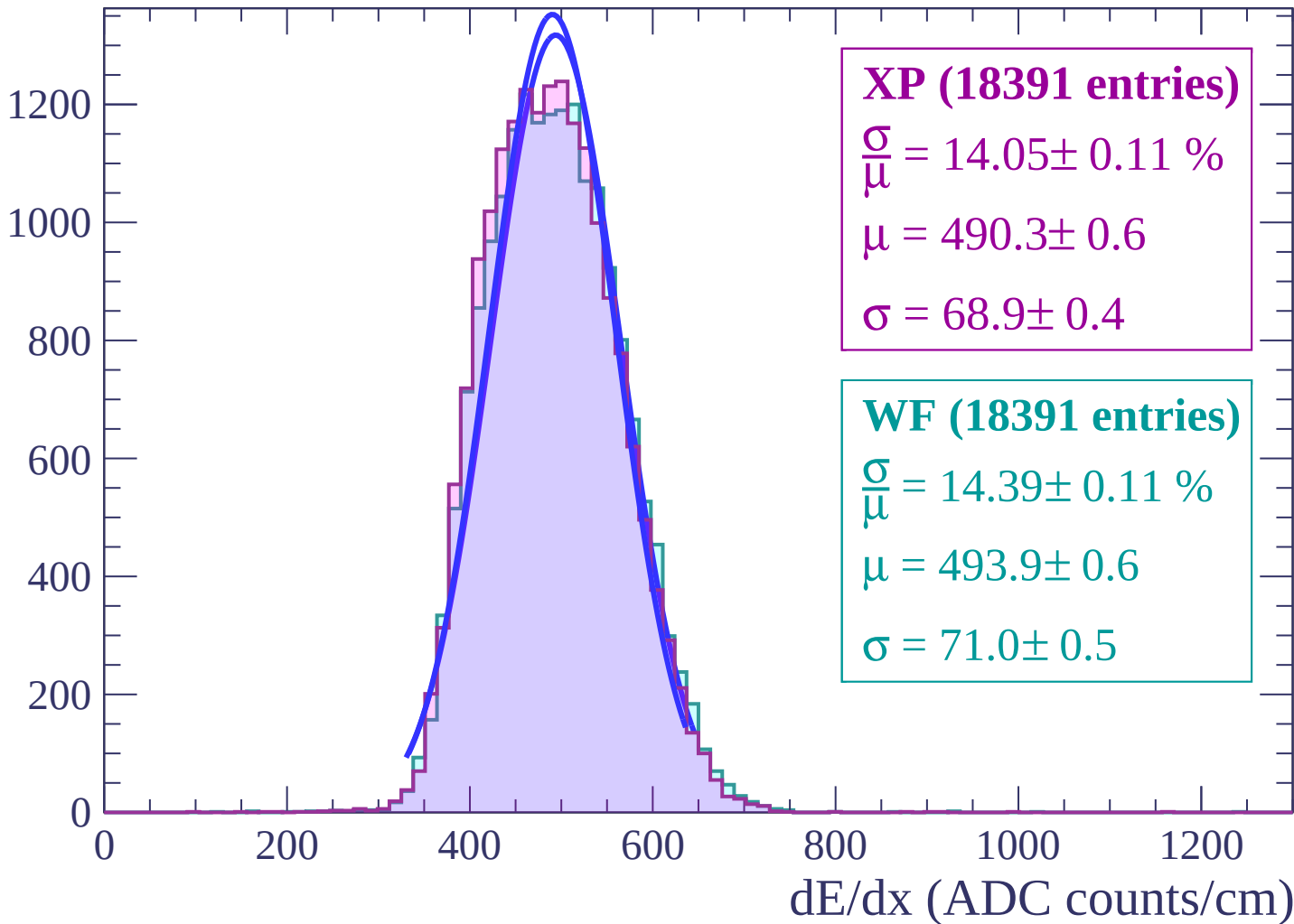


Energy loss | $-14 < \theta < -12$

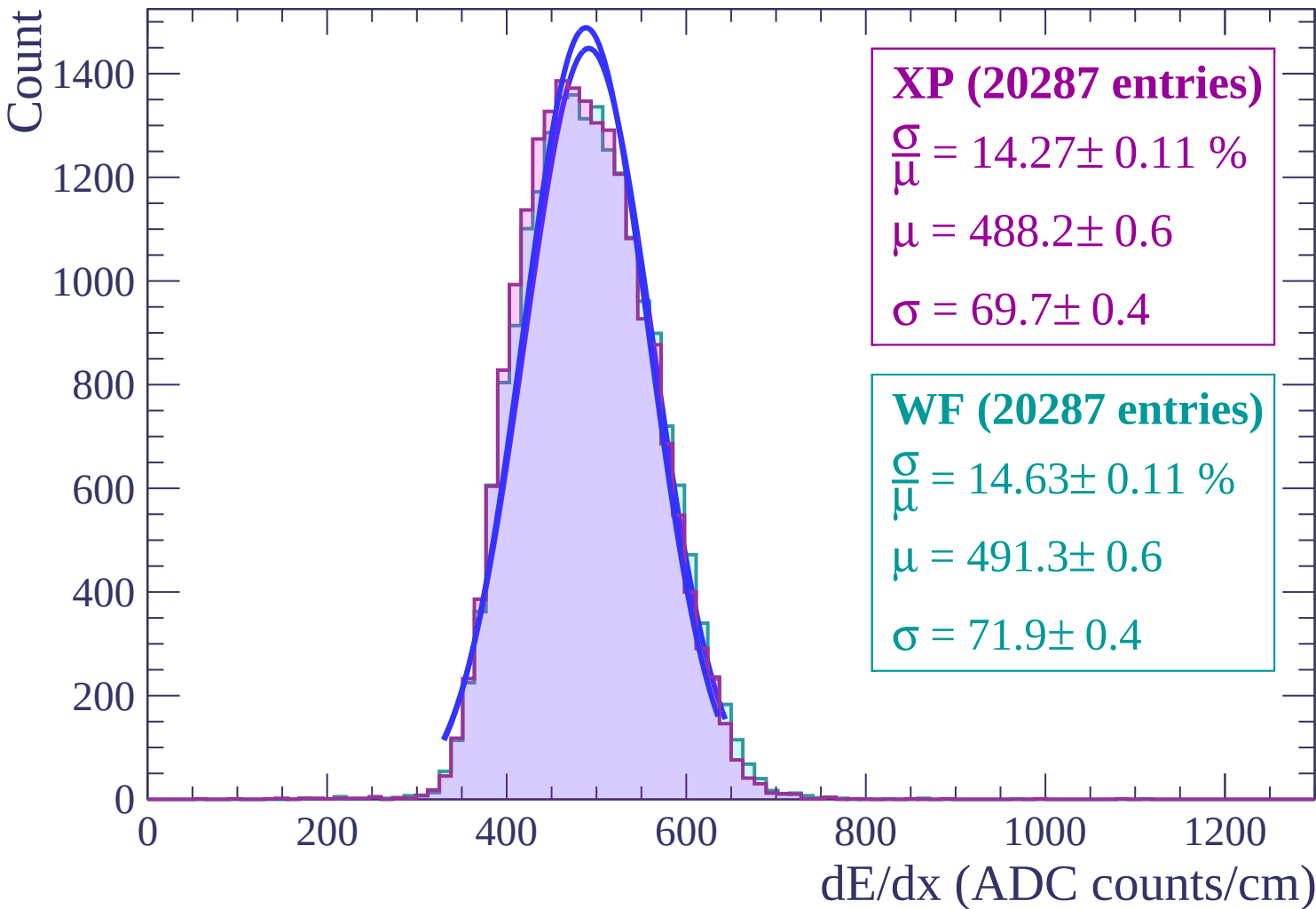


Energy loss | $-12 < \theta < -10$

Count



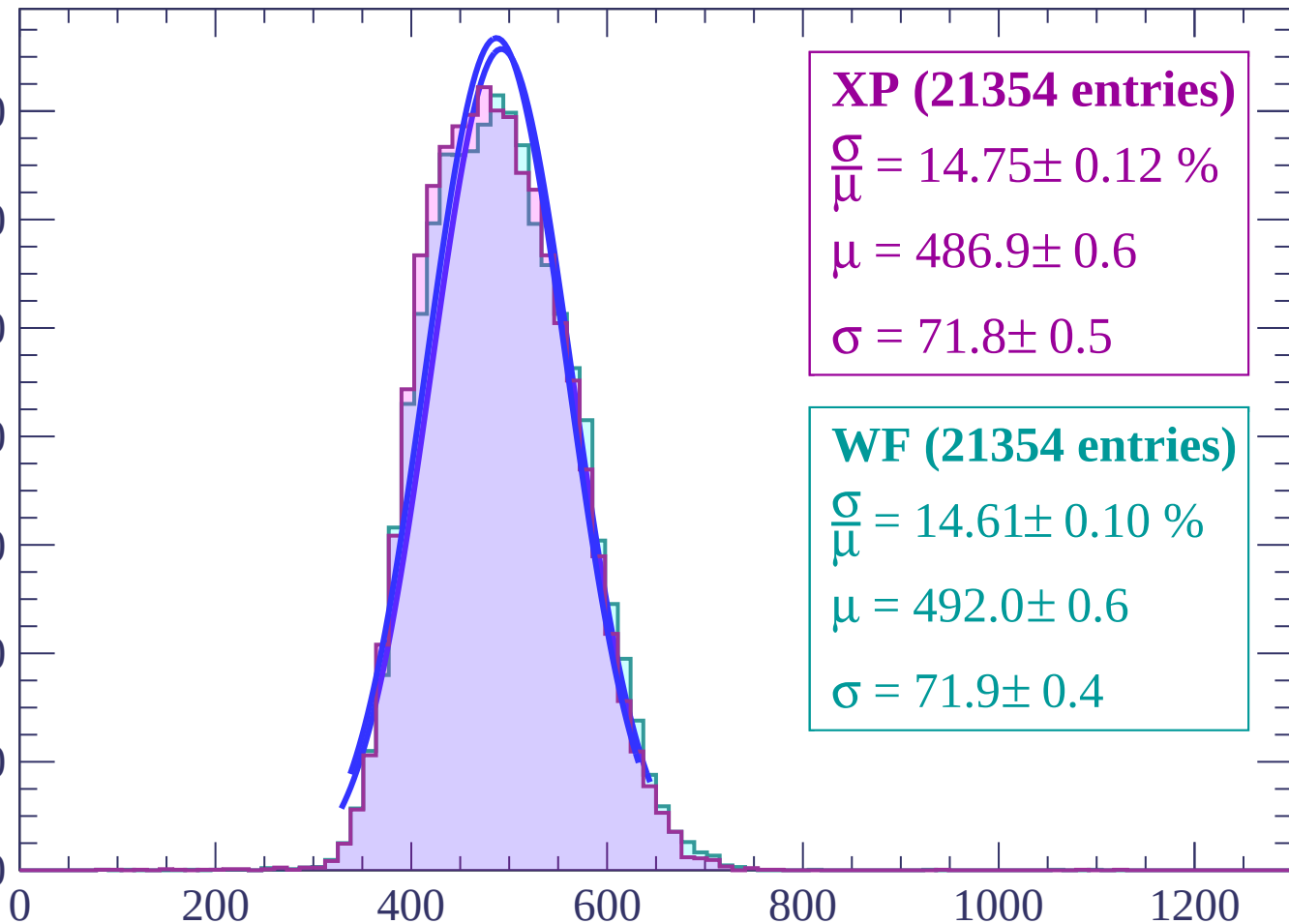
Energy loss | $-10 < \theta < -8$



Energy loss | $-8 < \theta < -6$

Count

1400
1200
1000
800
600
400
200
0



XP (21354 entries)

$\frac{\sigma}{\mu} = 14.75 \pm 0.12 \%$

$\mu = 486.9 \pm 0.6$

$\sigma = 71.8 \pm 0.5$

WF (21354 entries)

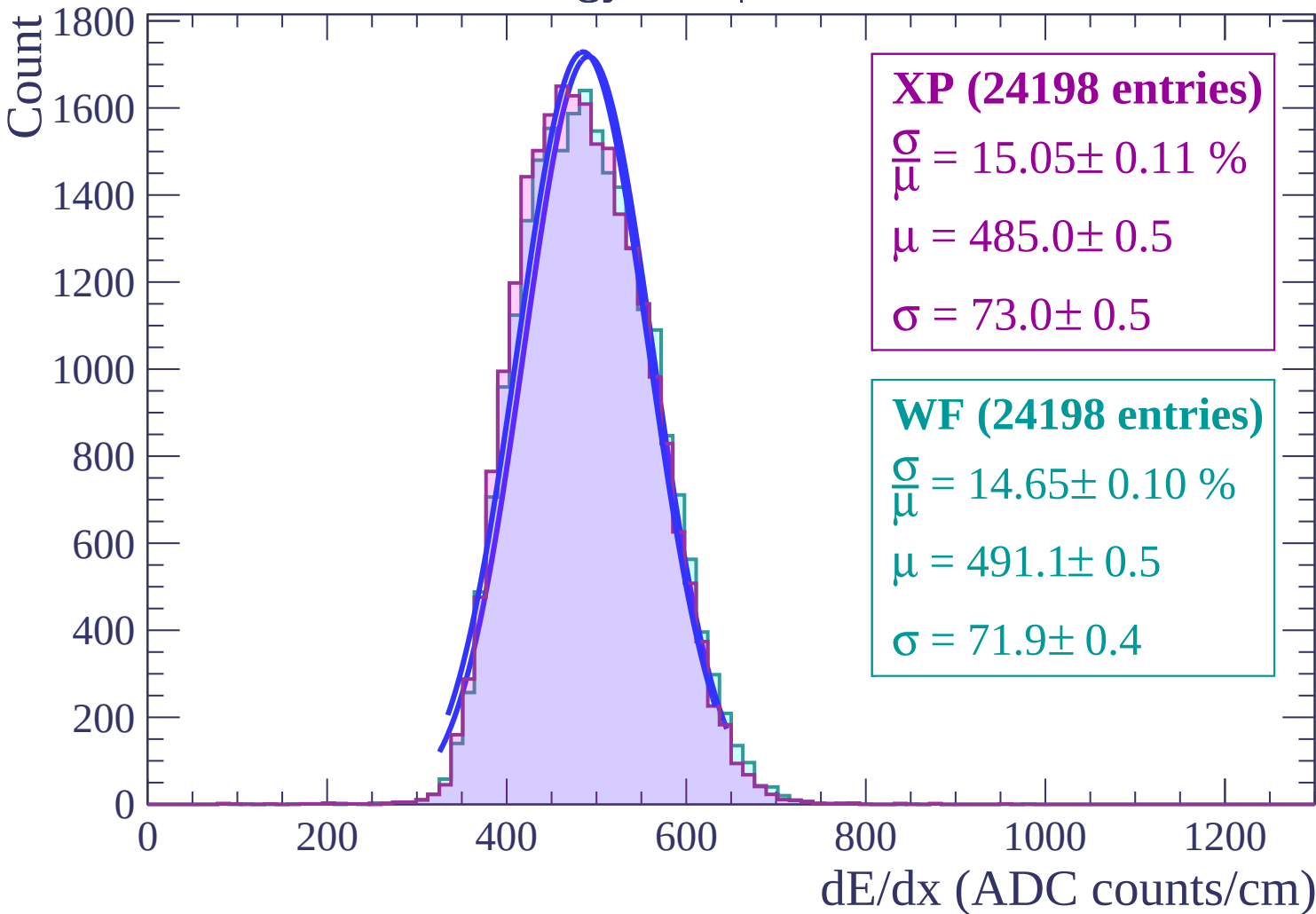
$\frac{\sigma}{\mu} = 14.61 \pm 0.10 \%$

$\mu = 492.0 \pm 0.6$

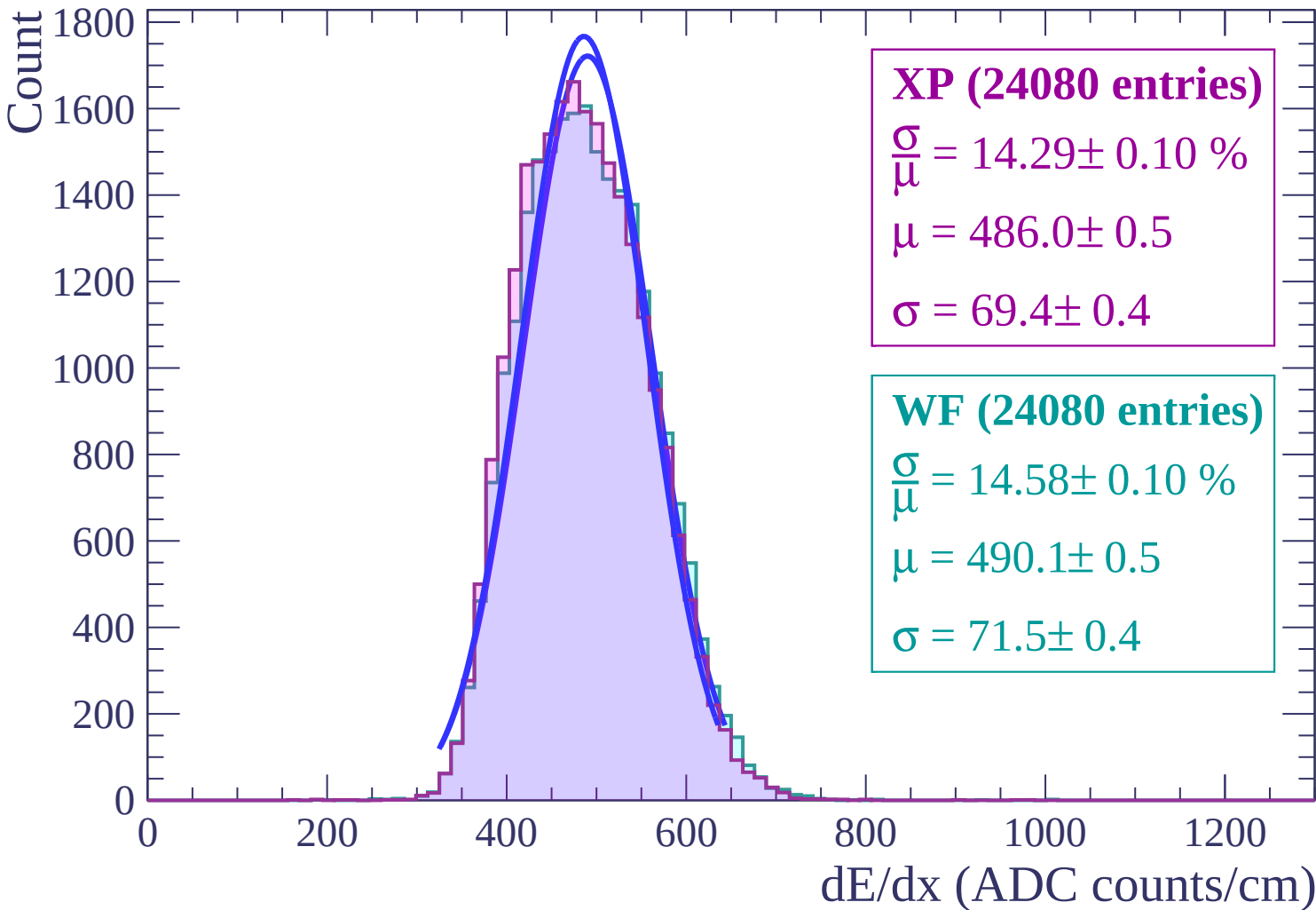
$\sigma = 71.9 \pm 0.4$

dE/dx (ADC counts/cm)

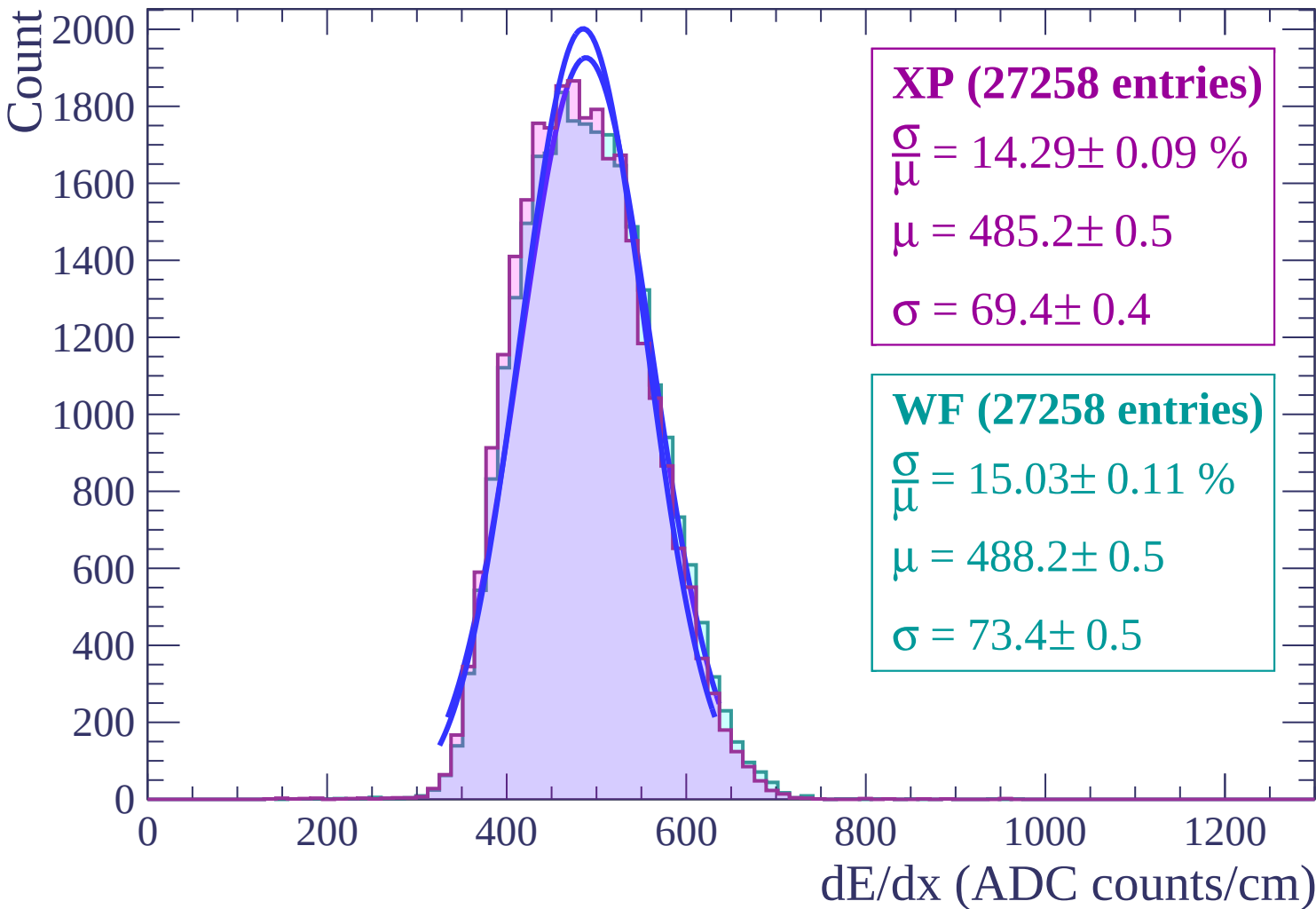
Energy loss | $-6 < \theta < -4$



Energy loss | $-4 < \theta < -2$



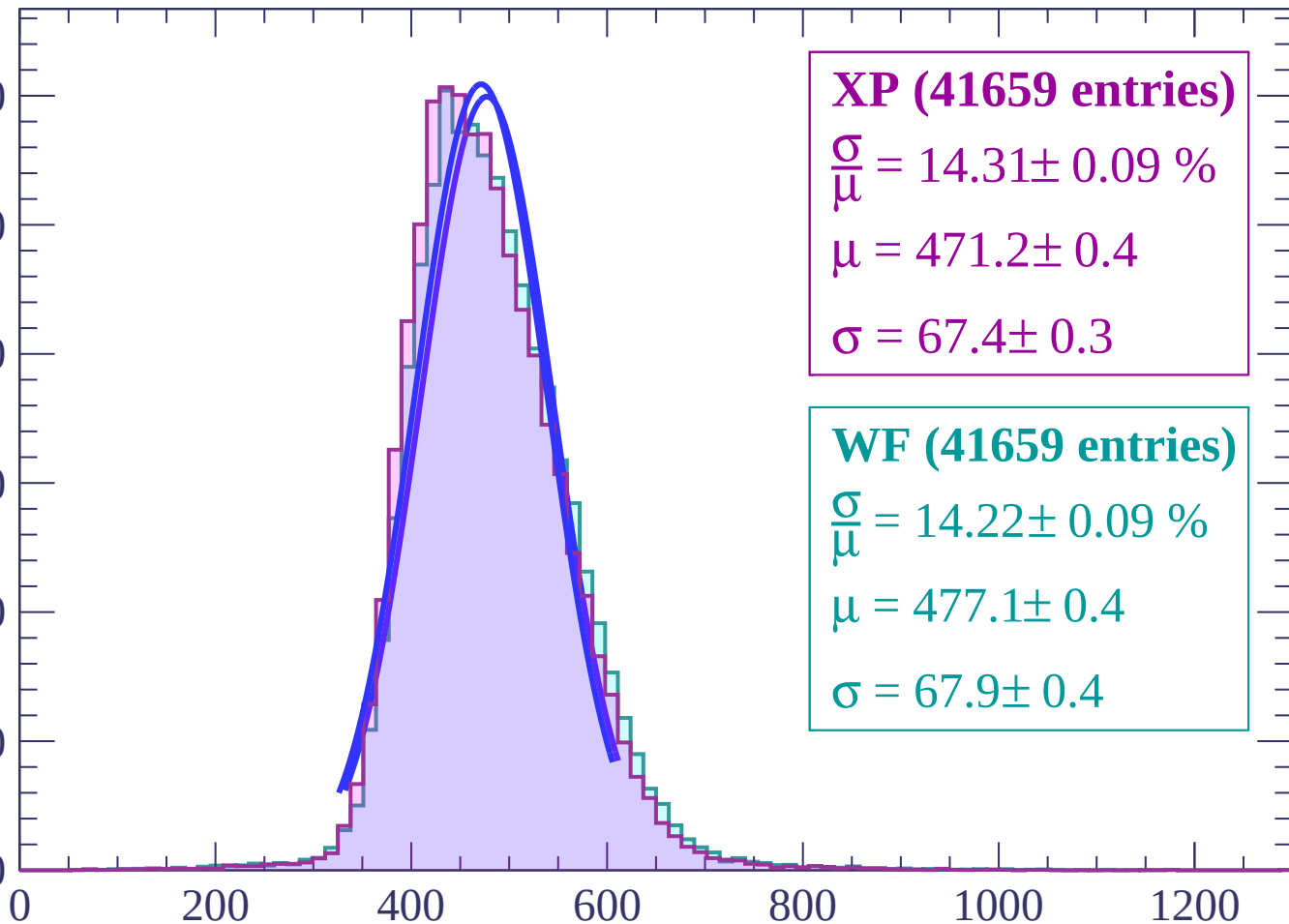
Energy loss $-2 < \theta < 0$



Energy loss | $0 < \theta < 2$

Count

3000
2500
2000
1500
1000
500
0



XP (41659 entries)

$\frac{\sigma}{\mu} = 14.31 \pm 0.09 \%$

$\mu = 471.2 \pm 0.4$

$\sigma = 67.4 \pm 0.3$

WF (41659 entries)

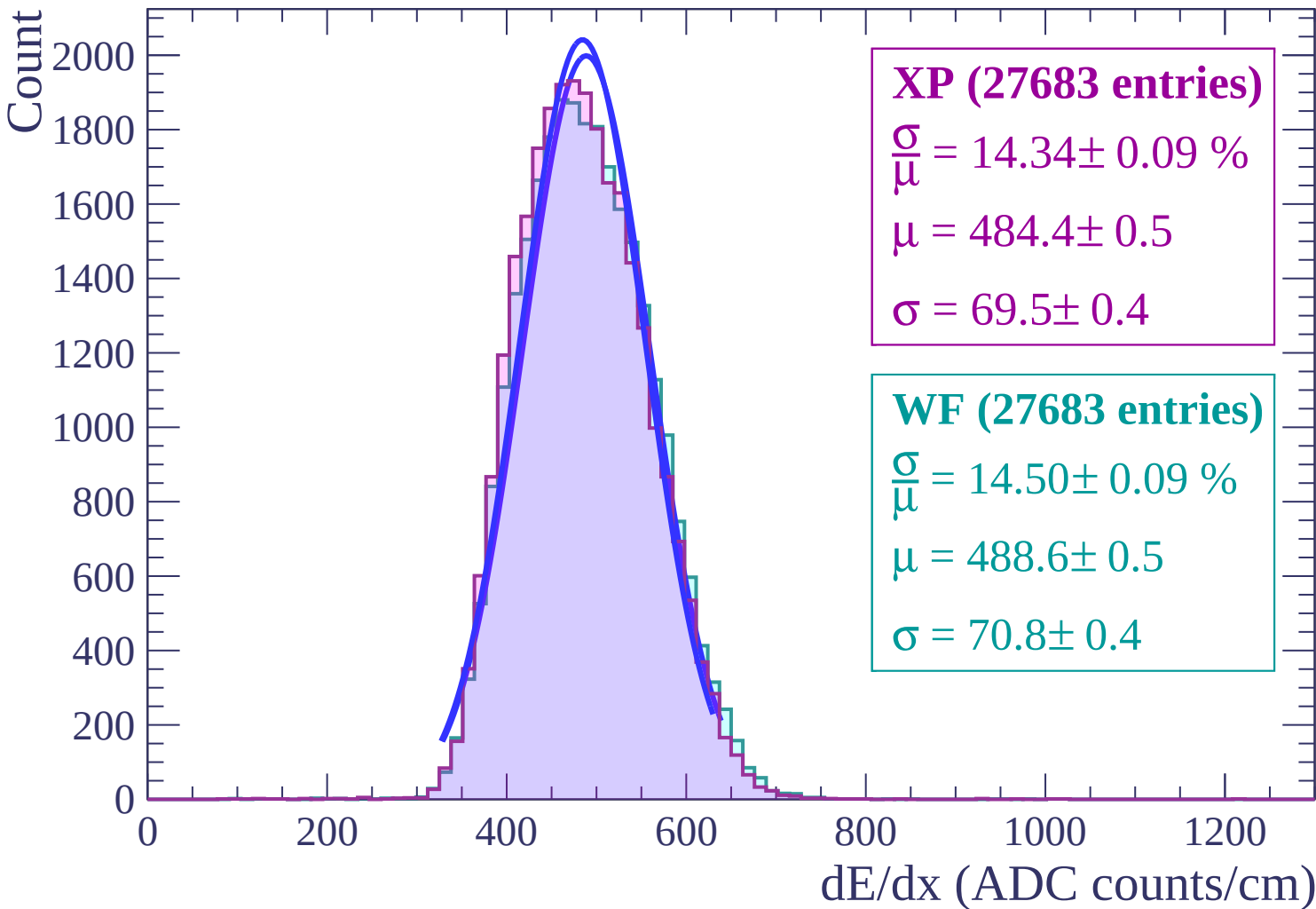
$\frac{\sigma}{\mu} = 14.22 \pm 0.09 \%$

$\mu = 477.1 \pm 0.4$

$\sigma = 67.9 \pm 0.4$

dE/dx (ADC counts/cm)

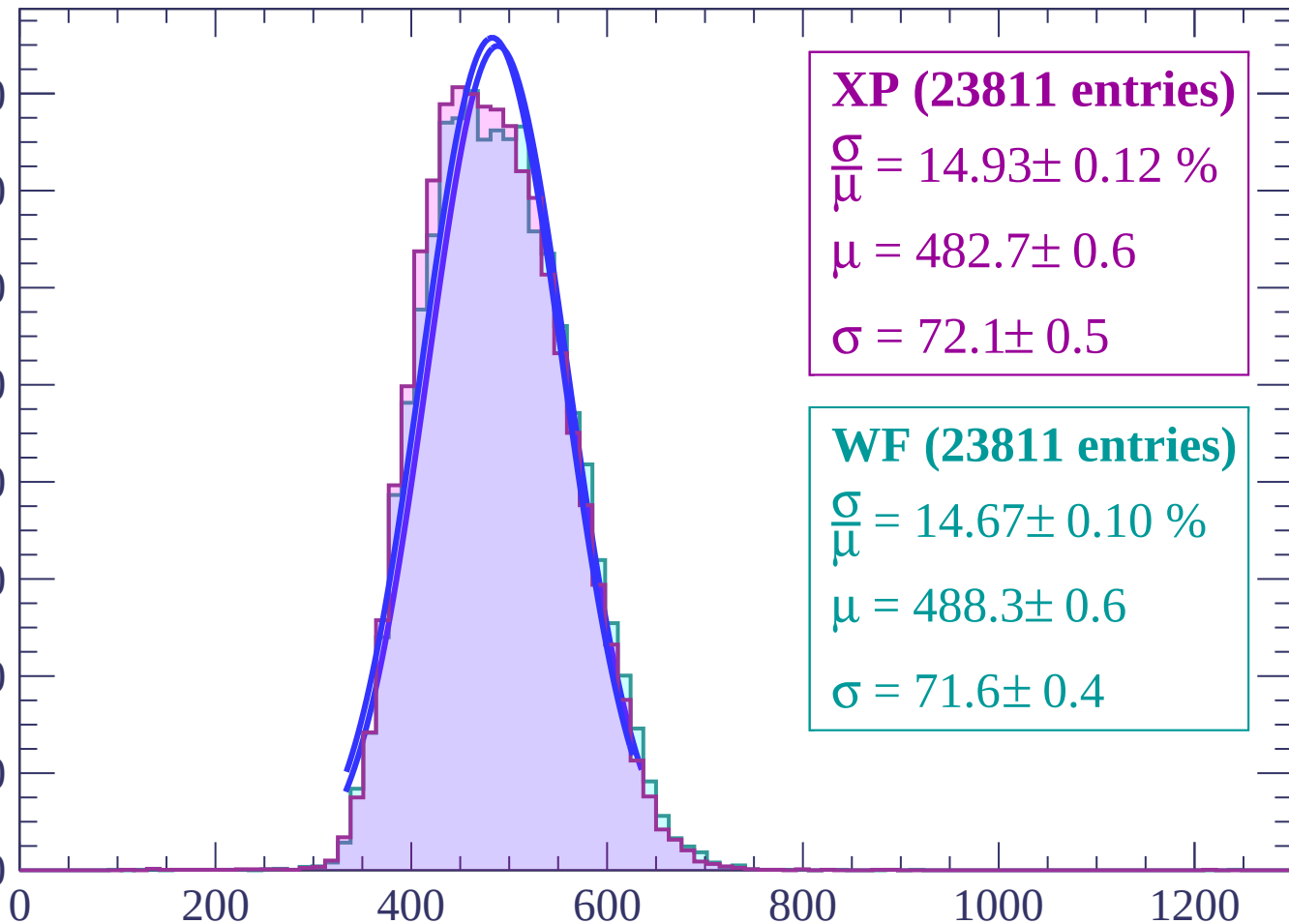
Energy loss | $2 < \theta < 4$



Energy loss | $4 < \theta < 6$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (23811 entries)

$\frac{\sigma}{\mu} = 14.93 \pm 0.12 \%$

$\mu = 482.7 \pm 0.6$

$\sigma = 72.1 \pm 0.5$

WF (23811 entries)

$\frac{\sigma}{\mu} = 14.67 \pm 0.10 \%$

$\mu = 488.3 \pm 0.6$

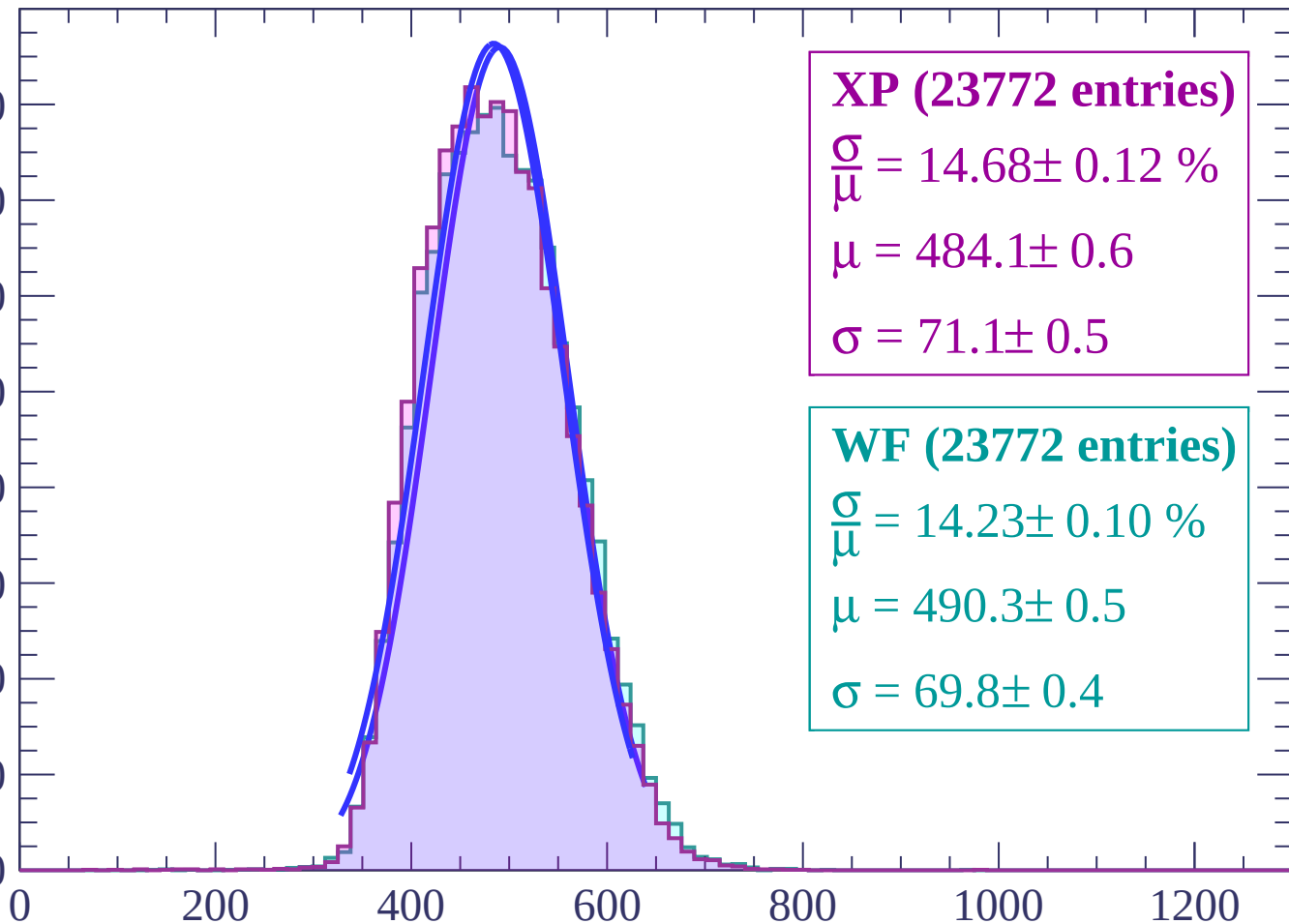
$\sigma = 71.6 \pm 0.4$

dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 8$

Count

1600
1400
1200
1000
800
600
400
200
0



XP (23772 entries)

$\frac{\sigma}{\mu} = 14.68 \pm 0.12 \%$

$\mu = 484.1 \pm 0.6$

$\sigma = 71.1 \pm 0.5$

WF (23772 entries)

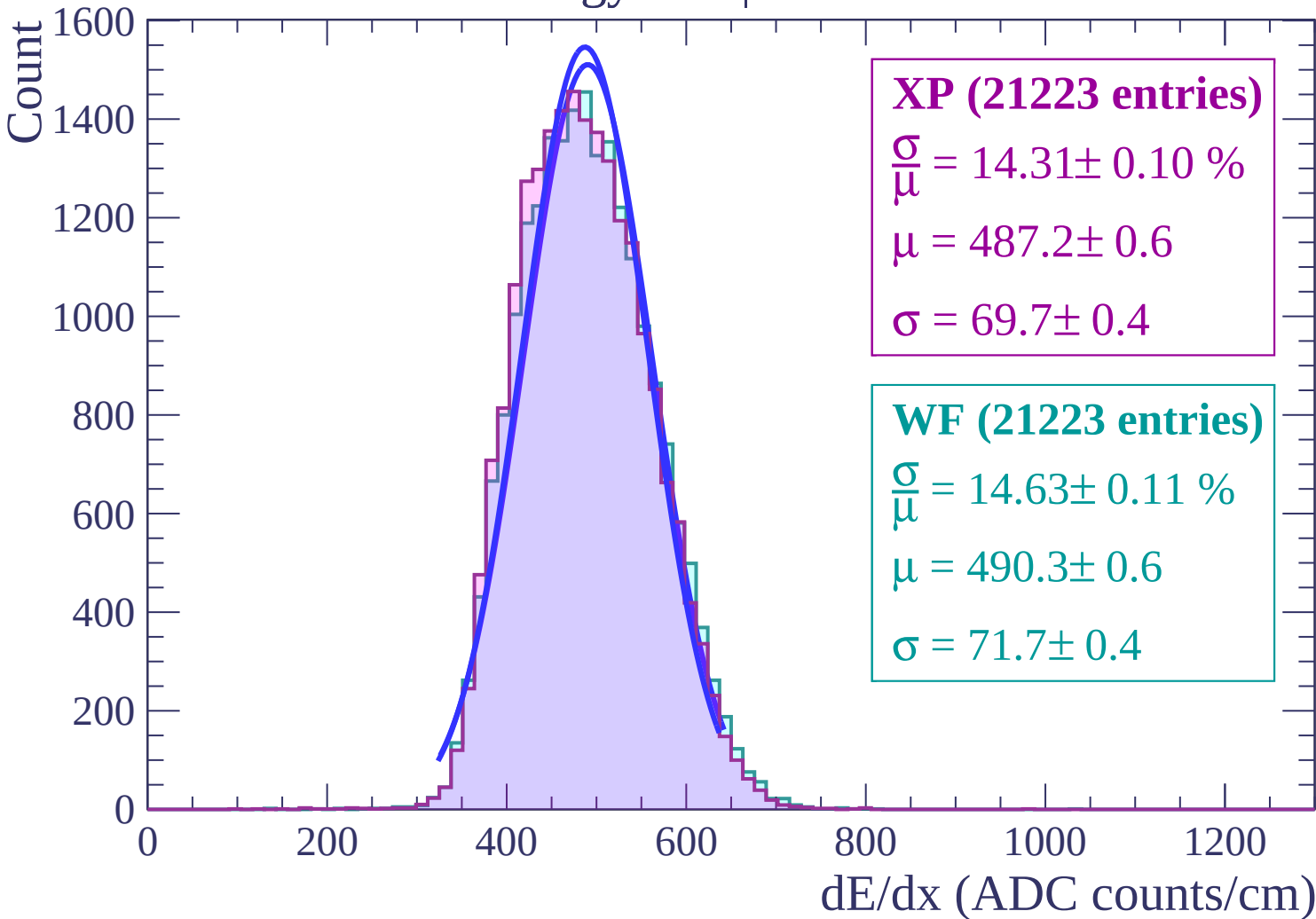
$\frac{\sigma}{\mu} = 14.23 \pm 0.10 \%$

$\mu = 490.3 \pm 0.5$

$\sigma = 69.8 \pm 0.4$

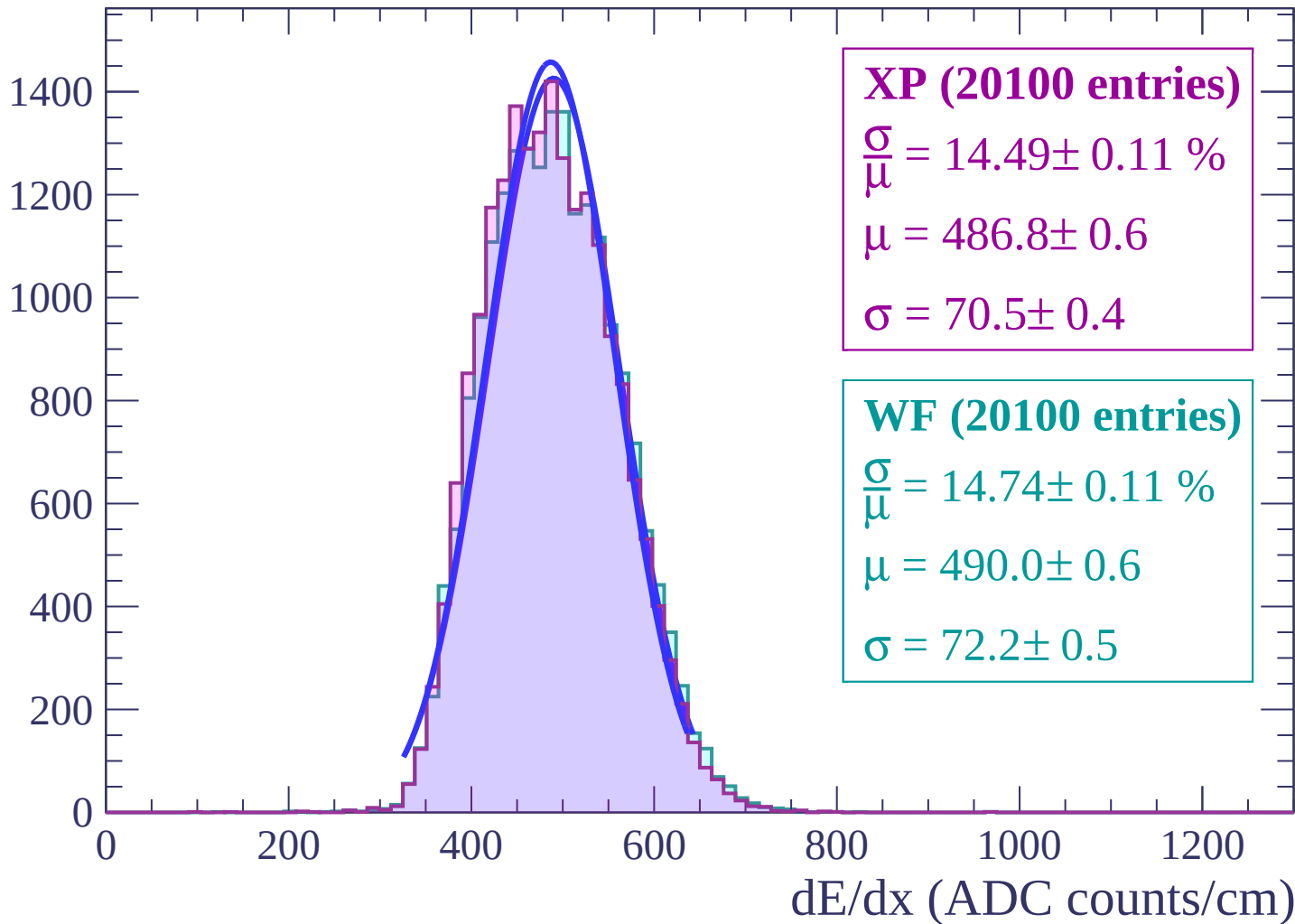
dE/dx (ADC counts/cm)

Energy loss | $8 < \theta < 10$



Energy loss | $10 < \theta < 12$

Count



Energy loss | $12 < \theta < 14$

Count

1200

1000

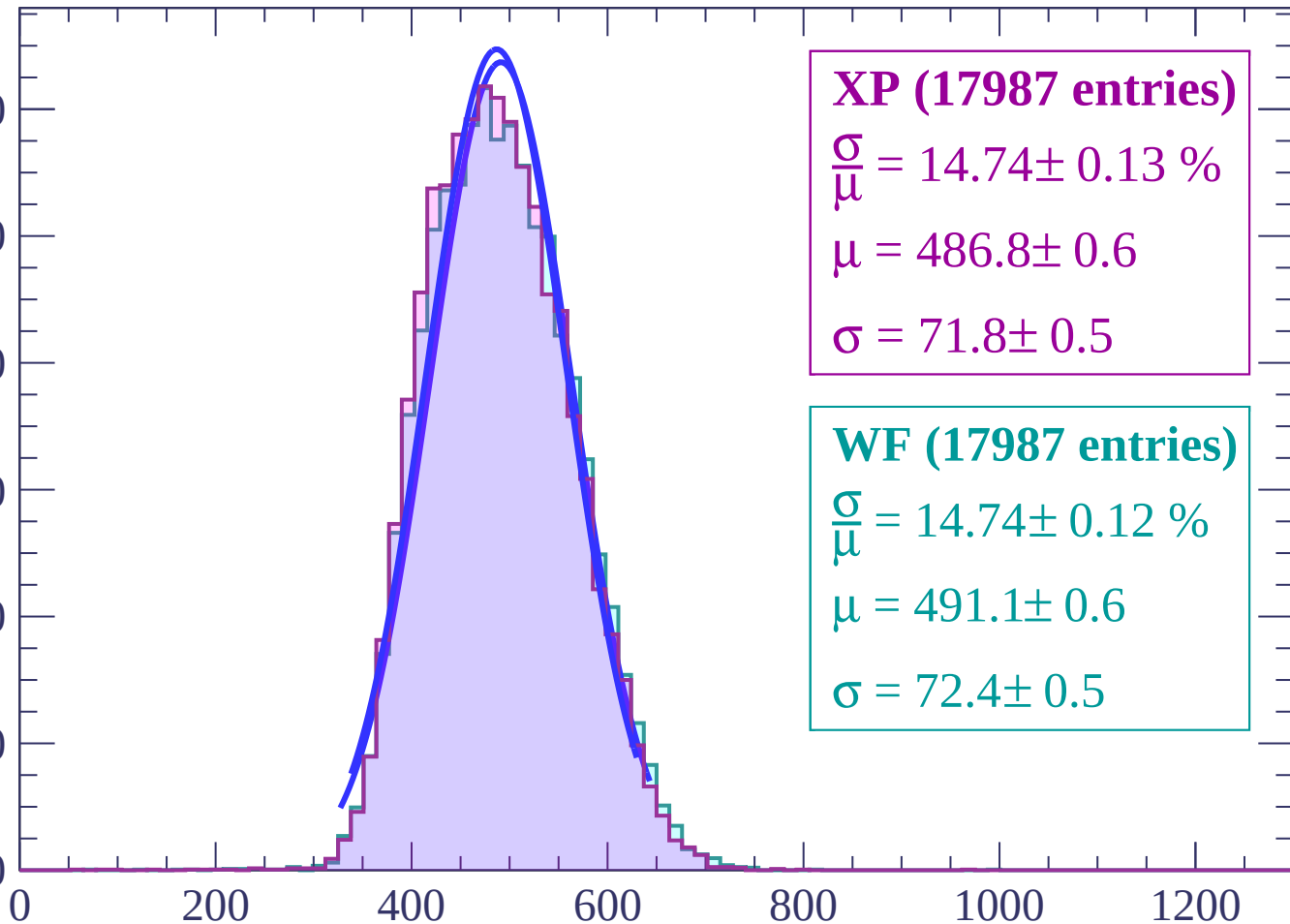
800

600

400

200

0



XP (17987 entries)

$\frac{\sigma}{\mu} = 14.74 \pm 0.13 \%$

$\mu = 486.8 \pm 0.6$

$\sigma = 71.8 \pm 0.5$

WF (17987 entries)

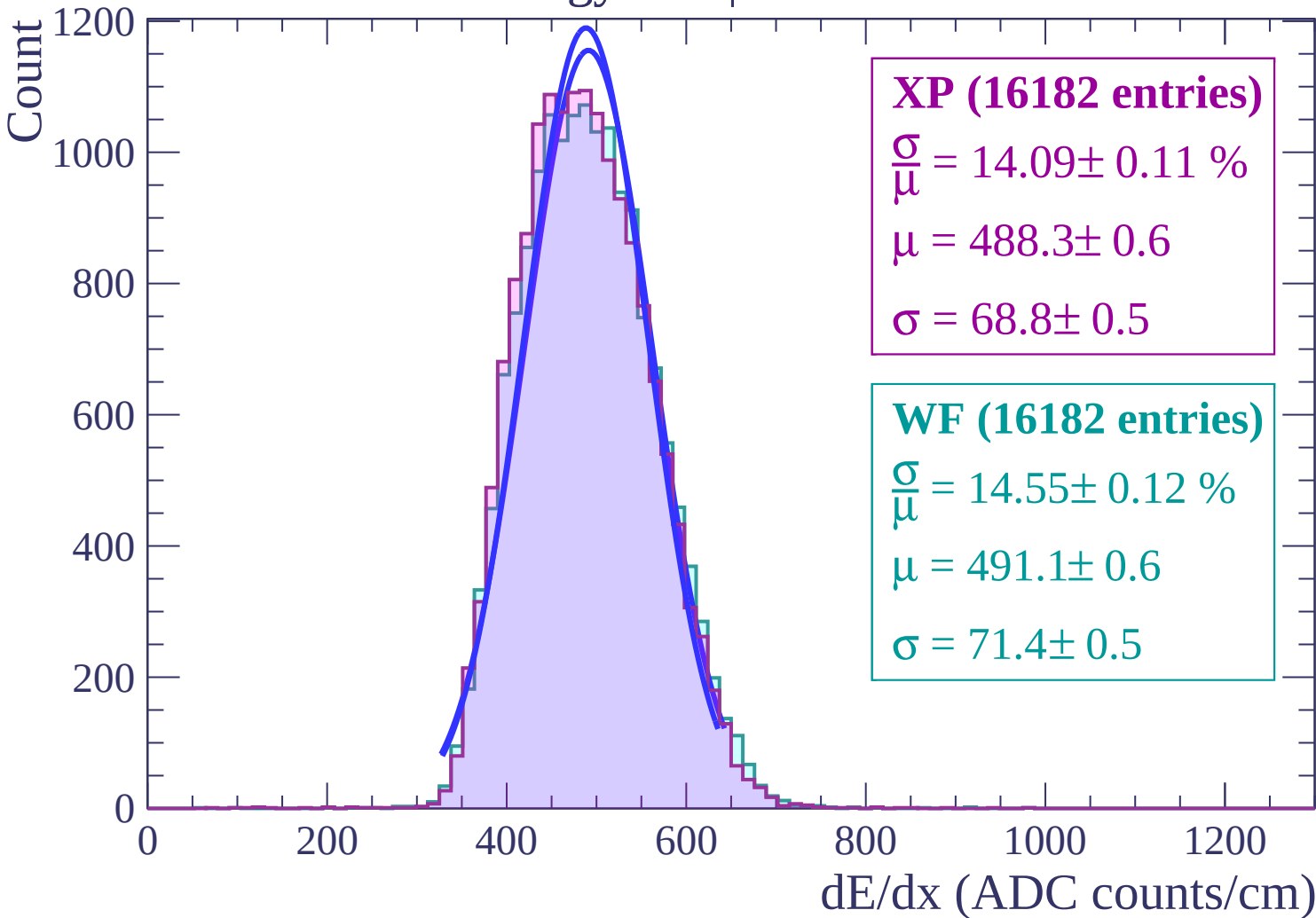
$\frac{\sigma}{\mu} = 14.74 \pm 0.12 \%$

$\mu = 491.1 \pm 0.6$

$\sigma = 72.4 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

Count

1000

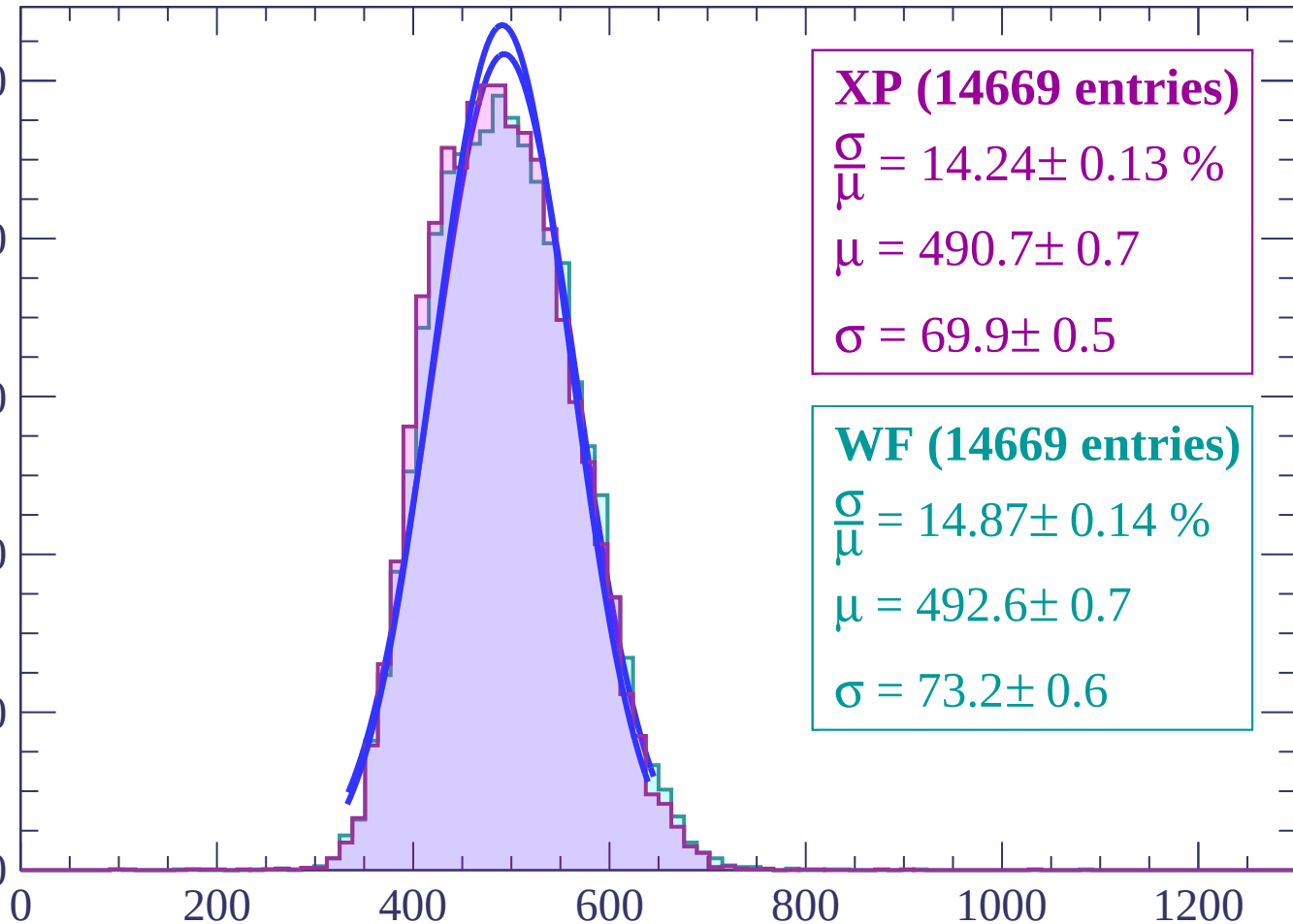
800

600

400

200

0



XP (14669 entries)

$\frac{\sigma}{\mu} = 14.24 \pm 0.13 \%$

$\mu = 490.7 \pm 0.7$

$\sigma = 69.9 \pm 0.5$

WF (14669 entries)

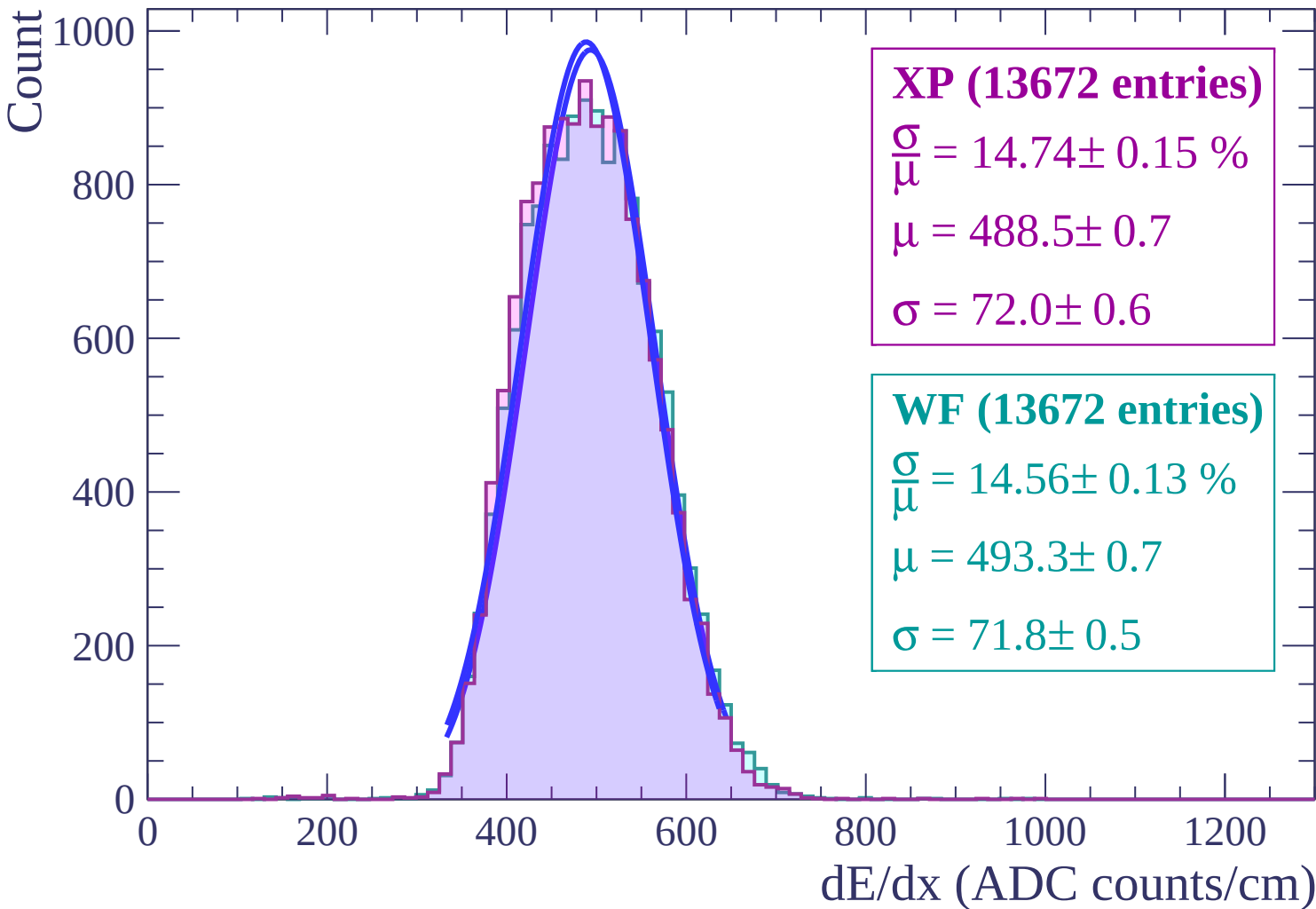
$\frac{\sigma}{\mu} = 14.87 \pm 0.14 \%$

$\mu = 492.6 \pm 0.7$

$\sigma = 73.2 \pm 0.6$

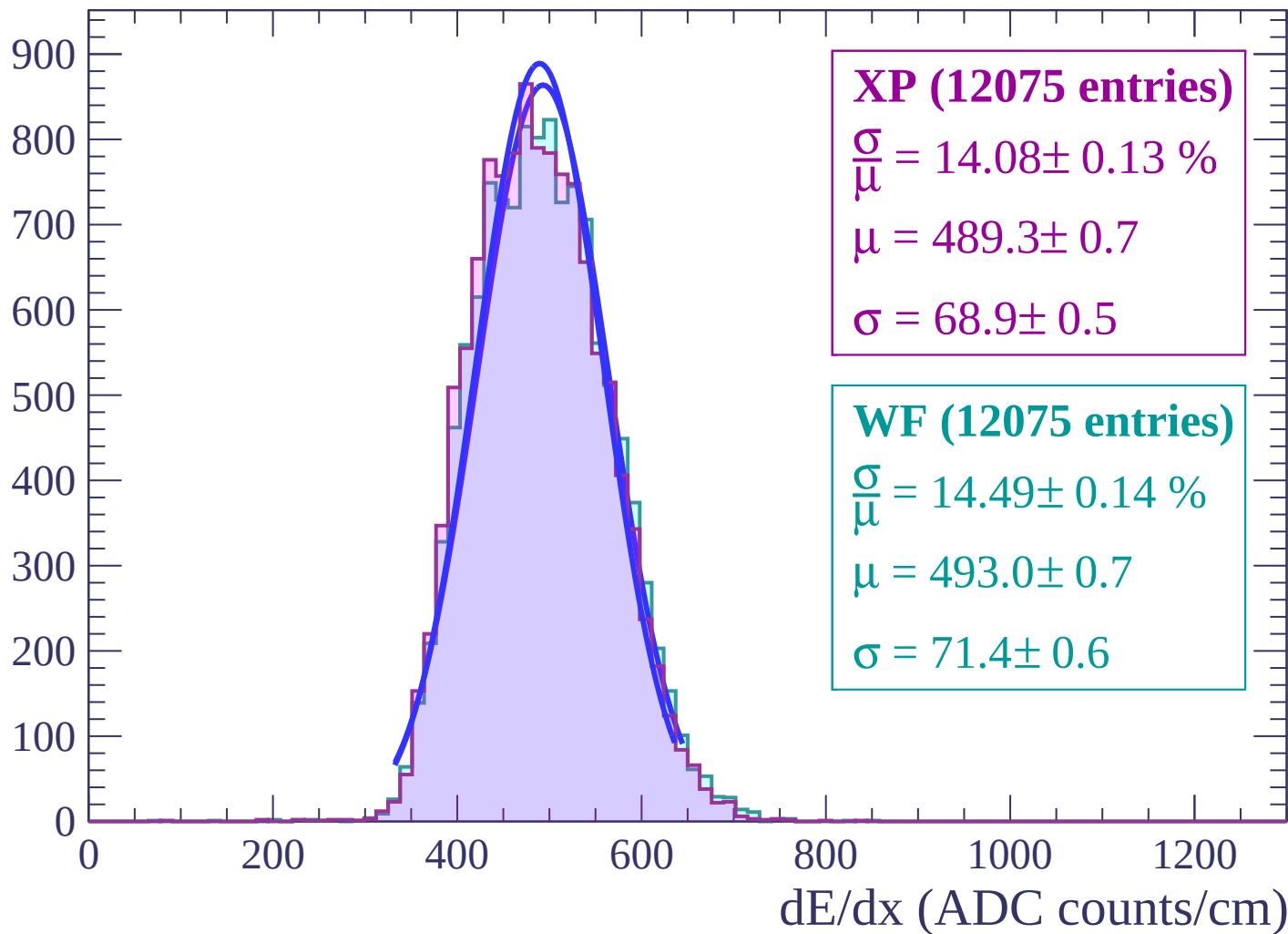
dE/dx (ADC counts/cm)

Energy loss | $18 < \theta < 20$



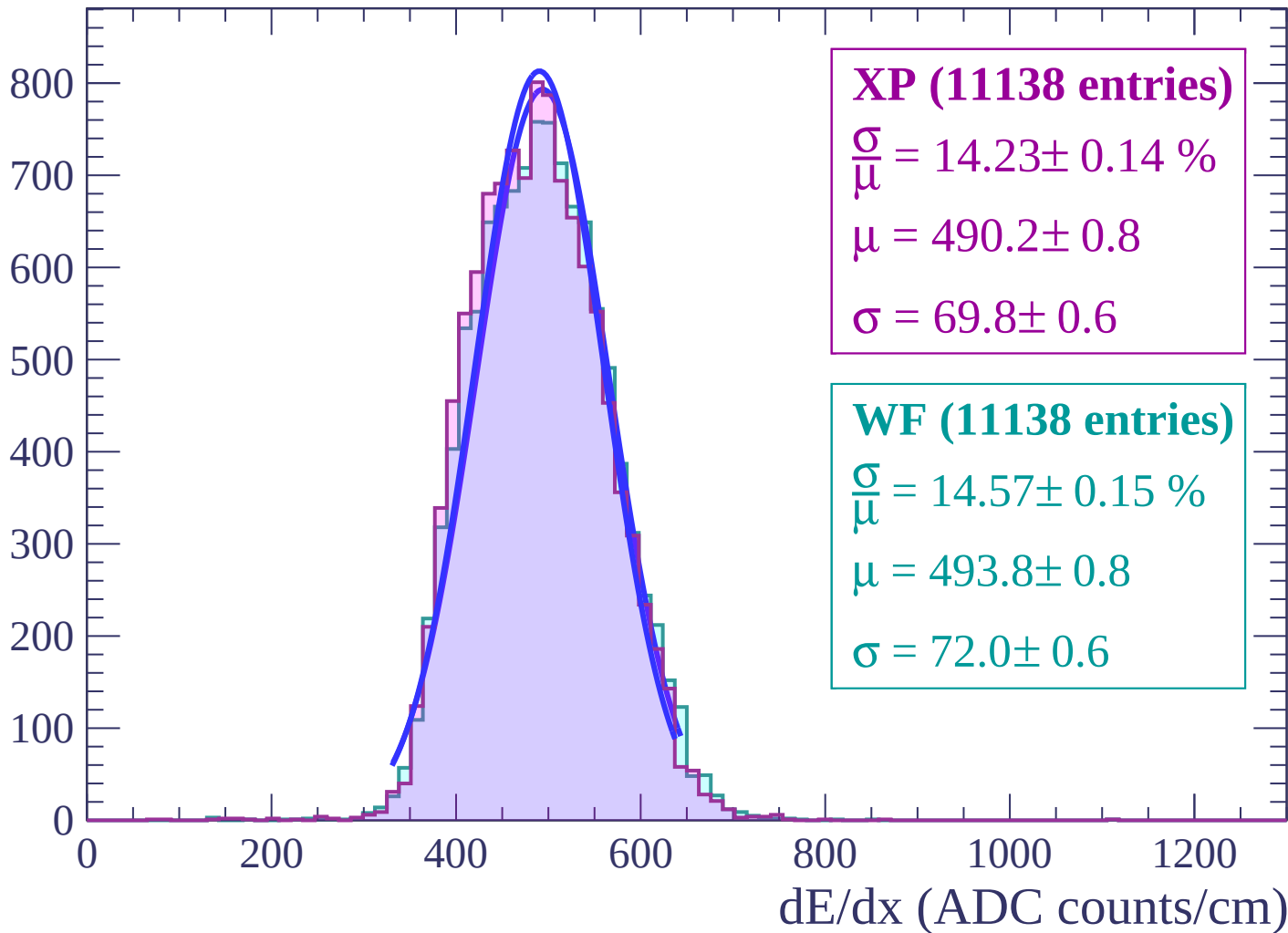
Energy loss | $20 < \theta < 22$

Count



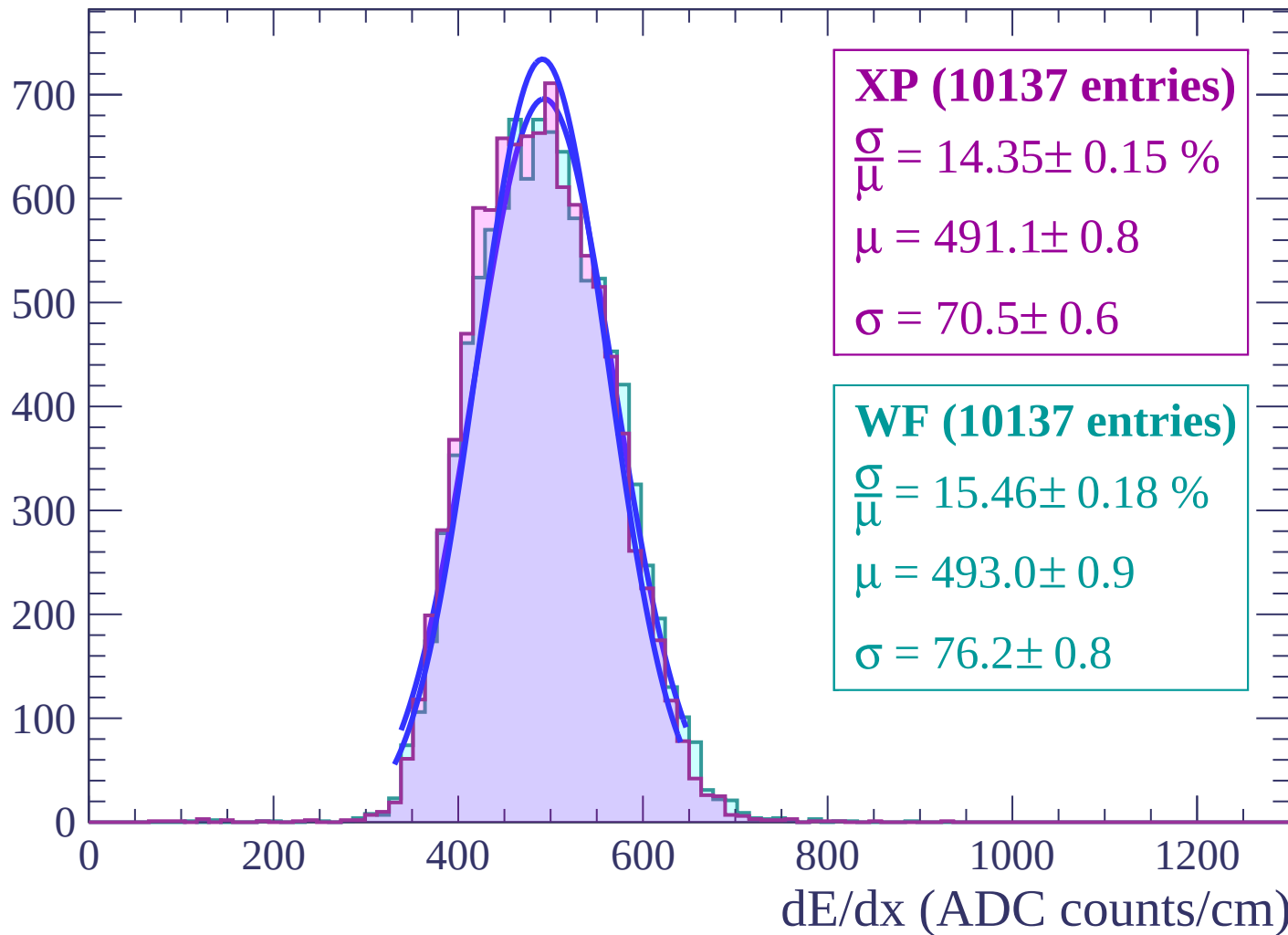
Energy loss | $22 < \theta < 24$

Count



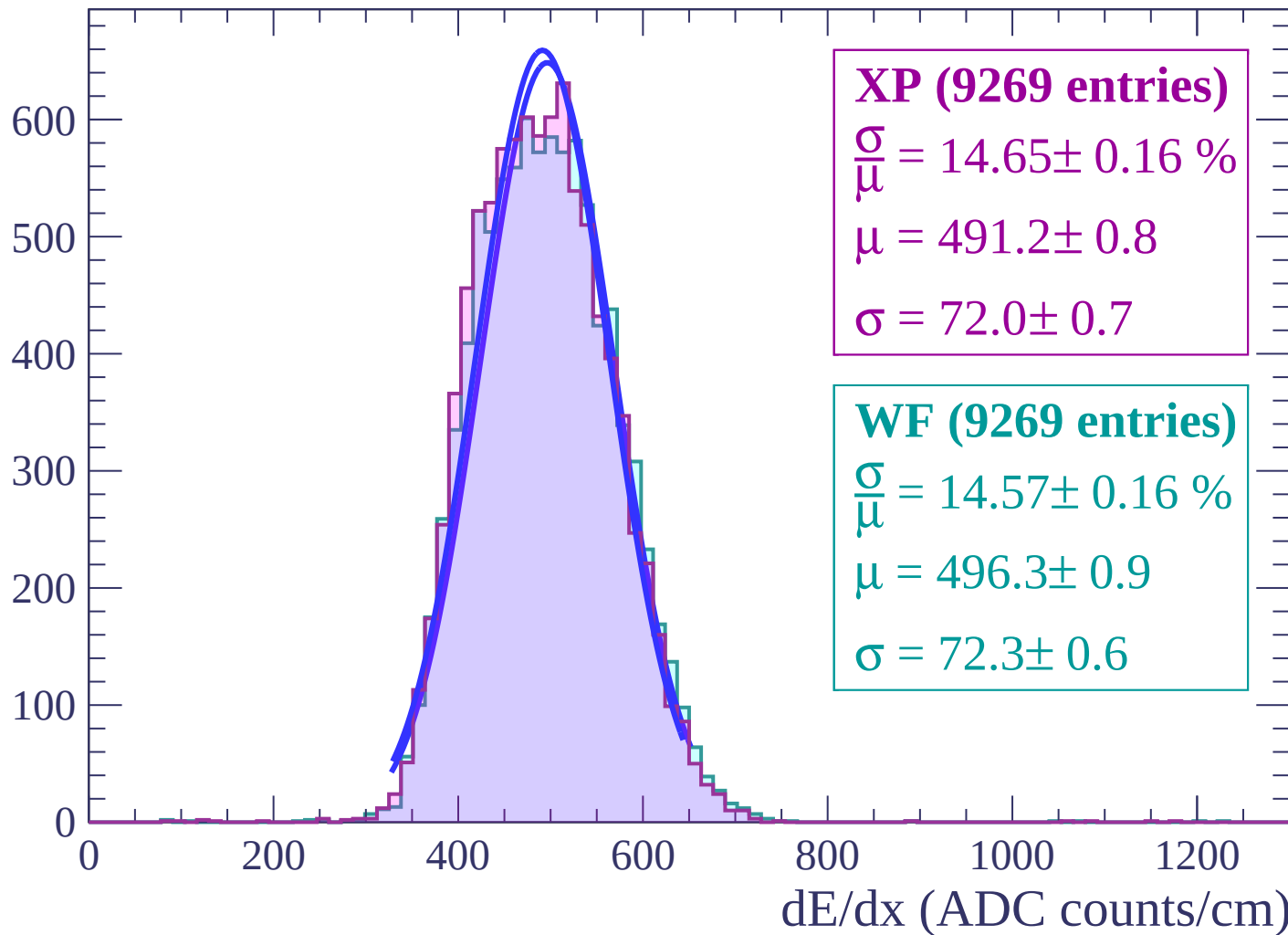
Energy loss | $24 < \theta < 26$

Count



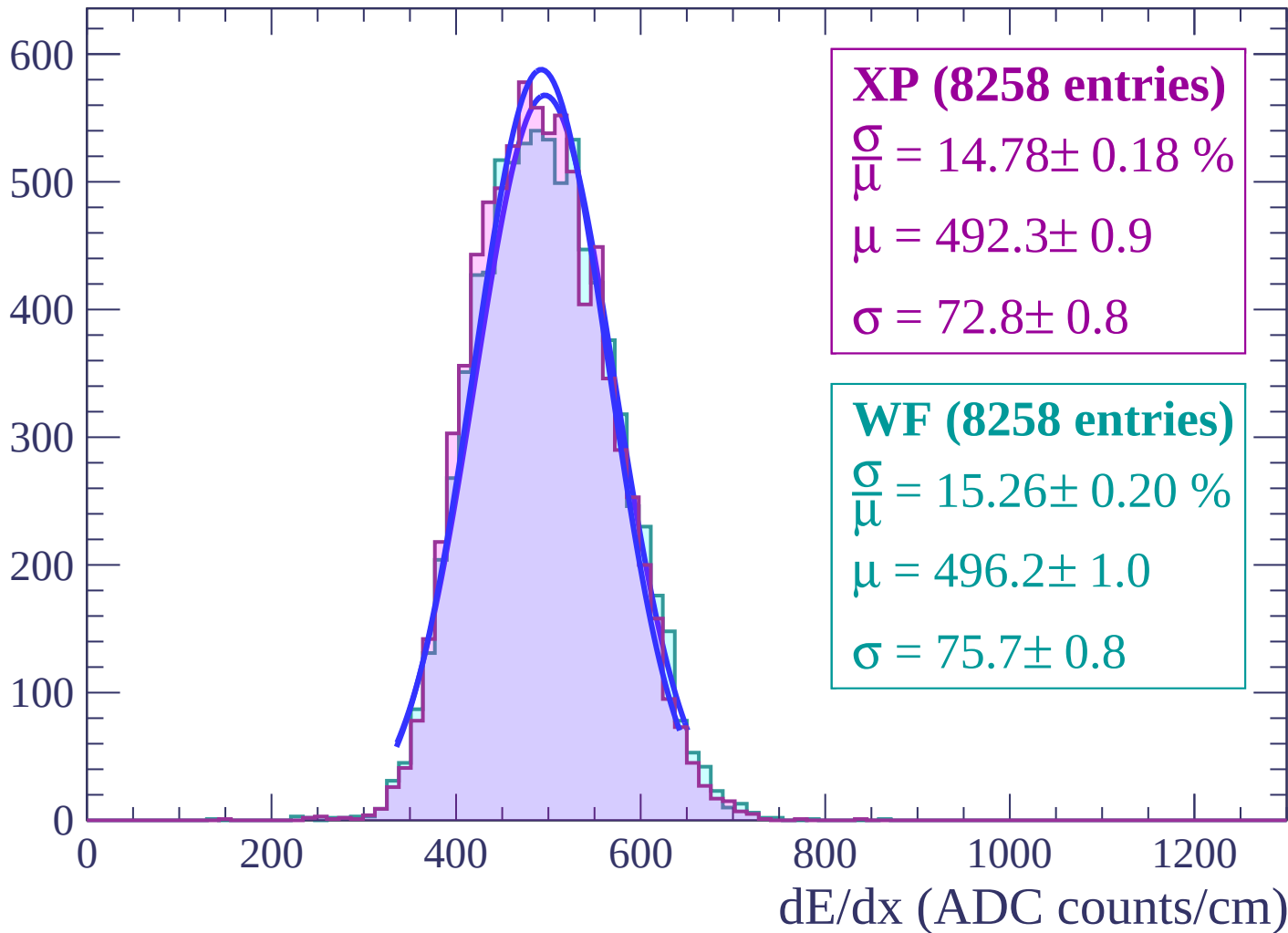
Energy loss | $26 < \theta < 28$

Count



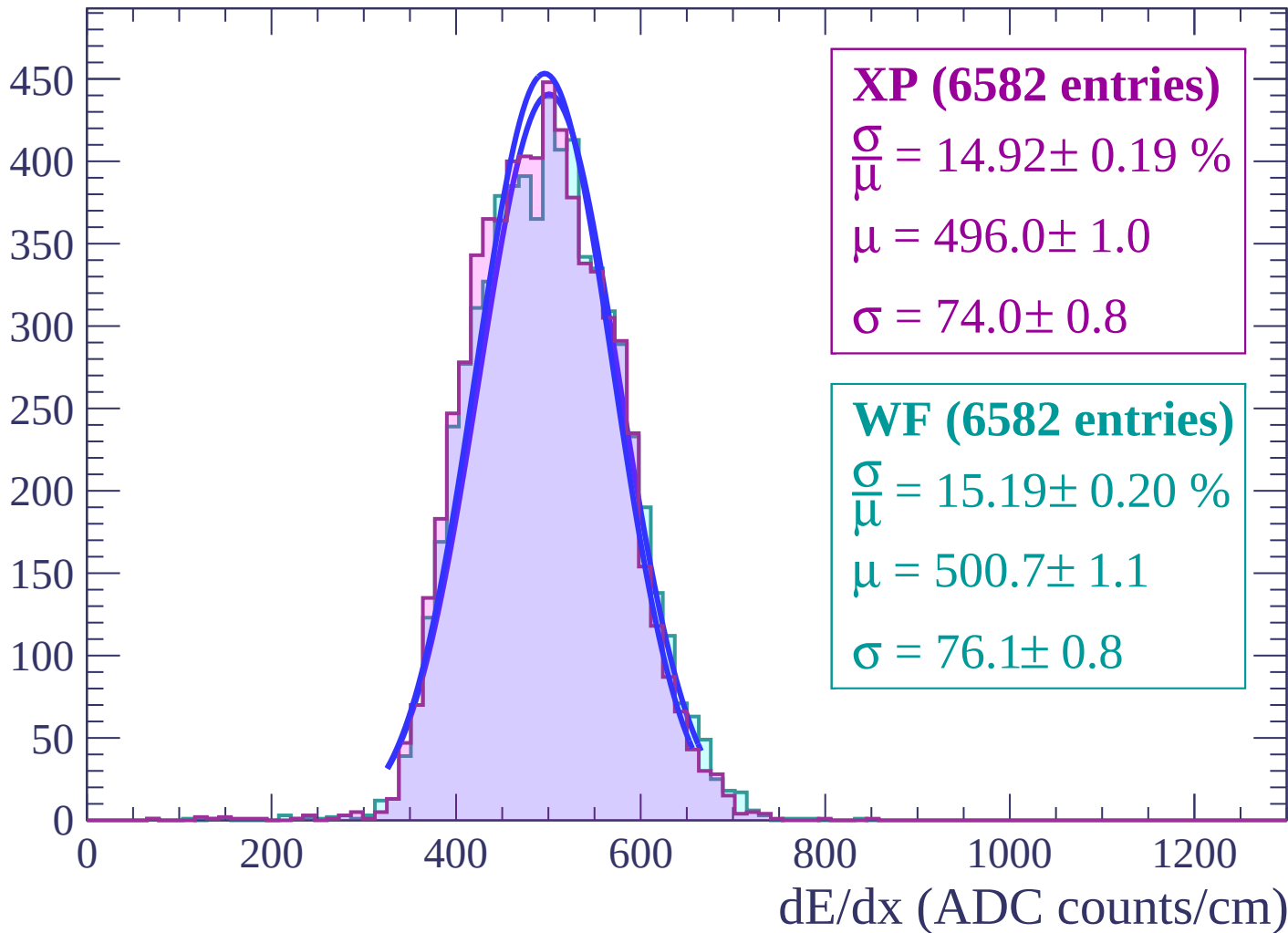
Energy loss | $28 < \theta < 30$

Count



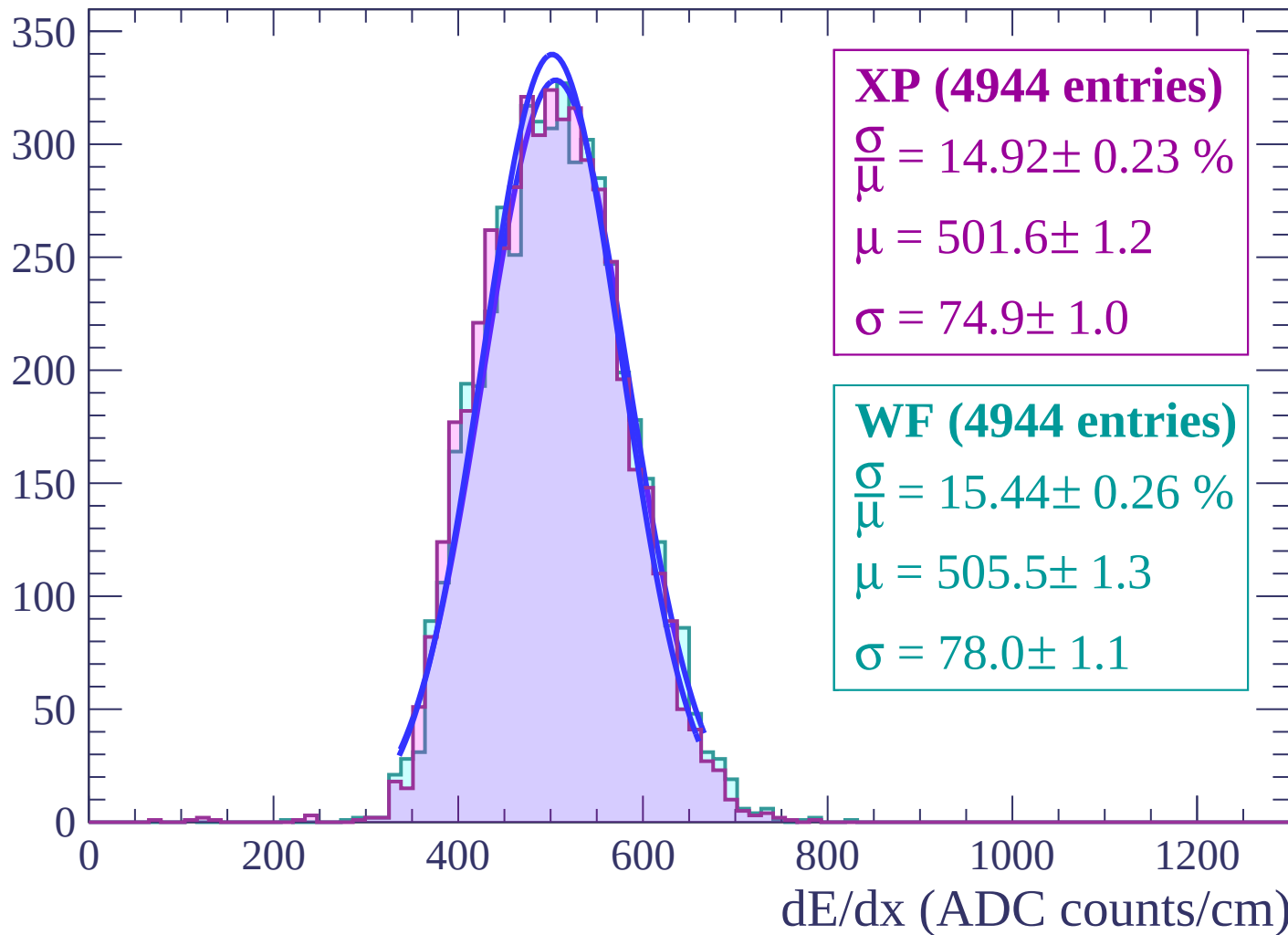
Energy loss | $30 < \theta < 32$

Count



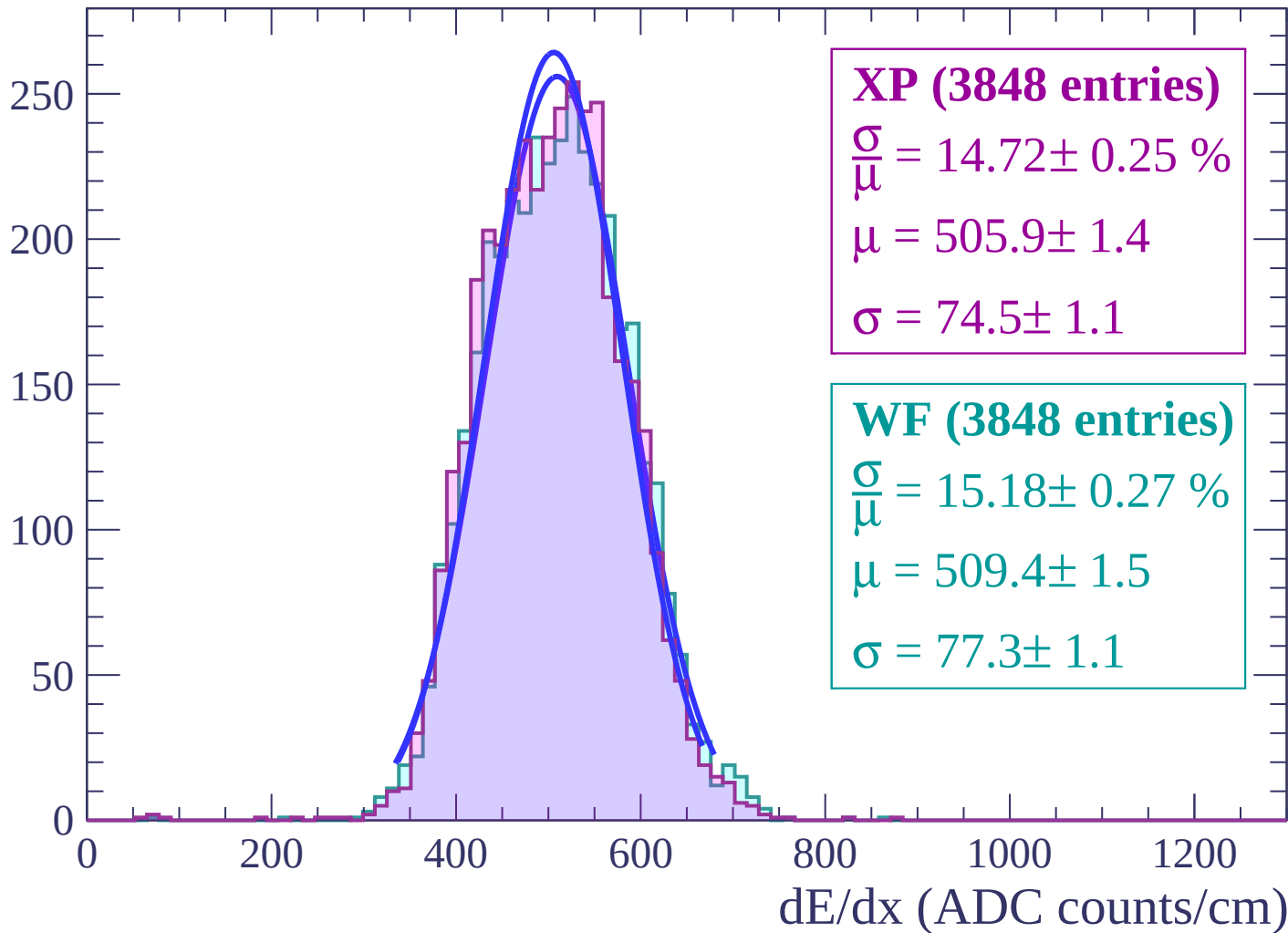
Energy loss | $32 < \theta < 34$

Count



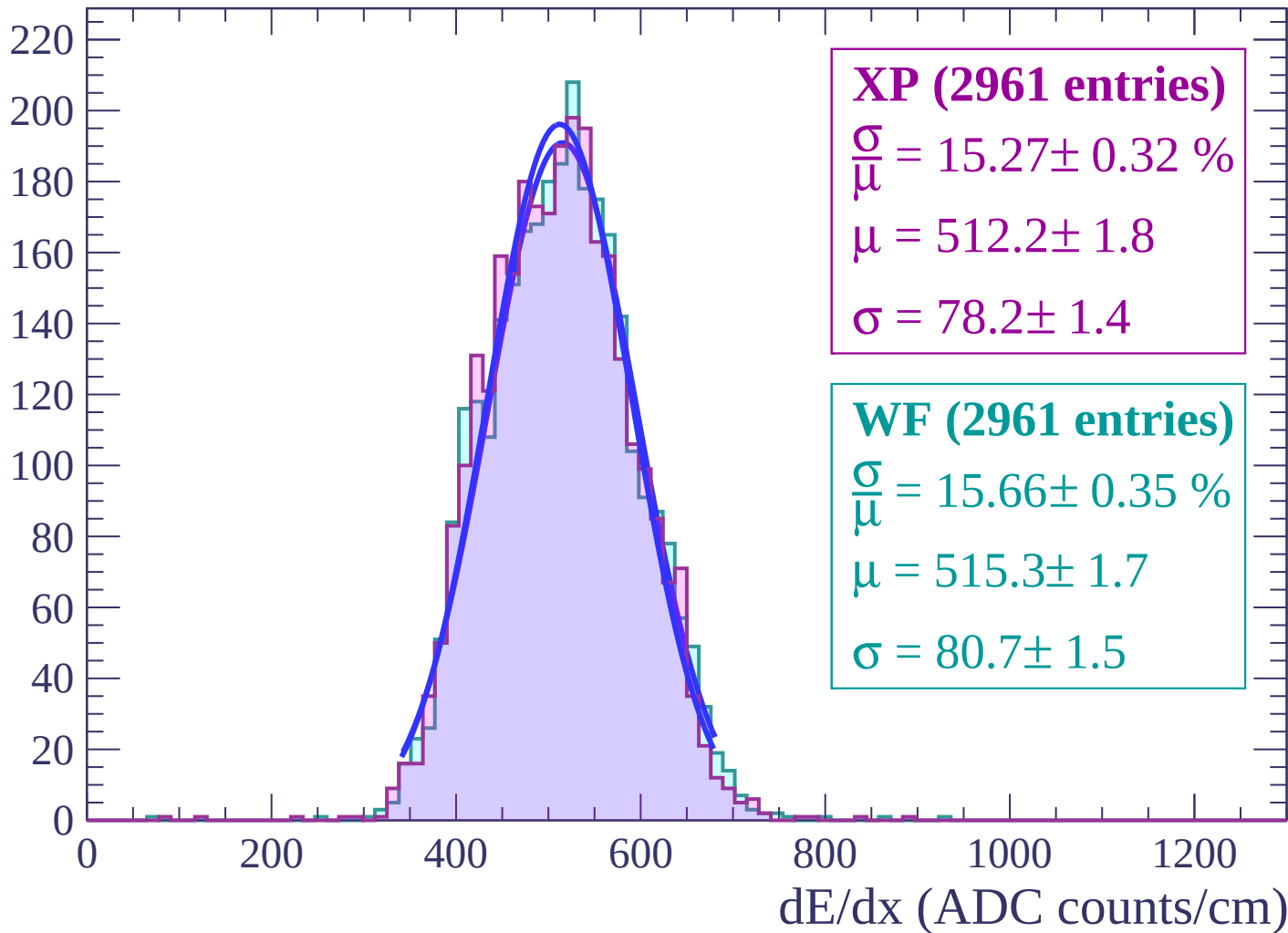
Energy loss | $34 < \theta < 36$

Count



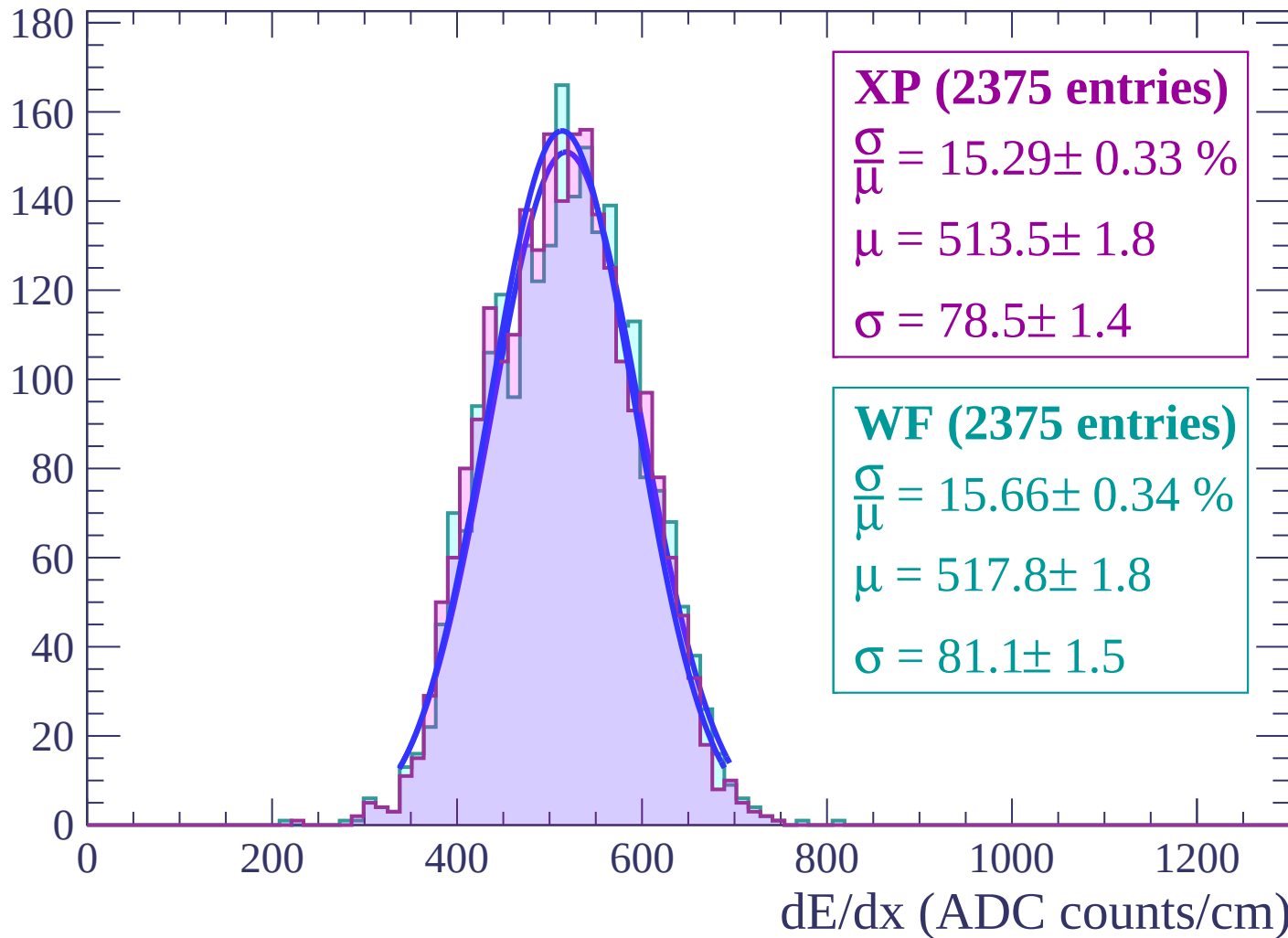
Energy loss | $36 < \theta < 38$

Count



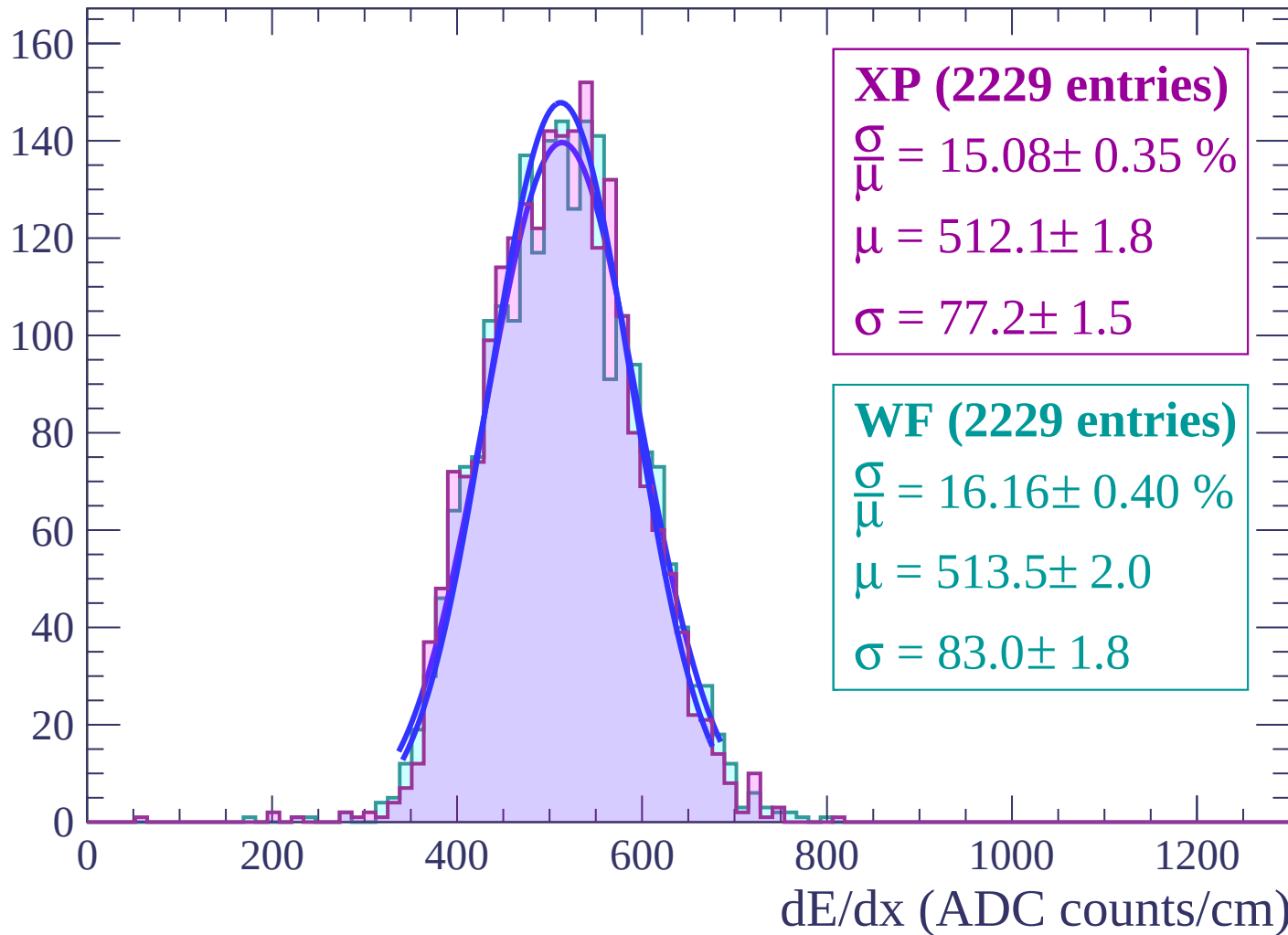
Energy loss | $38 < \theta < 40$

Count



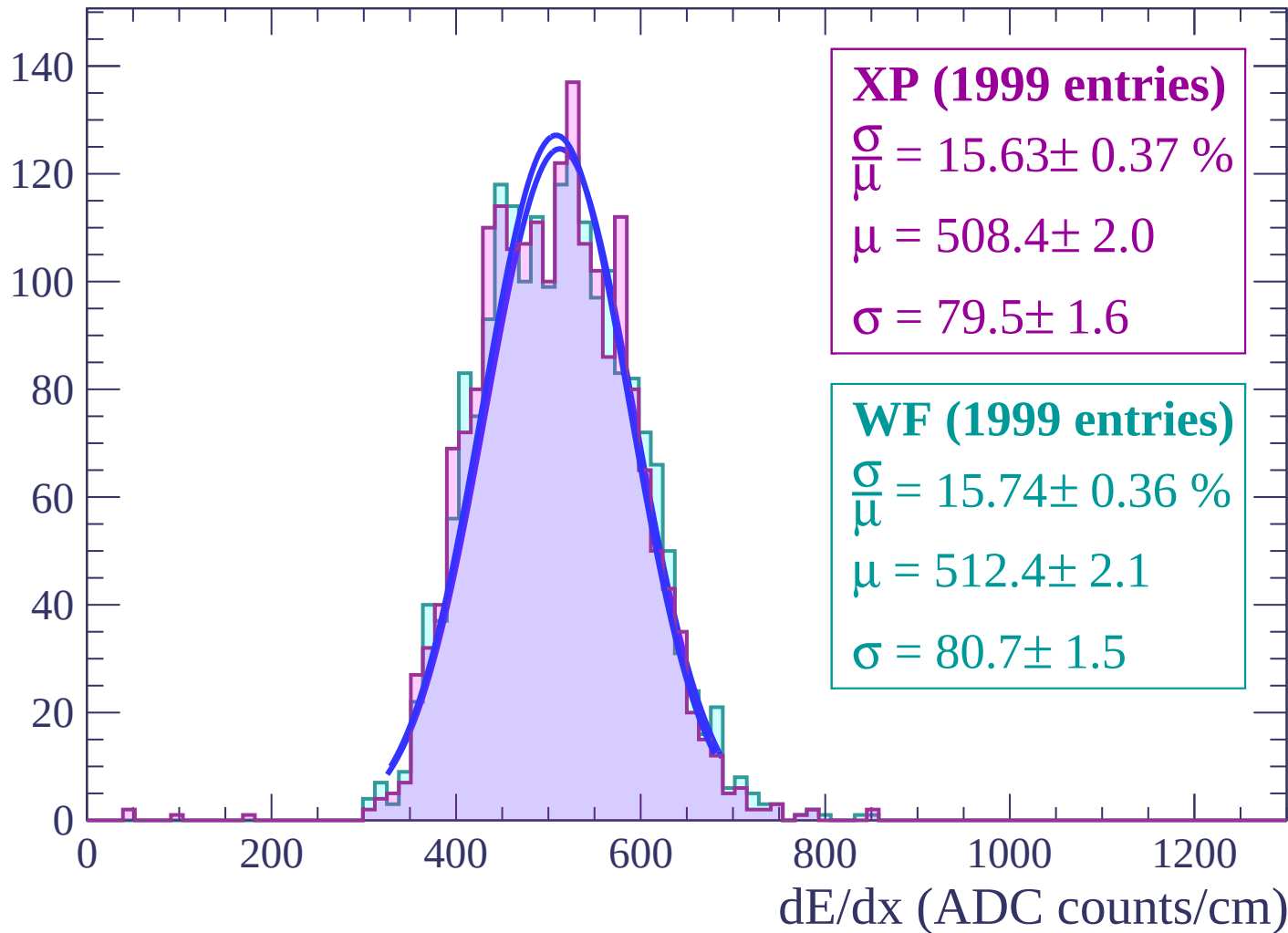
Energy loss | $40 < \theta < 42$

Count



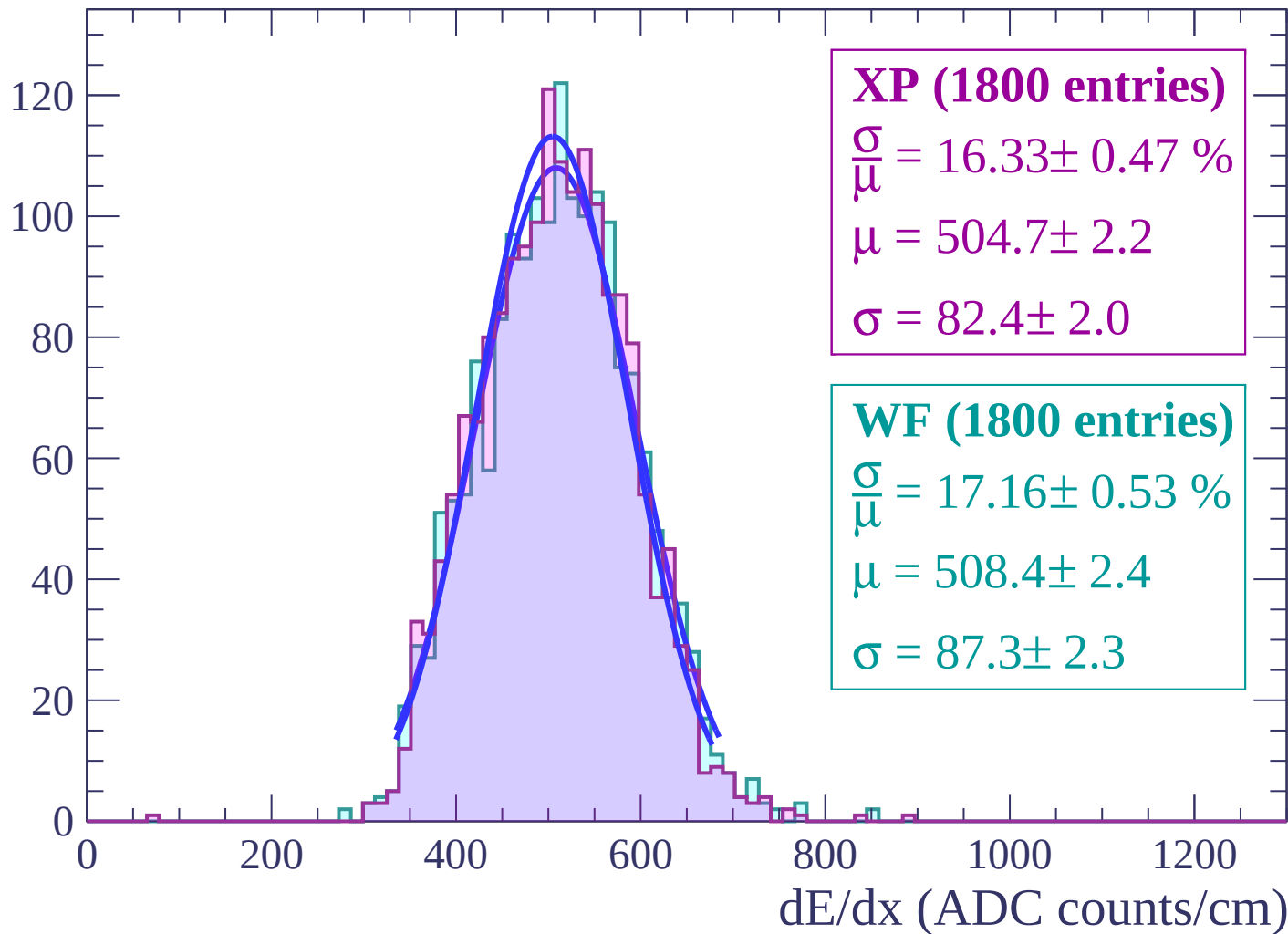
Energy loss | $42 < \theta < 44$

Count



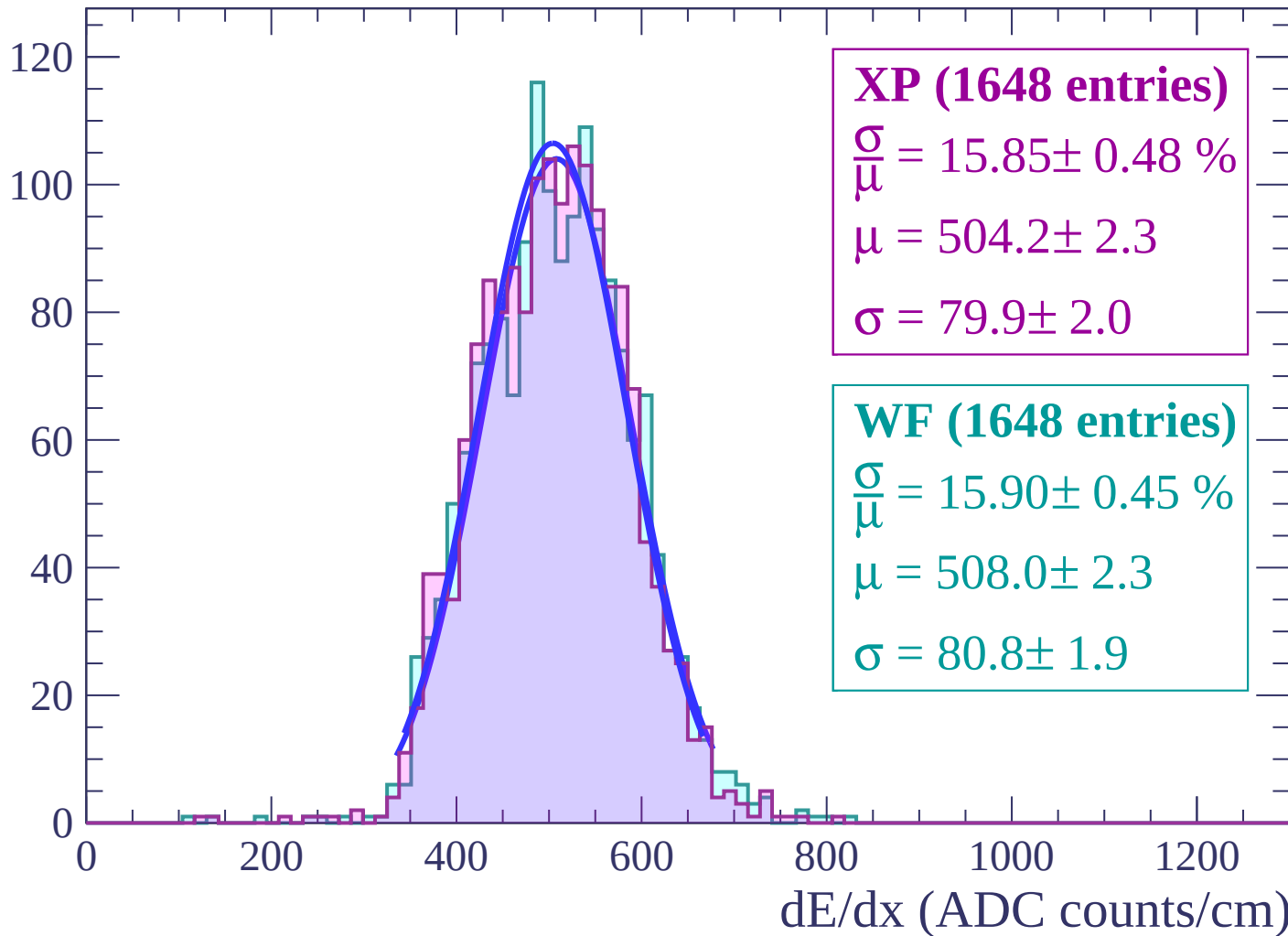
Energy loss | $44 < \theta < 46$

Count



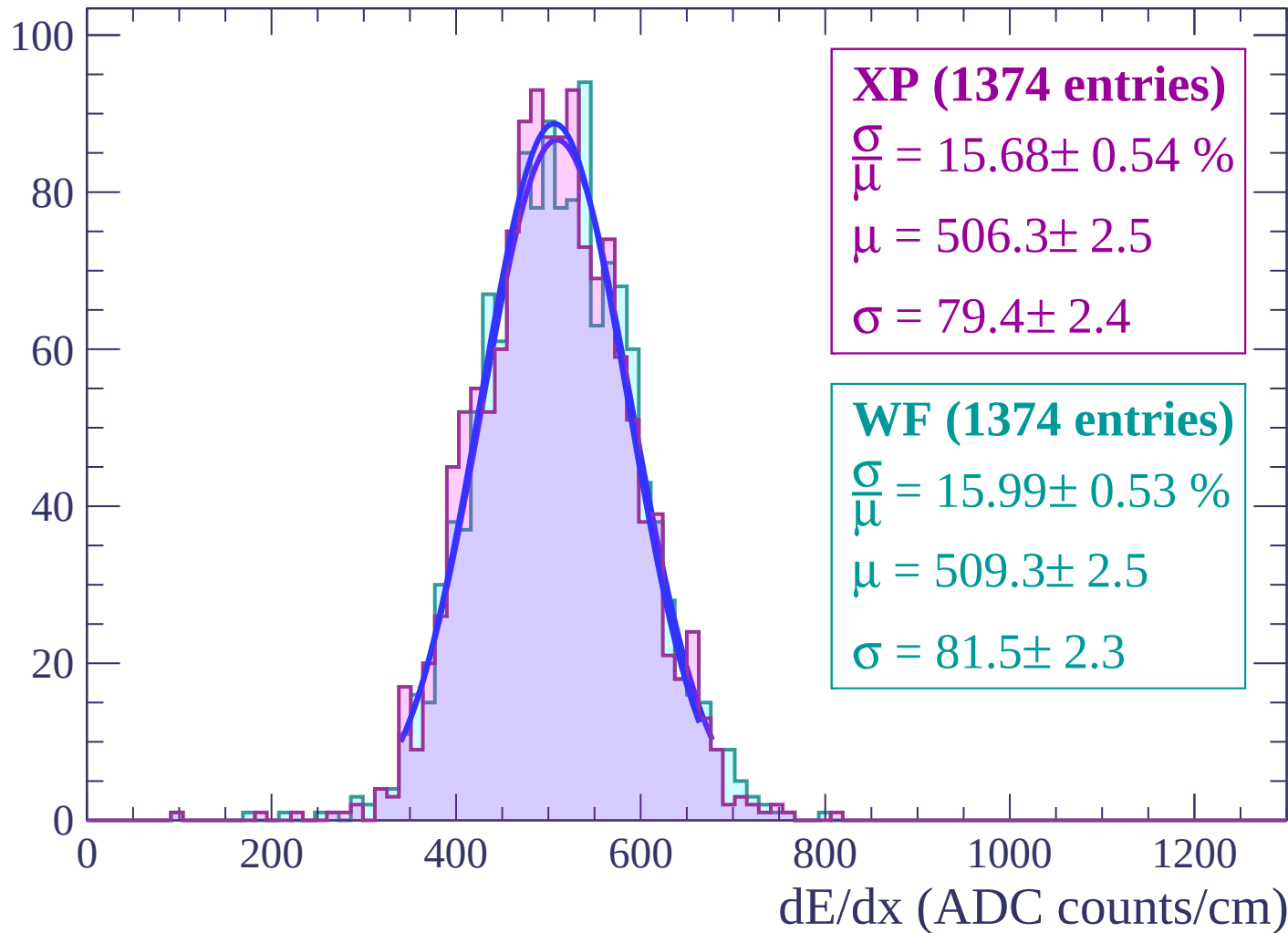
Energy loss | $46 < \theta < 48$

Count



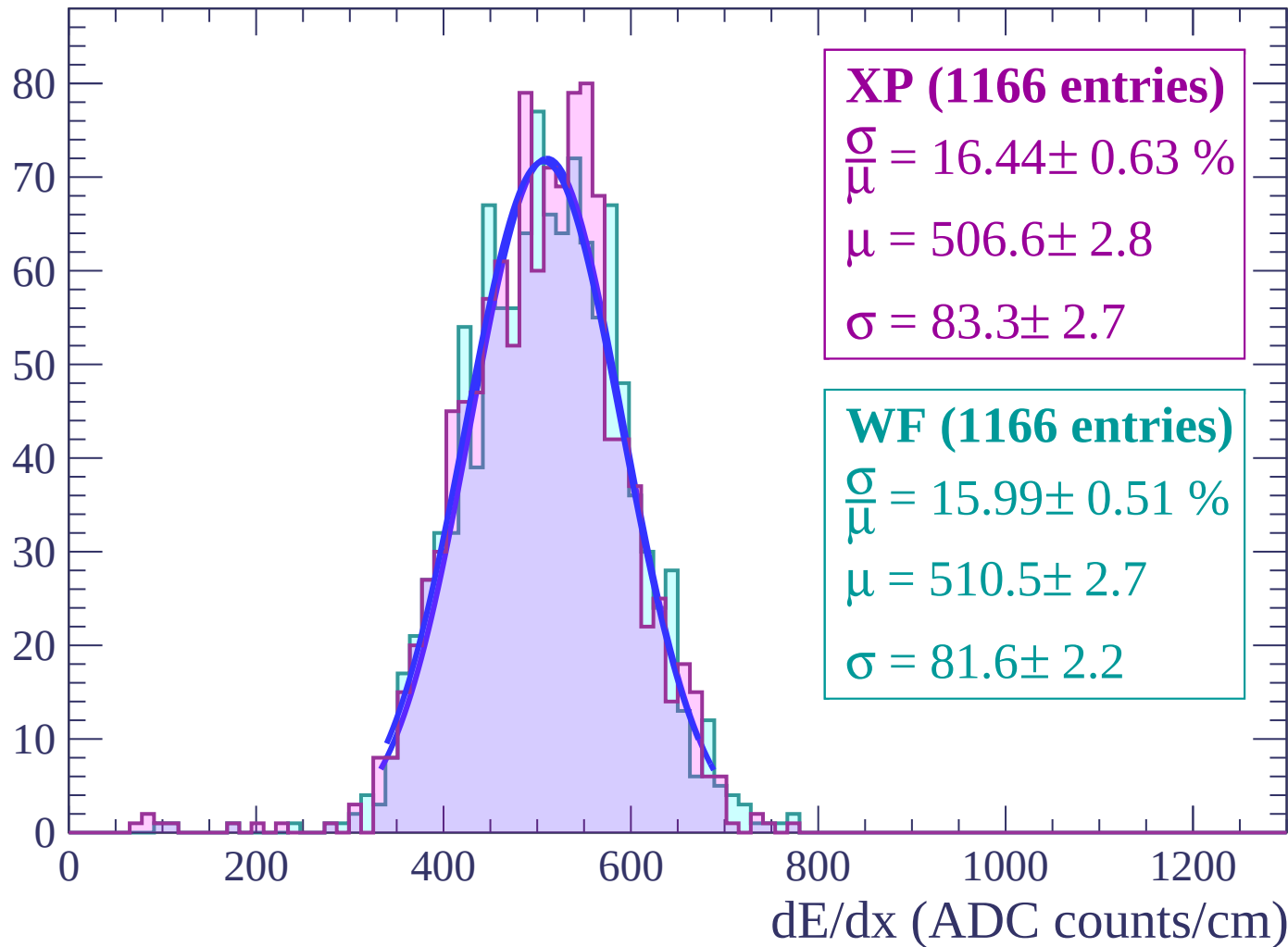
Energy loss | $48 < \theta < 50$

Count



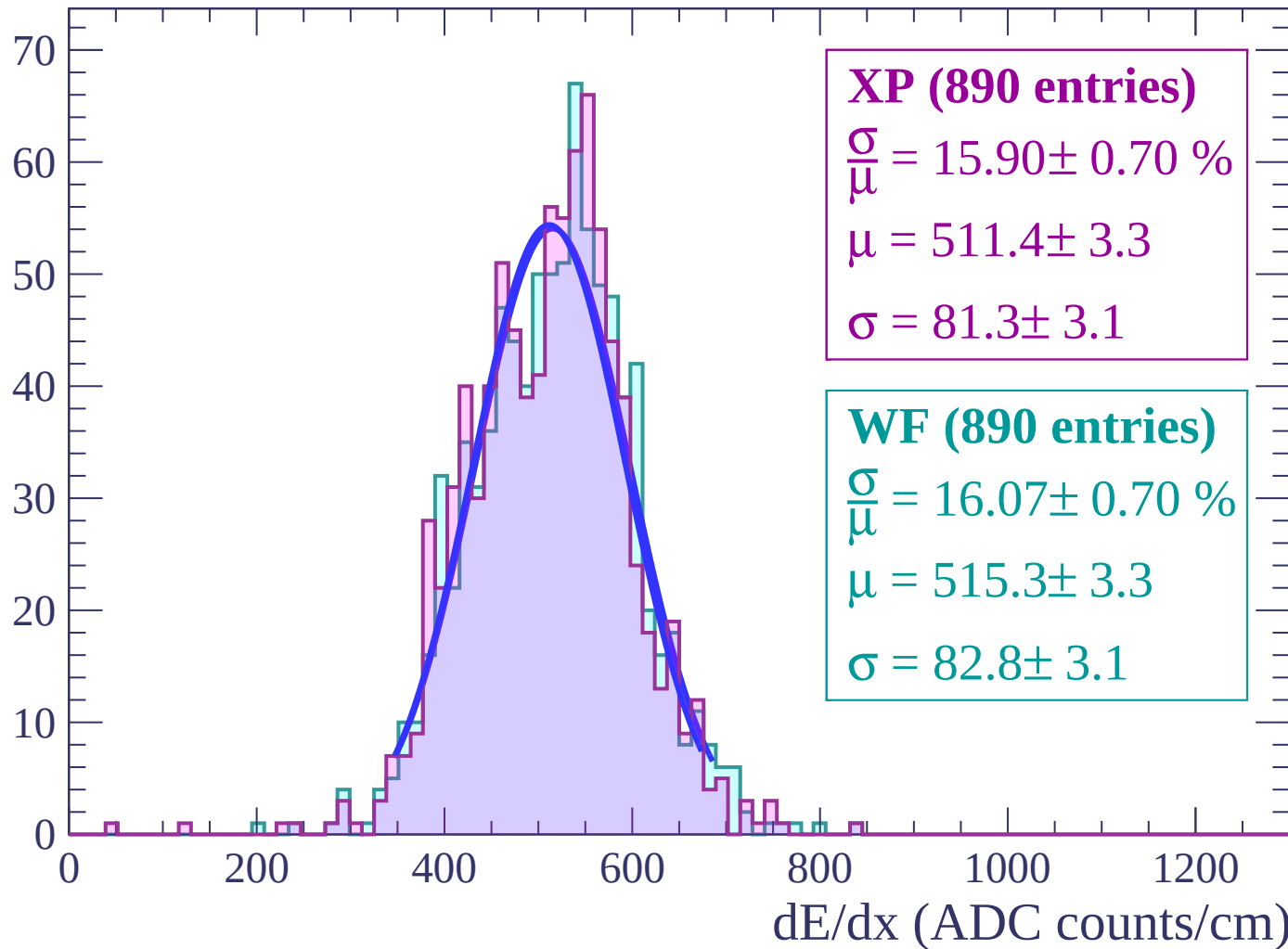
Energy loss | $50 < \theta < 52$

Count



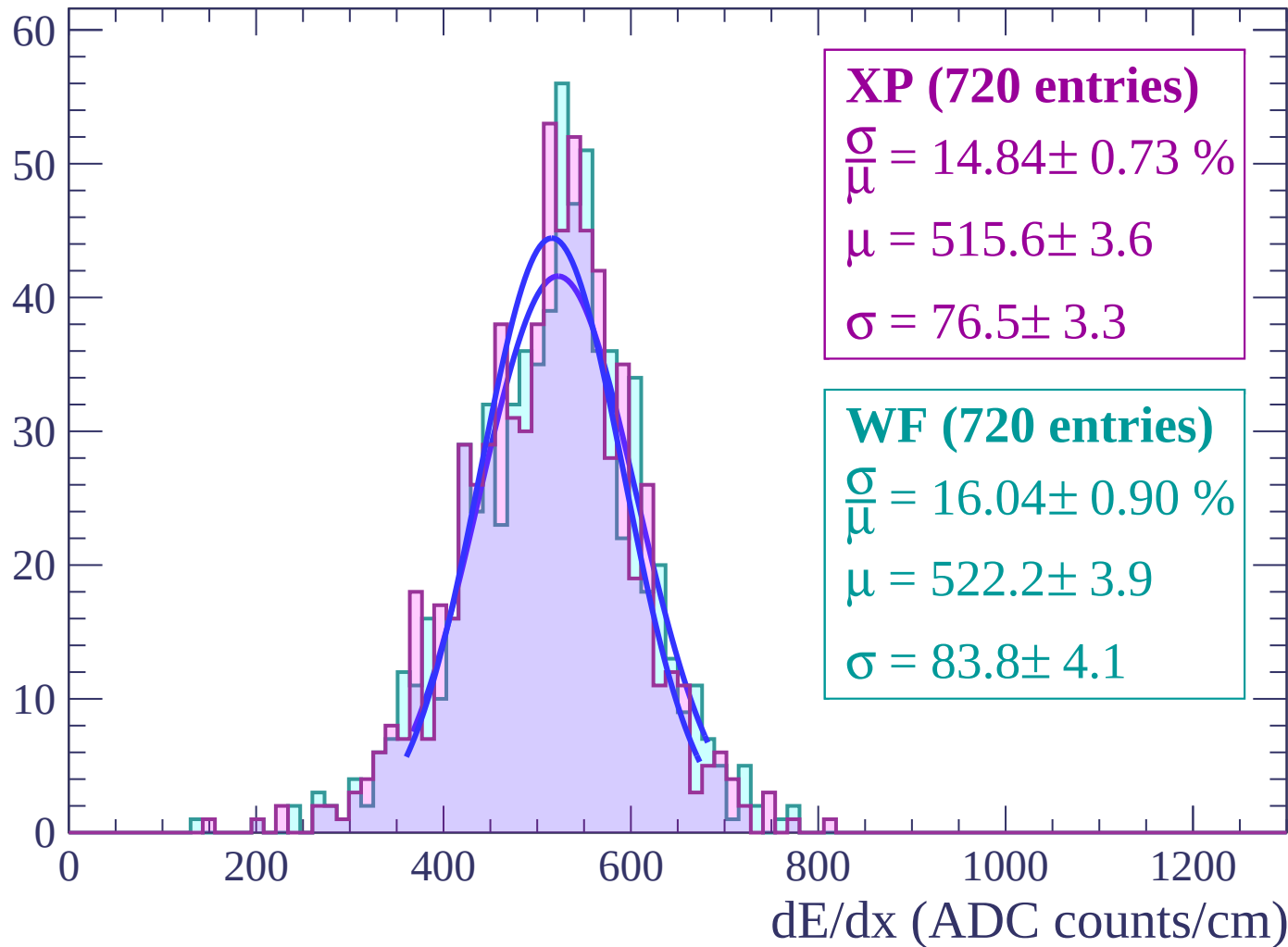
Energy loss | $52 < \theta < 54$

Count



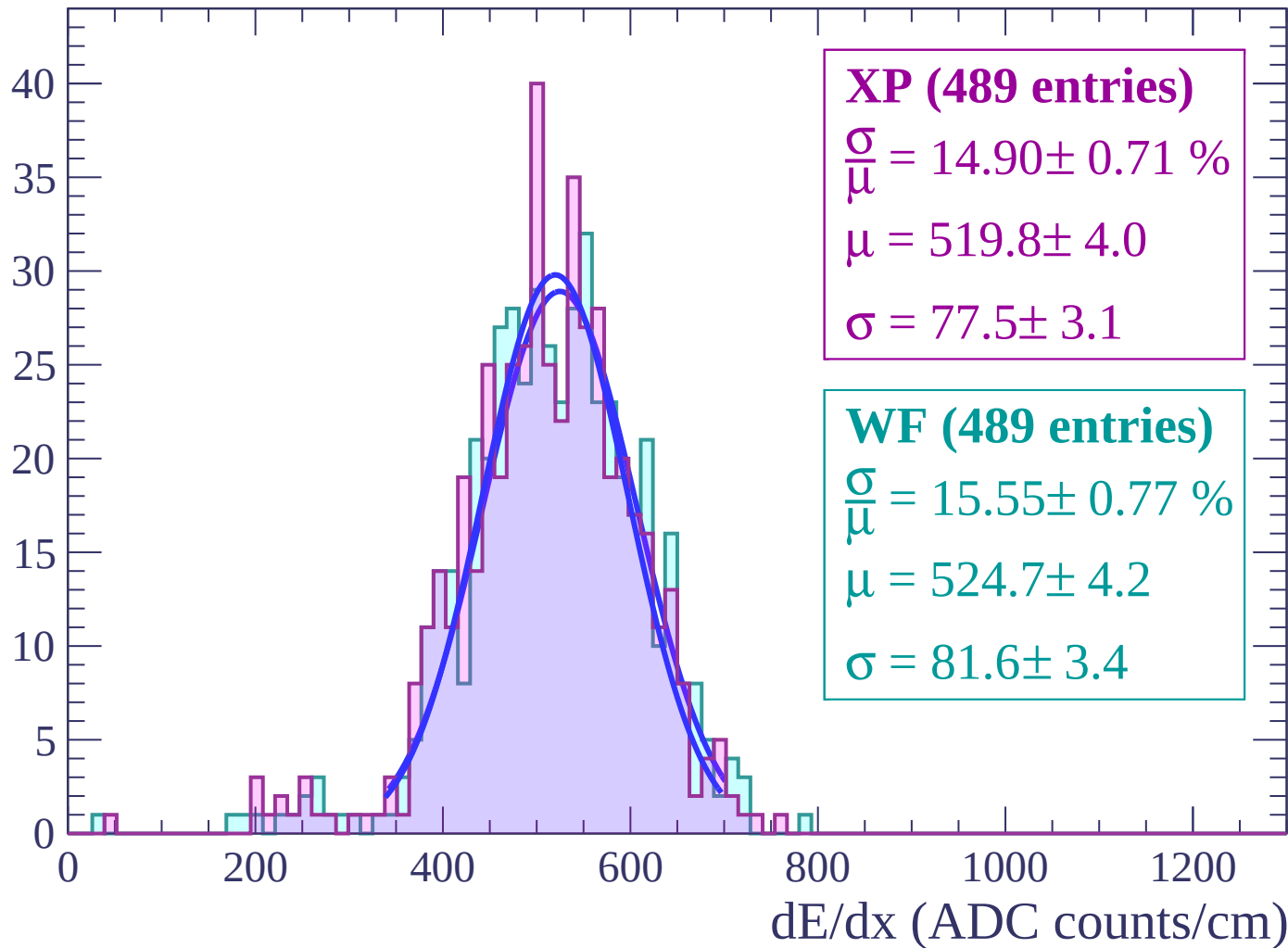
Energy loss | $54 < \theta < 56$

Count



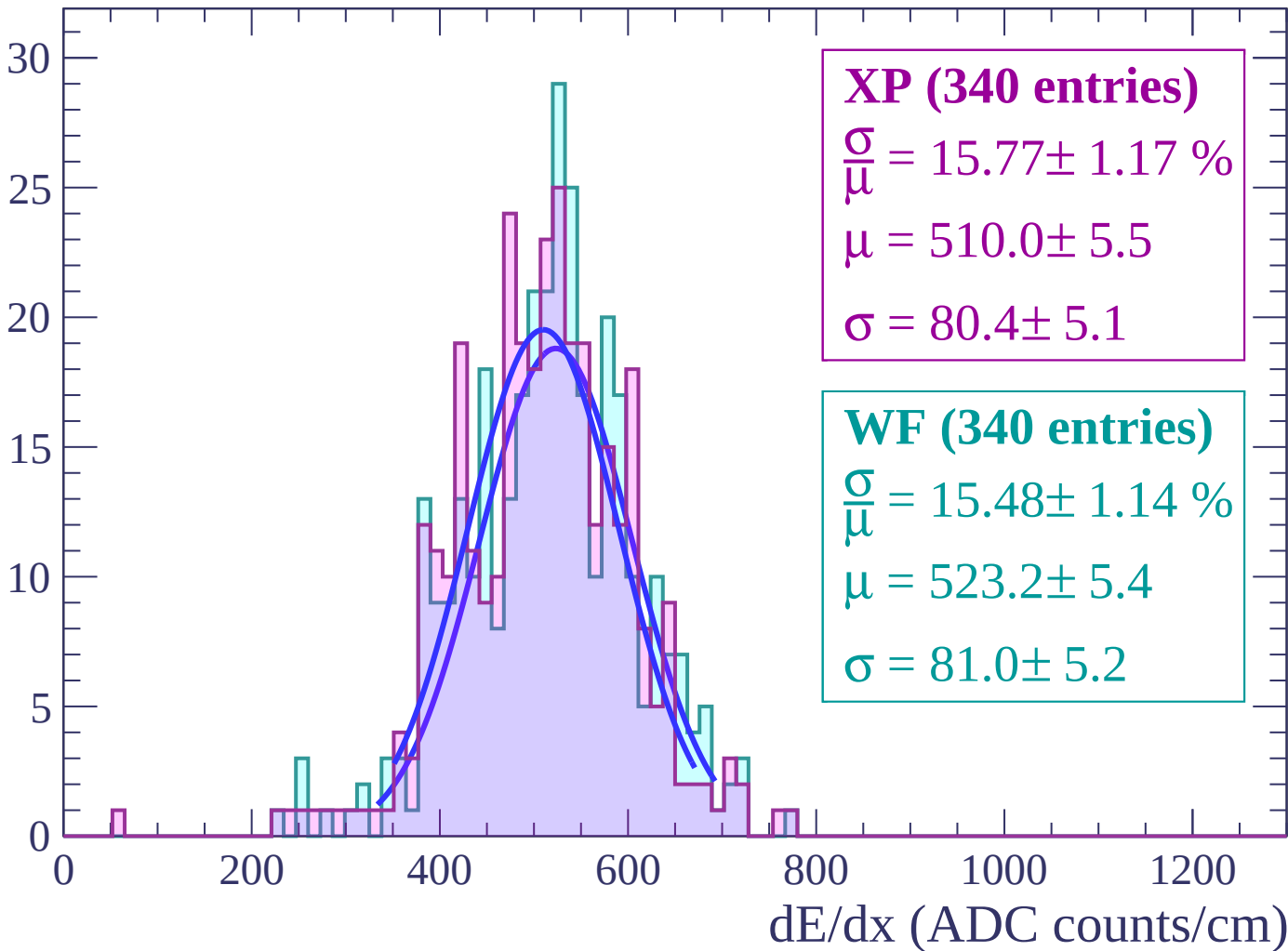
Energy loss | $56 < \theta < 58$

Count



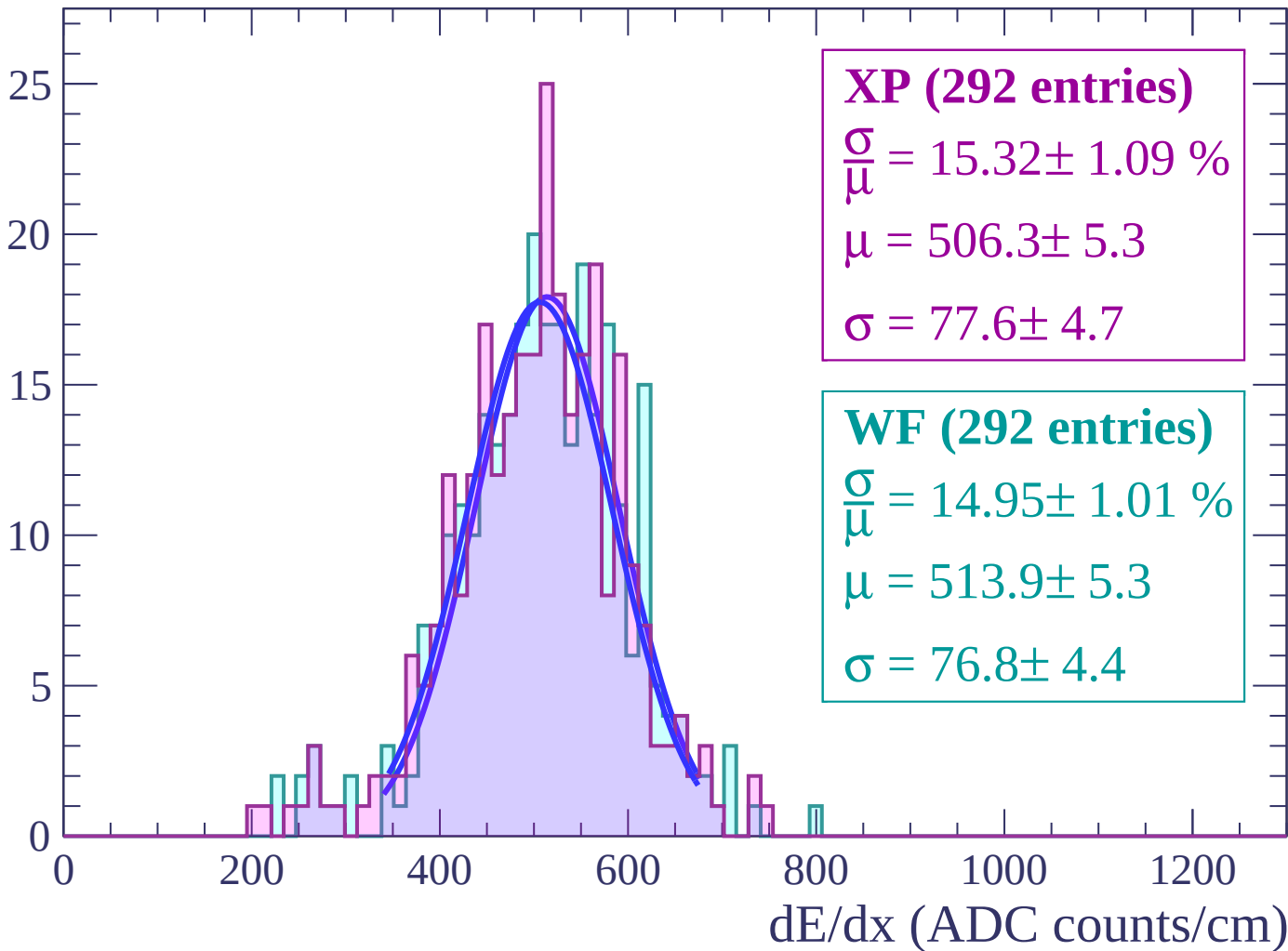
Energy loss | $58 < \theta < 60$

Count



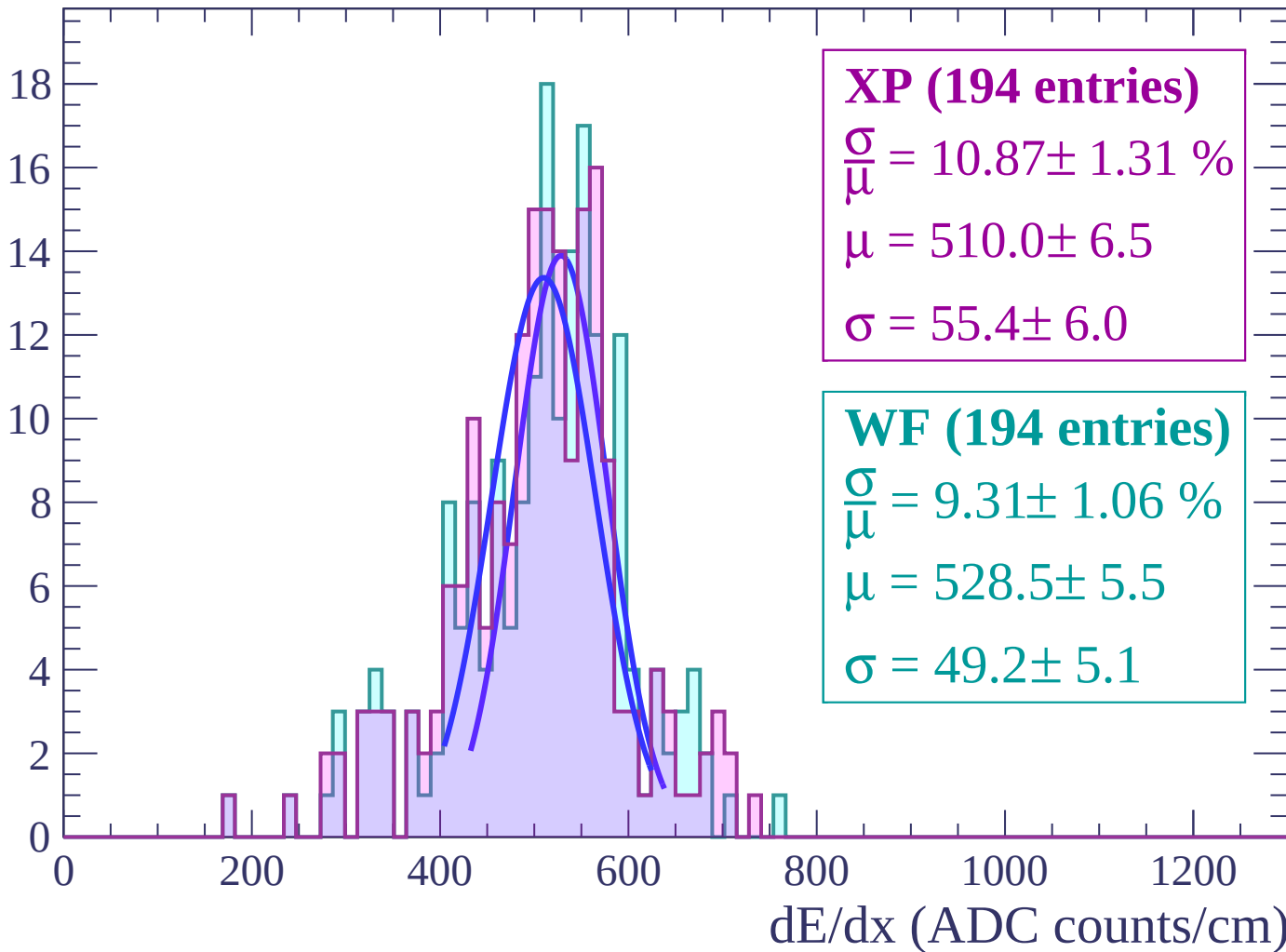
Energy loss | $60 < \theta < 62$

Count



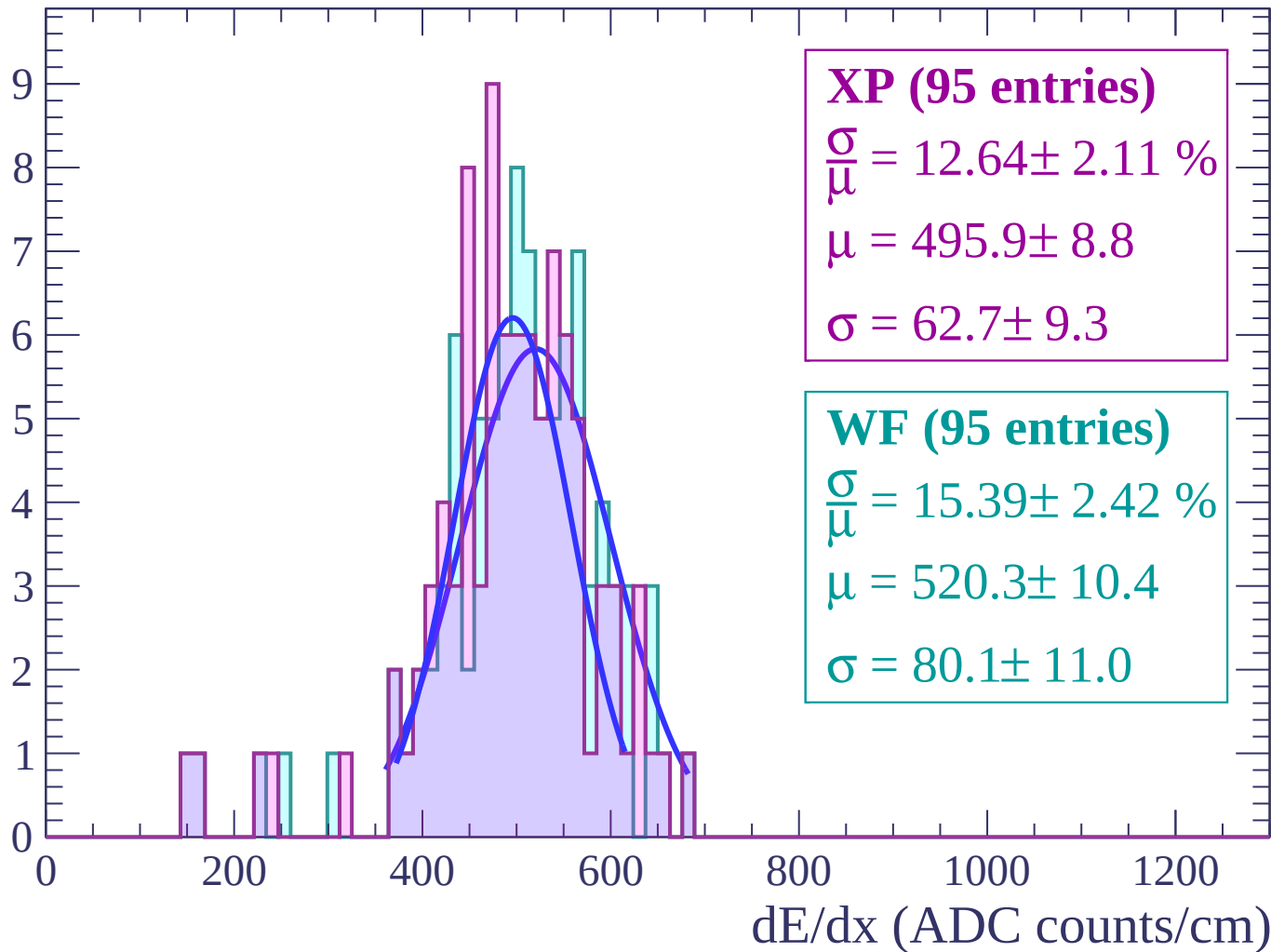
Energy loss | $62 < \theta < 64$

Count



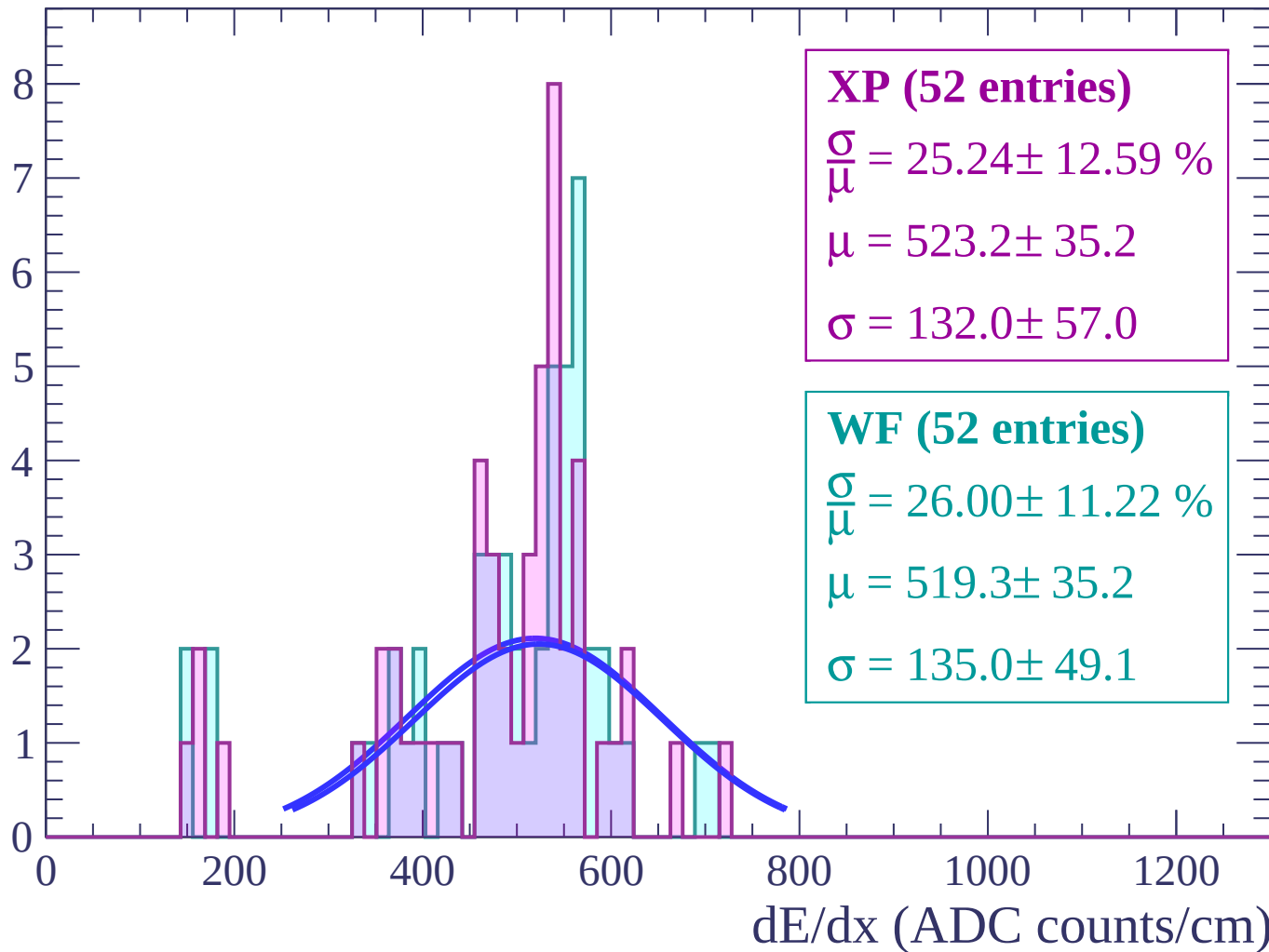
Energy loss | $64 < \theta < 66$

Count



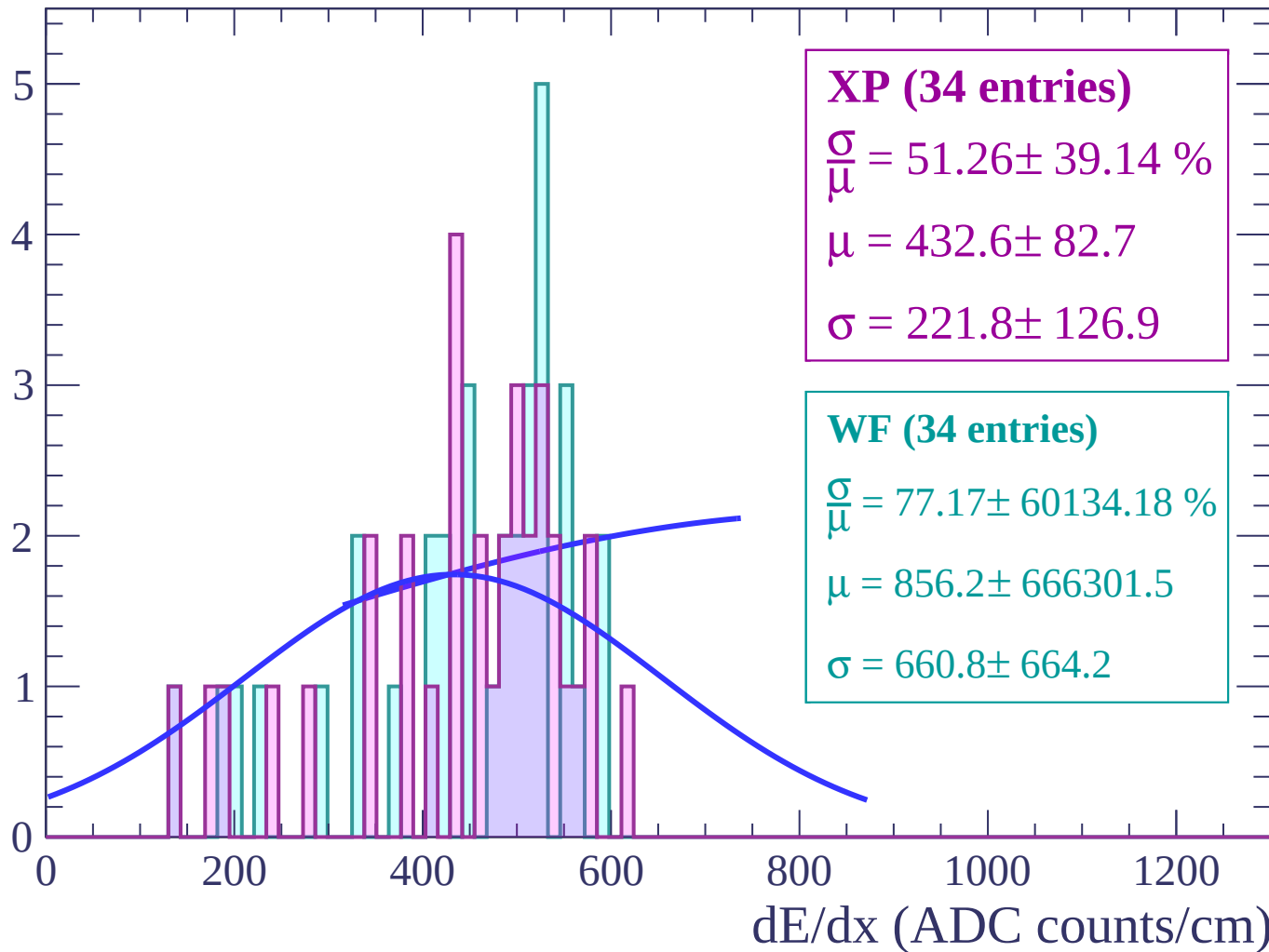
Energy loss | $66 < \theta < 68$

Count



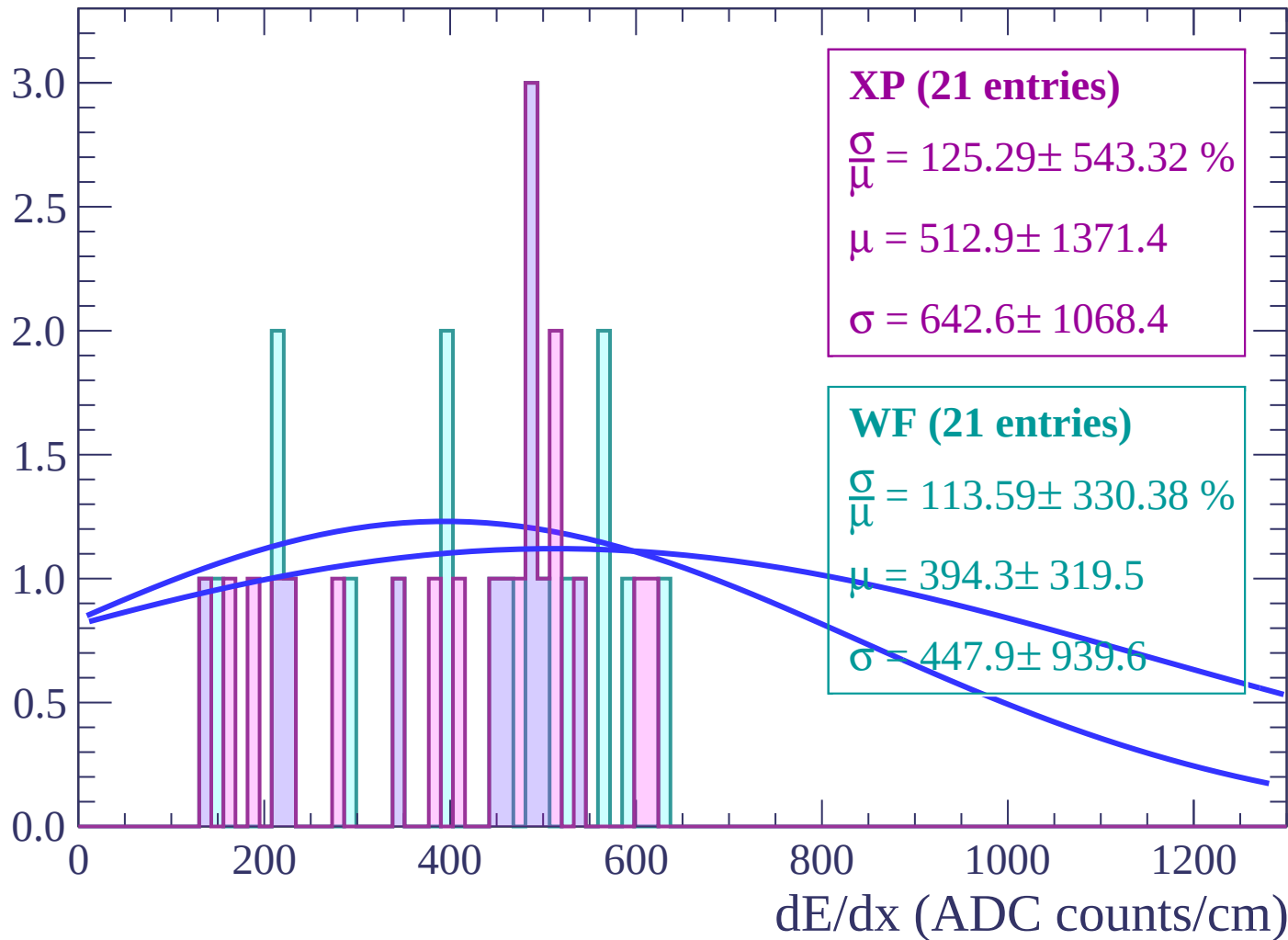
Energy loss | $68 < \theta < 70$

Count



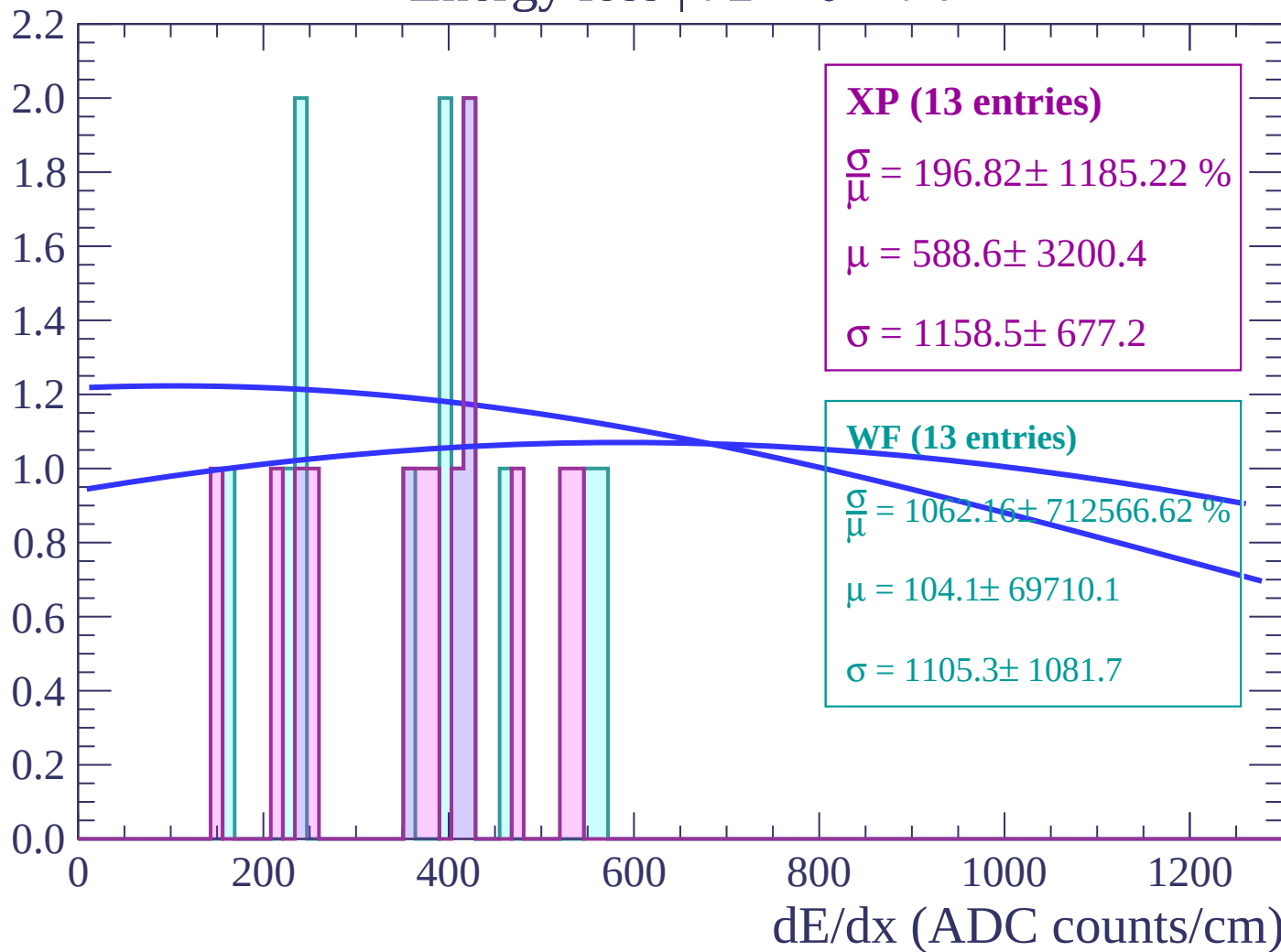
Energy loss | $70 < \theta < 72$

Count



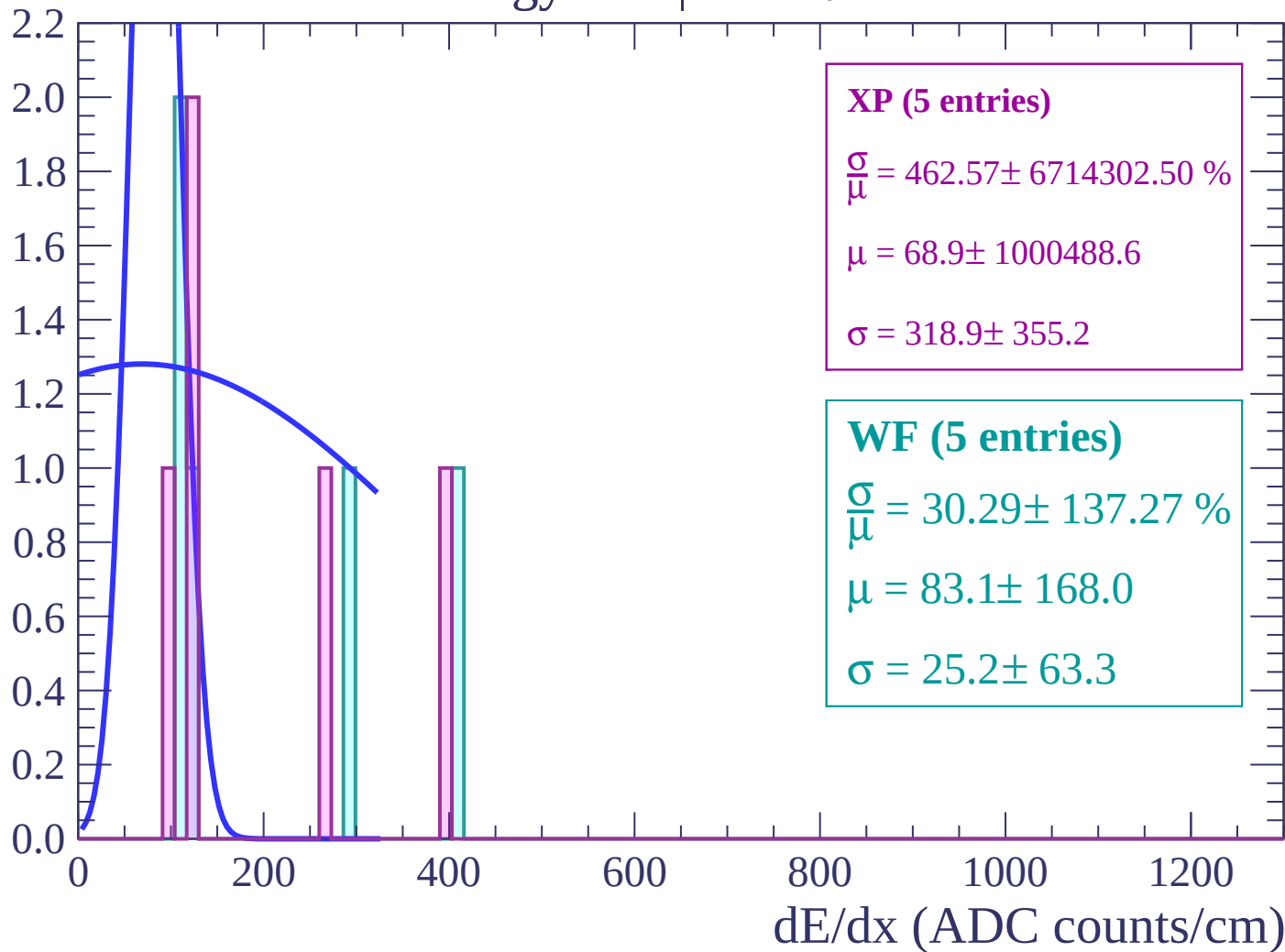
Energy loss | $72 < \theta < 74$

Count



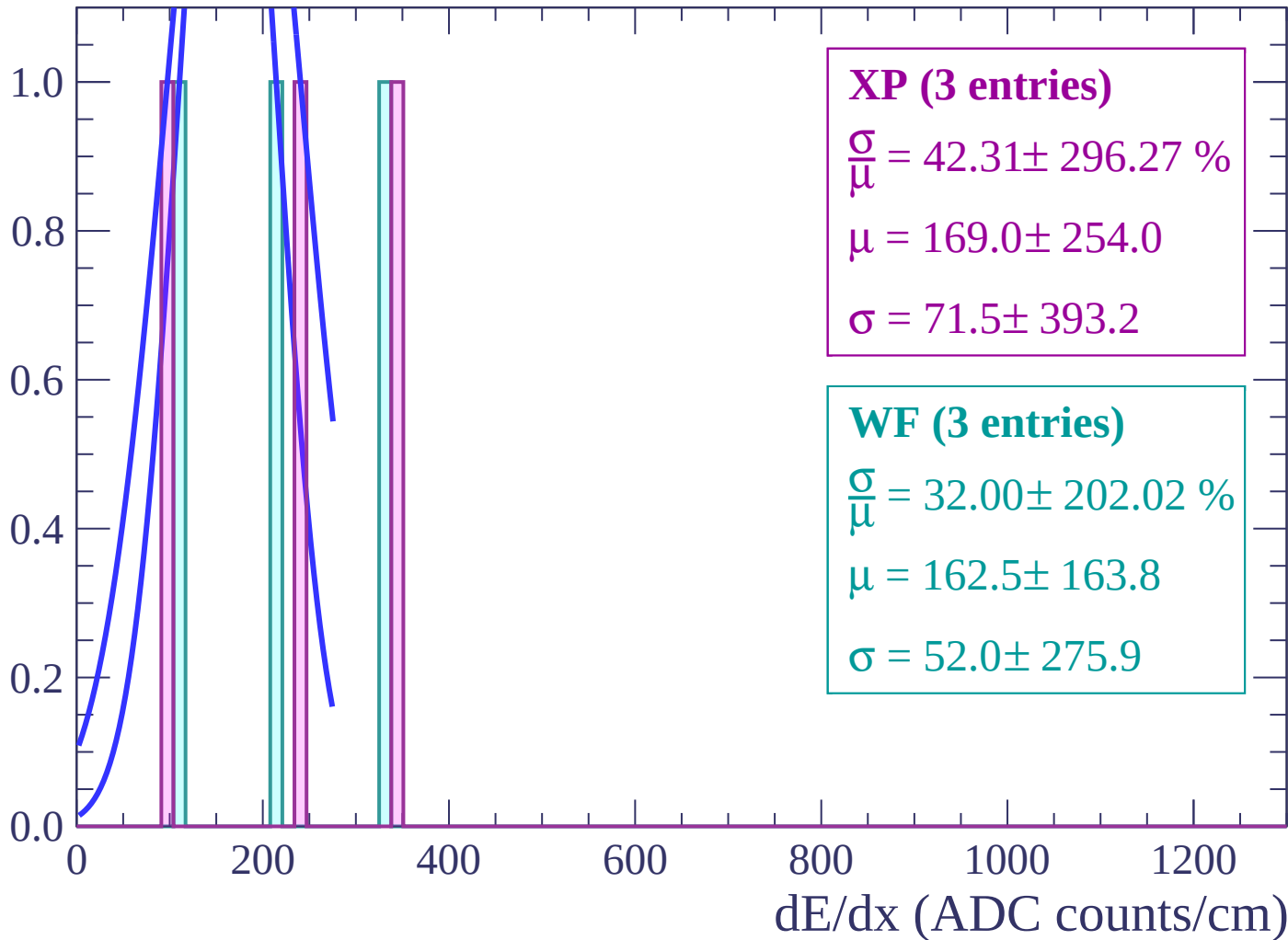
Energy loss | $74 < \theta < 76$

Count



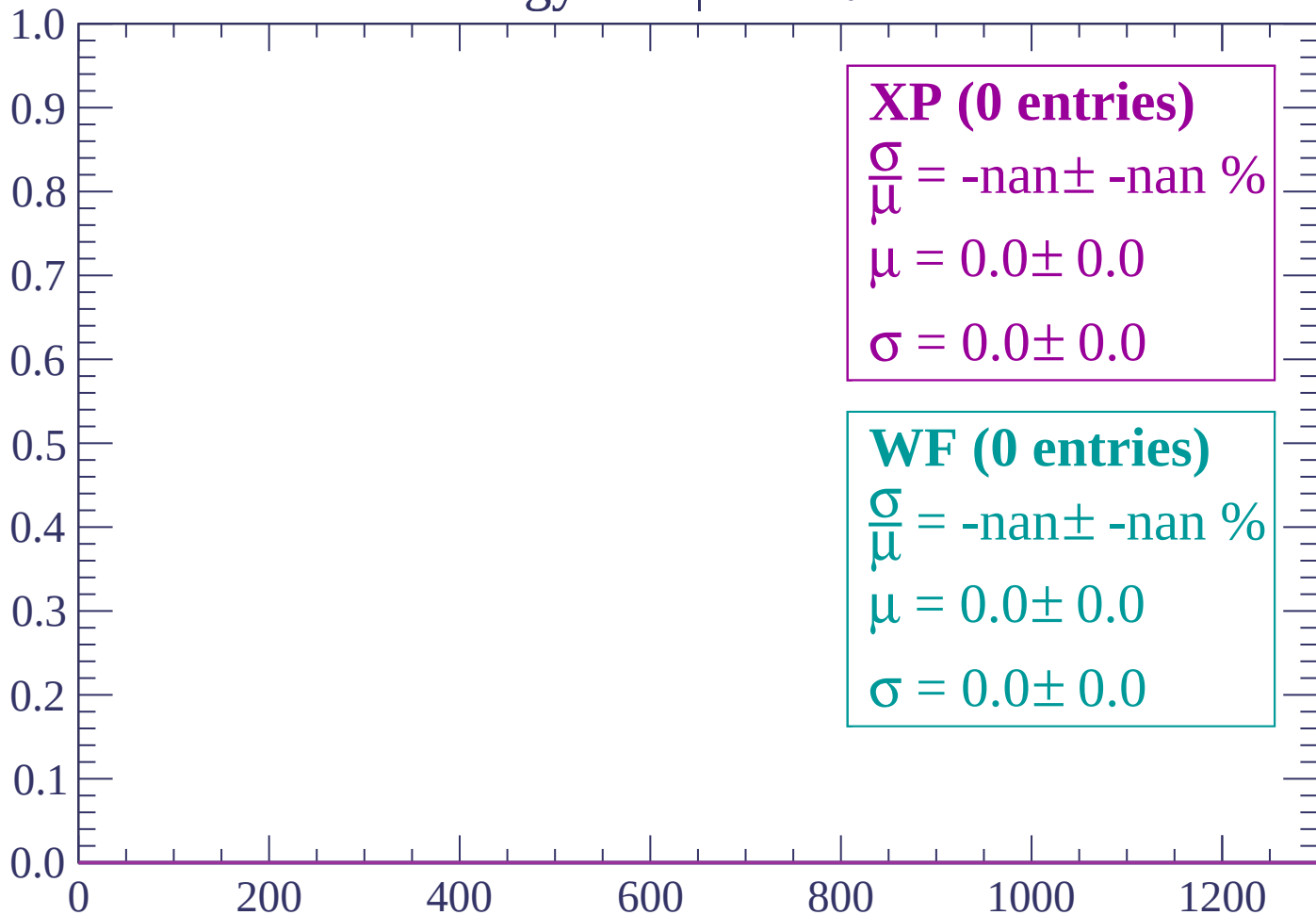
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

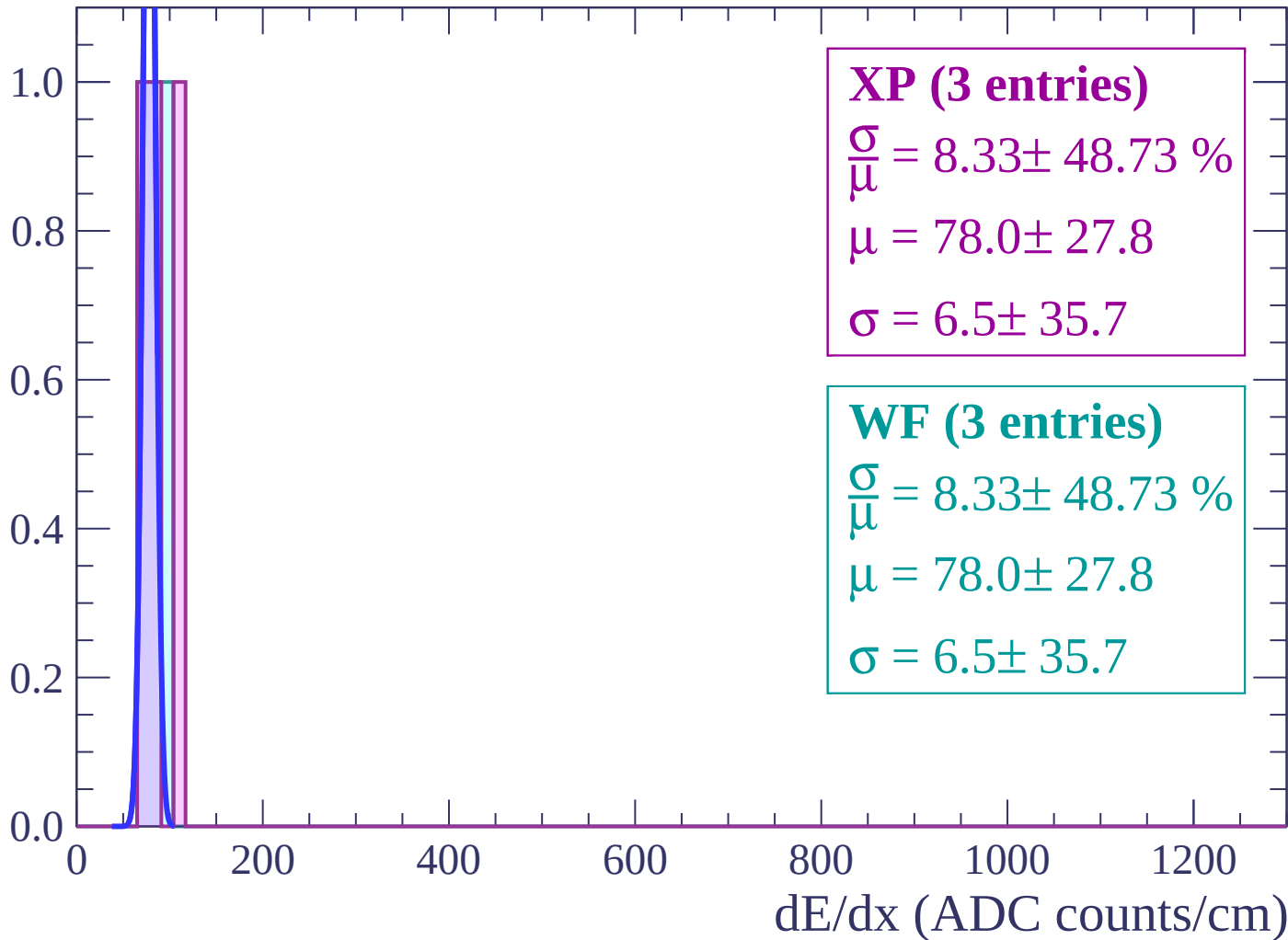
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

Count



Energy loss | $82 < \theta < 84$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $84 < \theta < 86$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $86 < \theta < 88$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $88 < \theta < 90$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF (0 entries)

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $90 < \theta < 92$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP (1 entries)
 $\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$
 $\mu = 0.0 \pm 0.0$
 $\sigma = 0.0 \pm 0.0$

WF (1 entries)
 $\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$
 $\mu = 0.0 \pm 0.0$
 $\sigma = 0.0 \pm 0.0$

